SUSTAINABLE DESIGN OF STUDENTS' ACCOMMODATION BLOCK

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Sustainable Environmental Design

MSc. Climate Resilience and Environmental Sustainability in Architecture

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Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design
Afua Ampomah-Amoako

1.0 PROJECT LOCATION : SEATTLE, WASHINGTON STATE, U.S.A.

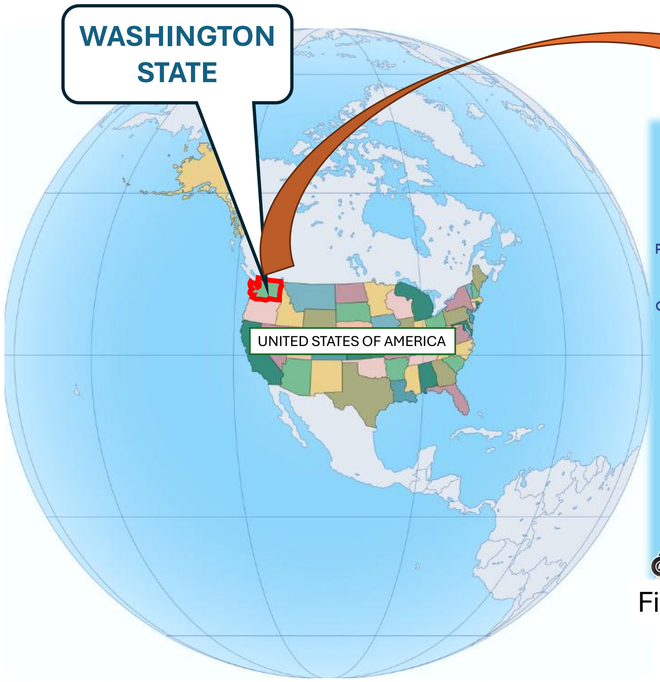


Figure 1:Location of United States of America (VectorStock, 2025)

Figure 2 : Map of Washington State (Tripsmap, 2018)

1.1 KOPPEN CLIMATE CLASSIFICATION

Six different climate types and Seattle falls within the Cold-summer mediterranean.

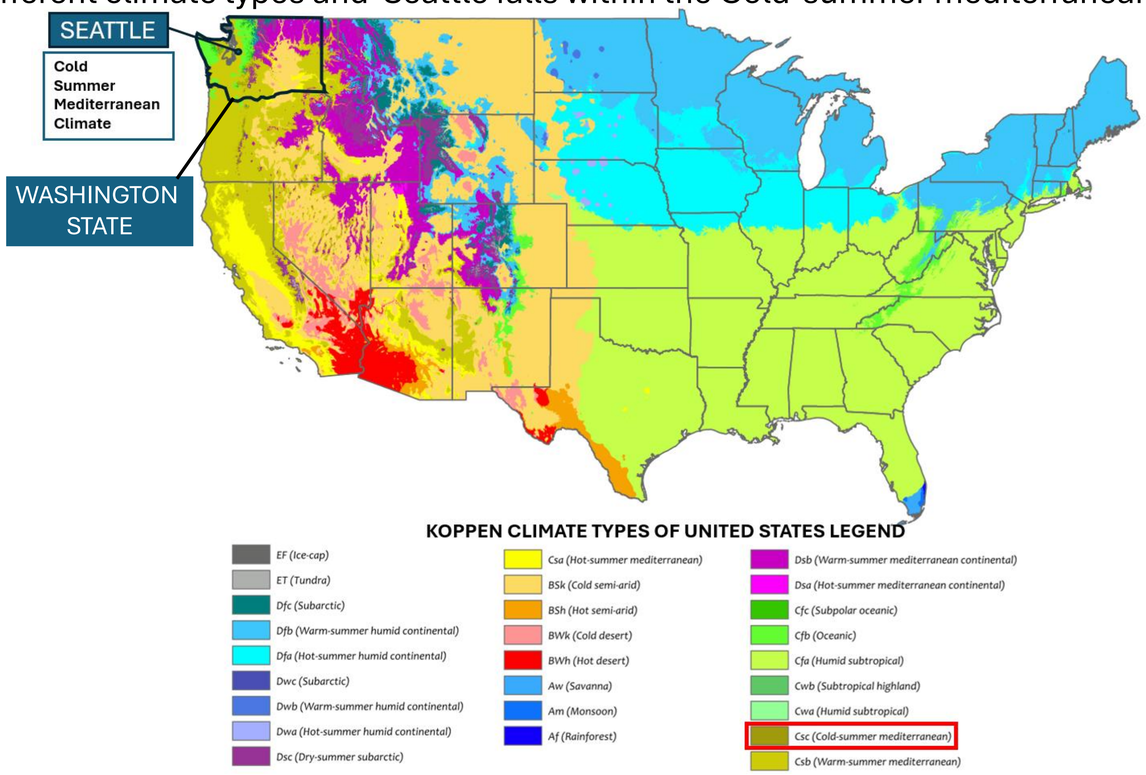


Figure 3 : Koppen Climate Classification of United States of America (Beck et al.,2023 p.3)

1.2 LOCAL GEOGRAPHIC CHARACTERISTICS OF SEATTLE

Seattle is an urban coastal city surrounded by **water, mountains, and parks**.

Figure 4 : Aerial view of Seattle City (Google Earth, 2025 with modifications by author)

1.3 TEMPERATURE RANGE OF SEATTLE

Temperature at Seattle range between -4 degrees Celsius and 31 degrees Celsius as seen in Figure 3. This indicates that at most times of the year (83.5%) Temperatures fall below the comfort zone requiring heating.

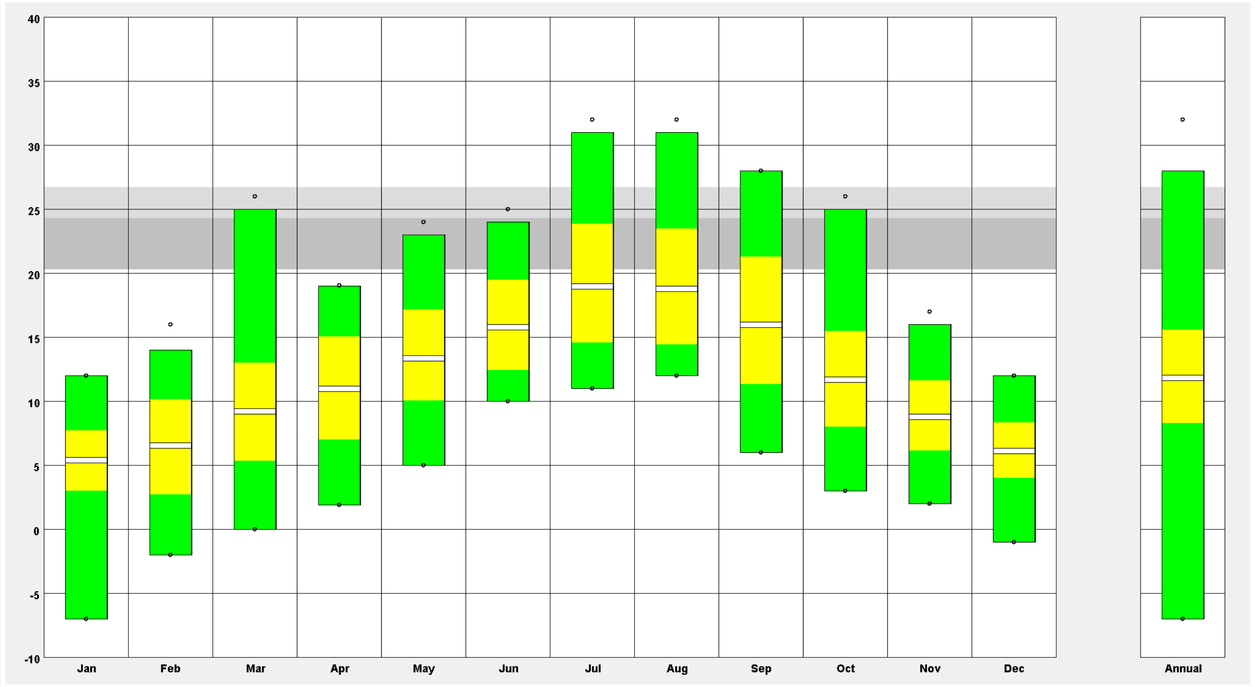


Figure 5 : Temperature range chart of climate (UCLA Energy Design Group, 2021)

1.3.1 RELATIVE HUMIDITY

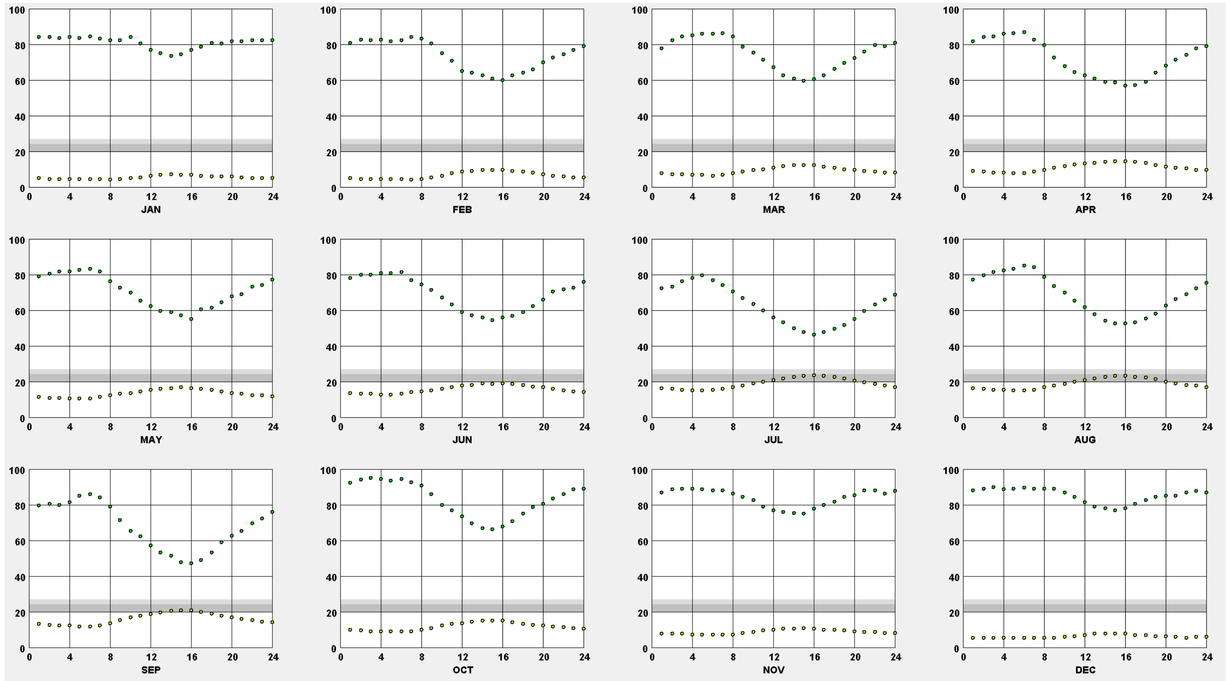
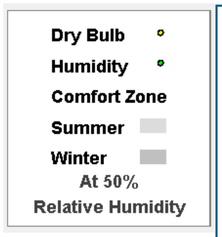


Figure 6: Dry Bulb and Relative Humidity (UCLA Energy Design Group, 2021)



Dry bulb temperatures below the comfort zone except during summer months which falls within the comfort range.
Average relative humidity of 70% but rises during the winter months above 80%.
Seattle had an average precipitation of 32.7 inches in 2024 which was below the yearly average of 39.34 inches indicating a drop in rainfall.

1.3.2 PRECIPITATION PATTERN OF SEATTLE

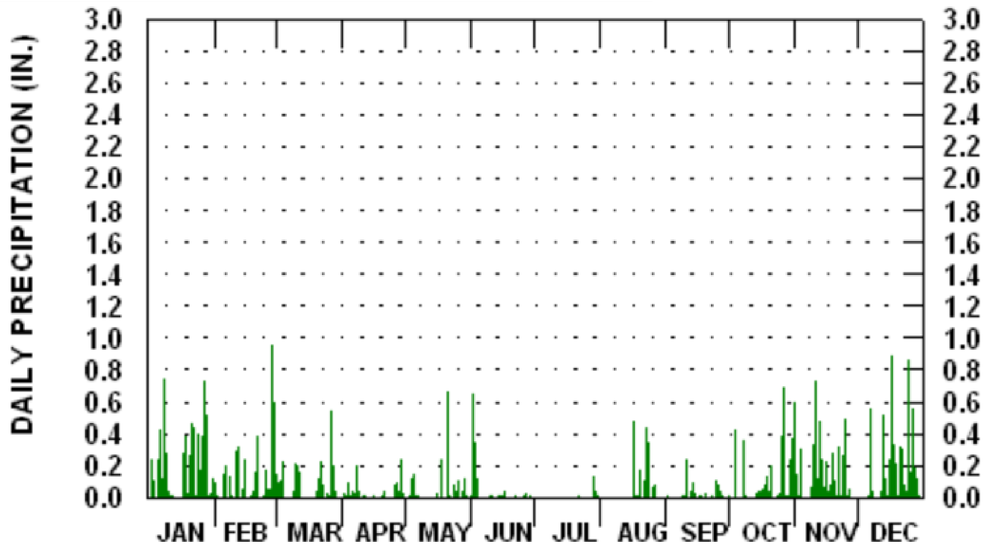


Figure 7: Monthly precipitation for 2024 year (OceanWP, 2025)

1.3.3 SUN SHADING CHART

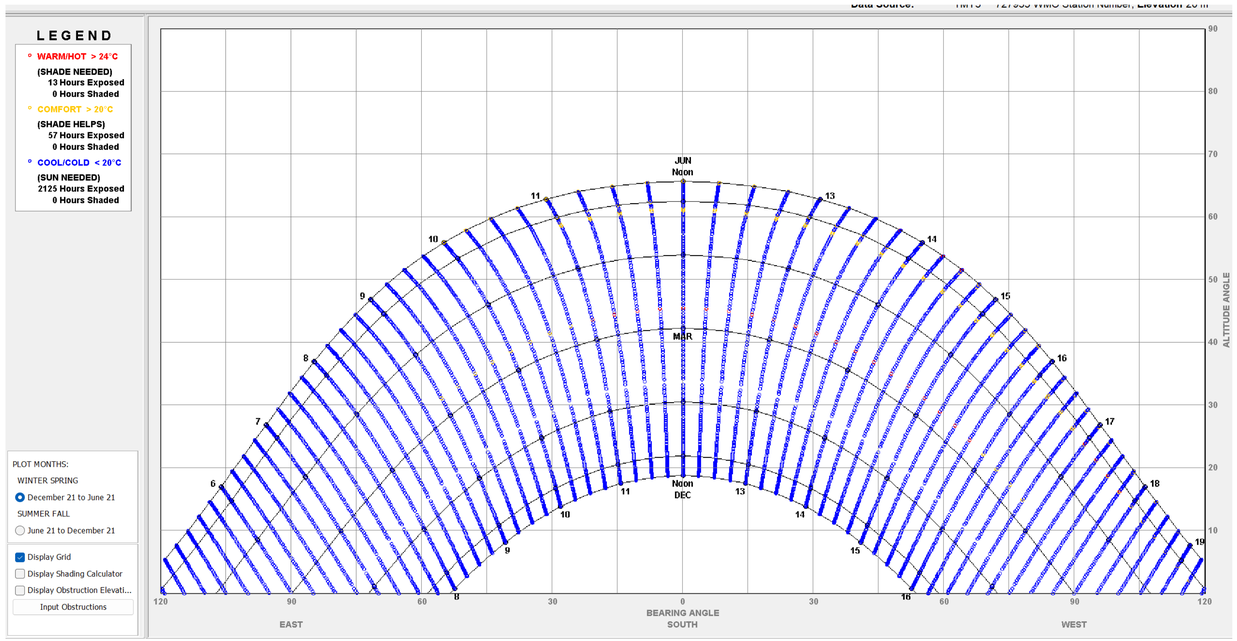


Figure 8: Sun shading chart from December to June
(UCLA Energy Design Group, 2021)

Shading is required from **June to September** because of the high exposure to sun specifically on the Southeast, South, Southwest and West facades. This will require both horizontal and vertical sun shading devices along the named facades.

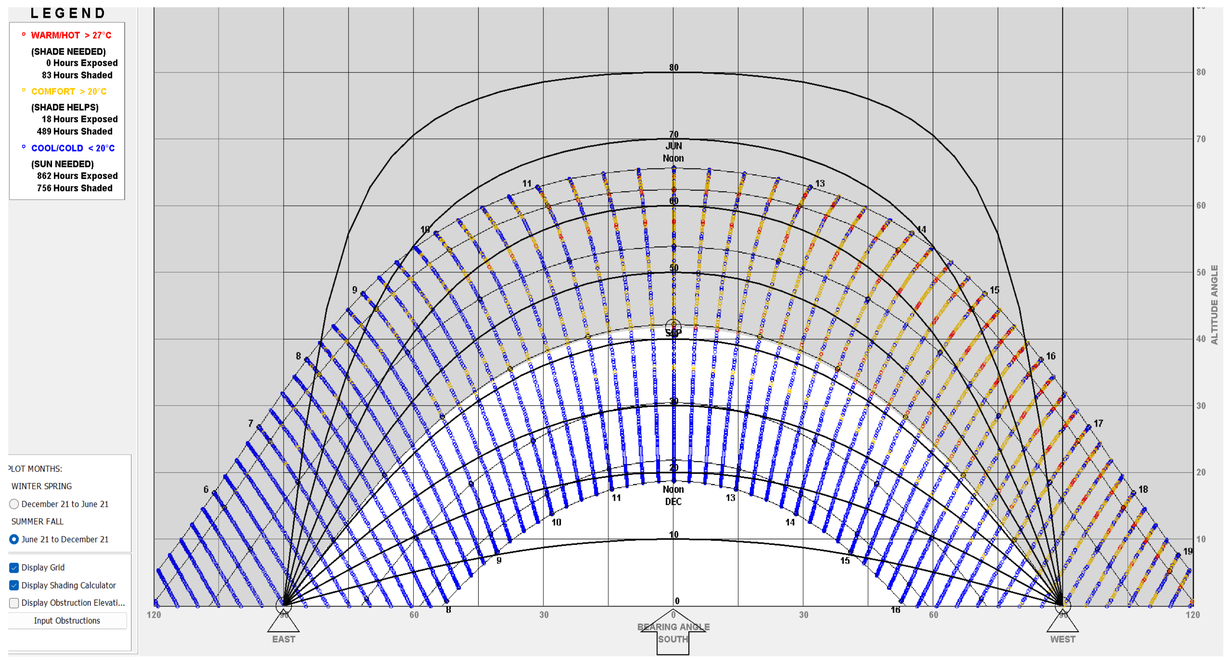


Figure 9: Sun shading chart from June to December
(UCLA Energy Design Group, 2021)

1.3.4 PSYCHROMETRIC CHART

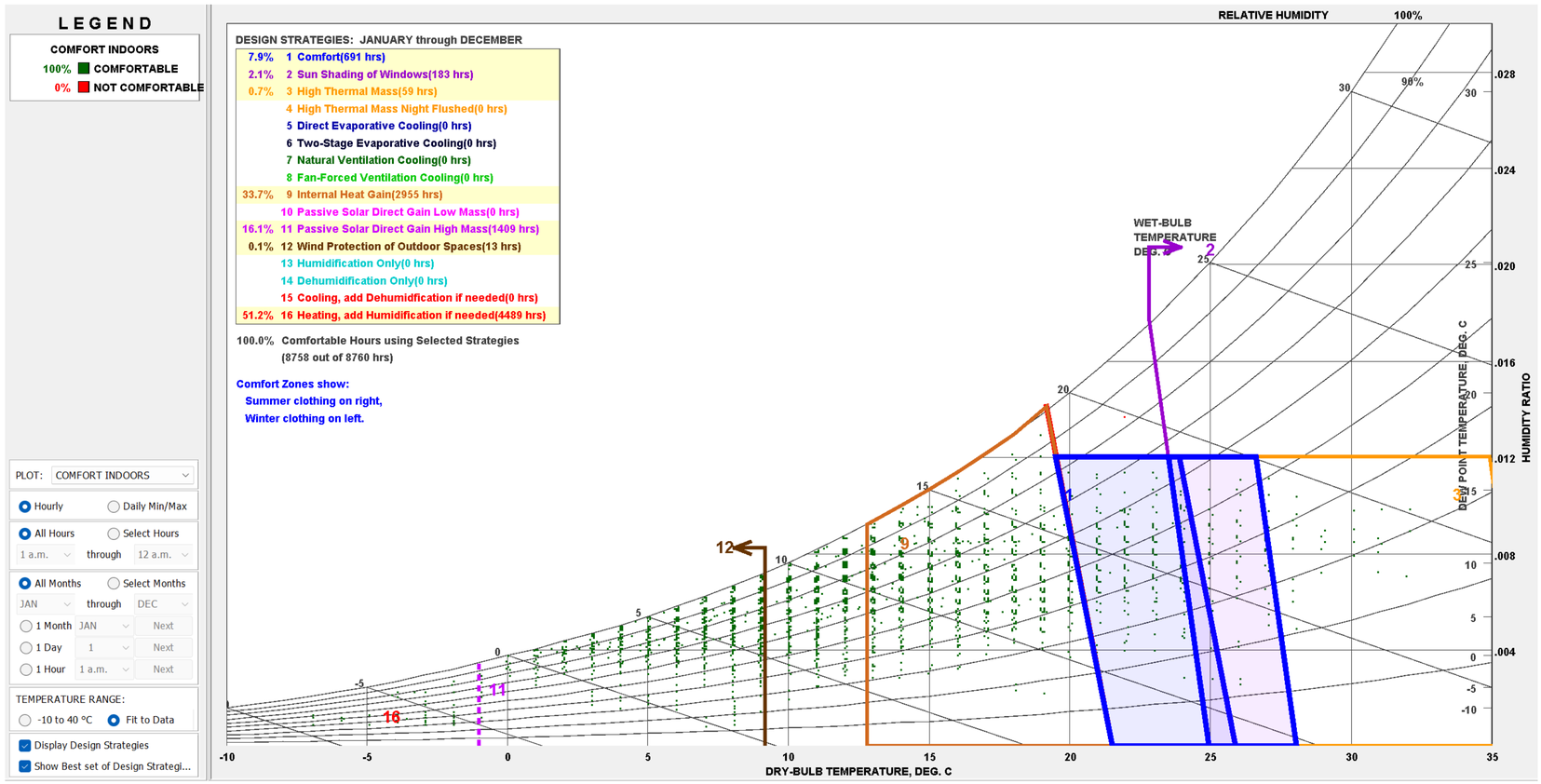


Figure 10: Psychrometric chart (UCLA Energy Design Group, 2021)

The psychrometric chart informs design strategies to bring optimum comfort as listed below:

- **Heating of spaces and internal heat gain as most months are cold below the comfort zone.**
- **Passive solar direct gain and incorporating high thermal mass.**
- **Sun shading of windows on South, Southeast and Southwest facades**
- **Wind protection of outdoor spaces**

The ground temperature chart (Figure 11) indicates a minimum temperature of 9 degrees and 15 degrees maximum at 4m depth.

The prevailing wind directions (Figure 12) are mainly from South, Southwest (10 m/s) and Northwest (8m/s).

1.3.5 GROUND TEMPERATURE (MONTHLY AVERAGE)

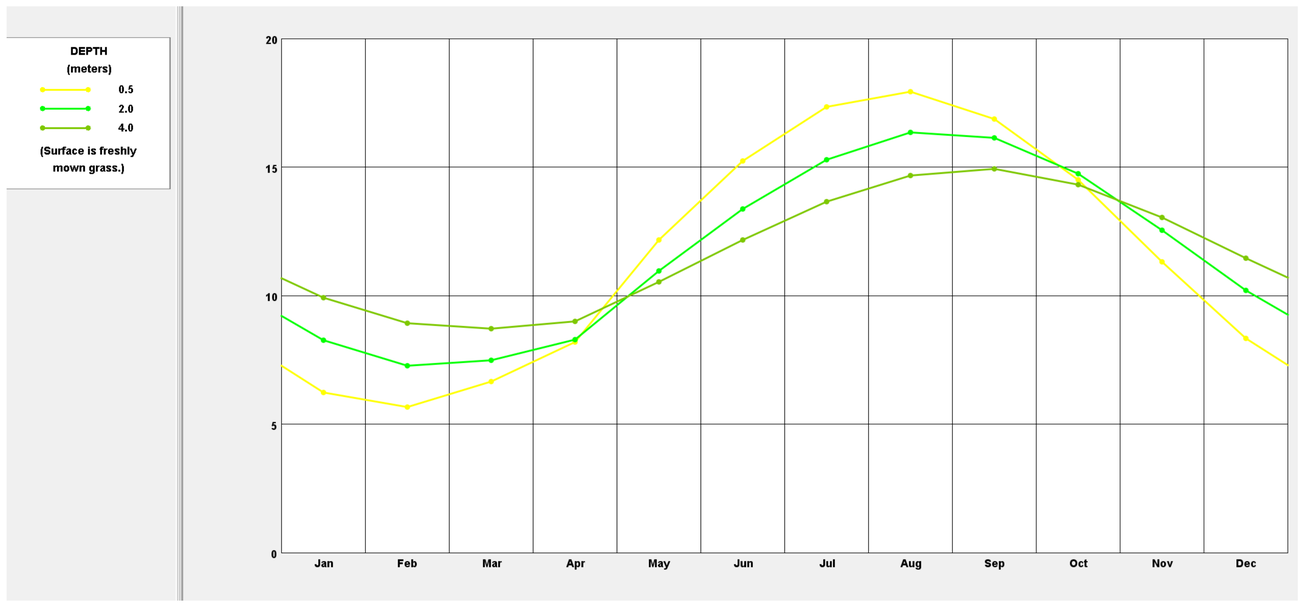


Figure 11: Ground Temperature (UCLA Energy Design Group, 2021)

1.3.6 WIND WHEEL

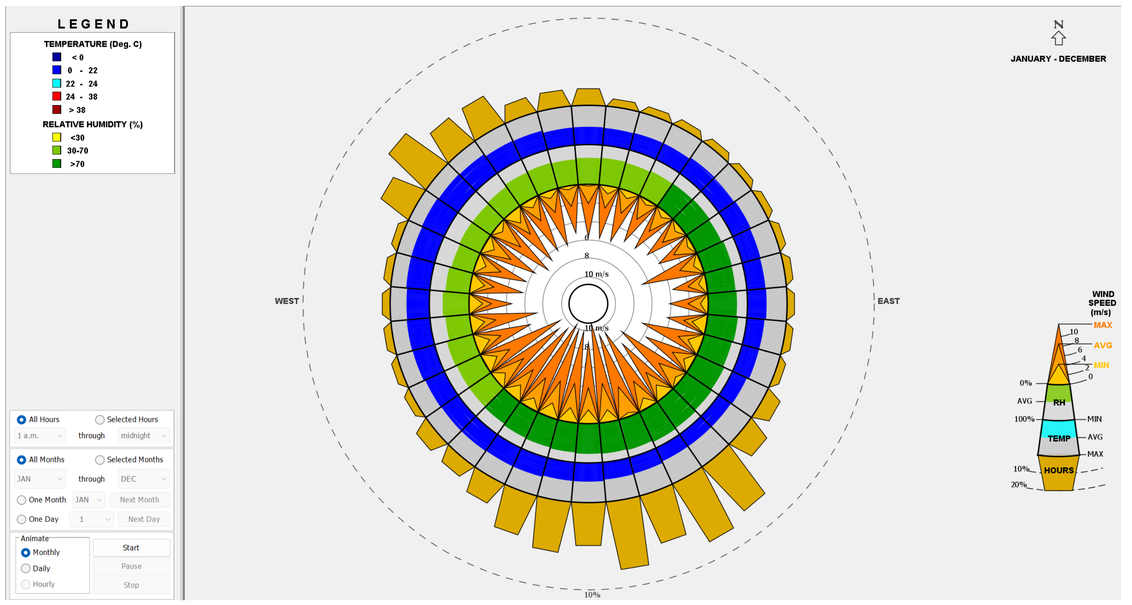


Figure 12: Wind Wheel (UCLA Energy Design Group, 2021)

1.4.0 CLIMATE CHANGE IMPACT – DROUGHT RISK

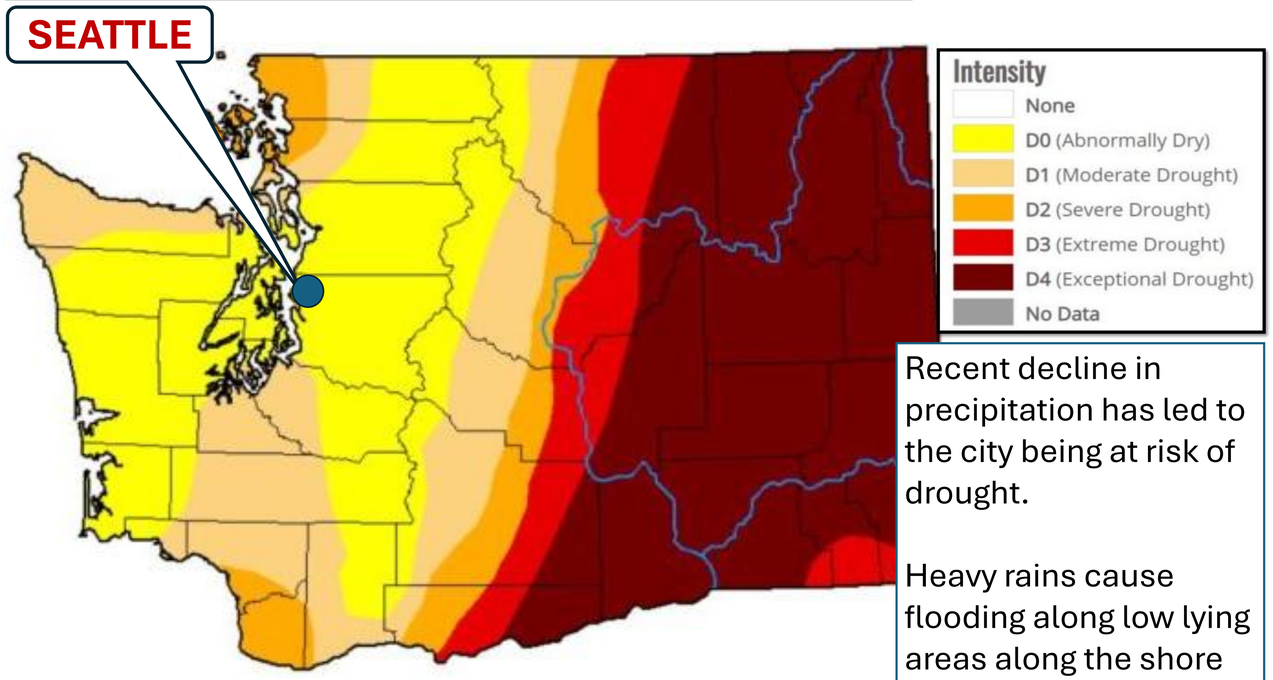


Figure 13: Drought risk
(Washington Water Trust, 2023)

Recent decline in precipitation has led to the city being at risk of drought.

Heavy rains cause flooding along low lying areas along the shore of River Duwamish.

1.4.1 FLOOD RISK

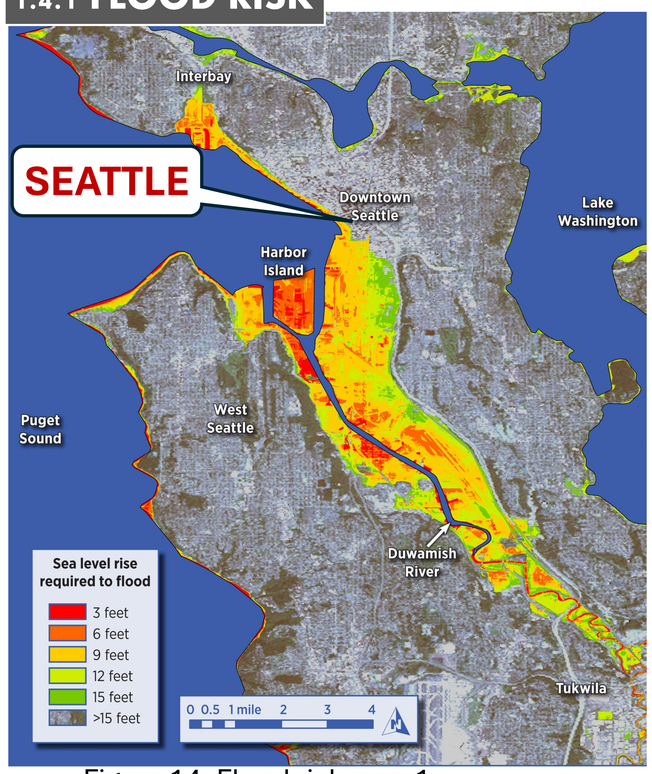
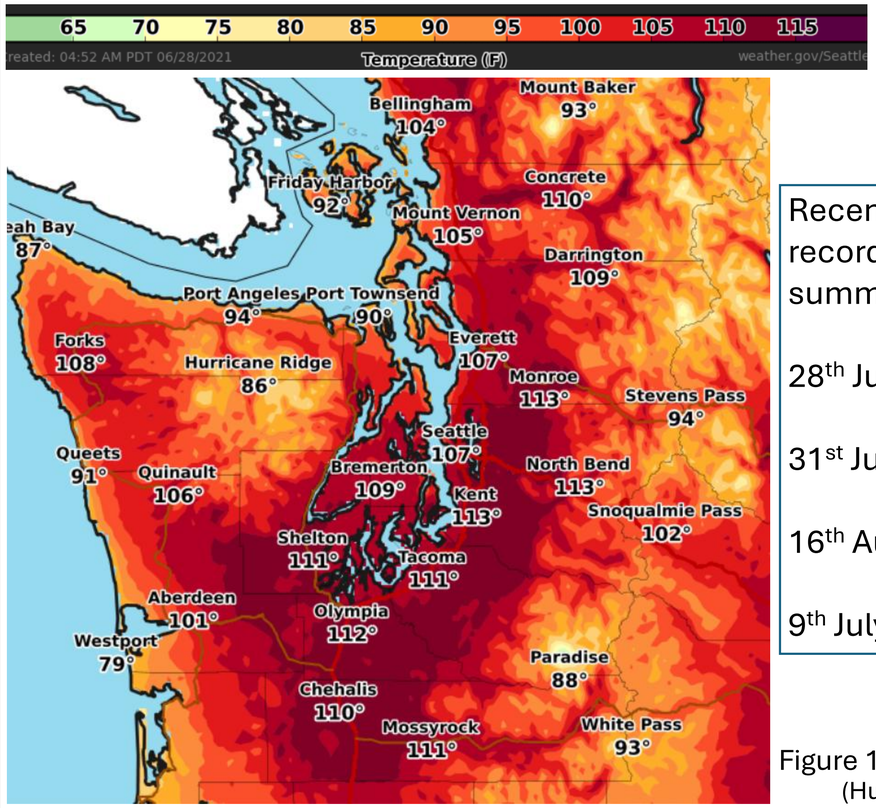


Figure 14: Flood risk map 1
(Mahr, 2019)

Figure 15: Flood risk map 2
(Schultz, 2021)

1.4.2 EXTREME HEAT



Recent temperature recording of Seattle in summer

28th June 2021 - 42 deg. C

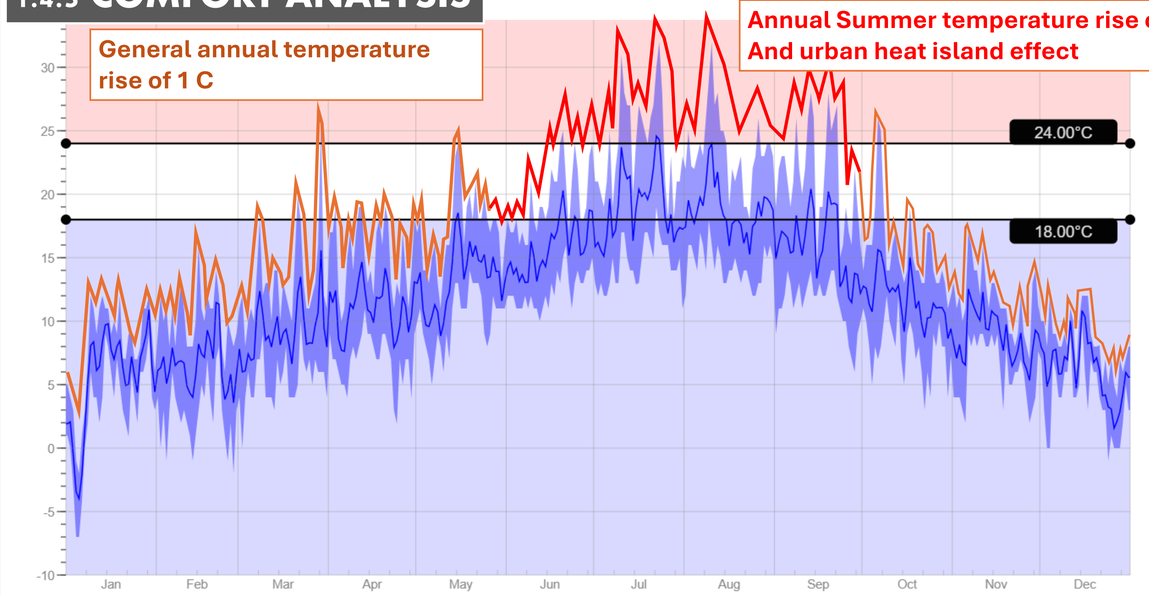
31st July 2022 – 35 deg. C

16th August 2023 – 35 deg. C

9th July 2024 – 36 deg. C

Figure 16: Flood risk map 2
(Hunziker, 2022)

1.4.3 COMFORT ANALYSIS



BELOW COMFORT			WITHIN COMFORT		ABOVE COMFORT		
Hrs	Deg.Hrs	Percent	Hrs	Percent	Hrs	Deg.Hrs	Percent
7312	58187.2	83.5%	1276	14.6%	172	495.2	2.0%

Figure 17 : Comfort Analysis and temperature rise (Marsh, 2025 with modifications by author)

1.5.0 SITE : SEATTLE UNIVERSITY CAMPUS

The proposed project will be situated at the Central campus of Seattle University close to the school of business.

The proposed project will supplement the limited number of students residence at the University.

P Car Park

Bus stop

Campus boundary

Central campus boundary

Existing Students' accommodation on central campus

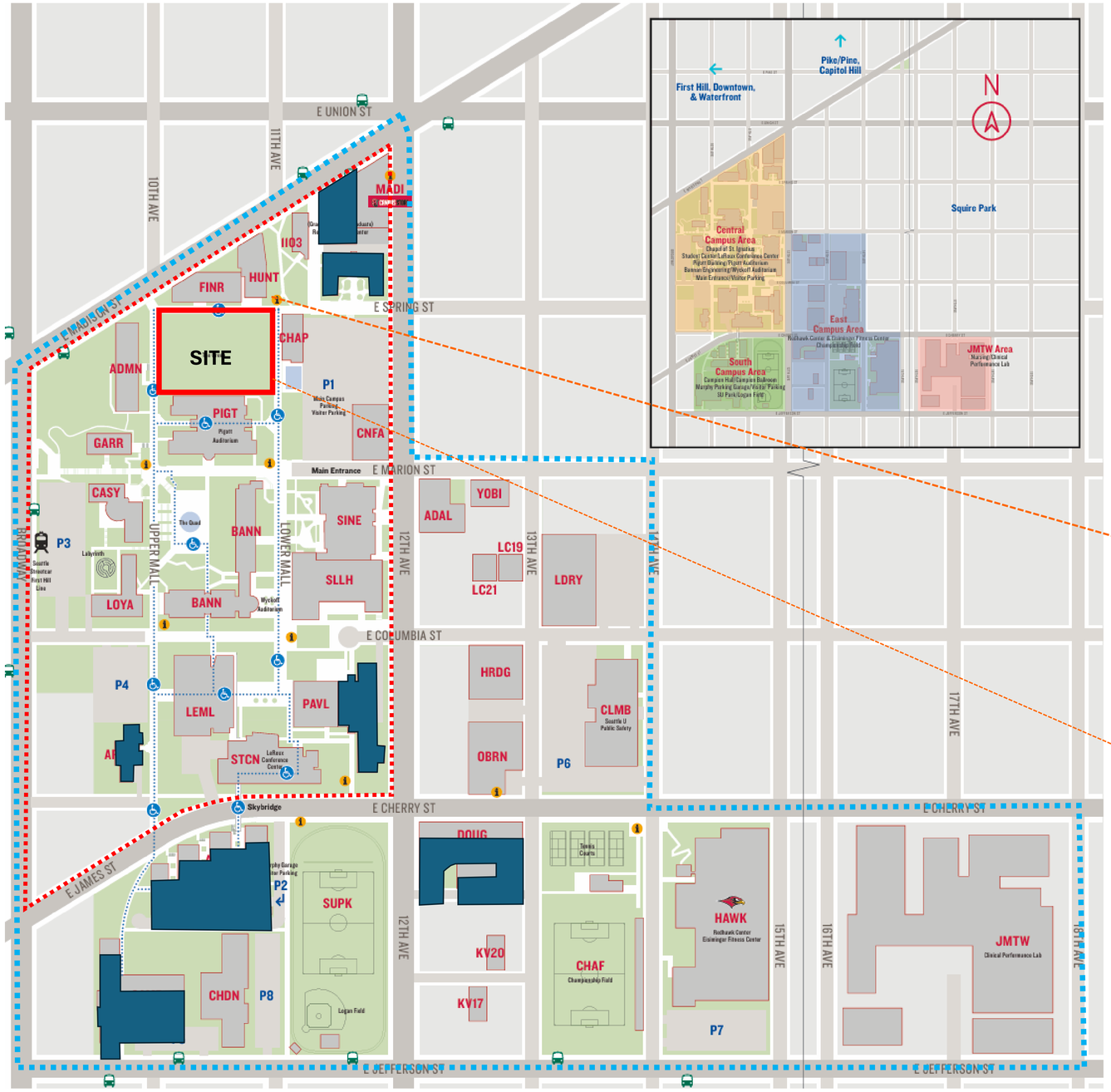


Figure 18 : Site Plan with different elevations (Seattle University, 2025)

1.5.1 STUDENT ACCOMMODATION CHALLENGE

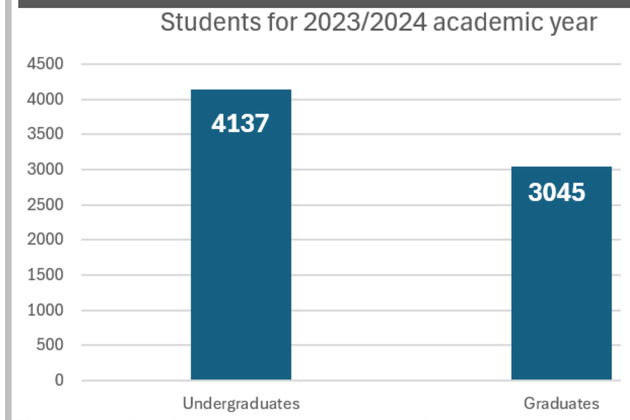


Figure 19 : Undergraduate versus graduate students' numbers

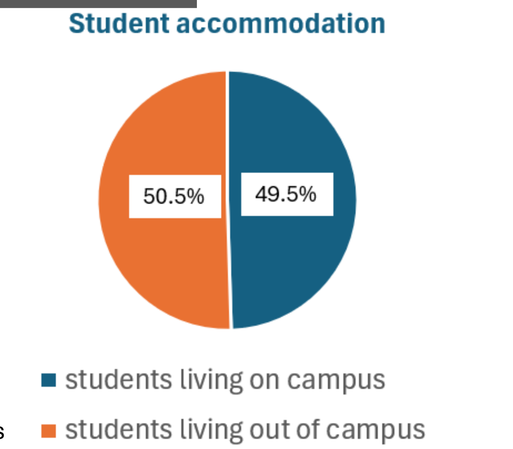


Figure 20 : residential versus non-residential Campus accommodation (Seattle University, 2025)

- Rising rent prices in Seattle
- Student population growth
- Limited On-Campus Housing Availability

Housing disparity: Graduate students and upperclassmen often face more difficulty in securing on-campus housing, as priority is typically given to first-year students

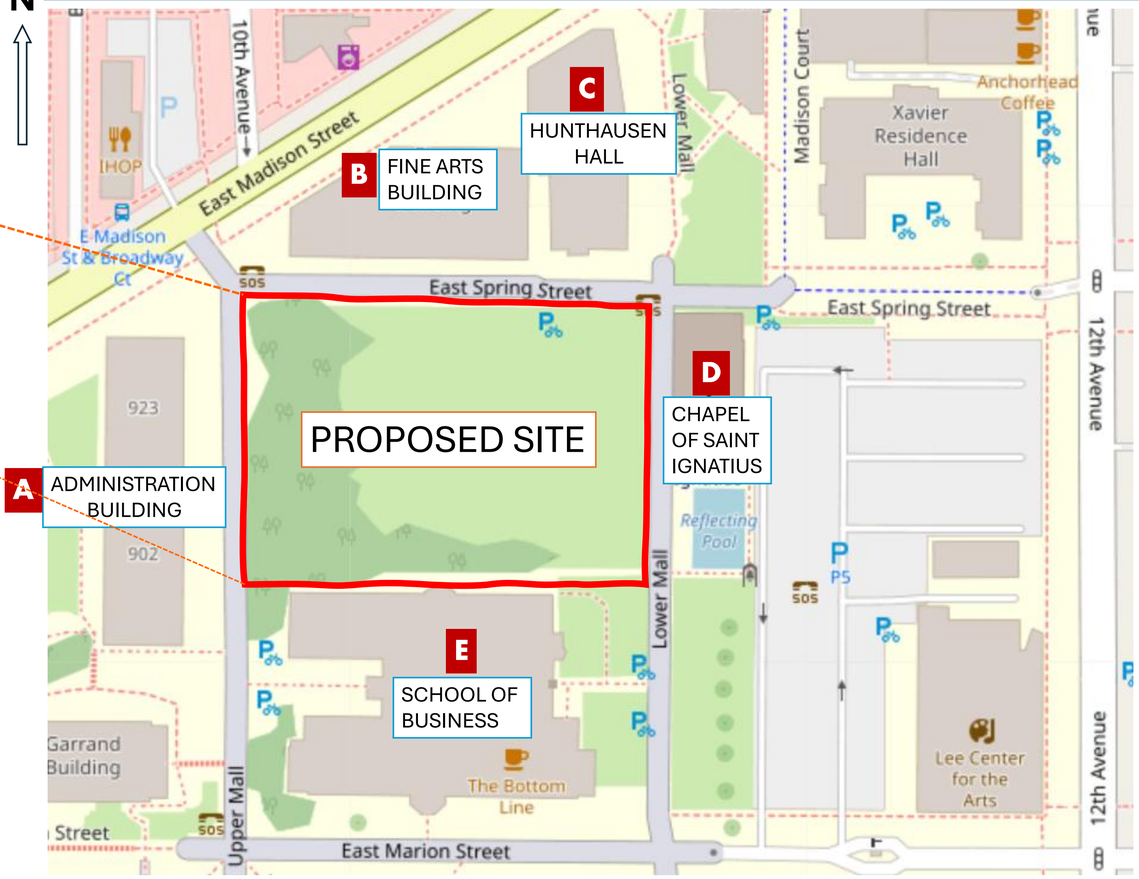


Figure 21 : Site Plan of proposed site (Google, 2025)

1.5.2 SITE TOPOGRAPHY AND VEGETATION

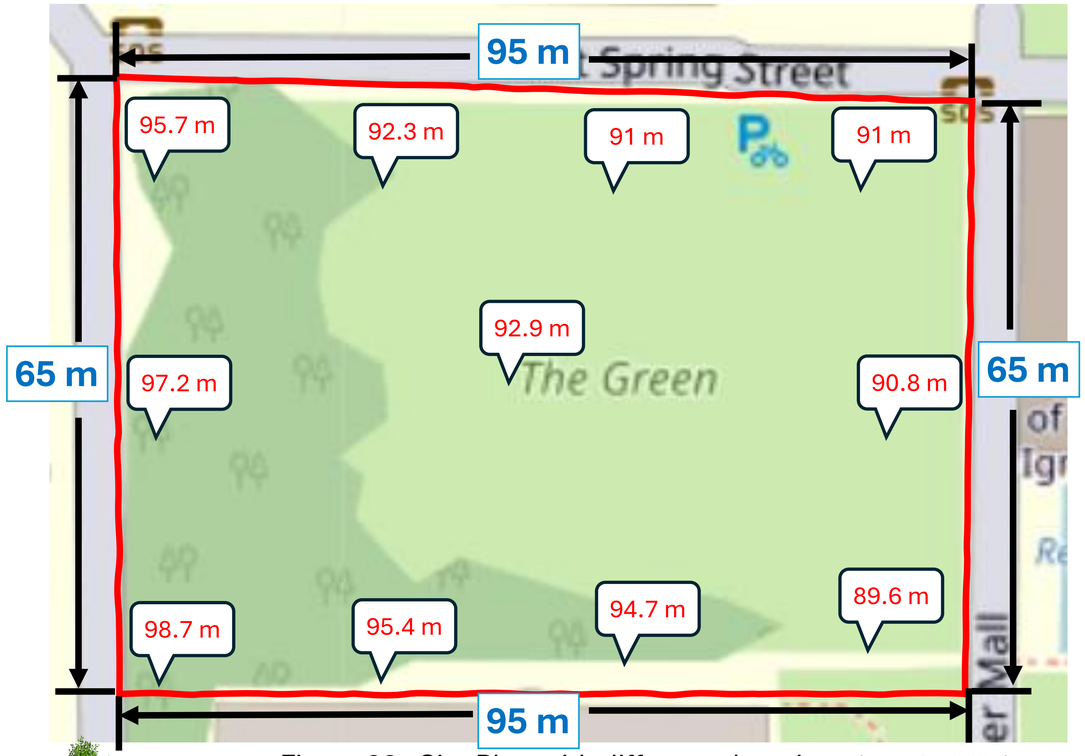


Figure 22 : Site Plan with different elevations (Google, 2025)

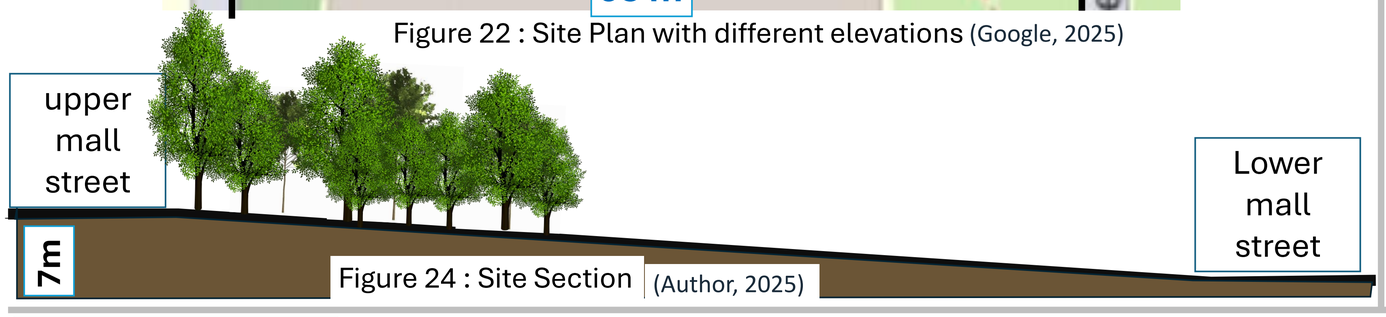


Figure 24 : Site Section (Author, 2025)

1.5.3 SITE ARIEL VIEW

Figure 23: Northeast view of proposed site (Google, 2025)

EVERGREEN TREES ON SITE

- ❑ Douglas fir
- ❑ Western Red Cedar

DECIDUOUS TREES ON SITE

- ❑ Red maple
- ❑ Bigleaf maple

1.5.4 SITE IMAGES

Figure 25 : Grass cover on site (Google, 2025)

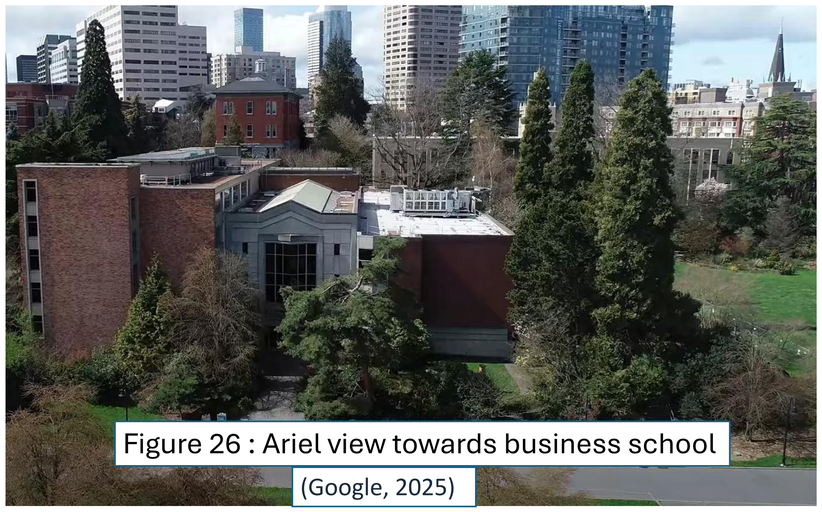


Figure 26 : Ariel view towards business school (Google, 2025)

Figure 27 : View from site towards chapel (Google, 2025)

1.5.5 SITE ANALYSIS : SUN PATH AND SHADOW STUDIES

SUN PATH – 21st JUNE : 9AM

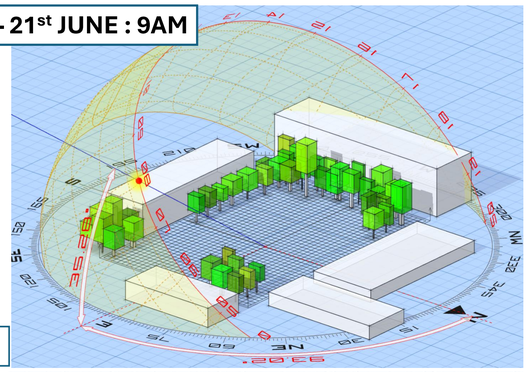


Figure 28 :

4PM

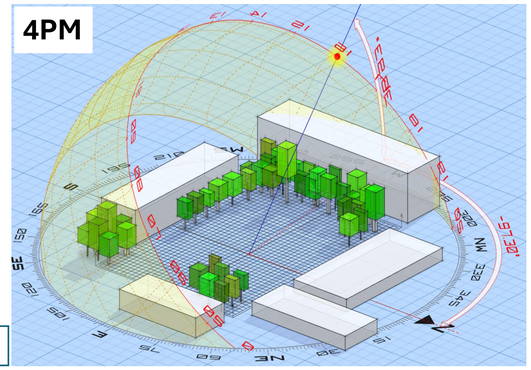


Figure 29 :

SUN PATH – 21st DECEMBER : 9AM

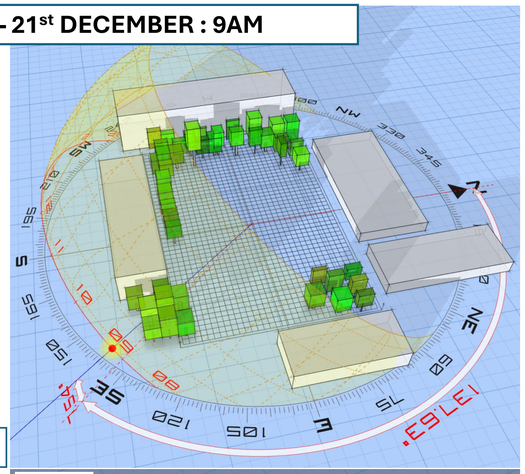


Figure 30 :

4PM

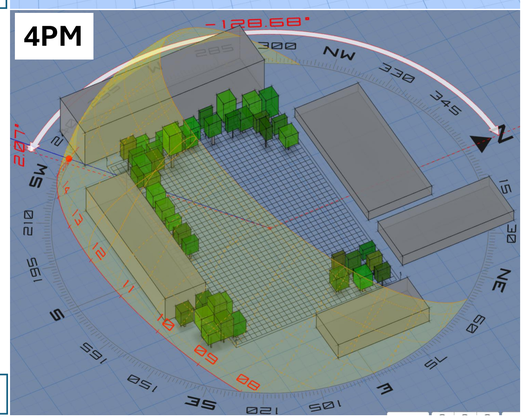


Figure 31 :

21ST JUNE : 9AM

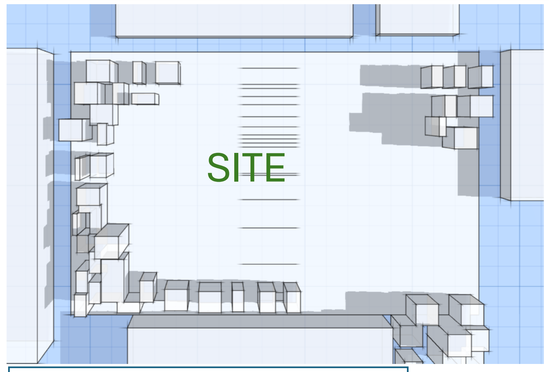


Figure 32 : Shadow studies at 9am

12 NOON

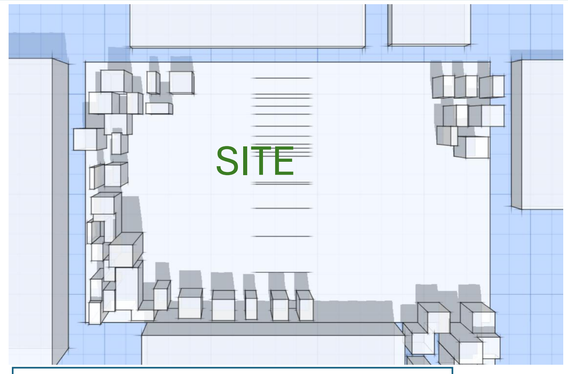


Figure 33 : Shadow studies at 12 noon

4PM

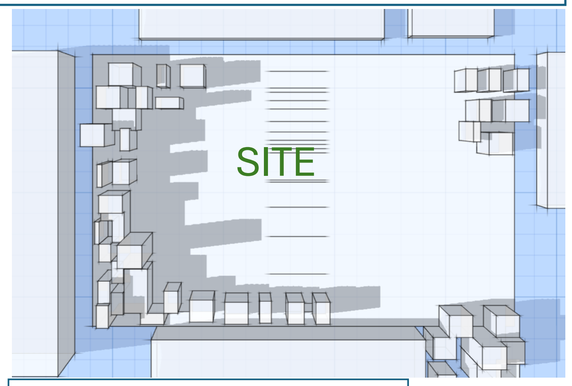


Figure 34 : Shadow studies at 4pm

22ND SEPTEMBER : 9AM

Figure 35: Shadow studies at 9am

12 NOON

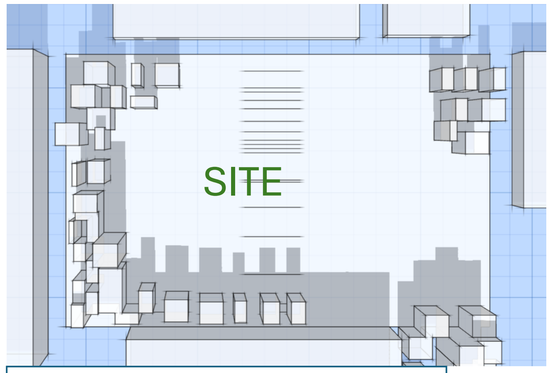


Figure 36 : Shadow studies at 12 noon

4PM

Figure 37 : Shadow studies at 4pm

21ST DECEMBER: 9AM

Figure 38 : Shadow studies at 9am

12 NOON

Figure 39 : Shadow studies at 12noon

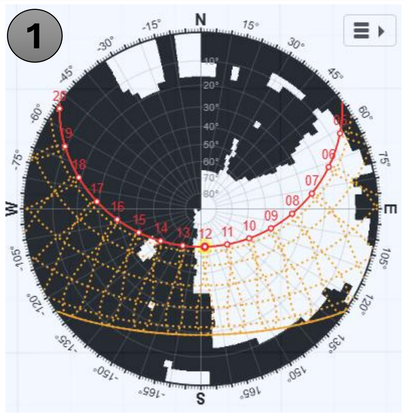
4PM

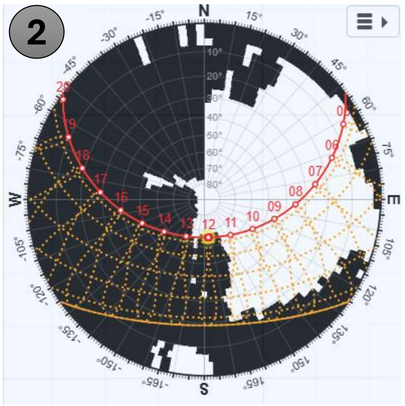
Figure 40 : Shadow studies at 4pm

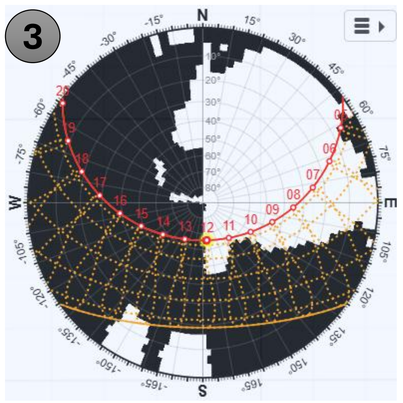
The above shadow studies of the site informs the longest and shortest daylight times. The duration of daylight is longest in summer at 16 hours and winter at 8 hours 25 minutes.
The site is partly shaded by adjacent buildings from Northwest and extended shading from the south and west due to the existing buildings and trees.

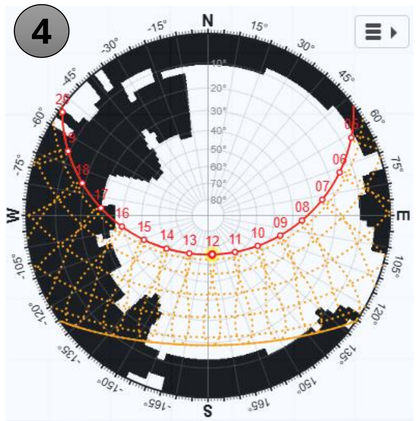
All figures on page: (Marsh, 2025)

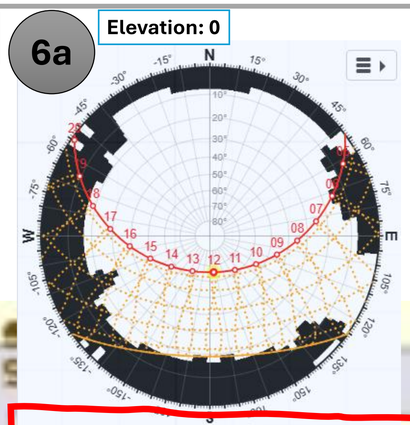
1.5.6 SITE ANALYSIS:SHADING MASK

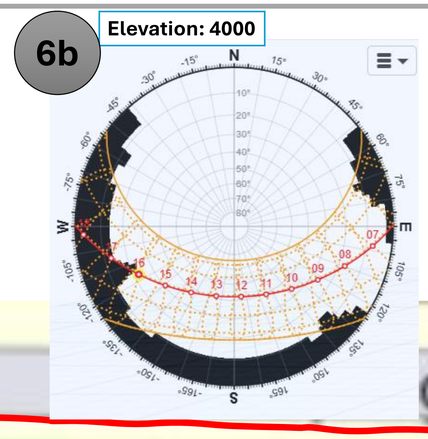


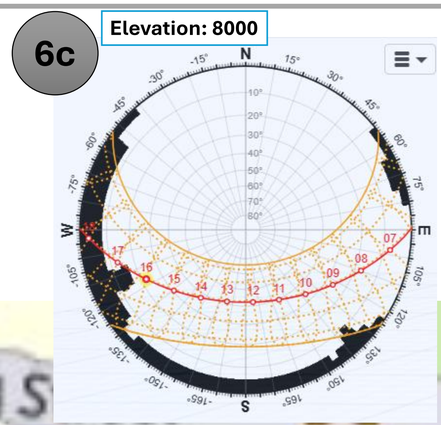












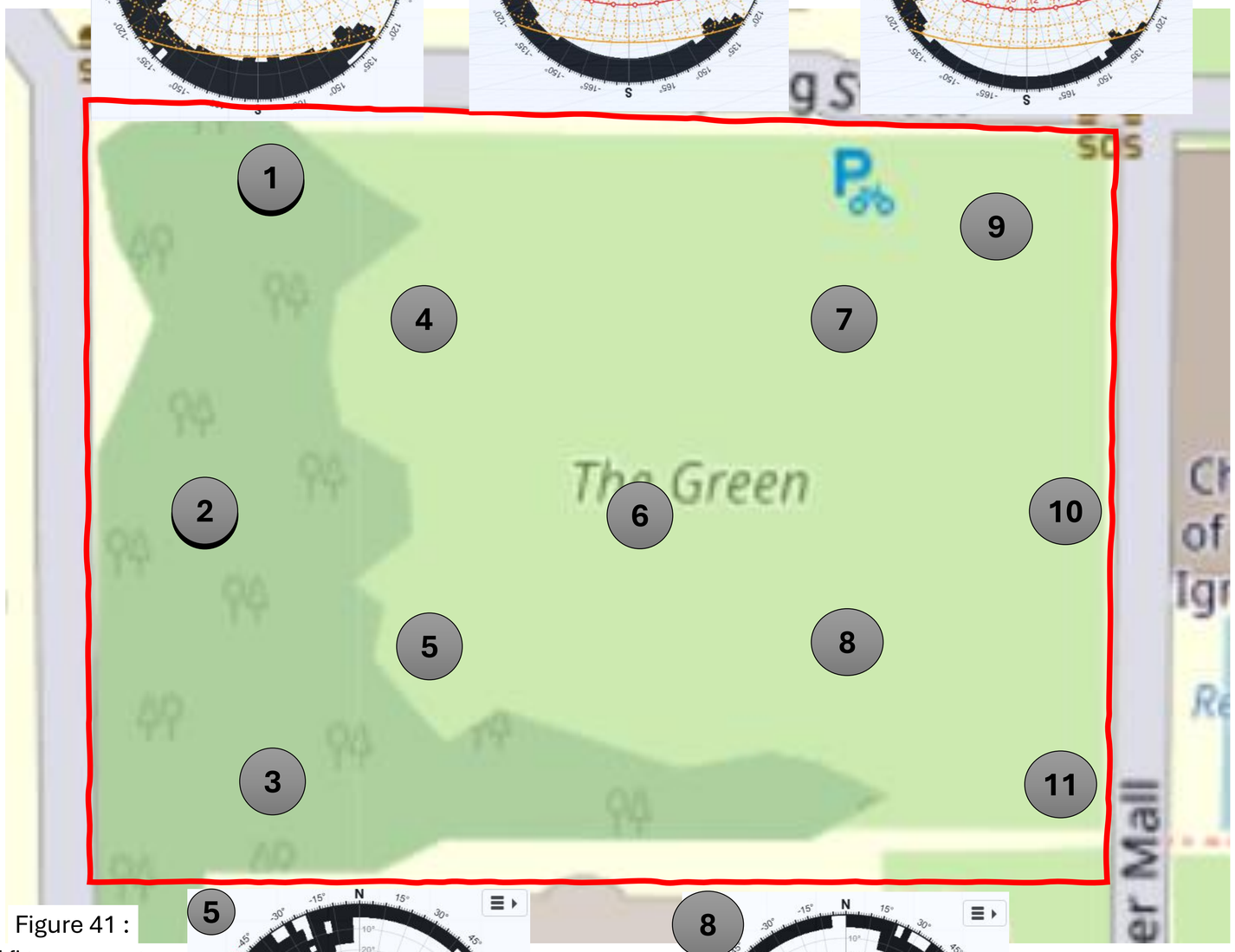
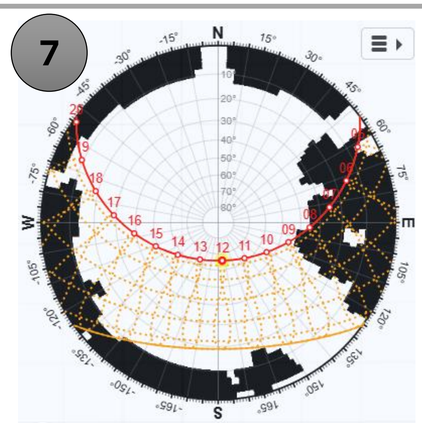
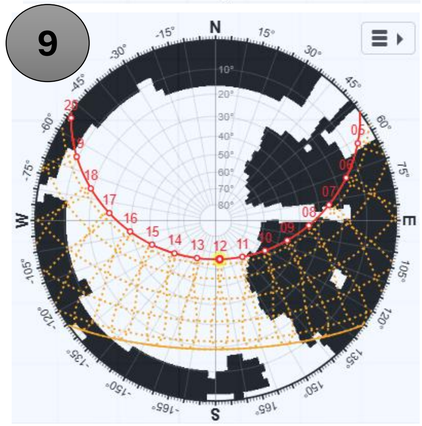
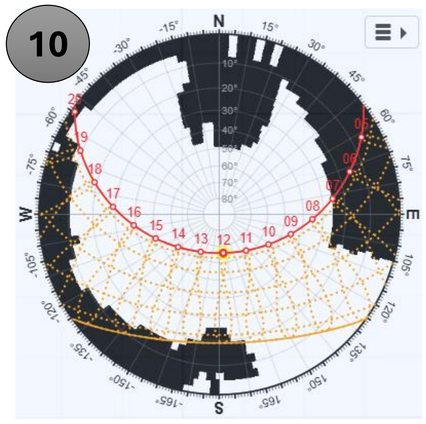
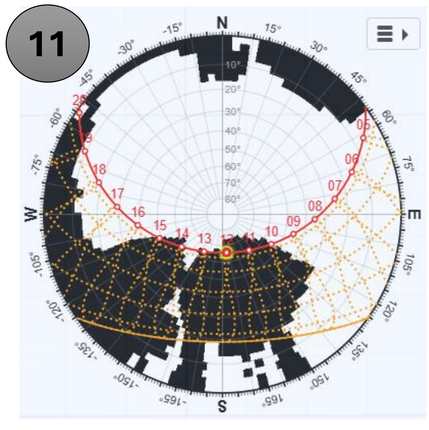


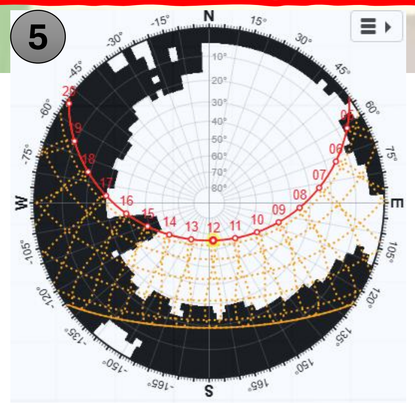
Figure 41 :
(All figures on page: Marsh, 2025)

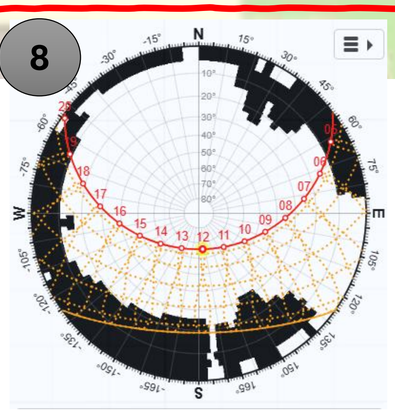












1.5.7 SITE ANALYSIS – AGGREGATE SOLAR ANALYSIS : DIRECT + DIFFUSE

ANNUAL AGGREGATE SOLAR ANALYSIS

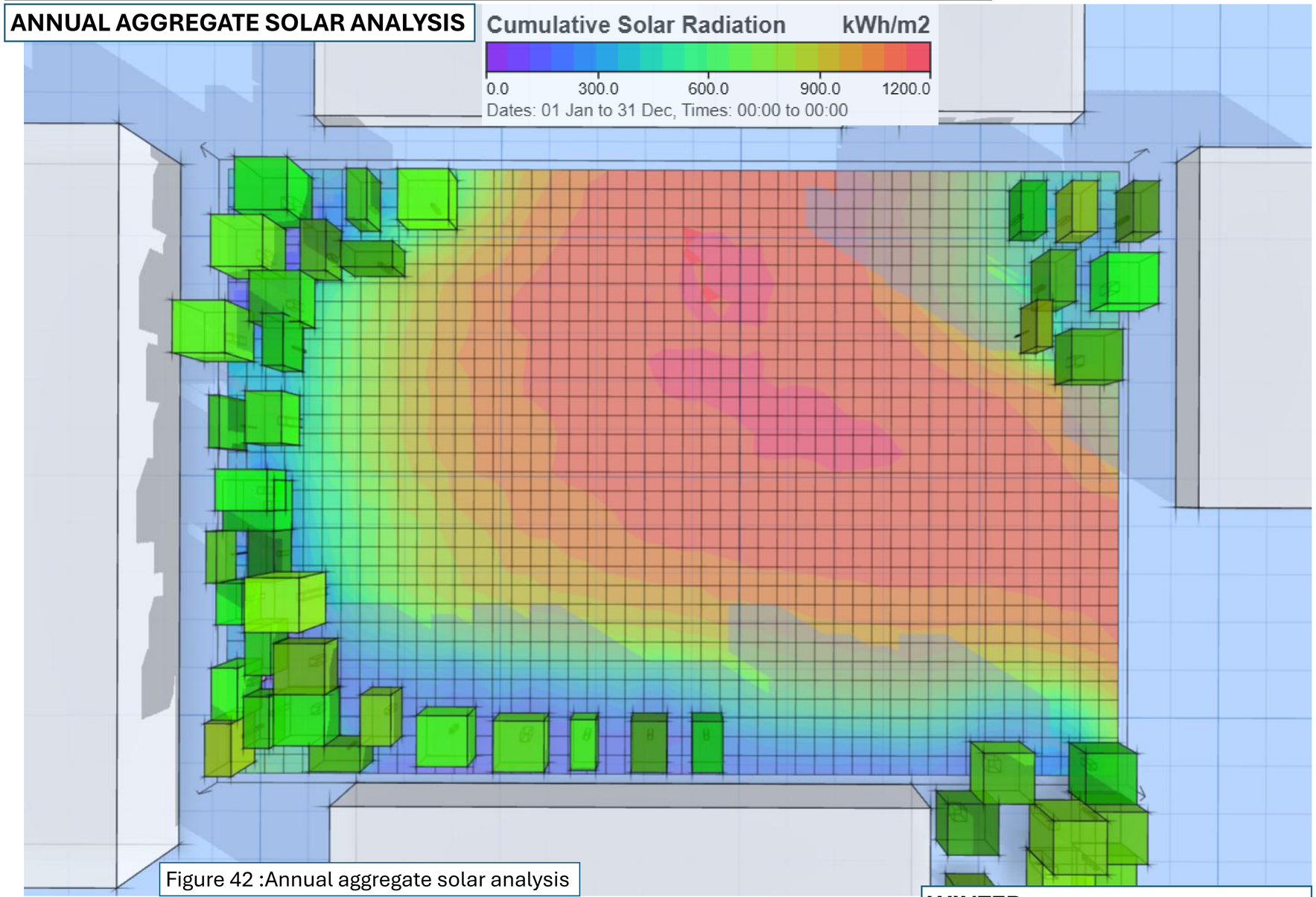


Figure 42 :Annual aggregate solar analysis

All figures on page (Marsh, 2025)

Highest solar radiation in summer : 500kWh/m2
Autumn : 200kWh/m2
Spring : 250kWh/m2
Highest in winter : 150kWh/m2

Solar radiation is very low for 6 months and moderate levels recorded in 3 months.
The highest recorded in summer is double the radiation recorded in spring.

SUMMER

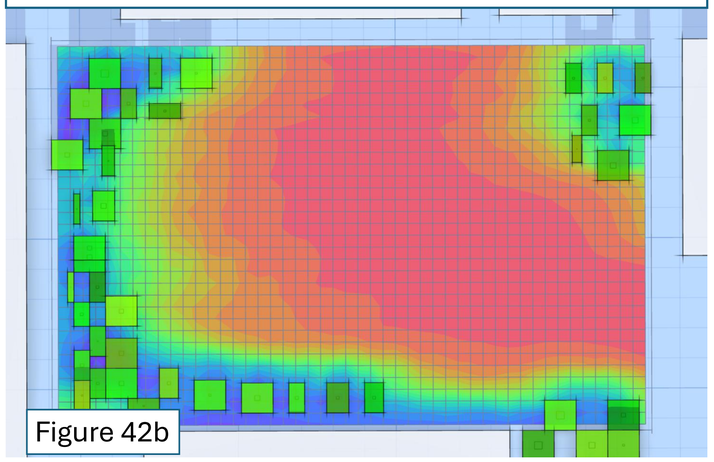


Figure 42b

AUTUMN

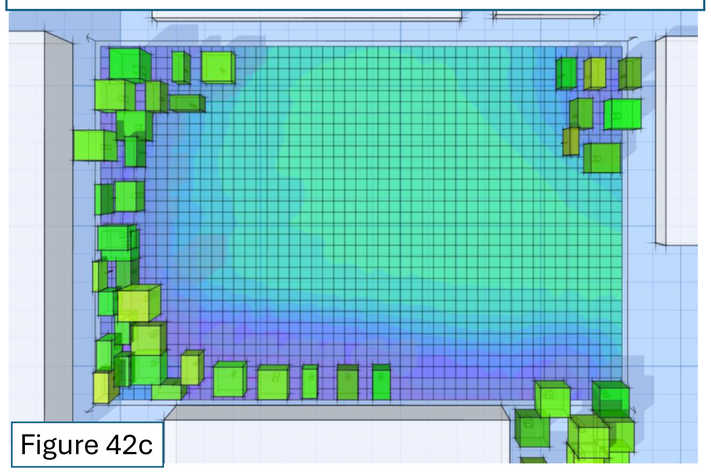


Figure 42c

WINTER

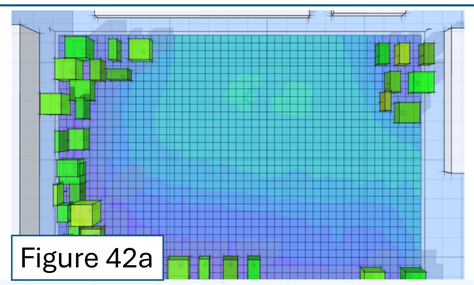
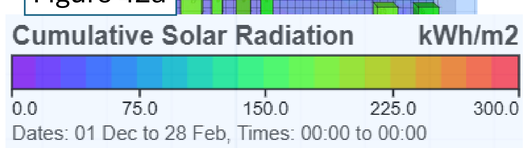


Figure 42a



SPRING

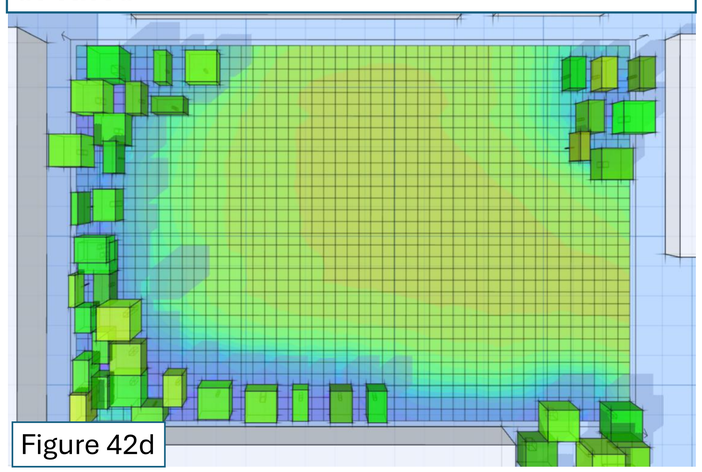
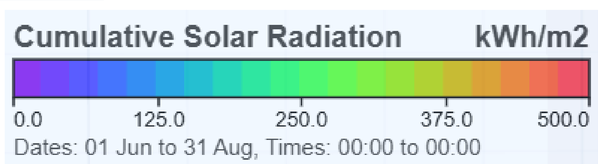


Figure 42d



1.5.8 AGGREGATE SOLAR ANALYSIS : PROPOSED BLOCK

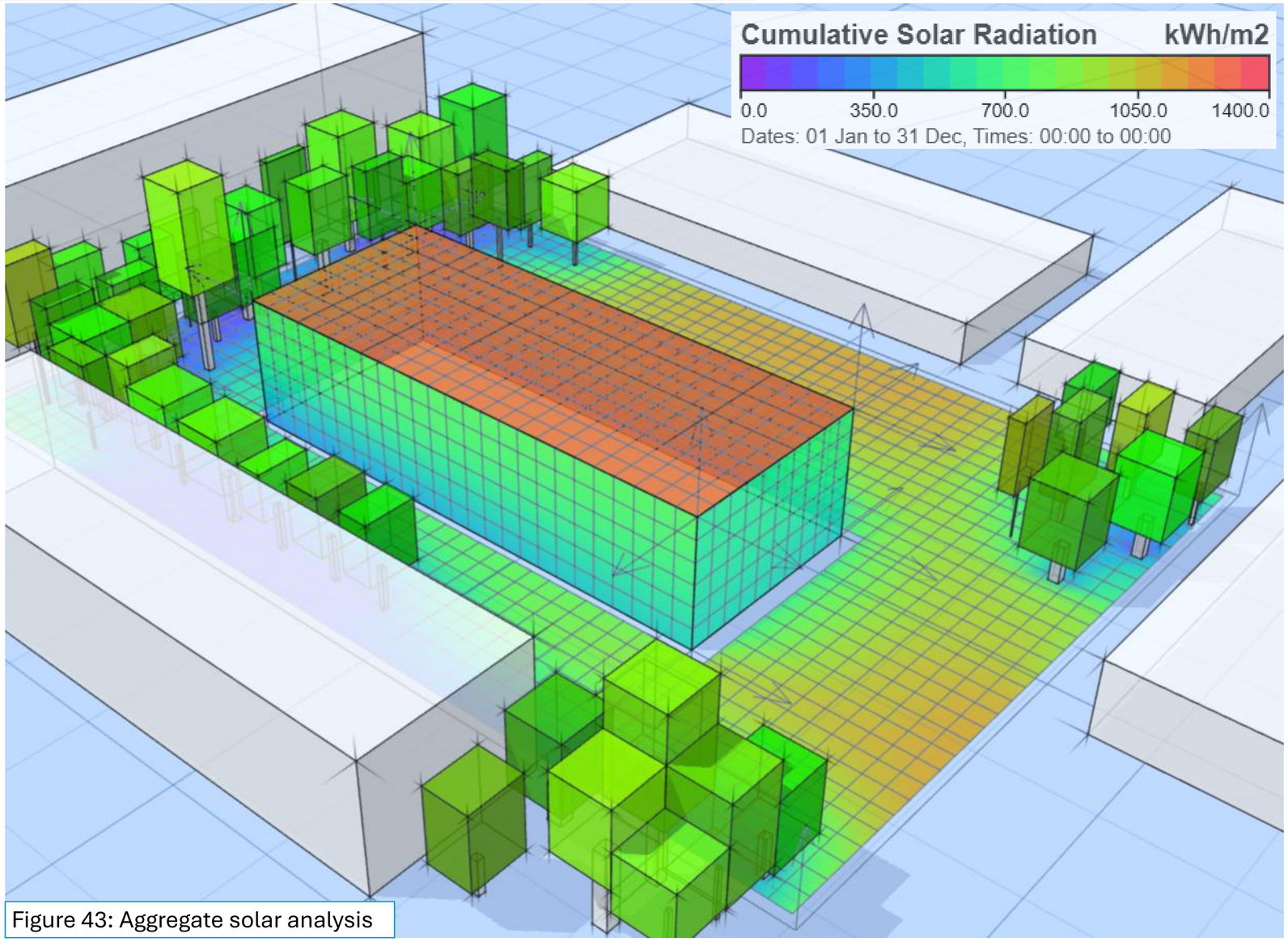


Figure 43: Aggregate solar analysis

All figures on page (Marsh, 2025)

- Minimal solar radiation from the West due to shading from vegetation and building

Recommendations for proposed project

- Shading for south facing windows from first floor upwards
- Create outdoor spaces at shaded areas without high solar radiation
- Studio rooms to face south for passive solar heat gain
- Solar radiation on roof is high which is ideal for solar photovoltaic array and solar water heater

WINTER

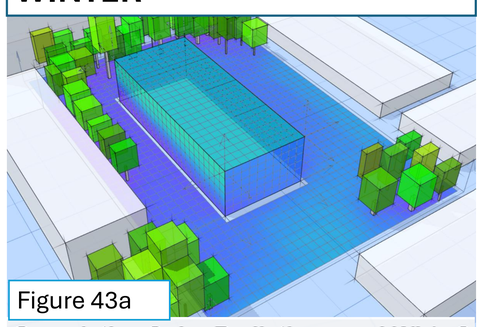
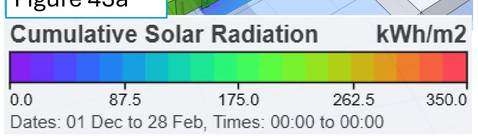


Figure 43a



SUMMER

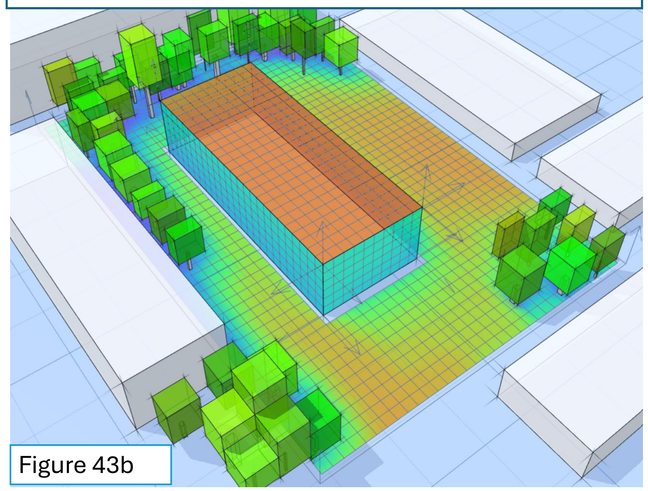


Figure 43b

AUTUMN

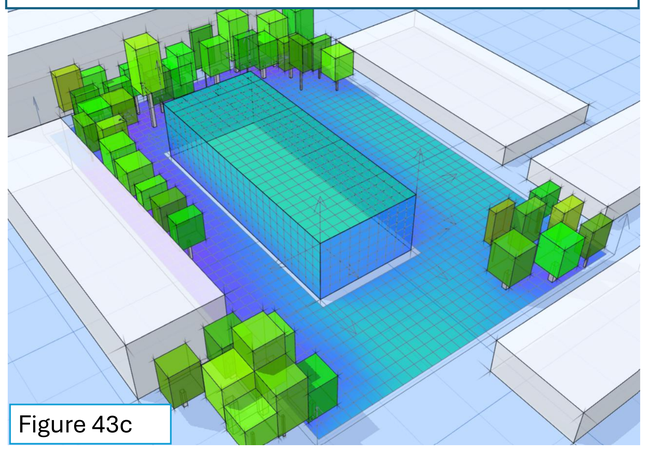


Figure 43c

SPRING

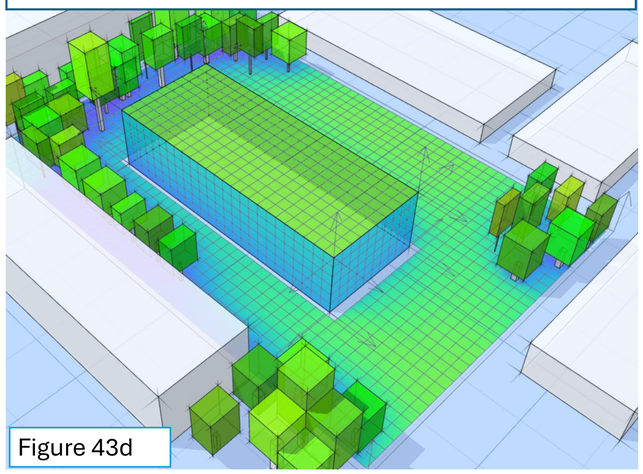
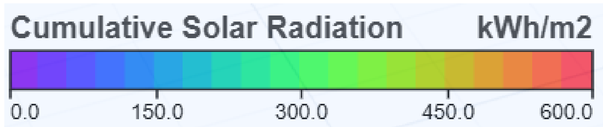


Figure 43d



1.5.9 SITE ANALYSIS-WIND PATTERN ON SITE

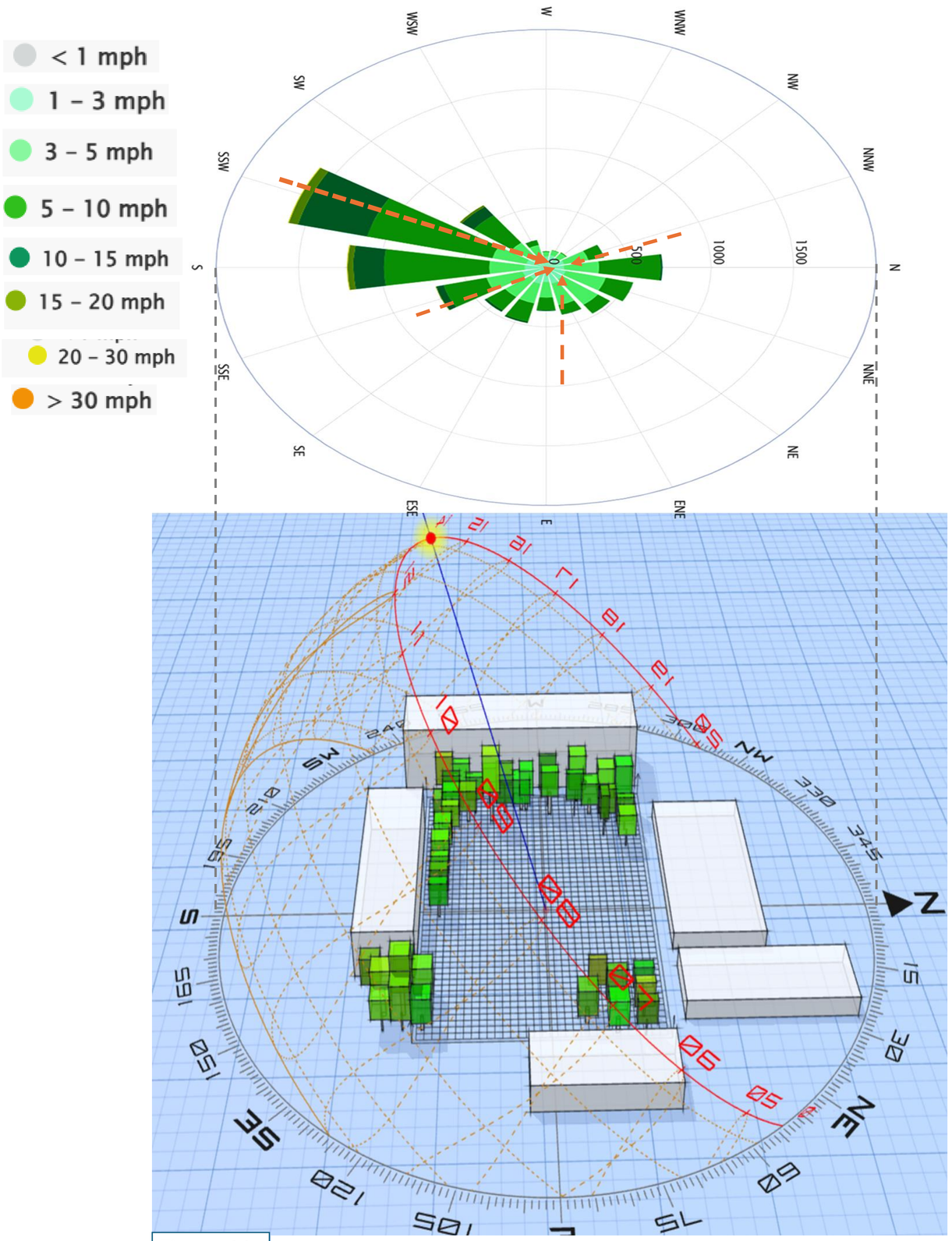


Figure 44 : (Marsh, 2025 and USDA, 2025) with modifications by author

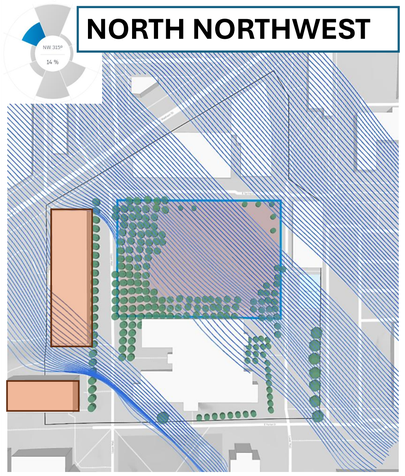


Figure 45: Wind direction from North Northwest

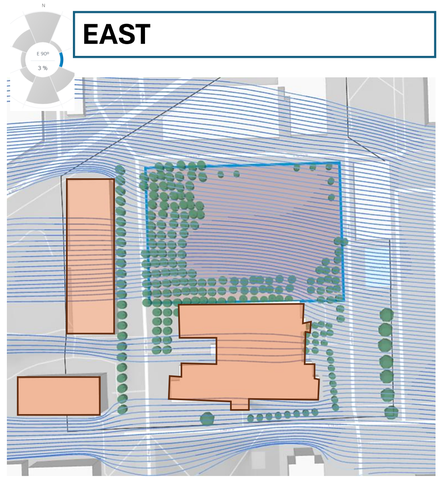


Figure 46: Wind direction from East

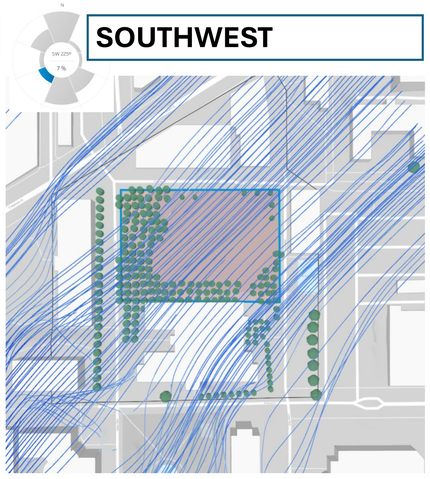


Figure 47: Wind direction from Southwest

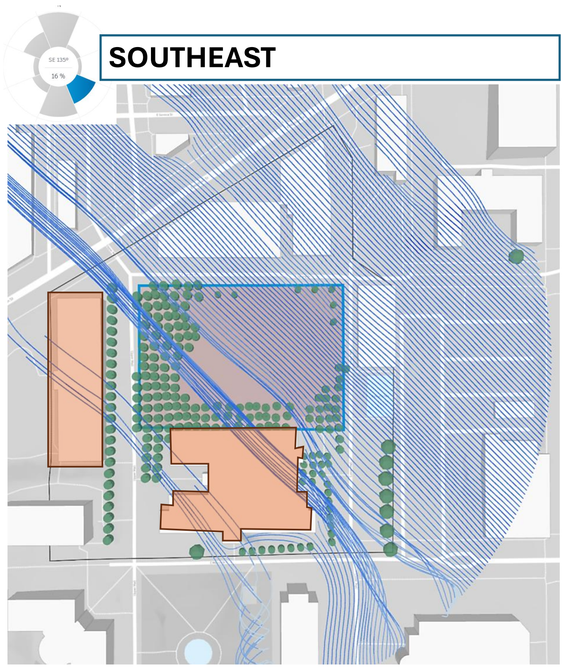


Figure 48: Wind direction from Southeast

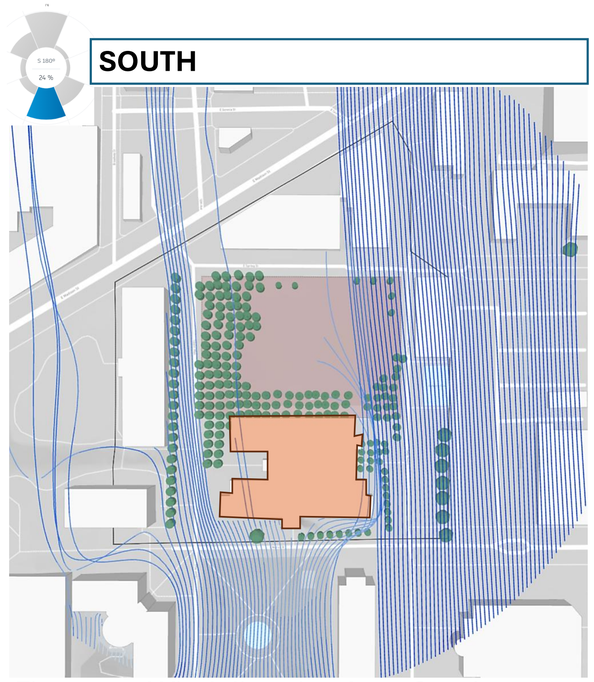


Figure 49: Wind direction from South

Figures 45 to 49: (Autodesk Forma, 2025)

Obstruction to wind flow

Prevailing winds from Northwest (6-8m/s) East(4-6 m/s) and Southeast offer the best wind directions to site.
Minimal south winds to site due to obstruction.

1.5.10 SITE ANALYSIS : STREET NOISE MAPPING

- < 50 db (quite rural background noise, library)
- 50-55 db (office average background noise)
- 55-60 db (conversational speech)
- 60-65 db (piano practice)
- 65-70 db (noisy restaurant)
- 70 > db (vacuum cleaner)

According to the noise mapping of the site, all three streets have acceptable sound levels of 59 decibels during the day and 52 decibels at night. This indicates the site will have any negative sound impact.

STREET NAME	SOUND LEVEL (decibels)		
East spring street	Avg. 58 dB	Day. 59 dB	Nig. 52 dB
Upper mall street	Avg. 58 dB	Day. 59 dB	Nig. 53 dB
Lower mall street	Avg. 58 dB	Day. 59 dB	Nig. 53 dB

(Open Street Map, 2025)

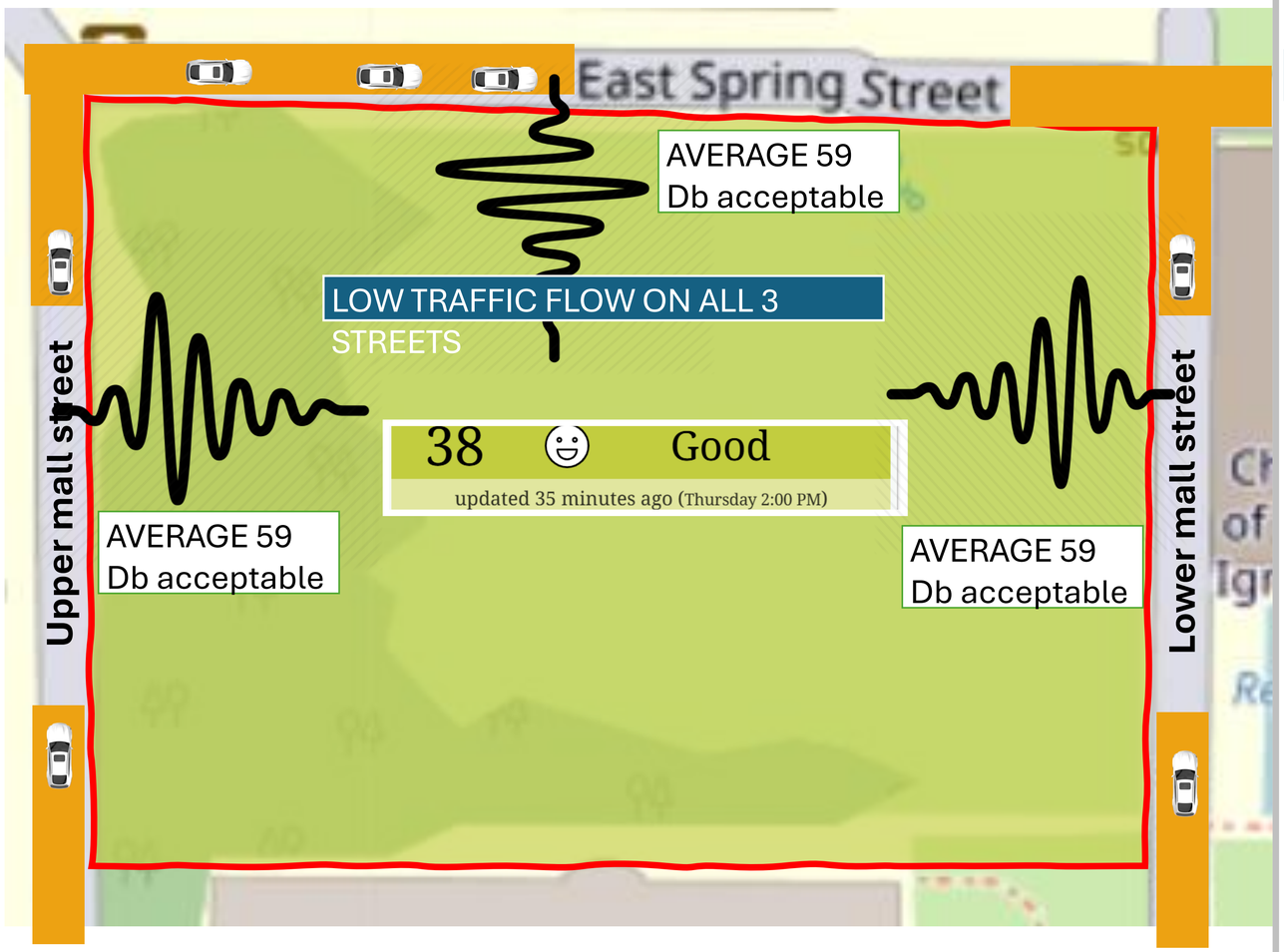
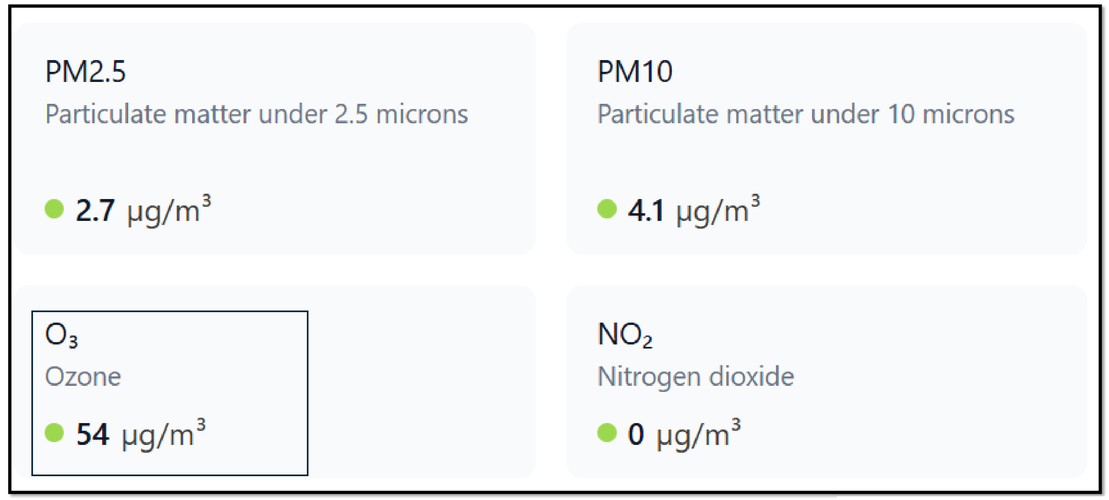


Figure 50 (Google, 2025) ; (IQAir, 2025) ; (Open Street Map, 2025) with modifications by author

1.5.11 AIR QUALITY FORECAST

The air quality index at the site is 38 which is generally considered acceptable as there are no high records of pollutants that will impact the health of people.



Source: (IQAir, 2025)

Table 1 : Air Quality definition scale by US-EPA (2016 Standard)

AQI	Air Pollution Level	Health Implications	Cautionary Statement (for PM2.5)
0 - 50	Good	Air quality is considered satisfactory, and air pollution poses little or no risk	None
51 -100	Moderate	Air quality is acceptable; however, for some pollutants there may be a moderate health concern for a very small number of people who are unusually sensitive to air pollution.	Active children and adults, and people with respiratory disease, such as asthma, should limit prolonged outdoor exertion.
101-150	Unhealthy for Sensitive Groups	Members of sensitive groups may experience health effects. The general public is not likely to be affected.	Active children and adults, and people with respiratory disease, such as asthma, should limit prolonged outdoor exertion.
151-200	Unhealthy	Everyone may begin to experience health effects; members of sensitive groups may experience more serious health effects	Active children and adults, and people with respiratory disease, such as asthma, should avoid prolonged outdoor exertion; everyone else, especially children, should limit prolonged outdoor exertion
201-300	Very Unhealthy	Health warnings of emergency conditions. The entire population is more likely to be affected.	Active children and adults, and people with respiratory disease, such as asthma, should avoid all outdoor exertion; everyone else, especially children, should limit outdoor exertion.
300+	Hazardous	Health alert: everyone may experience more serious health effects	Everyone should avoid all outdoor exertion

Source: (IQAir, 2025)

1.5.12 SITE ANALYSIS-CONCLUSION

View sphere from site : Fig. 51 to Fig. 53

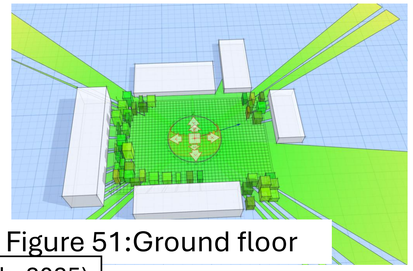


Figure 51:Ground floor (Marsh, 2025)

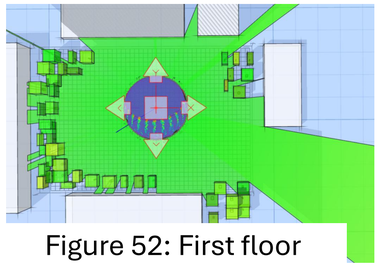


Figure 52: First floor

Figure 53: Second floor

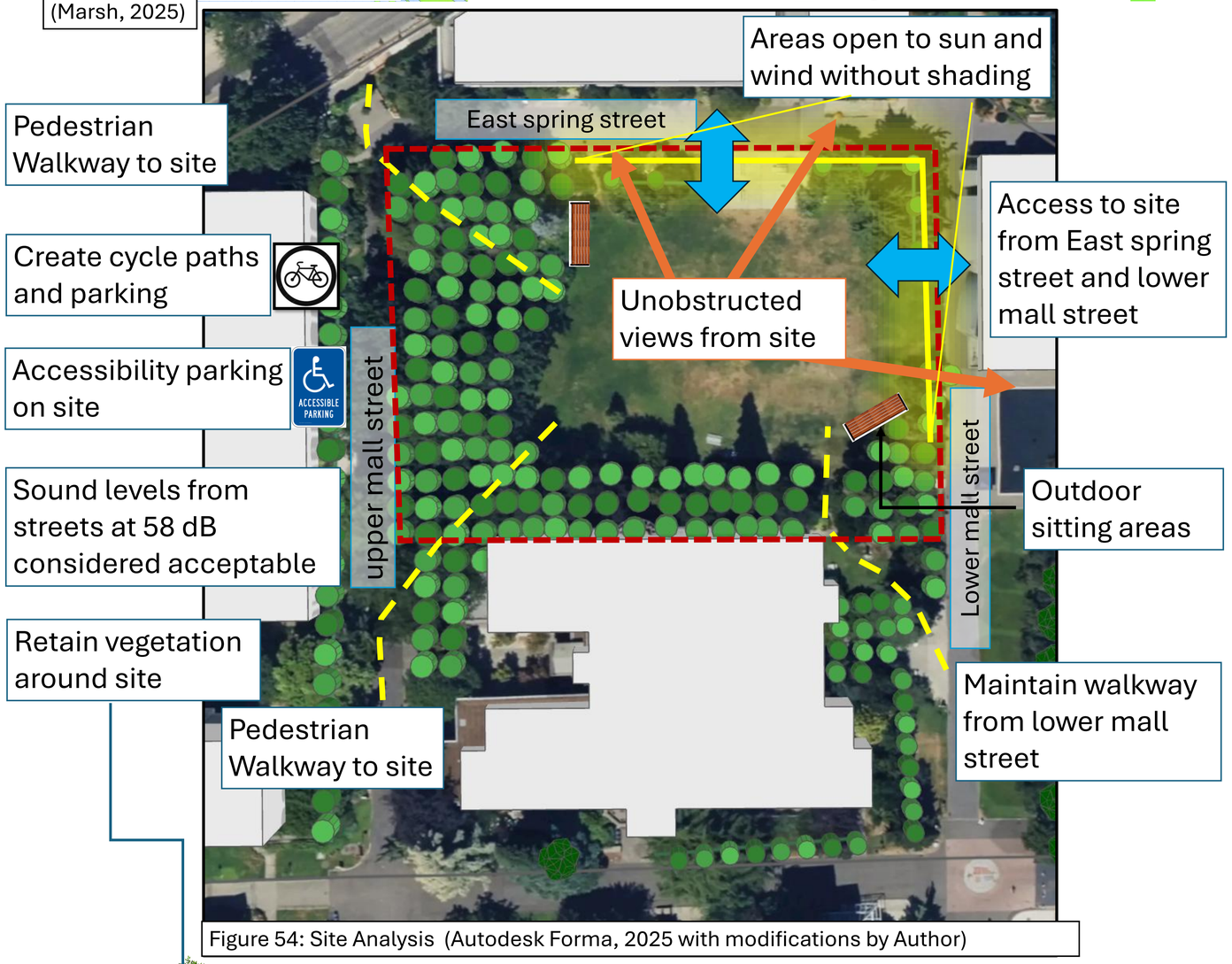


Figure 54: Site Analysis (Autodesk Forma, 2025 with modifications by Author)

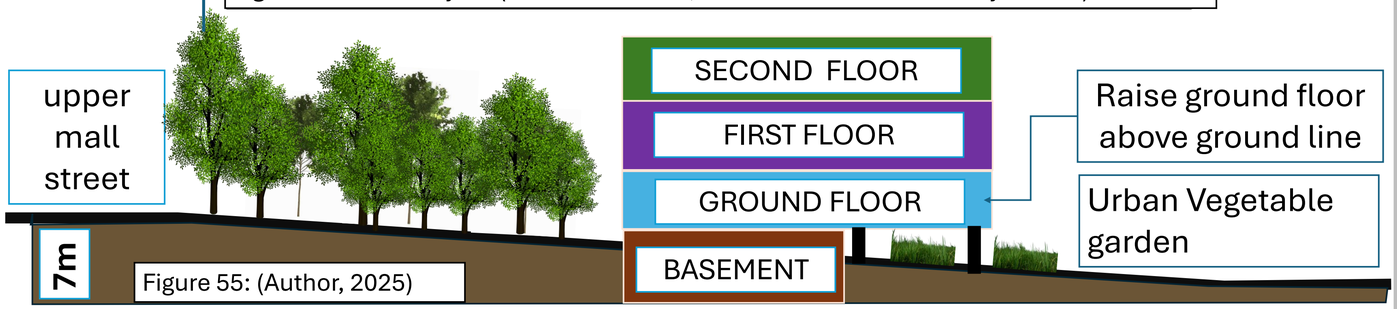


Figure 55: (Author, 2025)

Building orientation : South facing for passive solar heat gain

Shading : first, second and third floor south facing windows

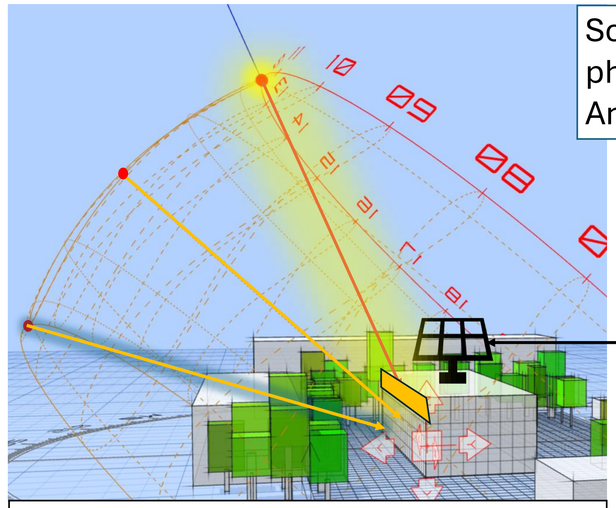


Figure 56:Solar angles and potential solar PV (Marsh, 2025) with modifications by author

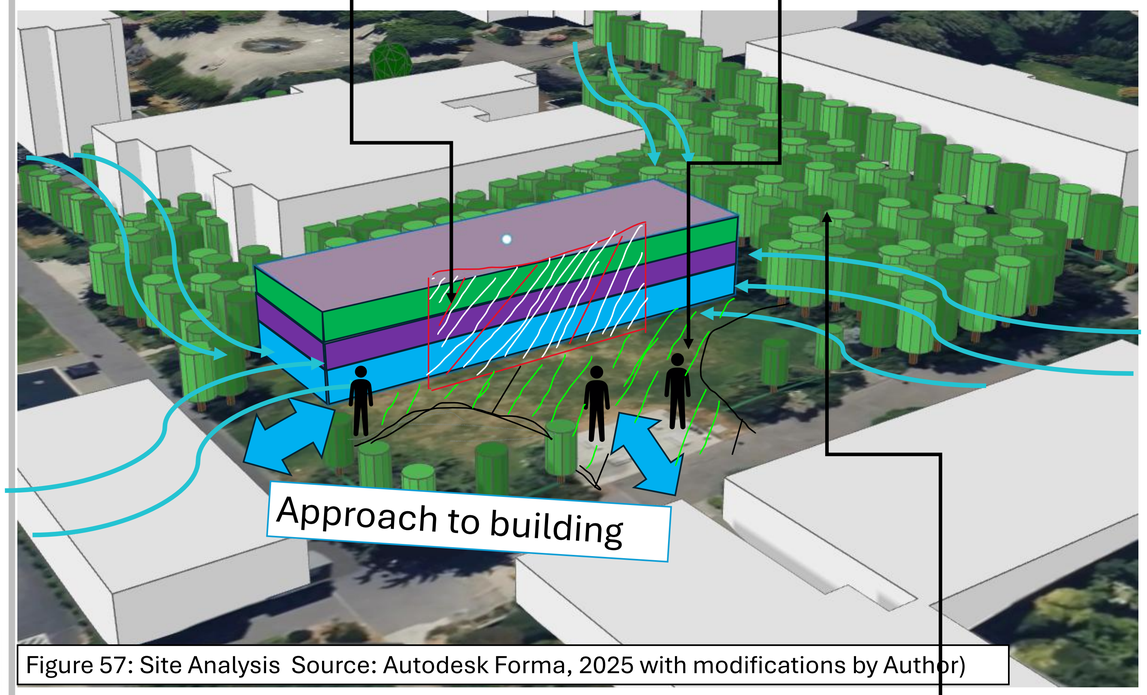


Figure 57: Site Analysis Source: Autodesk Forma, 2025 with modifications by Author)

Compact form : To reduce heat loss from the building and efficient use of space

Airflow: From different angles of site towards building for ventilation

Sustainable Design of Students' Accommodation Block

2.0 CASE STUDY 1 : THE HOUSE AT CORNELL TECH

LOCATION	Cornell University, New York
PROJECT TYPE	Residential – Student Housing
YEAR OF COMPLETION	2017
CERTIFICATIONS	Passive House LEED Platinum
NO. OF FLOORS	26
NO. OF UNITS	352

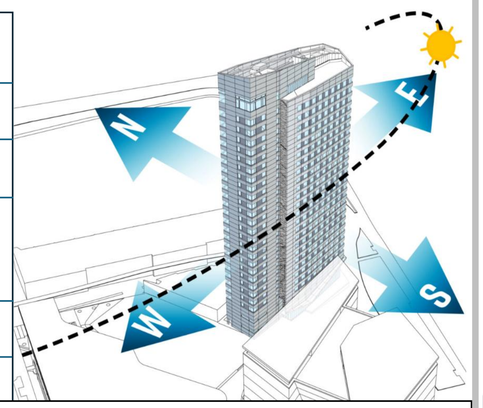


Figure 58: Building orientation(Handle Architects,2017)

Passive Design Strategies

Building orientation: East-west axis for passive solar heat gain.
Form: Compact to minimize heat losses
Insulation: thick insulated wall around the full height of the building
Envelope : Highly insulated prefabricated mega panels with R-value of 19 with windows fixed and sealed
Window to wall ratio: 23%

Overall Source Energy: 38.1 kBtu/ft/yr
Heating Energy: max 4.75 kBtu/ft/yr
Cooling Energy: Max 5.39 kBtu/ft/yr
Air changes per hour : 0.6 ACH
Exhaust and supply ventilation : with energy recovery

Active Systems

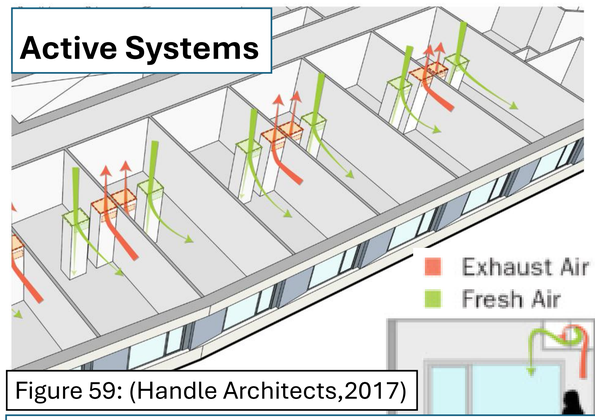


Figure 59: (Handle Architects,2017)

vertical supply and exhaust shafts provide a pathway for air to travel to one of two energy recovery ventilators that sit on the building's rooftop

Ventilation: central air distribution system with ERV on the rooftop

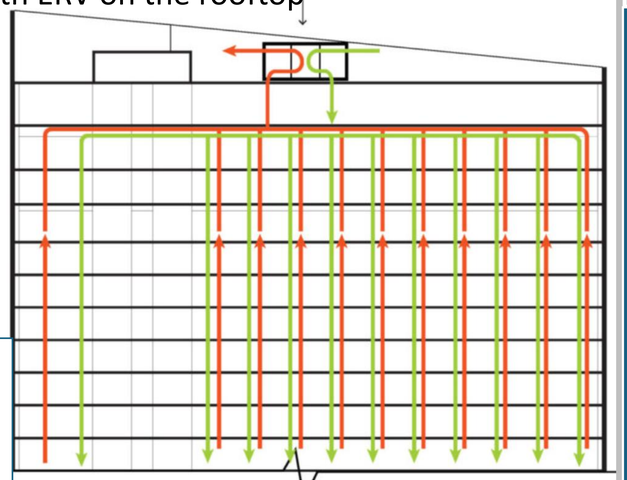


Figure 60: Energy Recovery System (Handle Architects,2017)

Heating and cooling system: Low-energy variable refrigerant flow (VRF) system
To extract heat from outside air to provide heat in cool months, or extract heat from inside air in warm months via an outdoor condenser.

2.1 CASE STUDY 2 : POMONA COLLEGE STUDENT HOUSING

LOCATION	Claremont, California
PROJECT TYPE	Residential – Student Housing
YEAR OF COMPLETION	2011
CERTIFICATIONS	LEED Platinum
NO. OF FLOORS	2 / 3
NO. OF UNITS	150
TOTAL AREA	81 sf

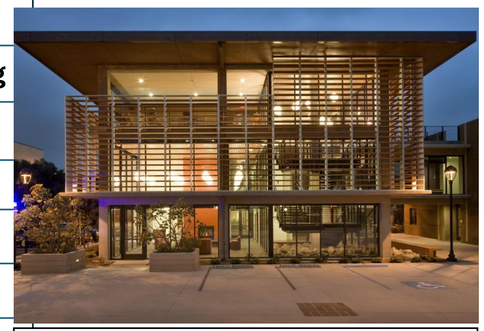


Figure 61: exterior view (Wyatt (2014))

Solar PV system offsets over 130,000 kWh (14%) of annual electricity consumption

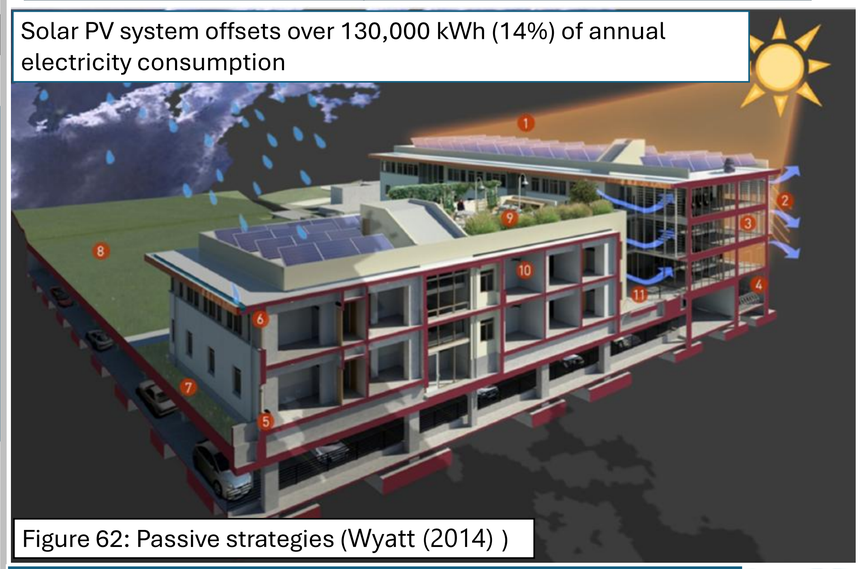


Figure 62: Passive strategies (Wyatt (2014))

Performance Results

Energy Use Intensity (EUI) - 53.47 kBTU/sqft/yr

Actual annual energy consumption reduced by 22% compared to California Building Code.

Performance 49% better than ASHRAE 90.1-2004 Standard

Daylighting
•Over 95% of occupied spaces have efficient daylight
Solar Control and Shading
•8-feet-deep roof overhangs
•Exterior sunscreens consisting of 2 by 8-inches deep horizontal terracotta planks
Thermal Mass
•11-inches thick precast concrete panels with 2 wythes (3-inch exterior and 5-inch interior) and 3-inches thick R-19 Polystyrene insulation
Materials
- 24% of materials were regionally selected within 500 miles of the project site

•72% of all wood products are FSC-certified
•18% recycled building materials

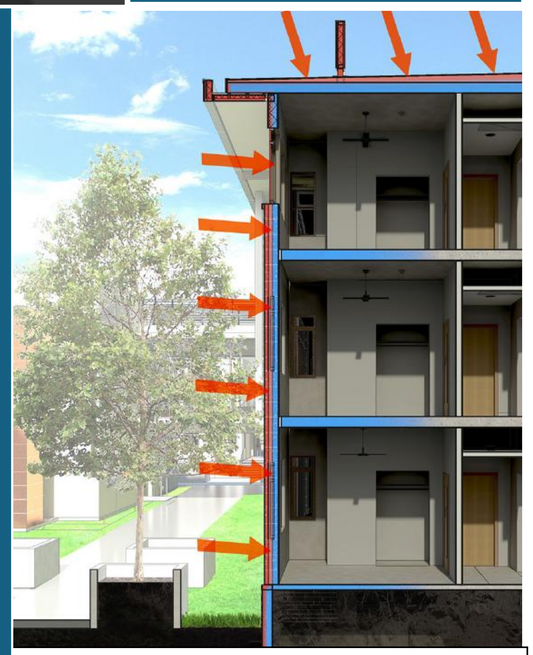


Figure 63: Thermal mass (Wyatt (2014))

3.0 DETAILED BRIEF WITH PERFORMANCE TARGETS

TARGET : GRADUATE STUDENTS

42

STAFF

8

Ground floor - Communal Spaces
Reception
Manager's office
Socialising/play area
Mailboxes zone
Bicycle storage
Circulation: Lift and stair

1ST, 2ND & 3RD FLOOR - Studios
Sleeping area
Study
Kitchenette
Toilet and shower
Laundry + utility store
Sun Space

Basement – Services

3.1 DETAILS OF SPACES AND ACTIVITY SCHEDULE

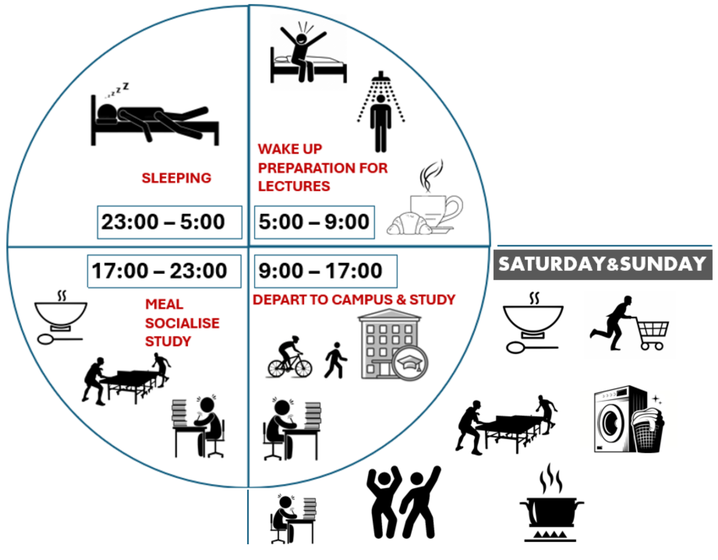


Figure 64: Activity Schedule
Source: (Google 2025 with modifications by Author)

3.2 TABLE 2: PERFORMANCE TARGETS

FLOOR	SPACE	AREA (m2)	NV THERMAL TARGETS	HVAC THERMAL TARGETS	DAYLIGHT TARGETS		CO2 TARGETS	HUMIDITY
			CIBSE A (°C)	LEED (°C)	LEED		LEED	LEED
TYPICAL FLOOR OF 1,2&3	STUDIO -sleeping area -study area -Kitchenette	10.18 (x 14no.) = 142.52	18 - 25	21-24	(sDA300,50 %) ≥55% space 300 lux	(ASE1000,250) <10% space 1,000 lux	≤ 1,000 ppm	30% to 60%
	Toilet and shower	2.06 (x 14no.) = 28.84	18-25	21-24	≥40% 300 lux	<10% 1,000 lux	≤ 1,000 ppm	30% to 60%
	Sun Space	3.43x14 = 48.02	-	-				
	Reading room	14	18 - 25	21-24	≥55% space 300 lux	<10% space 1,000 lux	≤ 1,000 ppm	30% to 60%
	Store	4.68	18-25	21-24	≥40% 300 lux	<10% 1,000 lux	≤ 1,000 ppm	30% to 60%
	Corridor	45.24	-					
TOTAL/floor		**321.31**						
GROUND FLOOR	Reception	103.18	18 - 25	21-24	≥55% space 300 lux	<10% space 1,000 lux	≤ 1,000 ppm	30% to 60%
	Entrance lobby	14.1	-	-	≥40% 300 lux	<10% 1,000 lux	≤ 1,000 ppm	-
	Manager's office	4.68	18 - 25	21-24	≥55% space 300 lux	<10% space 1,000 lux	≤ 1,000 ppm	30% to 60%
	Socialising area	85.58	18 - 25	21-24	≥55% space 300 lux	<10% space 1,000 lux	≤ 1,000 ppm	30% to 60%
	Stair 1	2	18-25	-	≥40% 300 lux	<10% 1,000 lux	≤ 1,000 ppm	30% to 60%
	Stair 2 with lobby	13.93	18-25	-	≥40% 300 lux	<10% 1,000 lux	≤ 1,000 ppm	30% to 60%
	Bicycle store	37.46	18-25	-			≤ 1,000 ppm	-
G. TOTAL		**260.93**						
BASEMENT	services	200	-					

Sources: LEED version 4.1
ASHRAE 90.1
CIBSE GUIDE A

TOTAL AREA OF BUILDING

TYPICAL FLOOR PLAN AREA = 321.31 msq. X 3 = 963.93 msq.
GROUND FLOOR = 260.93 msq.

TOTAL AREA OF BUILDING = 1,224.86 msq.

ENERGY USE INTENSITY TARGET

1.Seattle City Council: EUI target of average buildings = 90 kWh/m^2/yr

2.LEED version 4.1(refer ASHRAE 90.1) : 60-80 kWh/m^2/yr

TOTAL ENERGY USE TARGET

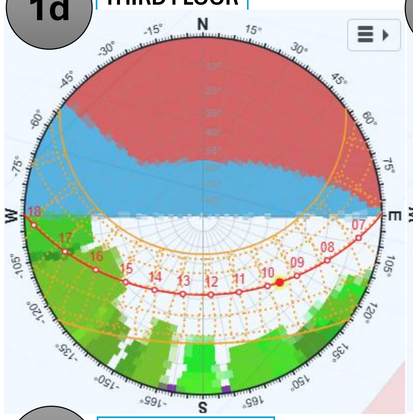
Annual energy use = EUI x floor area

If EUI = 60 kWh/m^2/yr

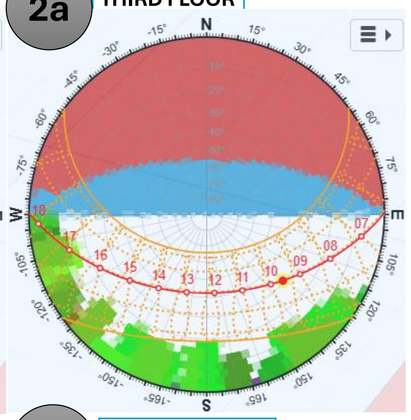
60 kWh/m^2/yr x 1224.86 m^2 = **73,469.6 kWh**

Sustainable Design of Students' Accommodation Block

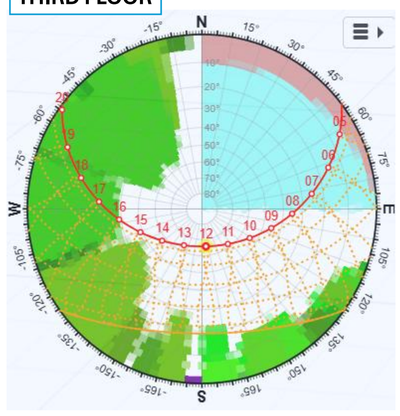
1d THIRD FLOOR



2a THIRD FLOOR



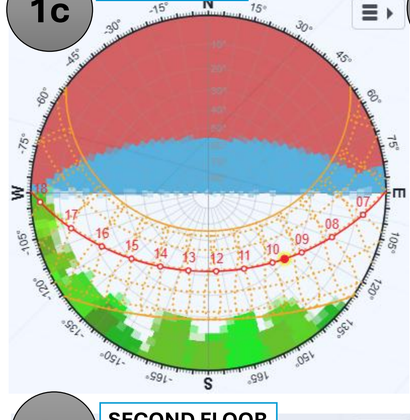
THIRD FLOOR



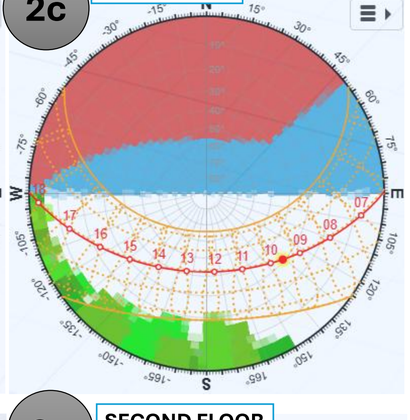
4.0 SHADING MASK

- Shading required for first, second and third floors in summer months except the west which receives shading from the trees from 14:00.
- All floors receive winter sun available for passive solar heating.

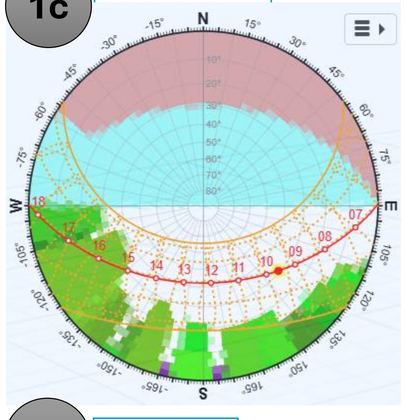
1c THIRD FLOOR



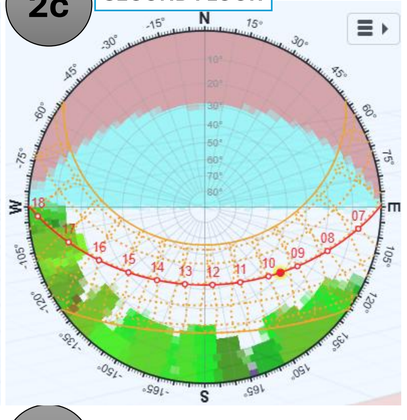
2c THIRD FLOOR

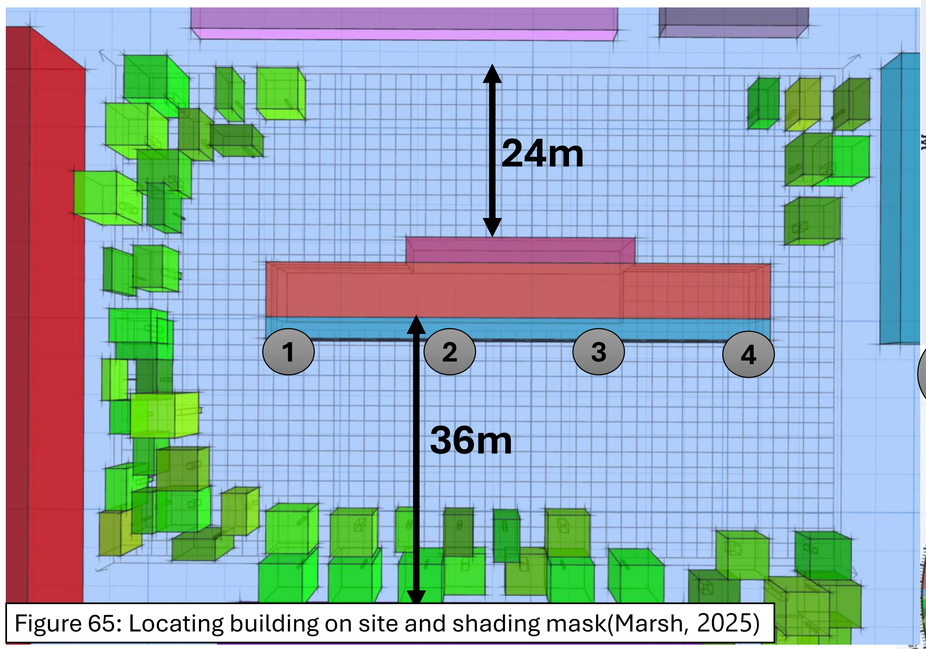


1c SECOND FLOOR

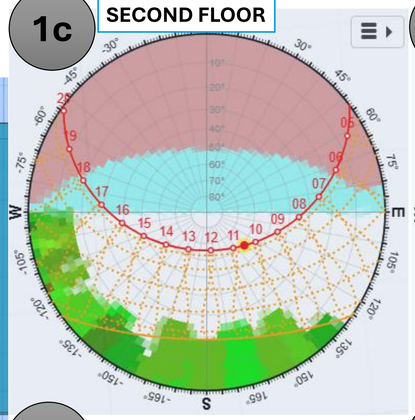


2c SECOND FLOOR

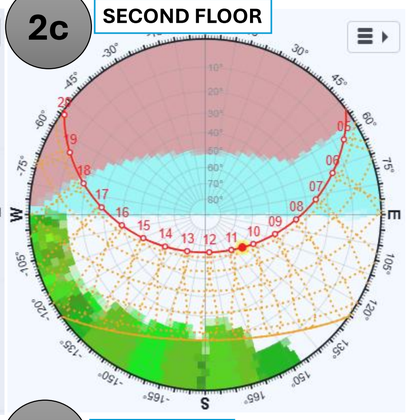




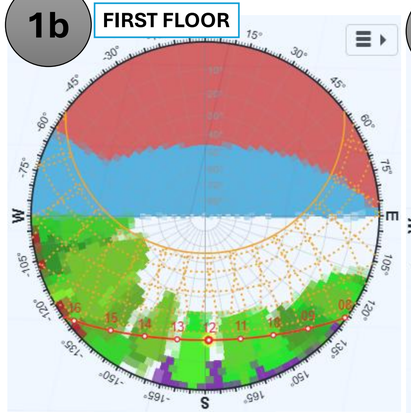
1c SECOND FLOOR



2c SECOND FLOOR



1b FIRST FLOOR



2b FIRST FLOOR

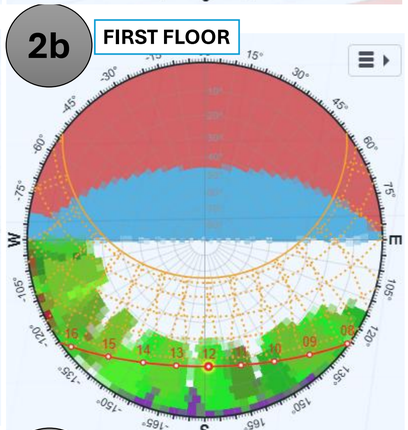
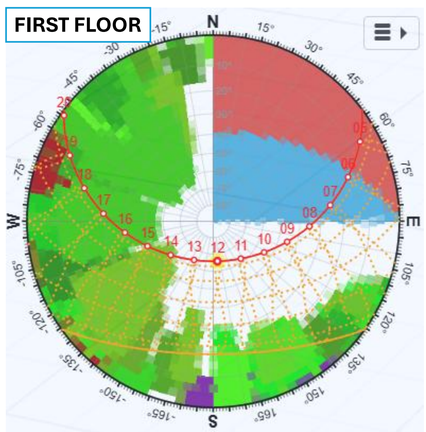
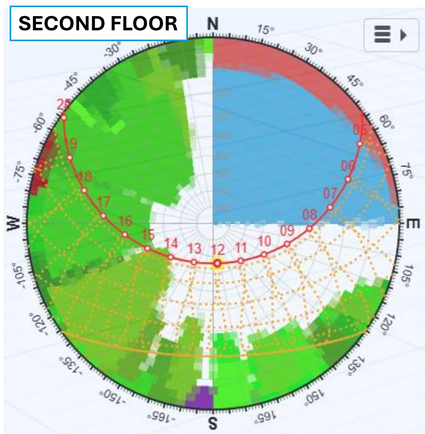


Figure 65: Locating building on site and shading mask(Marsh, 2025)

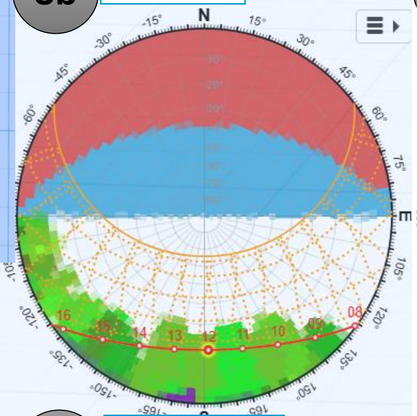
FIRST FLOOR



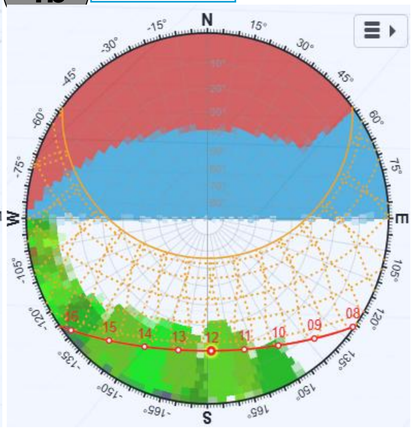
SECOND FLOOR



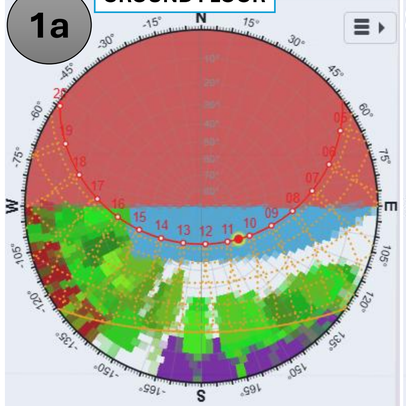
3b FIRST FLOOR



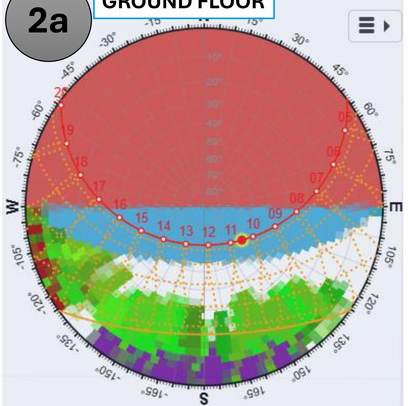
4b FIRST FLOOR



1a GROUND FLOOR

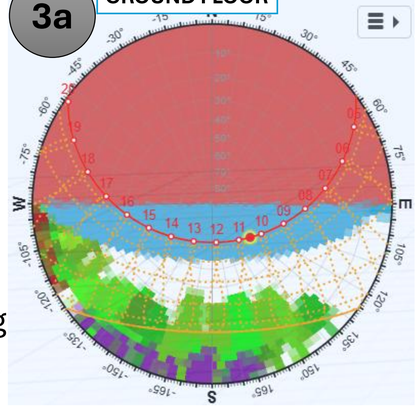


2a GROUND FLOOR



- Ground floor shaded by 1st floor cantilever in summer.
- Ground floor - winter sun available for passive solar heating except from 8-9am when south obstruction shades façade.

3a GROUND FLOOR



- Trees on site are a mixture of evergreen and deciduous trees.
- In winter, the sun will come through to the building when the trees shed their leaves.

4.1 SITE OPPORTUNITIES AND RESTRICTIONS

Simple and compact form: fewer thermal bridges and to minimize surface area of building to reduce heat loss.

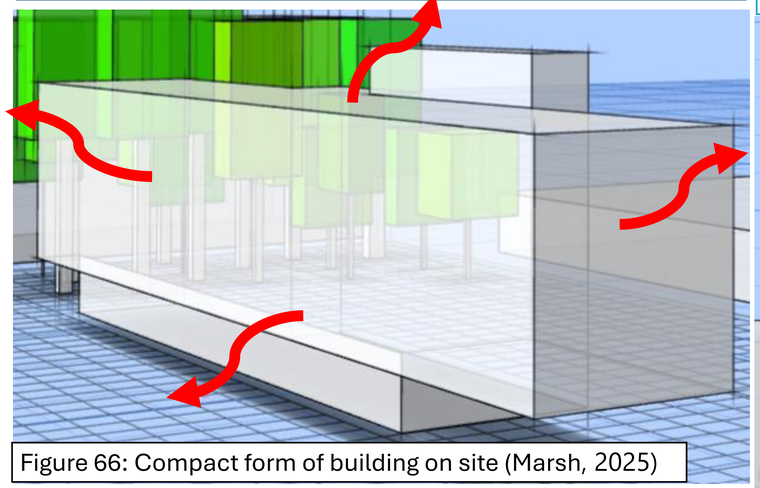


Figure 66: Compact form of building on site (Marsh, 2025)

To increase natural ventilation and daylight into spaces.

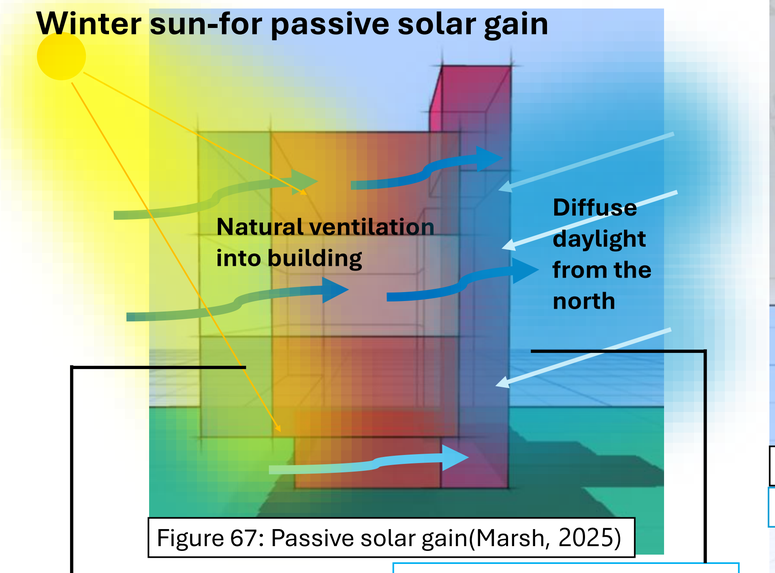


Figure 67: Passive solar gain(Marsh, 2025)

All students rooms to face South

- Sleeping and living area

Circulation space And less occupied spaces to north

- Entrance
- Laundry
- Stair and lift
- corridor
- Store

BUILDING FORM : Linear building to complement the shape of site.

Site gives an opportunity for the longest side of the building to face south for passive solar heating in winter.

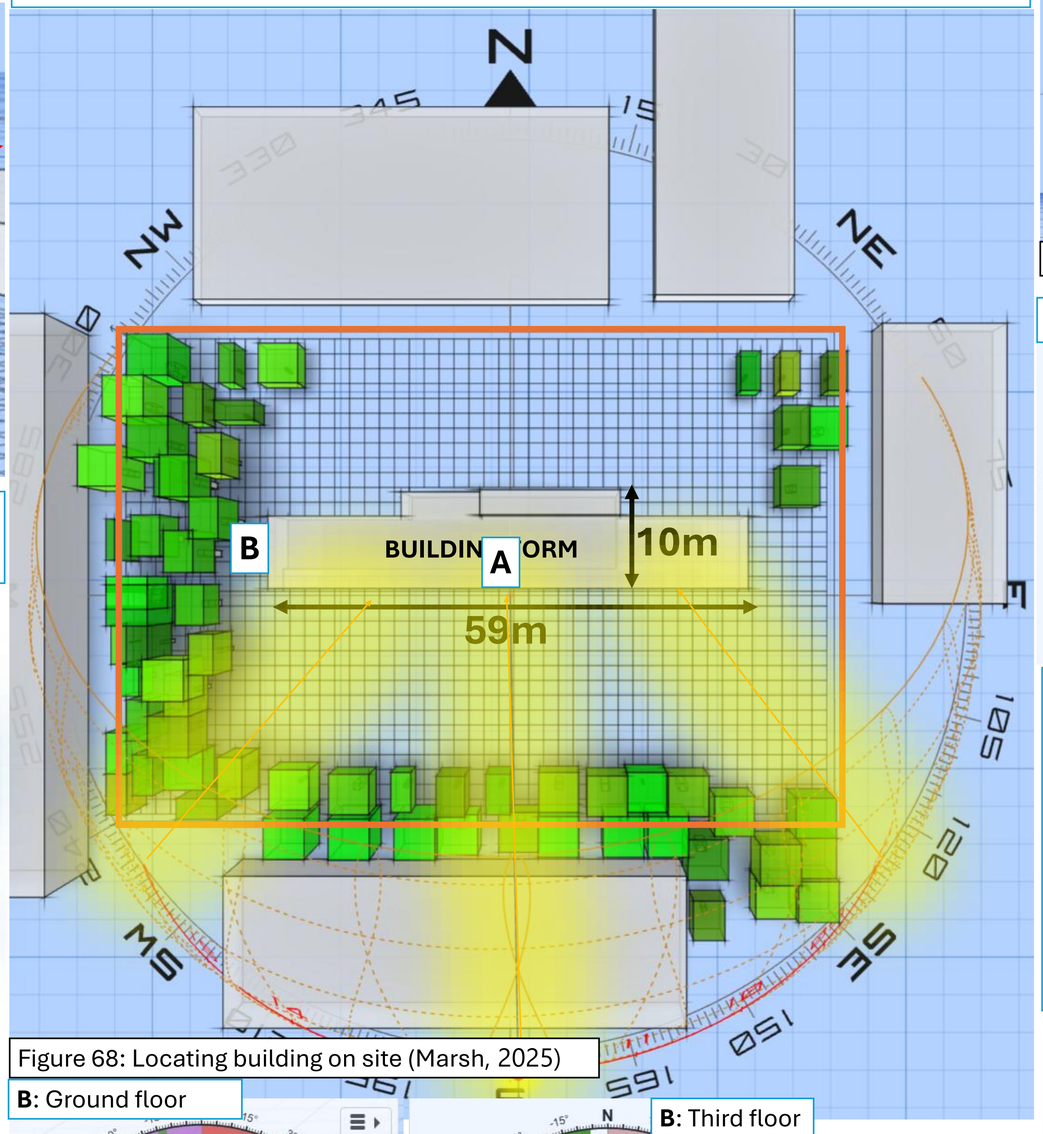
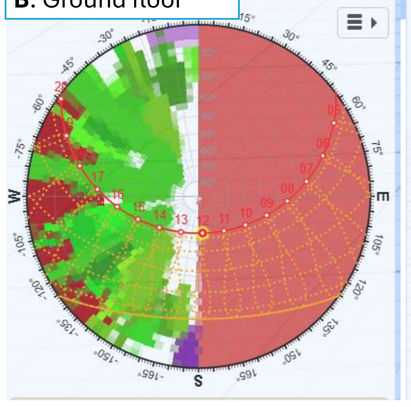
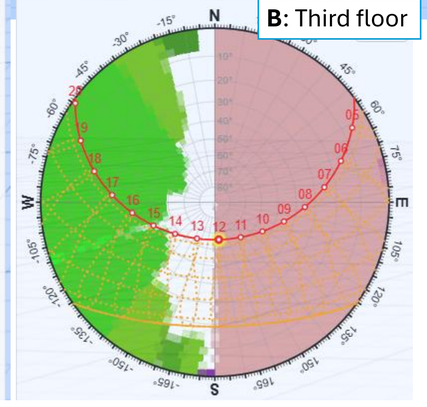


Figure 68: Locating building on site (Marsh, 2025)

B: Ground floor



B: Third floor



MAJOR CONSTRAINT: Building located at the south obstructs winter sun if building is close.

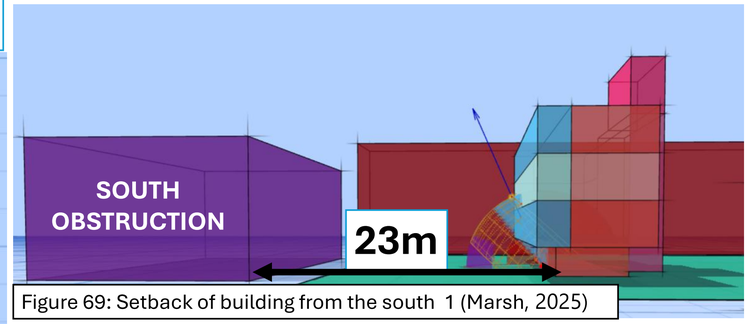
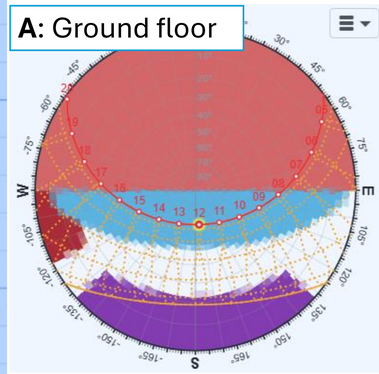
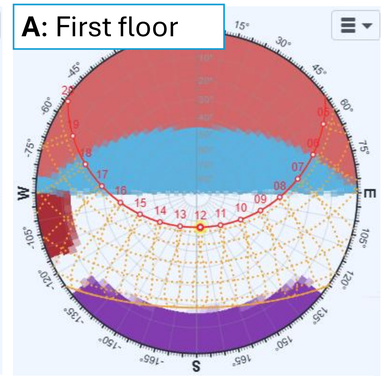


Figure 69: Setback of building from the south 1 (Marsh, 2025)

A: Ground floor



A: First floor



South

Obstruction prevents winter sun on the ground floor for all hours.

First floor receives winter sun from11am-13:00.

Recommendation:

Move building 36m away from south obstruction till ground floor receives winter sun.

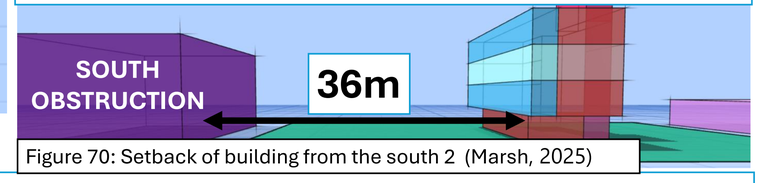


Figure 70: Setback of building from the south 2 (Marsh, 2025)

West

Trees around the site completely shades the west in Summer from 13.30 on ground floor so no shading required for that floor. The need for shading is for second and third floor from 14:00 and 14:30 in the afternoon.

Trees around the site completely shading the west will aid passive radiative cooling at night.

5.1 PASSIVE HEATING AND COOLING STRATEGIES

5.1.1 SUN SPACE + THERMAL MASS

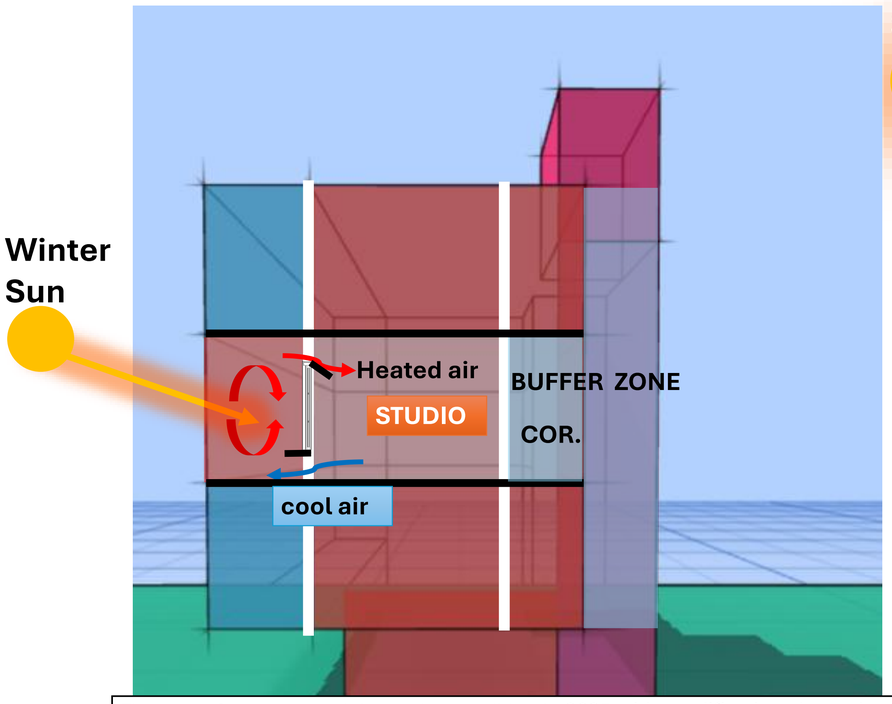


Figure 71: Sunspace and thermal mass (Marsh, 2025 with modifications by author)

5.1.2. THERMAL MASS – WINTER DAY

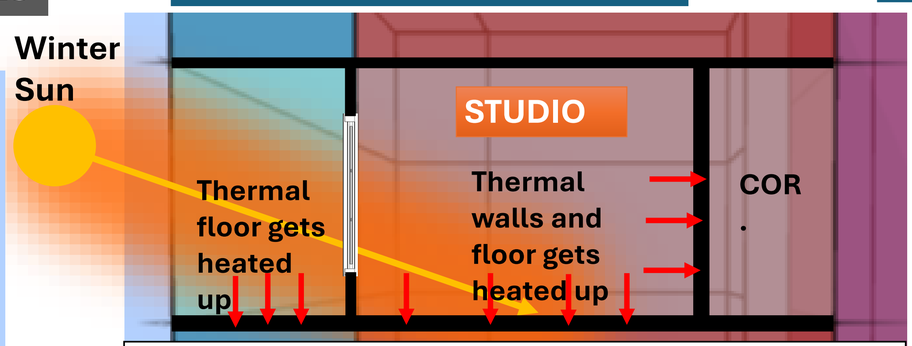


Figure 72: Thermal mass – winter day (Marsh, 2025 with modifications by author)

5.1.3. THERMAL MASS – WINTER NIGHT

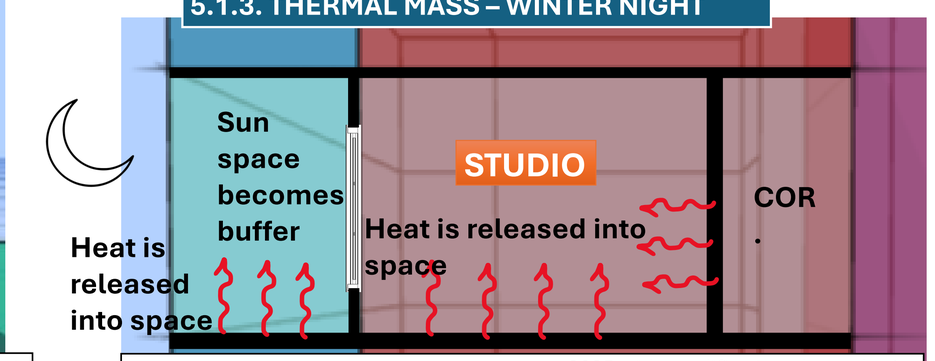


Figure 73: Thermal mass – winter night (Marsh, 2025 with modifications by author)

5.1.4.SUN SPACE – SUMMER VENTILATION AND COOLING

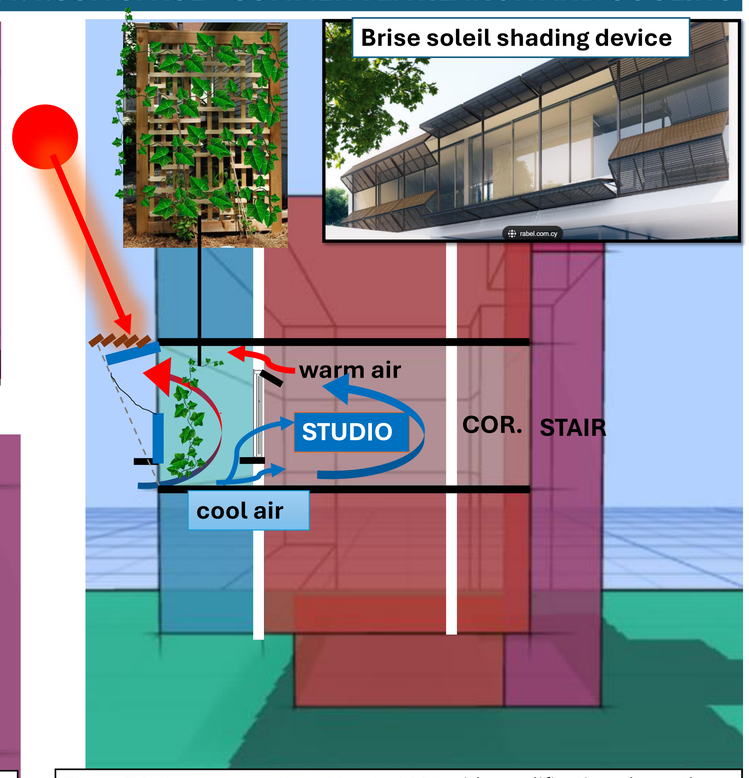


Figure 74: Summer cooling (Marsh, 2025 with modifications by author)

5.1.5 CORRIDOR VENTILATION

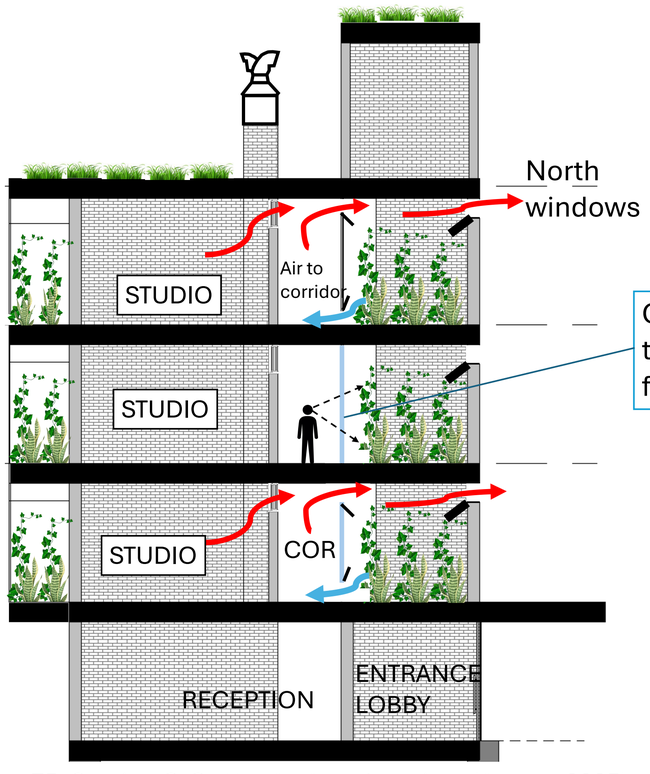
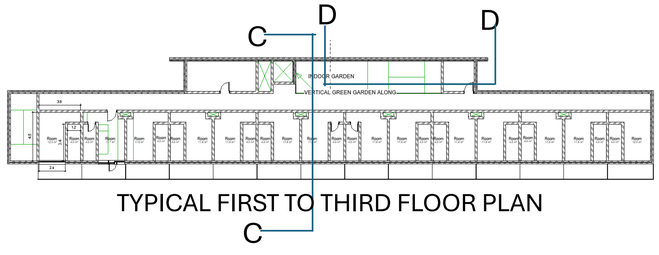


Figure 75: Section C-C : Ventilation of corridor (Author, 2025)



5.1.6.STAIR AND INDOOR GARDEN

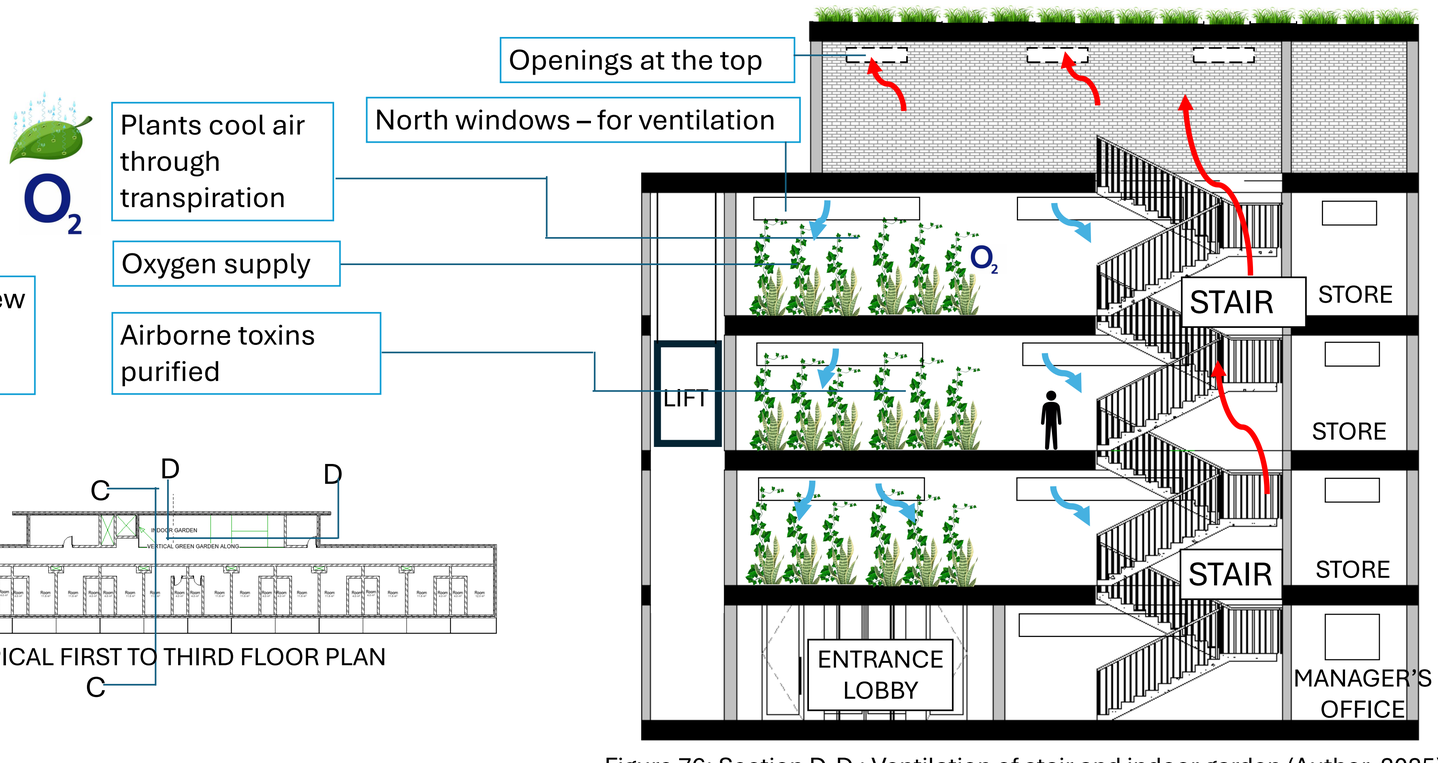
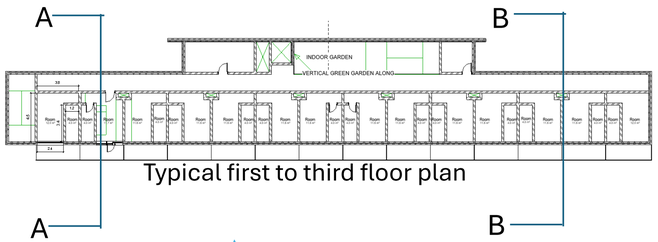


Figure 76: Section D-D : Ventilation of stair and indoor garden (Author, 2025)

5.2 PASSIVE COOLING STRATEGIES

5.2.1. CROSS VENTILATION



Typical first to third floor plan

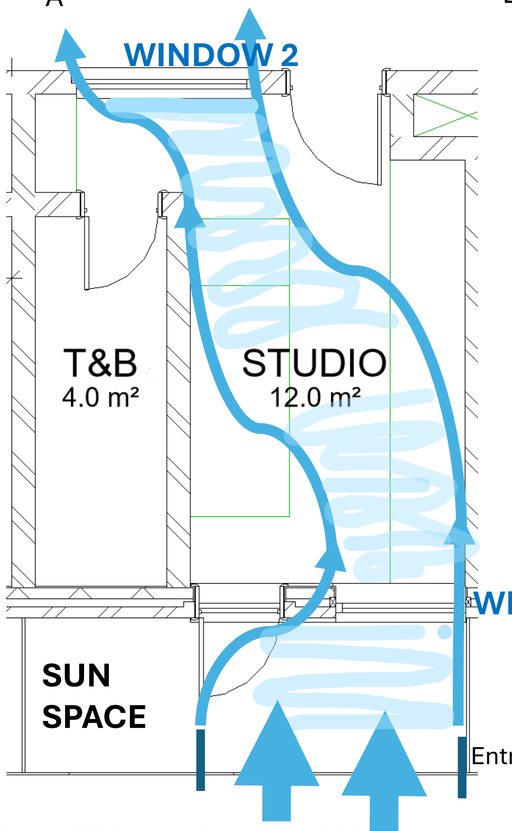


Figure 77: Typical Floor Plan Of Studio (Author, 2025)

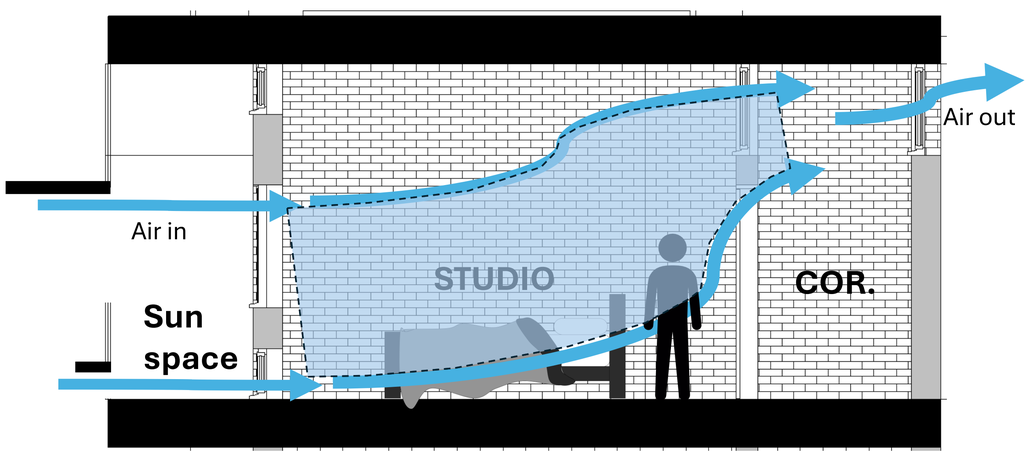


Figure 78: Section A-A – This is a typical windy day (Author, 2025)

5.2.2 EVAPORATIVE COOLING IN SUMMER

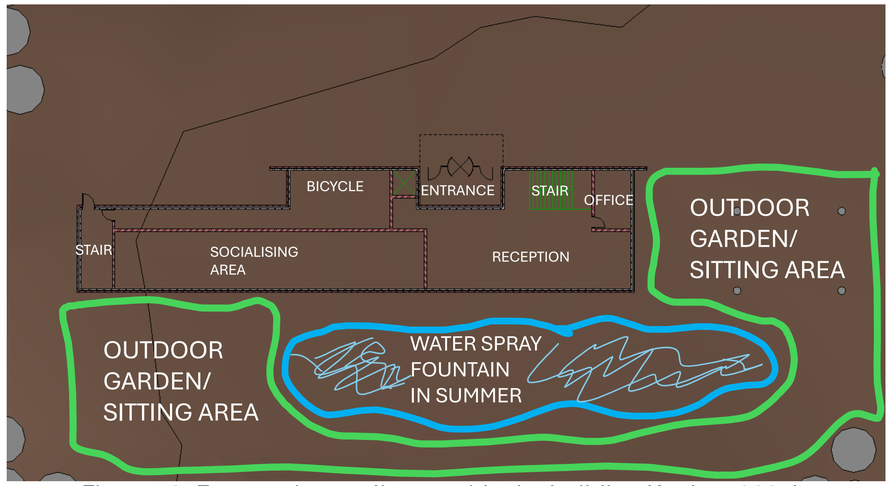
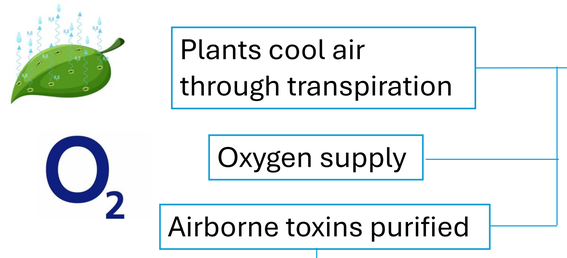
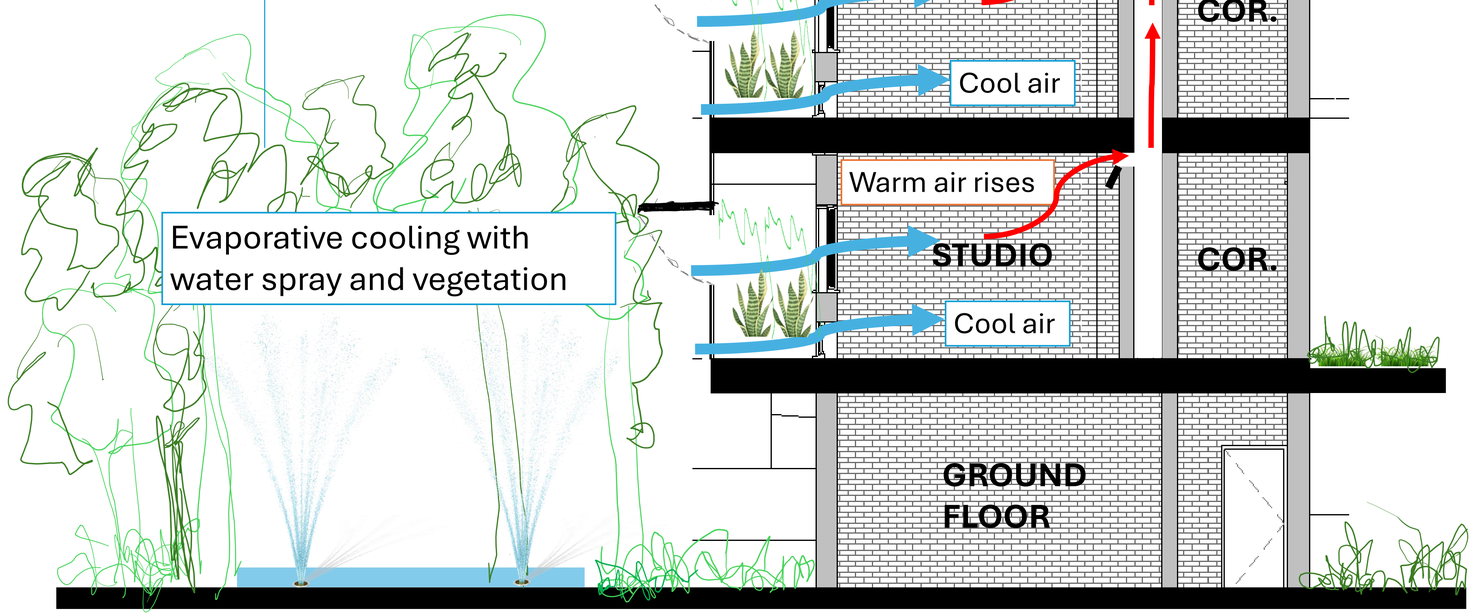


Figure 79: Evaporative cooling outside the building (Author, 2025)

Evaporative cooling with water spray and vegetation to change microclimate of the site in summer to be cooler.





5.2.3. WIND DRIVEN VENTILATION – WIND COWLS

Figure 80: Wind Cowls (Inhabitat, 2008)

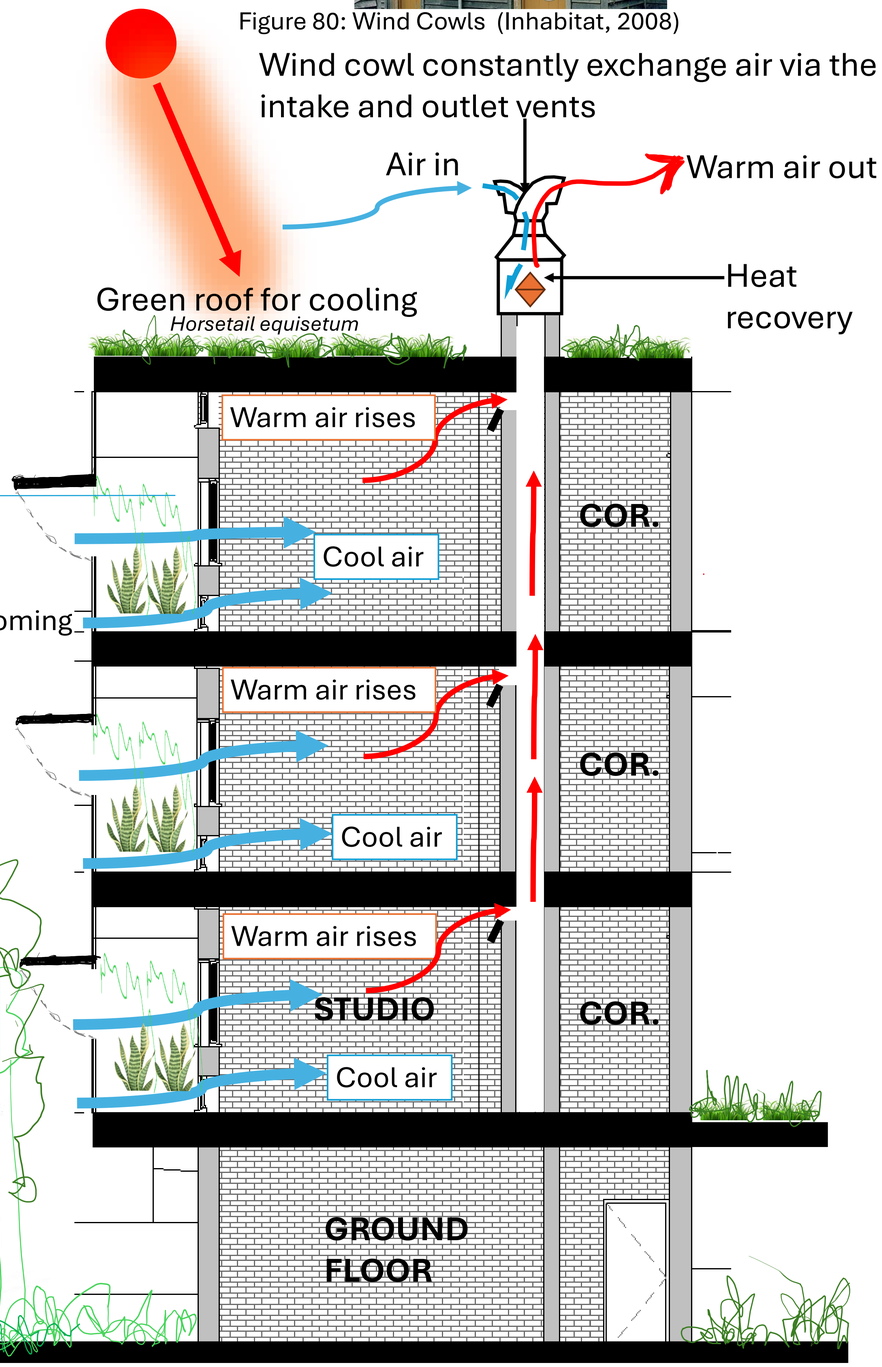


Figure 81: Section B-B – This is a typical still day (Author, 2025)

6.0 FLOOR PLANS

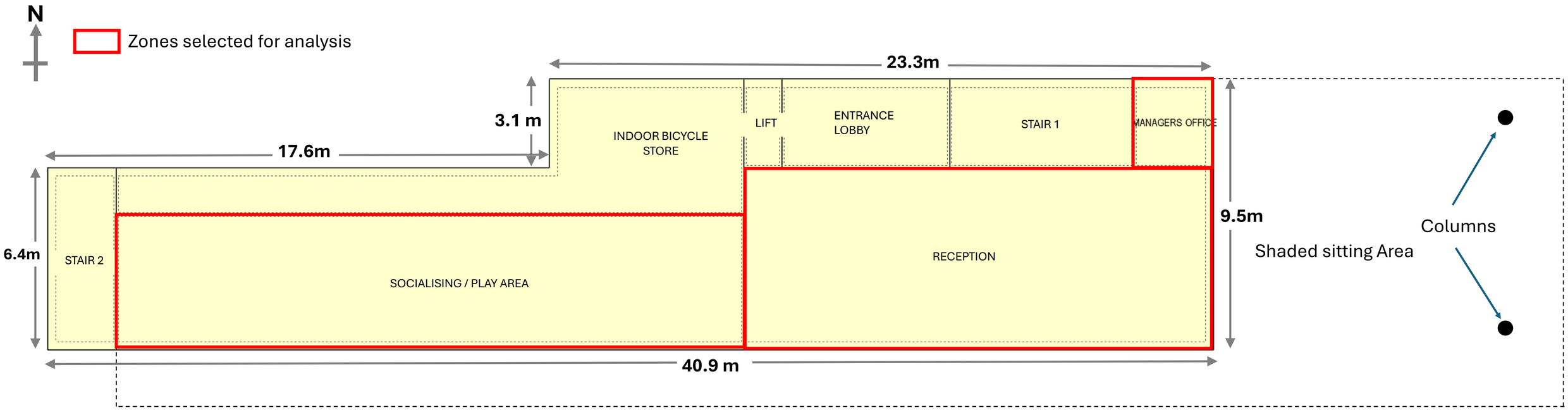


Figure 82: GROUND FLOOR PLAN (Author, 2025)

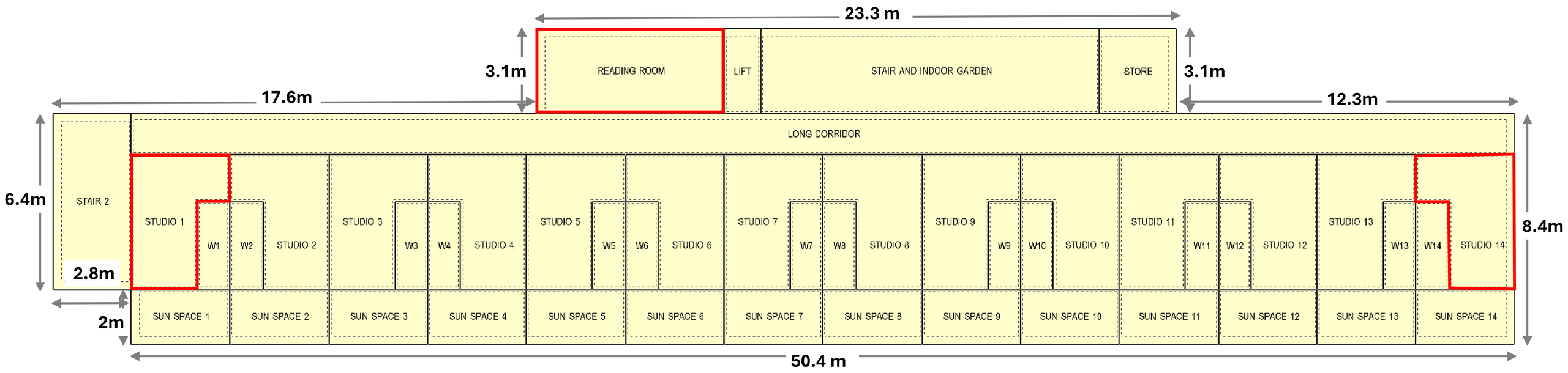


Figure 83: TYPICAL FIRST TO THIRD FLOOR PLAN (Author, 2025)

6.1 FINAL MODEL OF STUDENTS ACCOMMODATION BLOCK

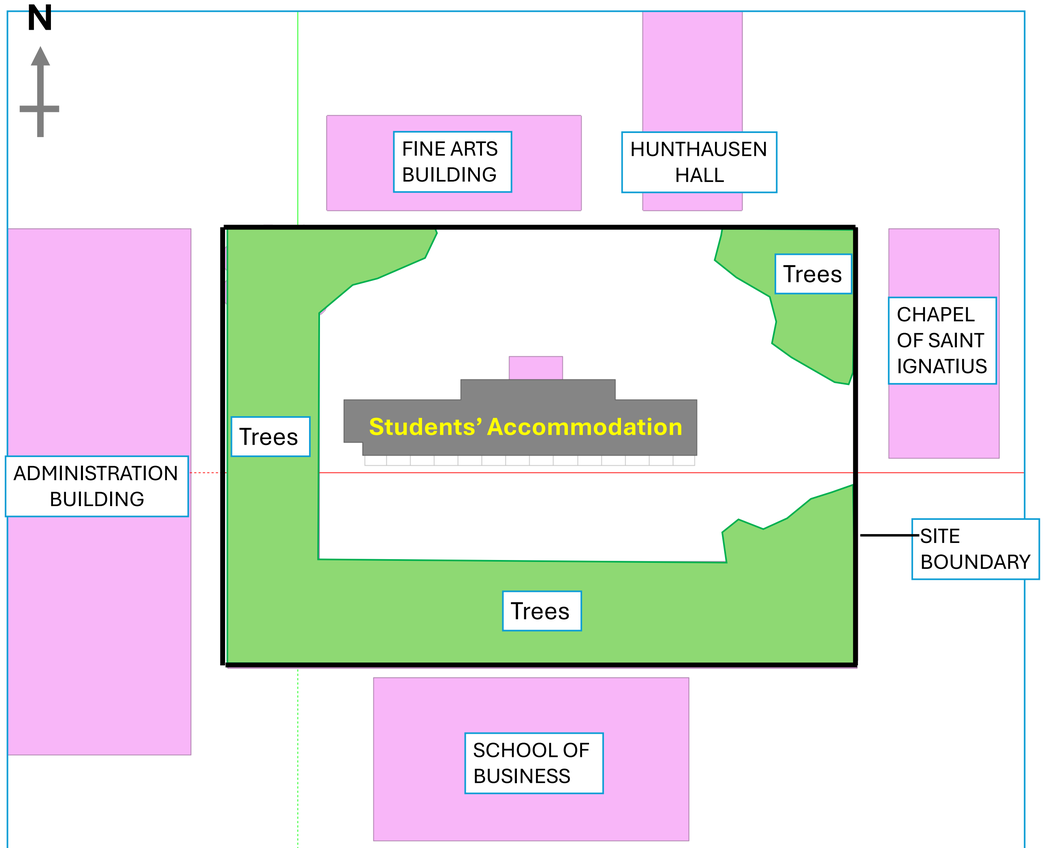


Figure 84: Site Layout of Students accommodation block (Author, 2025)

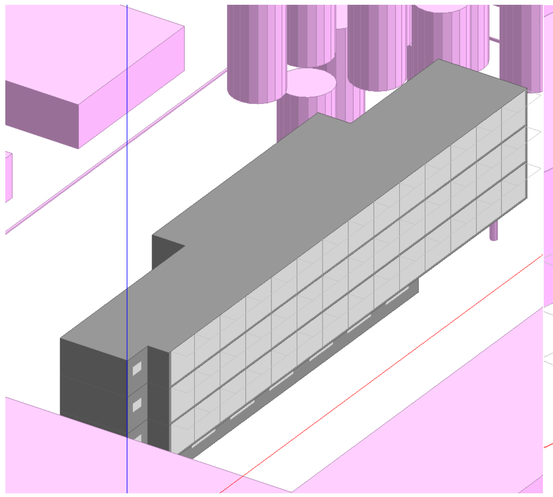


Figure 85: South-West View (Author, 2025)

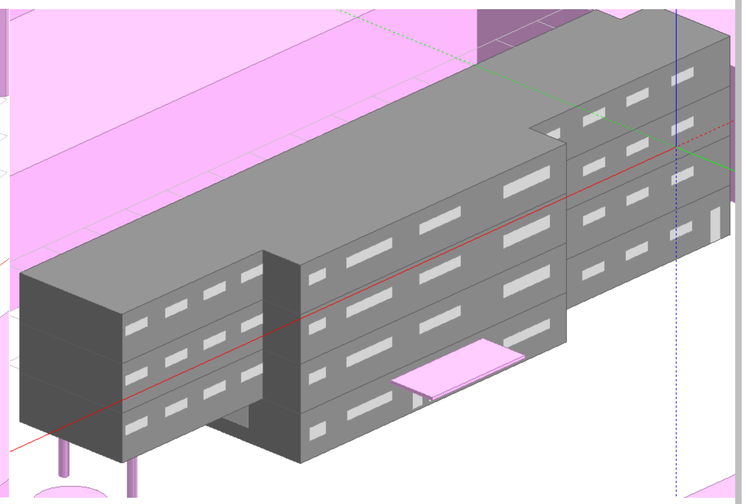


Figure 86: North-East View (Author, 2025)

6.2 THREE-DIMENSIONAL IMPRESSIONS OF BUILDING

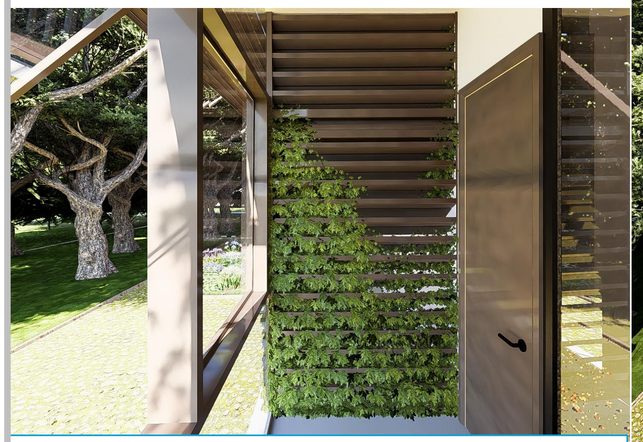


Figure 87 : View from sun space towards southern side (Author, 2025)

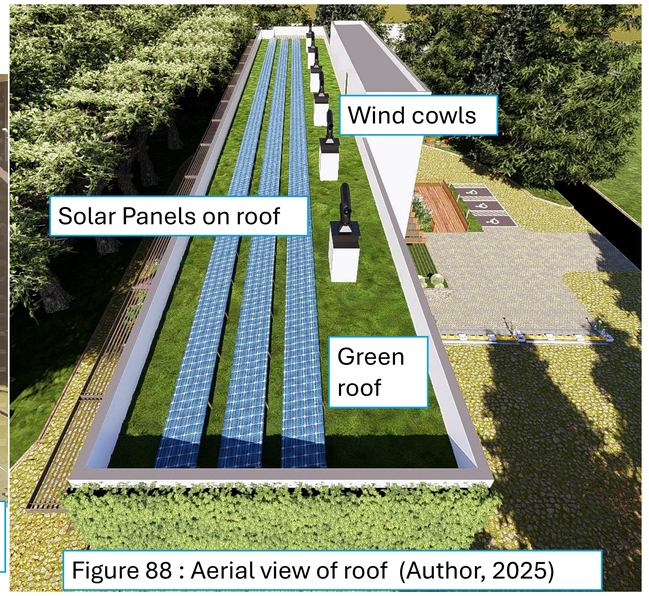


Figure 88 : Aerial view of roof (Author, 2025)

Figure 89 : South-East view of building (Author, 2025)

Figure 90 : North-West view of building (Author, 2025)

6.3 INTERIOR DETAILS OF STUDIO APARTMENT

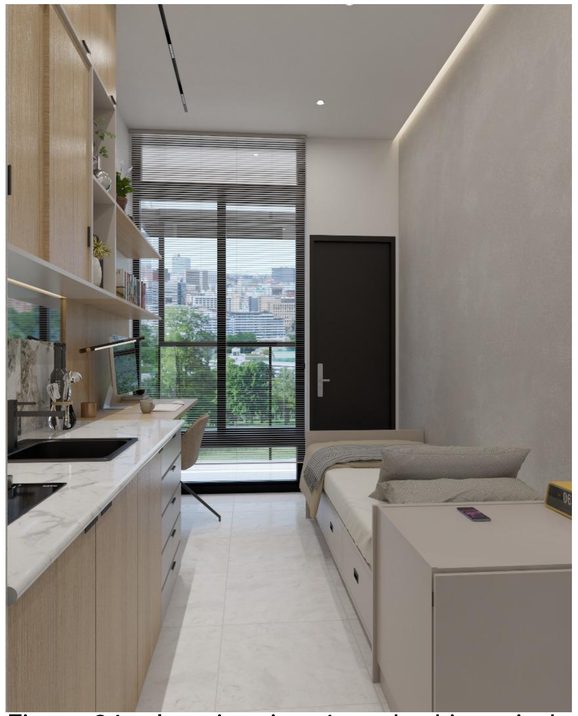


Figure 91a: Interior view 1overlooking window and door to sun space (Author, 2025)

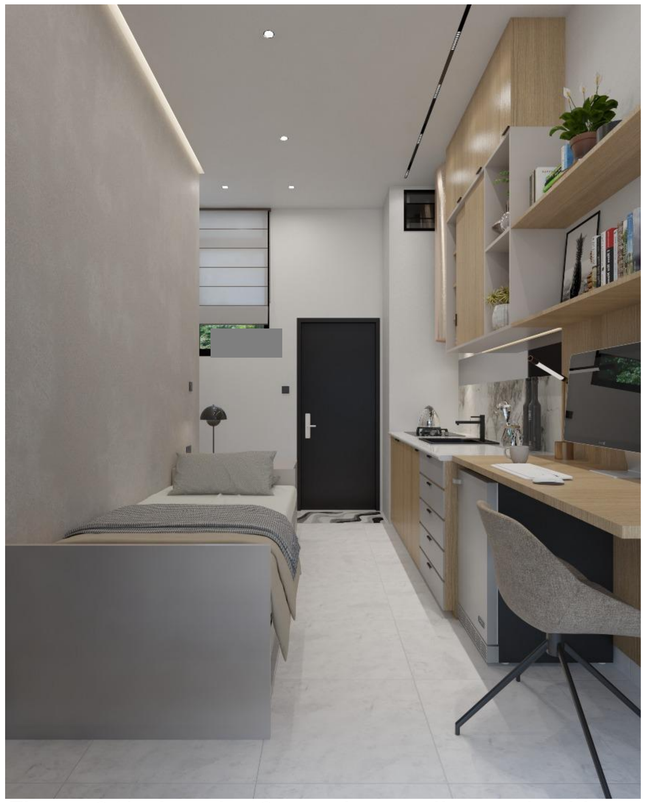


Figure 91b: Interior 2 – entire studio overlooking Sleeping and study area (Author, 2025)

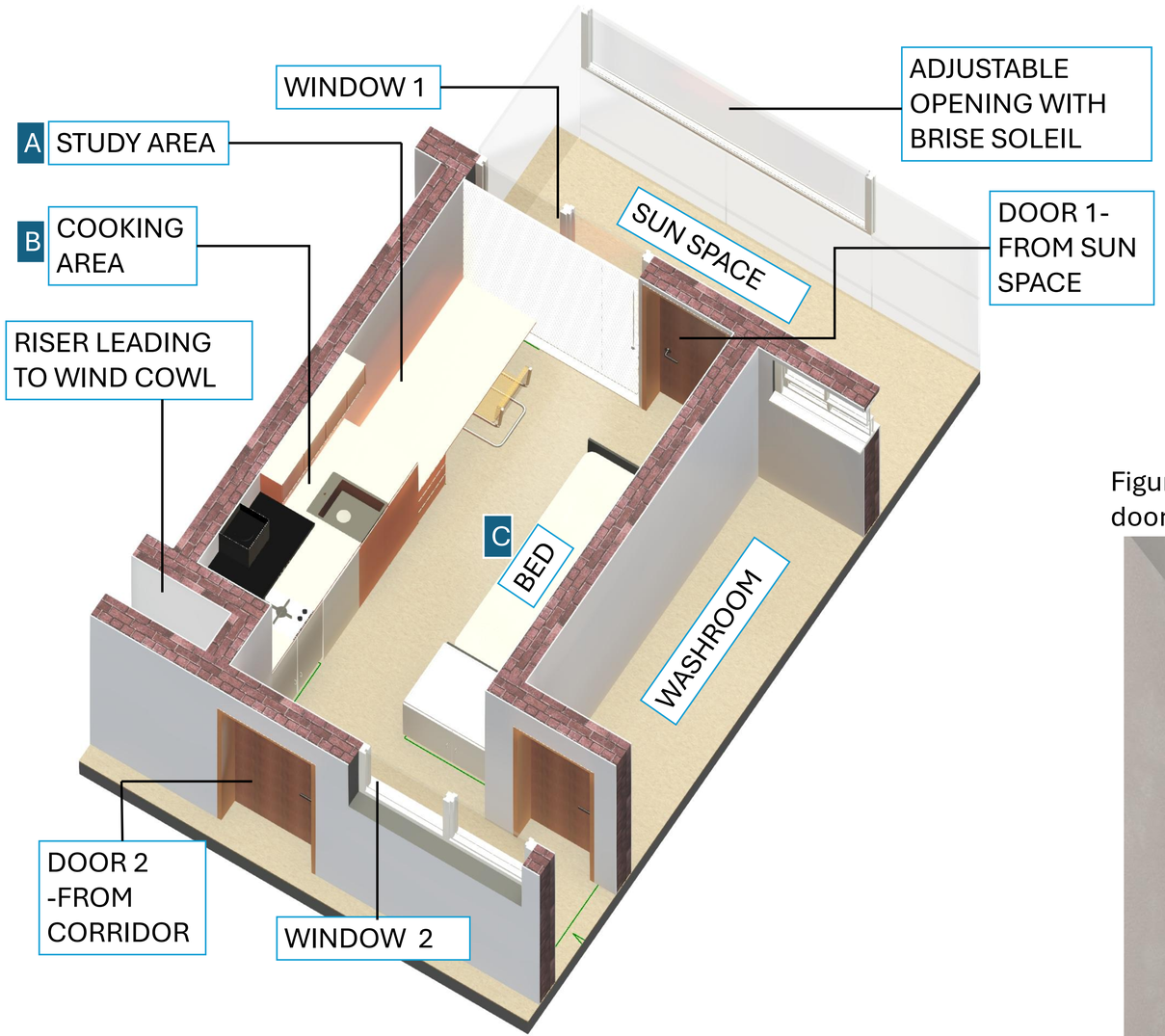


Figure 91: Axonometric view of a Typical Studio Apartment (Author, 2025)

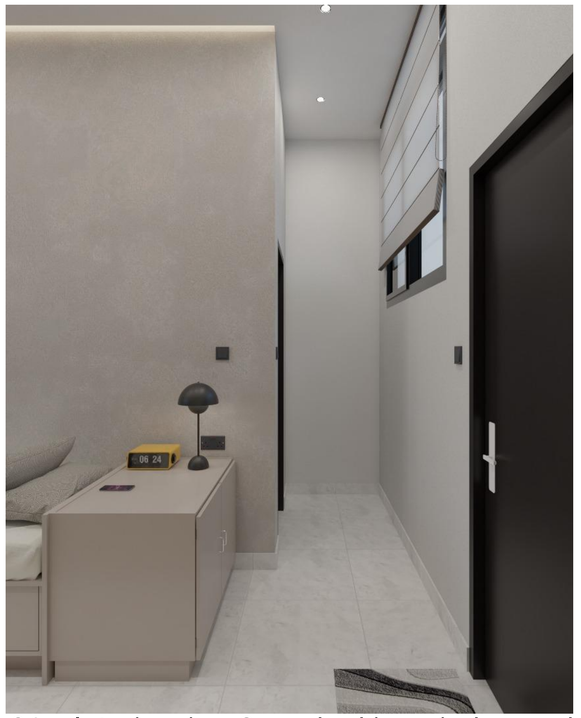


Figure 91c: Interior view 3 overlooking window and door leading to corridor (Author, 2025)

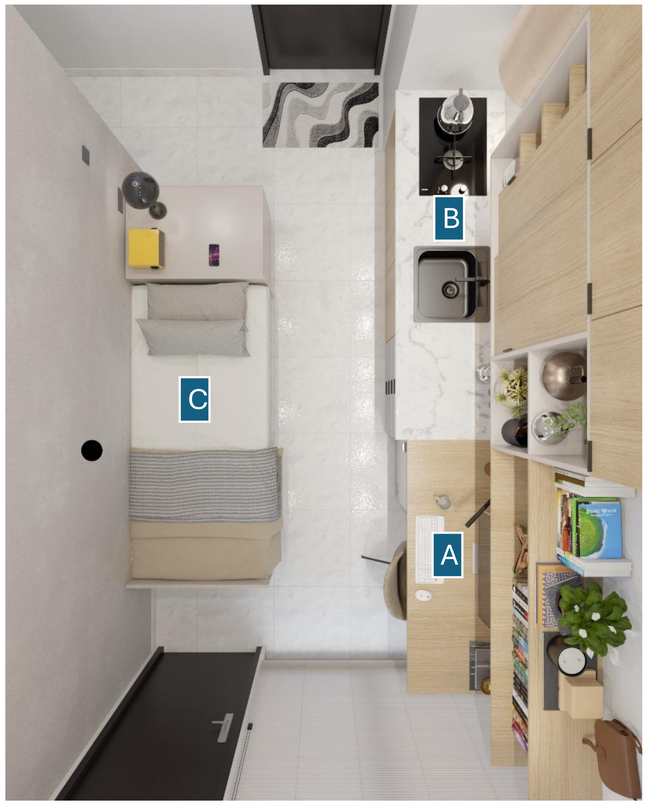
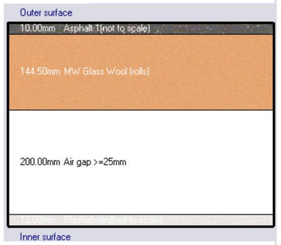


Figure 91d: Aerial view of Studio (Author, 2025)

Sustainable Design of Students' Accommodation Block

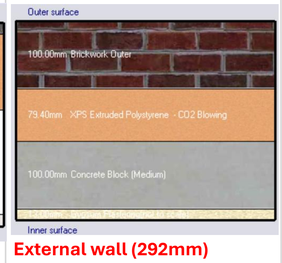
7.1.1 BASE CASE – ITERATION 1



Roof (367.5mm)

10mm asphalt
144.5mm glass wool
200mm air gap
13mm plasterboard

U-value(0.25)



External wall (292mm)

100mm brickwork
79.4mm XPS extruded polystyrene insulation
100mm concrete block (medium)
13mm gypsum plastering

U-value(0.351)

Ground floor slab (300mm)

100mm UREA fomaldehyde foam
100mm cast concrete
70mm floor screed
30mm timber flooring

U-value(0.314)

Window:Double Clear 3mm/13mmAir
Frame:painted wood

G-value: 0.764
VT: 0.812

U-value(2.7)

Table 3 : WINDOW SIZES (NO SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1m x 0.7m	2	North
STUDIO 1	1m x 0.9m	1	South
STUDIO 14	1m x 0.9m	1	South
RECEPTION	(South)1.5mx1m ; 1.5x1(East)	(South)4 ; (East)1	South & East
MANAGER'S OFFICE	1m x 0.7m	1	North
SOCIALISING AREA	1.5m	5	South

RESULT: BASE CASE PASSES TM AND ASE BUT EXTREMELY LOW SDA VALUES

ACHIEVE DAYLIGHTING TARGET BY INCREASING WINDOW SIZES AND CHANGING GLASS TYPE

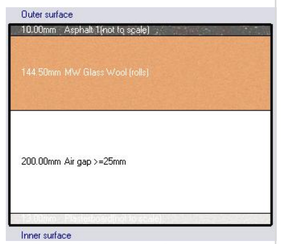
BASE CASE PASS CIBSE TM59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	0.28	N/A	Pass
SOCIALISINGAREA	0.00	N/A	Pass
READINGROOM	0.00	0.00	Pass
STUDIO1	0.00	0.00	Pass
STUDIO14	0.00	0.00	Pass

BASE CASE PASS ASE BUT NOT SDA

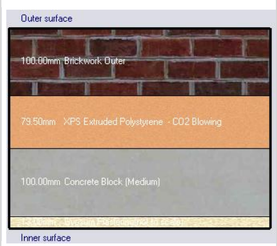
Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	17.7	0.00	0.00	17.66	100.0
STUDIO 1	12.5	0.00	0.00	12.55	100.0
STUDIO 14	11.5	0.11	0.95	11.51	100.0
RECEPTION	113.7	56.42	49.63	113.49	99.8
MANAGERS OFFICE	6.6	0.00	0.00	6.65	100.0
SOCIALISING AREA	95.8	8.76	9.14	95.74	99.9
	257.9	**65.28**	**25.32**	**257.59**	**99.9**

7.1.2 ITERATION 2



Roof (367.5mm)
10mm asphalt
144.5mm MW glass wool
200mm air gap
13mm plasterboard

U-value(0.25)

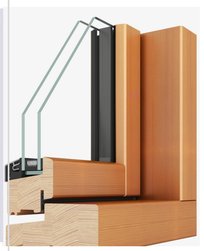


External wall (292mm)
100mm brickwork
79.5mm XPS insulation
100mm concrete block
13mm gypsum plastering

U-value(0.350)

Ground floor slab (300mm)
30mm timber flooring
70mm floor screed
100mm cast concrete
132.7mm UREA fomaldehyde
Foam

U-value(0.25)



Window:Double Clear
3mm/13mmAir
Frame:painted wood
G-value: 0.691
VT: 0.744

U-value(1.96)

Table 4: WINDOW SIZES (1m SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.78m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	1.7m x 0.9m	1	North
SOCIALISING AREA	1.9mx2.1m	9	South

RESULT: OVERHEATING OF SPACES AFTER INCREASING WINDOW SIZES. IMPROVED SDA BUT ASE SHOULD BE IMPROVED

ADDRESS OVERHEATING AND IMPROVE ASE. INCREASE U-VALUE OF BUILDING FABRIC.
ADDRESS OVERHEATING, INCREASE WINDOW SHADING AND CHANGE GLASS TYPE TO IMPROVE ASE.

ITERATION 2 – TWO SPACES FAIL CIBSE TM59

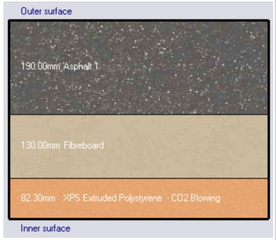
Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	5.68	N/A	Fail
SOCIALISINGAREA	6.01	N/A	Fail
READINGROOM	0.00	4.00	Pass
STUDIO1	0.57	112.50	Fail
STUDIO14	0.00	6.00	Pass

ITERAION 2 PASS BOTH ASE AND SDA

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	17.7	14.37	81.38	17.66	100.0
STUDIO 1	12.5	9.10	72.50	11.08	88.3
STUDIO 14	11.5	7.68	66.67	10.09	87.6
RECEPTION	113.7	113.21	99.59	101.29	89.1
MANAGERS OFFICE	6.6	4.77	71.83	6.65	100.0
SOCIALISING AREA	95.8	95.17	99.30	40.73	42.5
	257.9	**244.30**	**94.73**	**187.51**	**72.7**

Source of all images and tables : DesignBuilder version 7 and EnergyPlus

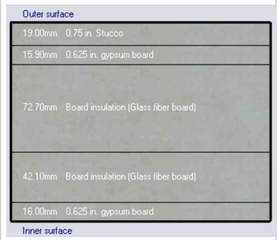
7.1.3 ITERATION 3



Roof (402mm)

190mm asphalt
130mm fibreboard
82.3mm XPS

U-value(0.20)



External wall (166mm)

19mm Stucco
16mm Gypsumboard
72.7mm glass fibre insulation
42.1mm glass fibre insulation
16mm gypsumboard

U-value(0.28)

Ground floor slab (358mm)

50mm floor screed
182.4mm EPS insul.
100mm concrete
25mm brick slips
Clay underfloor

U-value(0.20)

Window: LoE Clr
3mm double – 13mm Air

G-value: 0.65
VT: 0.76

U-value(1.78)

Table 5: WINDOW SIZES (1.5m SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.78m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	1.7m x 0.9m	1	North
SOCIALISING AREA	1.9mx2.1m	9	South

RESULT: SDA ACHIEVED BUT ASE IN SOME SPACES HAS TO BE IMPROVED AND OVERHEATING REMEDIED FOR RECEPTION

IMPROVE U-VALUE OF FABRIC AND CHANGE GLASS TYPE

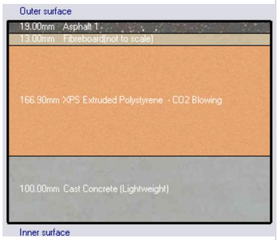
ITERATION 3 – ONE SPACE FAIL CIBSE TM59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.41	N/A	Pass
RECEPTION	3.34	N/A	Fail
SOCIALISINGAREA	2.95	N/A	Pass
READINGROOM	0.00	1.00	Pass
STUDIO1	0.00	6.50	Pass
STUDIO14	0.00	2.50	Pass

ITERAION 3 PASES BOTH ASE AND SDA

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	18.8	15.73	83.51	18.84	100.0
STUDIO 1	12.5	8.05	64.17	11.08	88.3
STUDIO 14	12.1	6.87	56.67	10.70	88.3
RECEPTION	117.2	112.85	96.27	99.07	84.5
MANAGERS OFFICE	7.3	4.12	56.34	7.32	100.0
SOCIALISING AREA	98.6	95.77	97.13	36.74	37.3
	266.7	**243.39**	**91.28**	**183.75**	**68.9**

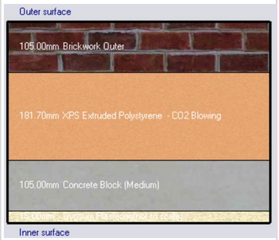
7.1.4 ITERATION 4



Roof (300mm)

19mm asphalt
13mm fibreboard
167mm XPS
100mm concrete

U-value(0.18)



External wall (407mm)

105mm brickwork
182mm XPS insulation
105mm concrete
15mm gypsum plaster

U-value(0.17)

Ground floor slab (440mm)

30mm timber floor
70mm floor screed
100mm concrete
240mm fomaldehyde foam

U-value(0.15)

Window:LoE Elec Abs bleached 6mm double Argon

G-value: 0.48
VT: 0.657

U-value(1.3)

Table 6: WINDOW SIZES(1.5m SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.78m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	1.7m x 0.9m	1	North
SOCIALISING AREA	1.9mx2.1m	9	South

RESULT: OVERHEATING SOLVED BUT THE SDA FOR MANAGER'S OFFICE AND ASE FOR SOCIALISING AREA SHOULD BE IMPROVED

IMPROVE U-VALUE OF FABRIC TO IMPROVE CIBSE VALUES FOR RECEPTION, SOCIALISING AREA AND STUDIO 1.
CHANGE WINDOW SIZES FOR MANAGER'S OFFICE AND SOCIALISING AREA.
CHANGE GLASS TYPE TO IMPROVE SDA AND ASE

ITERATION 4 – ALL SPACES PASS CIBSE TM59

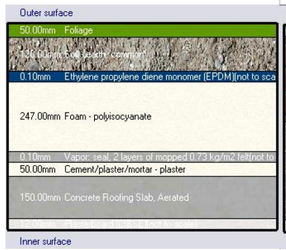
Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	2.70	N/A	Pass
SOCIALISINGAREA	2.62	N/A	Pass
READINGROOM	0.00	0.00	Pass
STUDIO1	0.00	3.00	Pass
STUDIO14	0.00	0.00	Pass

ITERAION 4 PASES BOTH ASE AND SDA – IMPROVEMENT REQUIRED

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	16.6	9.95	59.88	16.62	100.0
STUDIO 1	12.5	6.69	53.33	11.40	90.8
STUDIO 14	11.0	5.54	50.48	10.34	94.3
RECEPTION	110.5	104.16	94.26	107.37	97.2
MANAGERS OFFICE	6.1	2.98	49.09	6.07	100.0
SOCIALISING AREA	93.3	87.86	94.14	48.01	51.4
	250.0	**217.18**	**86.86**	**199.81**	**79.9**

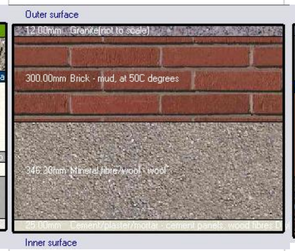
Source of all images and tables : DesignBuilder version 7 and EnergyPlus

7.1.5 ITERATION 5



Green Roof (629mm)

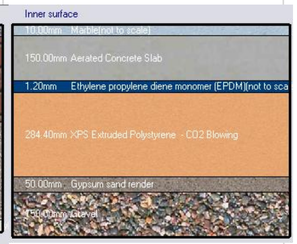
50mm vegetation
120mm soil
Waterproofing
247mm polysiocyanate insulation
Vapour seal
50mm screed
150mm concrete
12mm plasterboard
U-value(0.1)



External wall (683mm)

glass-fiber-reinforced concrete panels
300mm brick mud
346.3mm mineral fibre wool
12mm white granite

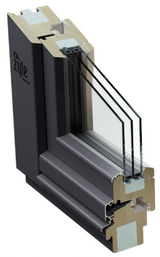
U-value(0.1)



Ground floor slab (540mm)

10mm marble
150mm aerated concrete
1.2mm damp proof
284.4mm extruded polystrene
50mm blinding
150mm hardcore

U-value(0.149)



Window: Triple lowE clear 3mm/13mm Argon
Frame: Wood

G-value: 0.47
VT: 0.66

U-value(0.78)

Table 7: WINDOW SIZES (1.5m SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.65m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	2m x 0.9m	1	North
SOCIALISING AREA	1.7mx1.65m	9	South

RESULT: MANAGER'S OFFICE SDA AND SOCIALISING AREA ASE ACHIEVED
RESULT: CIBSE TM 59 VALUES IMPROVED, SDA AND ASE TARGETS ACHIEVED.

ITERATION 5 – ALL SPACES PASS CIBSE TM59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	0.99	N/A	Pass
SOCIALISINGAREA	0.10	N/A	Pass
READINGROOM	0.00	0.00	Pass
STUDIO1	0.00	0.00	Pass
STUDIO14	0.00	0.00	Pass

ITERATION 5 ACHIEVES ASE AND SDA

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	14.0	7.36	52.52	14.02	100.0
STUDIO 1	10.2	5.79	56.84	9.64	94.7
STUDIO 14	8.7	4.40	50.62	8.49	97.5
RECEPTION	103.2	83.46	80.88	101.19	98.1
MANAGERS OFFICE	4.7	2.62	56.10	4.68	100.0
SOCIALISING AREA	85.6	58.38	68.22	80.94	94.6
	226.3	**162.02**	**71.58**	**218.95**	**96.7**

Source of all images and tables : DesignBuilder version 7 and EnergyPlus

7.1.6 SUMMARY OF ALL ITERATIONS

The base case passed CIBSE with extremely low window-to-wall ratio but gave poor daylighting results.

The following changes were made to finally achieve all requirements for CIBSE, SDA and ASE.
- **Increasing glazing areas** to maximize passive solar heat gain in cold months
- **Thermal mass** (natural stone flooring, concrete and brick) to absorb, store and release the heat
- **Increasing insulation** with extruded polystyrene(XPS) and making the envelope more airtight to prevent infiltration
- **Windows**: Some of the above-mentioned steps affected daylight targets. Solution was changing the windows' glass type from double 3mm clear glass to triple low-E clear 3mm with13mm Argon, achieving good U-value whiles attaining SDA and ASE daylighting targets.

Combining the above strategies also improved the operative temperatures comparing outside dry bulb temperature, operative temperature of the base case till iteration 5.

8.0 TEMPERATURE RANGE COMPARISON OF ITERATIONS

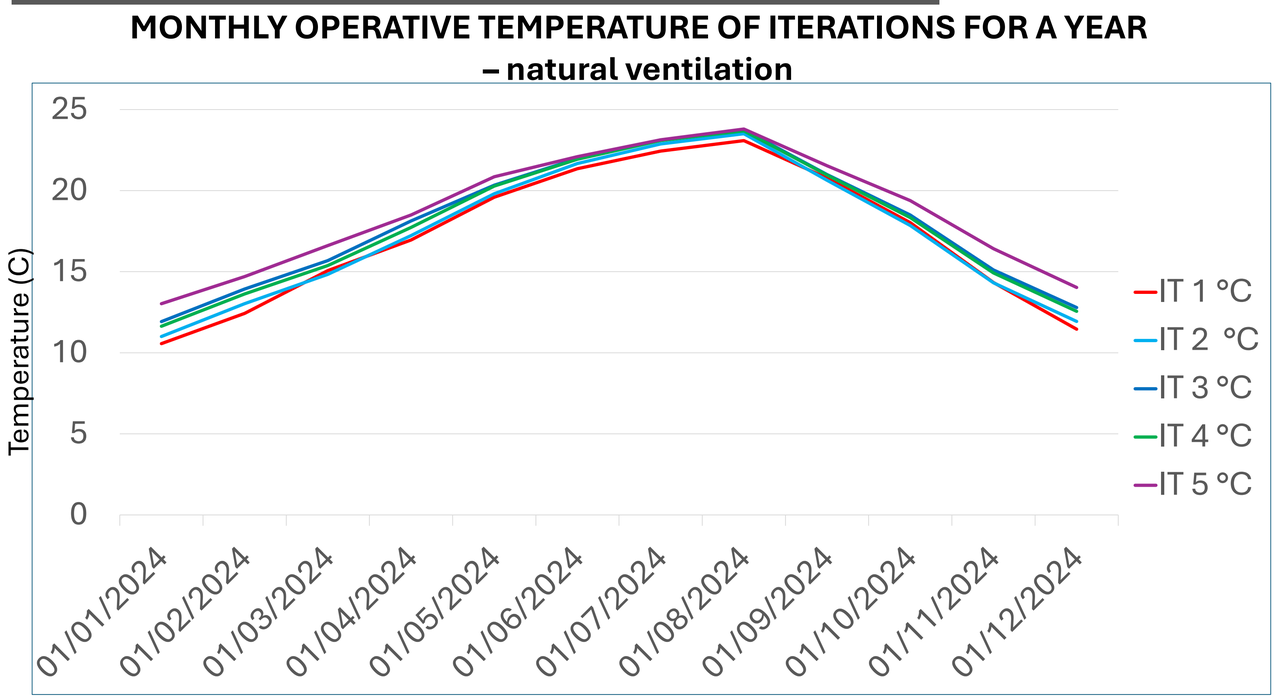


Figure 92: monthly operative temperature of all iterations for one year

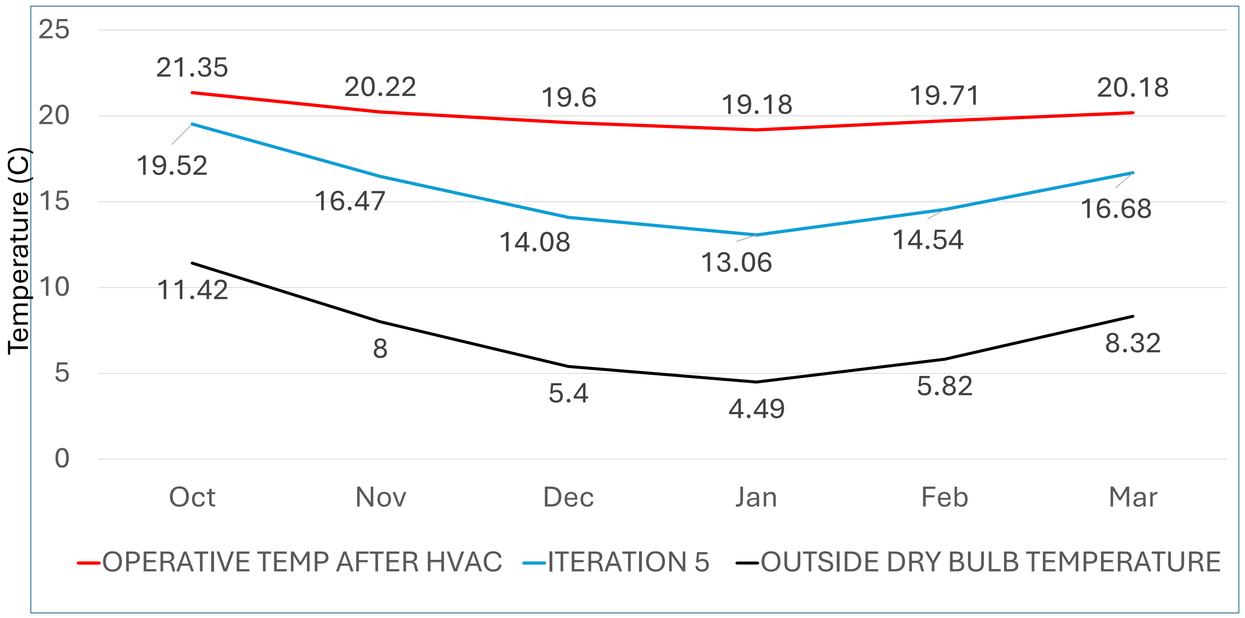


Figure 94: monthly operative temperatures from October to March

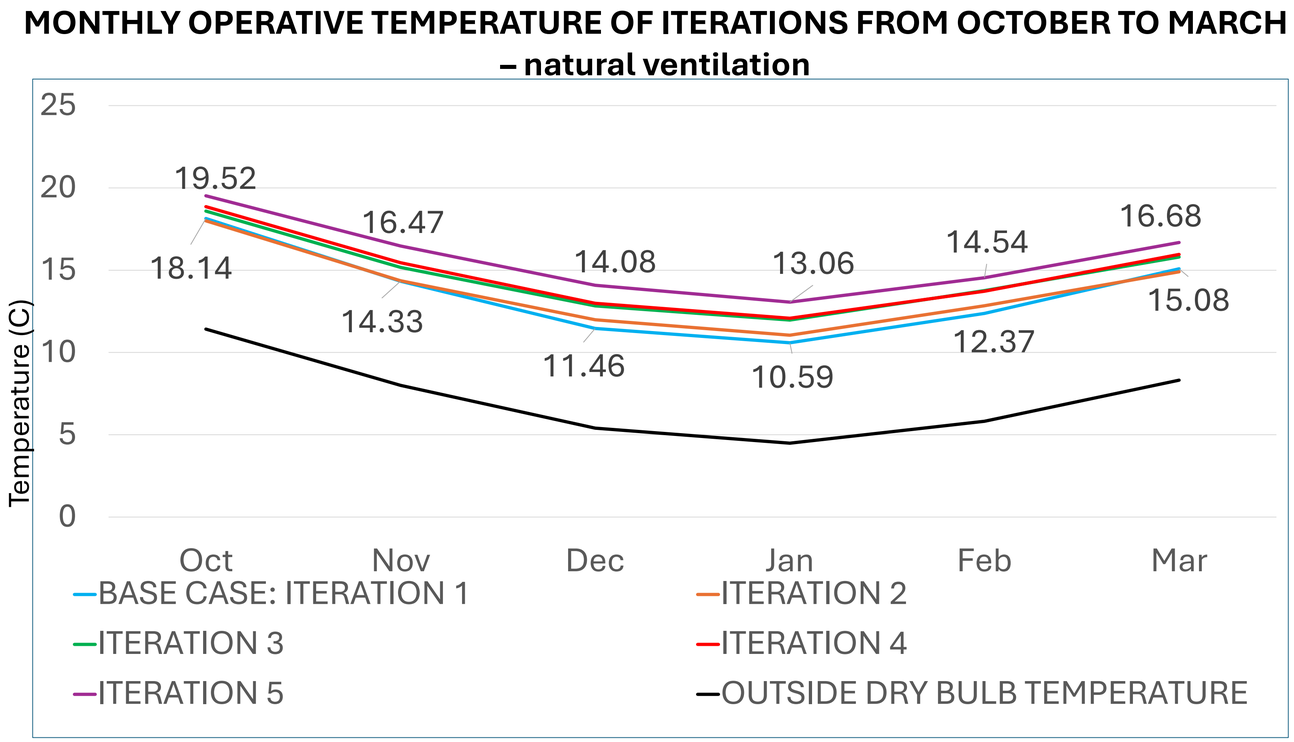


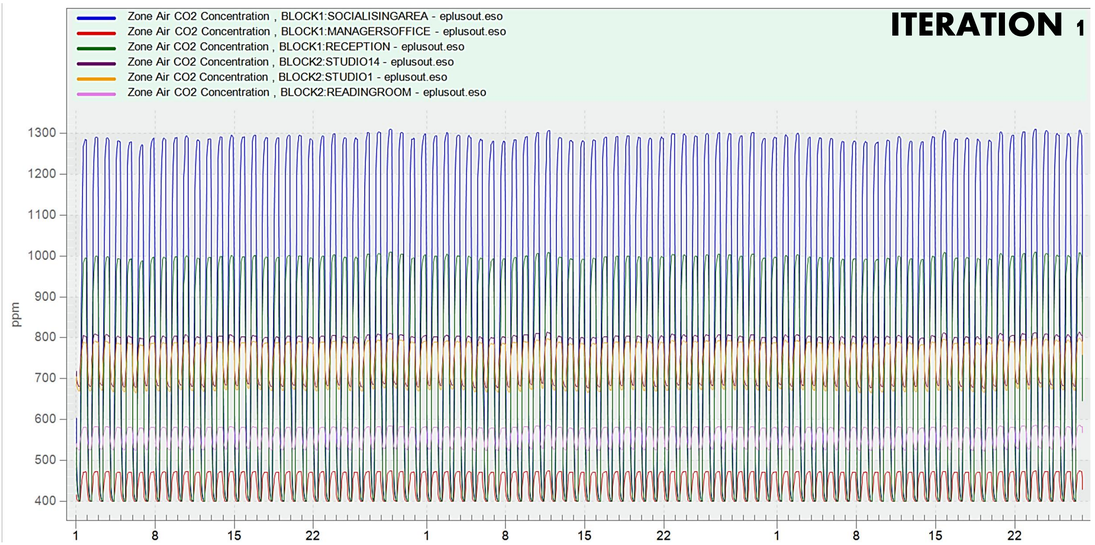
Figure 93: monthly operative temperatures from October to March

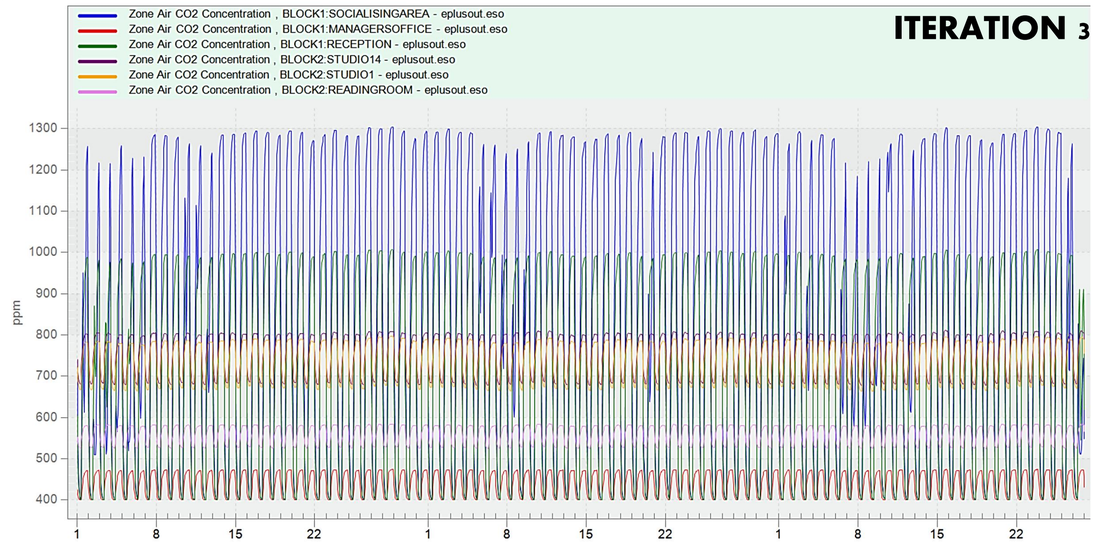
The operative temperatures of the project increased from iteration 1 (Base Case) to 5 due to passive strategies adopted. Comfort levels are achieved between the months of April and October where operative temperatures are between 18 and 25 degrees. Natural ventilation will be employed during these months to achieve indoor thermal comfort for occupants as seen in Figure 92.

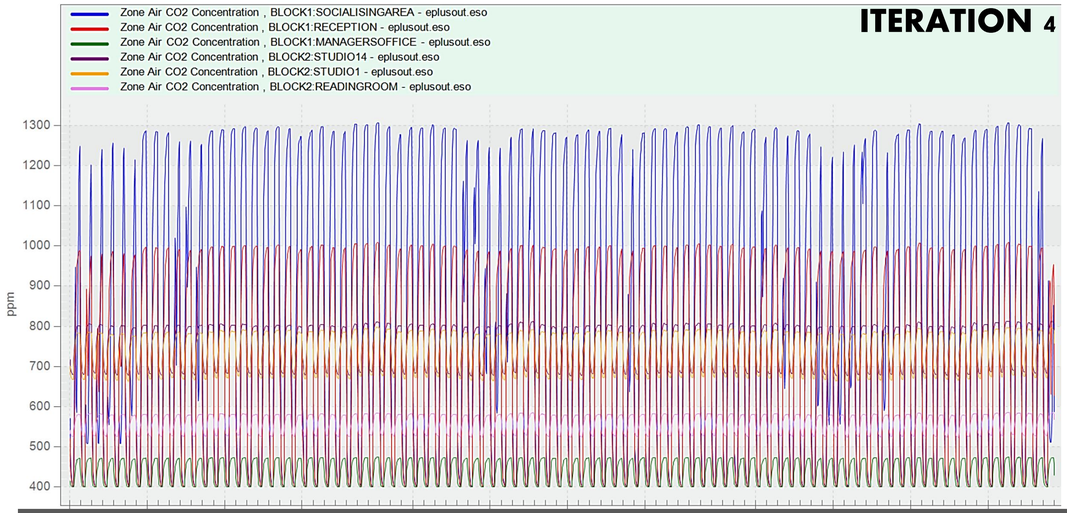
Figure 93 shows improvements in operative temperatures compared to the base case. The coldest month in January improved from 10.59 degrees(base case) to 13.06 degrees (Iteration 5) which shows a significant increase in operative temperature of 2.47 degrees, to increase the indoor comfort temperature.

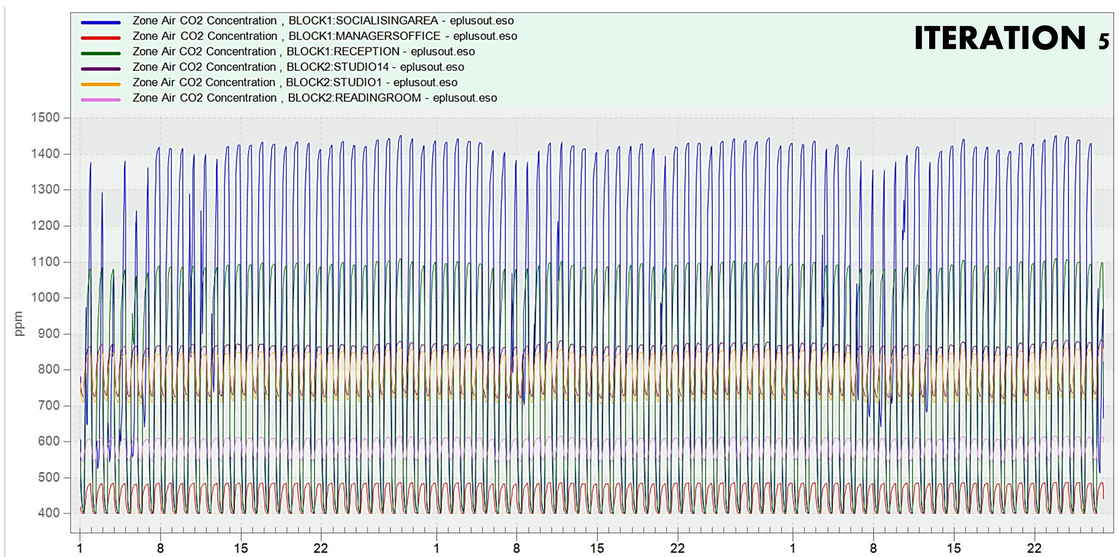
Figure 94 shows the difference between the outdoor dry bulb temperature, the improvements made for Iteration 5 and the operative temperature drastically improved after a mechanical system was introduced to keep comfort levels within 21 to 24 degrees as the set point temperatures.

8.1 HOURLY C_{02} LEVELS OF ZONES FROM DECEMBER TO FEBRUARY - NATURAL VENTILATION MODE









8.1.1 ACCEPTABLE C_{02} LEVELS AFTER MECHANICAL VENTILATION FOR ITERATION 5

8.2 PERFORMANCE TARGETS VERSUS ACTUAL OUTCOMES

TOTAL ANNUAL COMFORT HOURS VERSUS DISCOMFORT HOURS FOR BUILDING

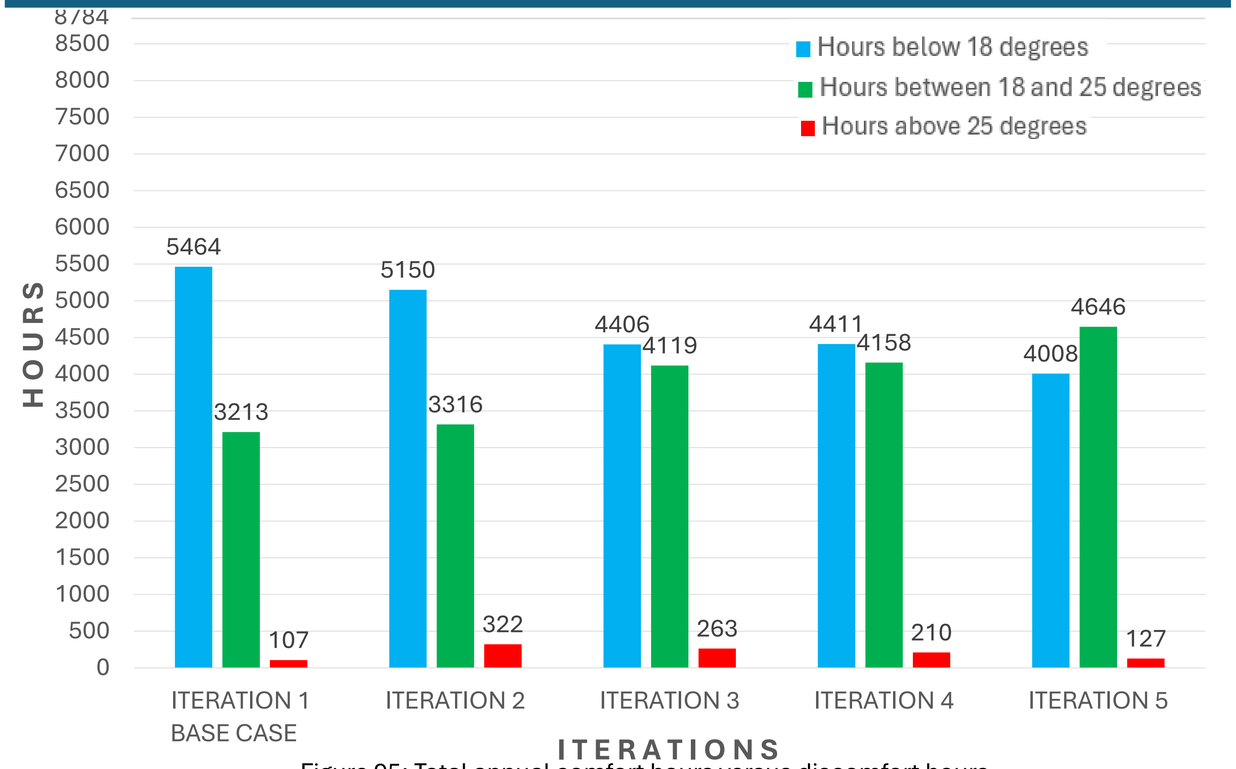


Figure 95: Total annual comfort hours versus discomfort hours

SDA, ASE AND CIBSE TM 59 TARGETS FOR 6 ZONES

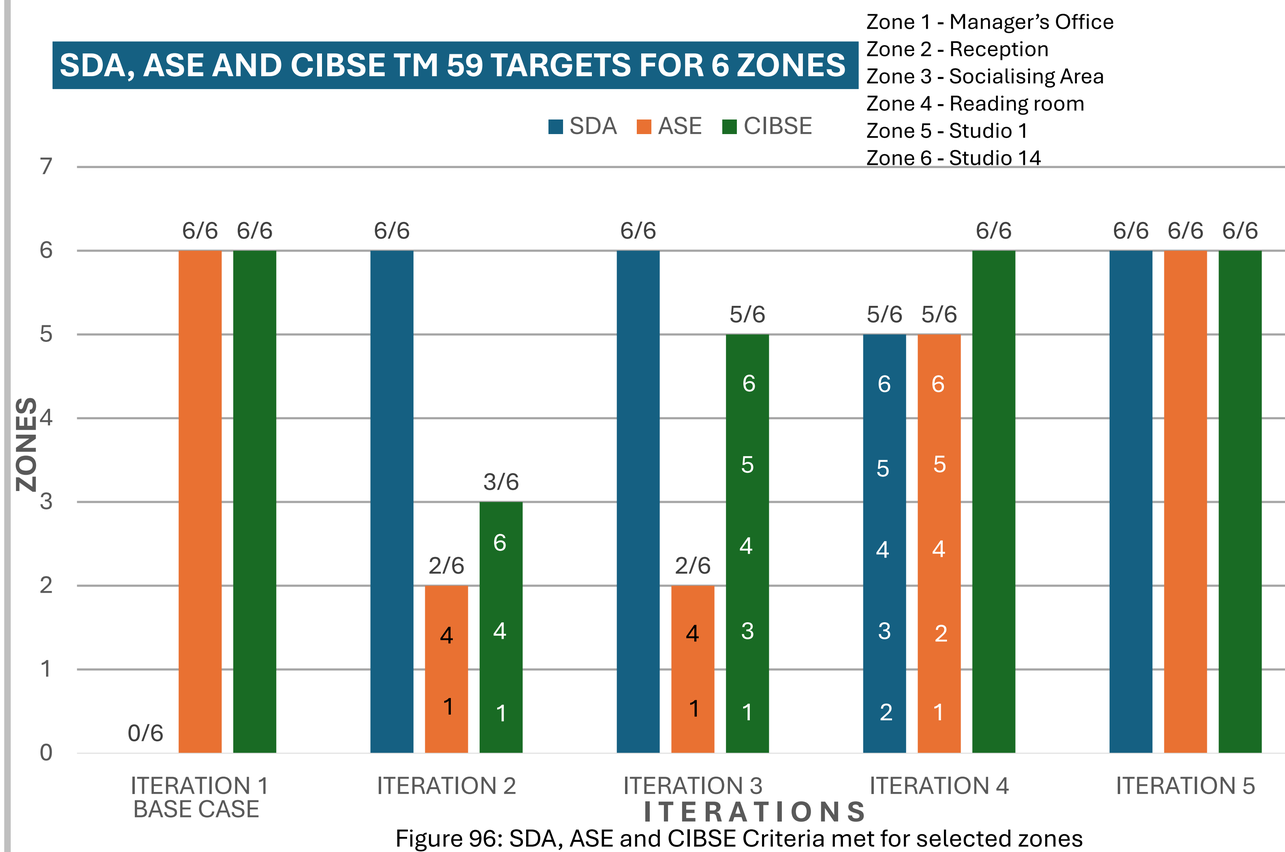


Figure 96: SDA, ASE and CIBSE Criteria met for selected zones

Table 3: Natural ventilation comfort range based on CIBSE GUIDE A.

	Temperature Range	Implication
Below Comfort	T<18 degrees	Too cold. Requires heating to achieve comfort
Within Comfort	18 degrees ≥ T ≤ 25 degrees	Ideal range. Natural Ventilation acceptable
Above Comfort	T > 25 degrees	Too warm. Requires cooling to maintain comfort

Conclusion of design optimization from Base Case to Iteration 5

1. **Hours below 18°C decreased** by **26.7%** (from 5,464 hours to 4,008 hours).
2. **Hours between 18–25°C increased** by **44.6%** (from 3,213 hours to 4,646 hours).
3. **Hours above 25°C decreased** by **8.5%** (from 107 hours to 127 hours).

Iteration 5 no more has summer overheating because it passed the CIBSE criteria for overheating. The temperature above 25 degrees is 1.4% of the total hours, which is below the 3% stated by CIBSE TM 59, therefore it is acceptable.
52.8% of total hours in the year is within the comfort range and 45.6% below comfort.
The design requires a heating system to increase the temperature below comfort to the acceptable comfortable range.

INDOOR CO2 LEVELS BELOW 1000ppm FOR 6 ZONES

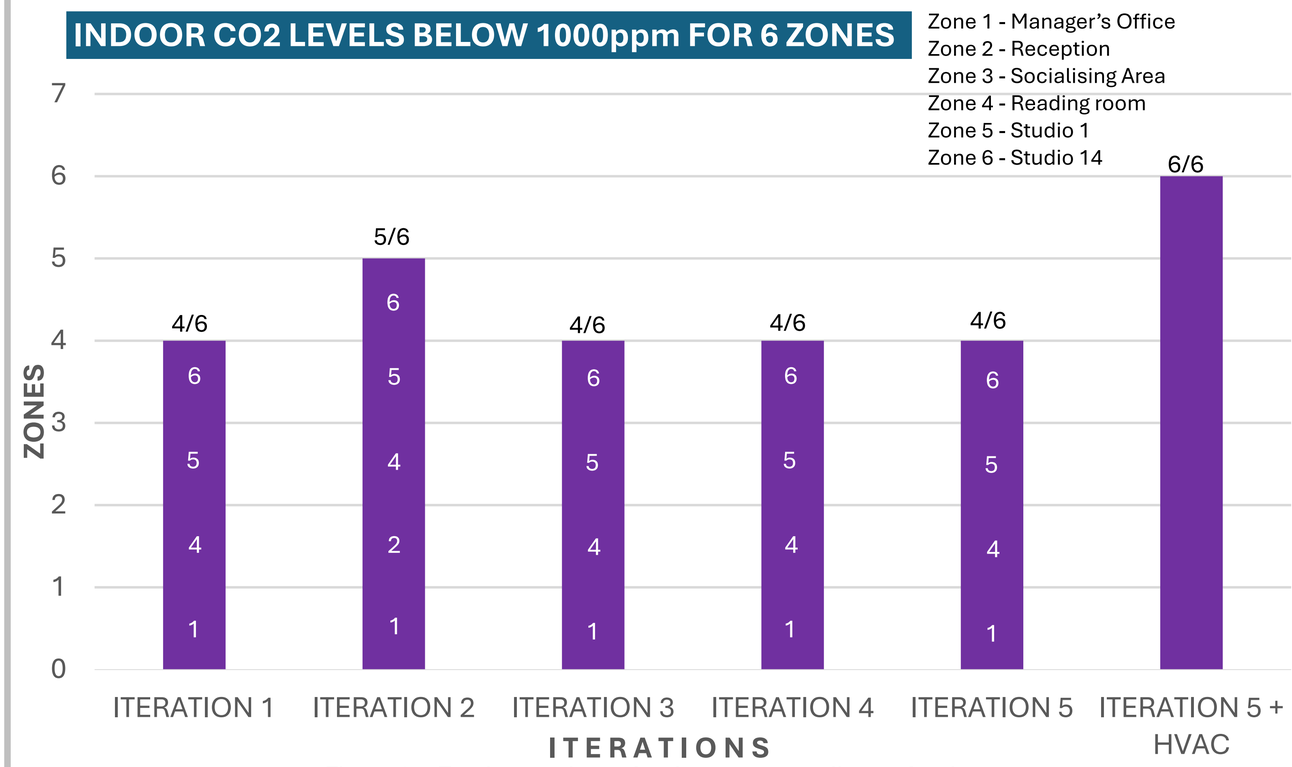


Figure 97: Total annual comfort hours versus discomfort hours

8.3 MECHANICAL SYSTEMS

	Total energy of building (kWh) if HVAC system heating only	Energy per total building area (kWh/m2)	**Total energy of building (kWh) if HVAC system heating and cooling**	Energy per total building area (kWh/m2)
1. Variable Refrigerant Flow (VRF) (air cooled) with heat recovery and Dedicated Outside Air System	69096.29	56.41	91550.22	74.74
2. Ground Source Heat Pump (GSHP): water to water heat pump with under floor heating	56591.96	46.20	69610.89	56.83

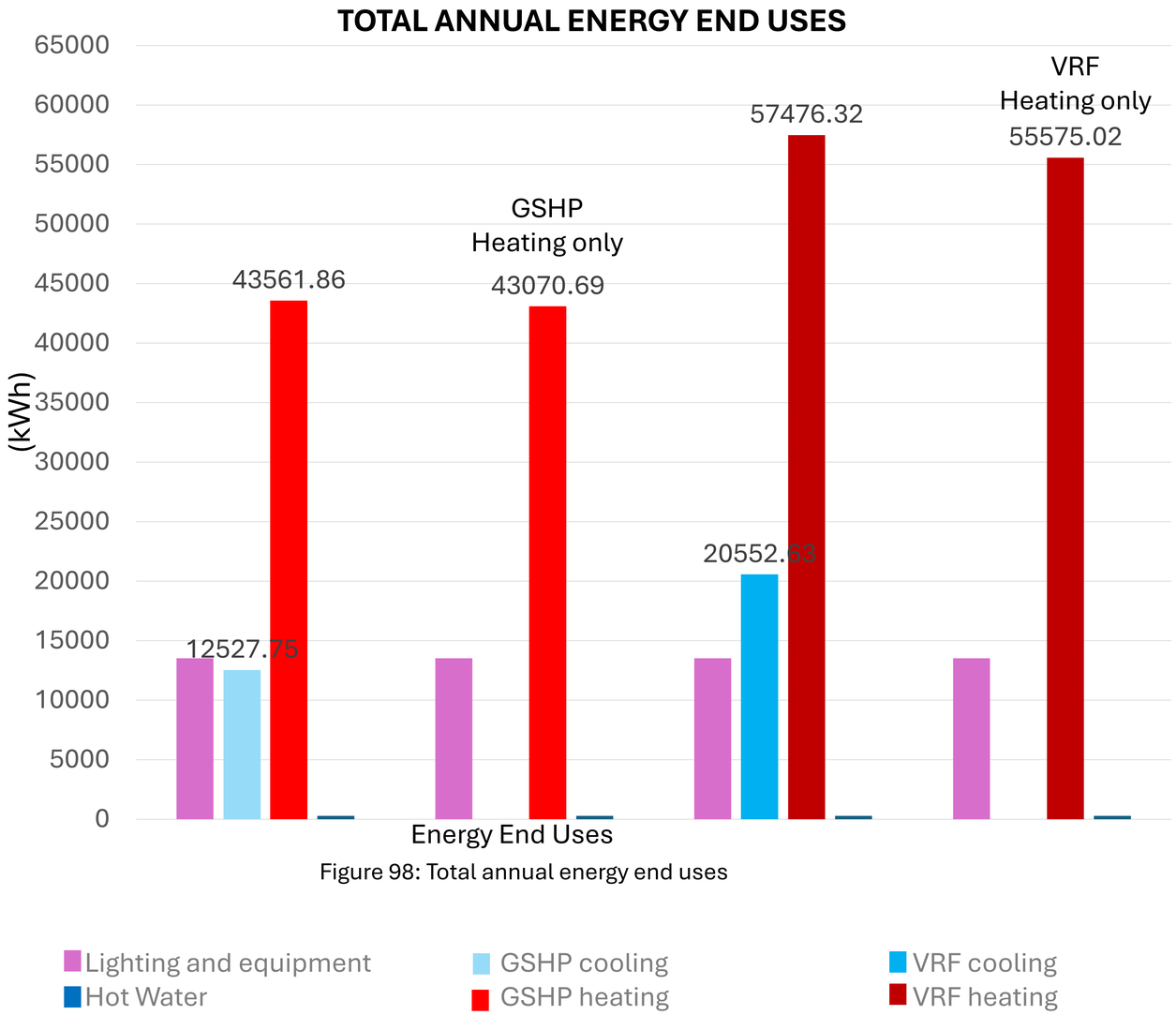


Figure 98: Total annual energy end uses

According to figure 98, In heating only mode, the GSHP uses 22.5% less energy compared to the VRF (ie. 43,070.69kWh versus 55,575.02kWh).

When heating and cooling is combined, the GSHP energy use is 24.2% less than the VRF.

In cooling mode, the GSHP outperforms the VRF because it consumes 64% less energy than the VRF (ie. 12,527.75kWh versus 20,552.63 kWh).

SUMMARY: The GSHP presents better efficiency and more energy savings and is the recommended mechanical system.

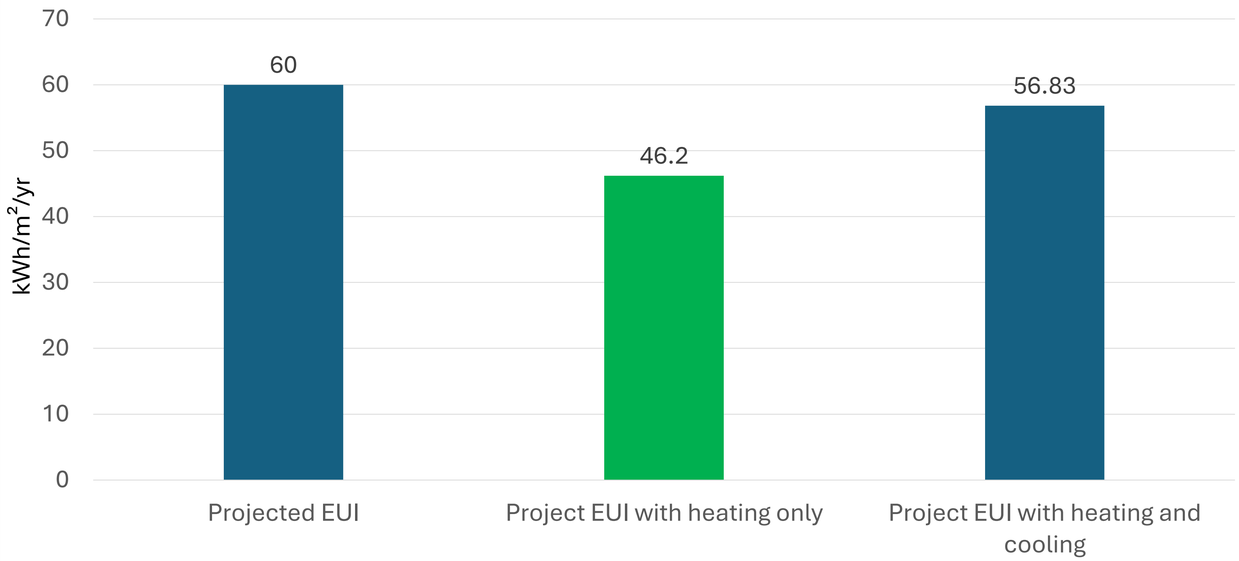


Figure 99: Projected EUI versus actual for project

8.4 RENEWABLE ENERGY

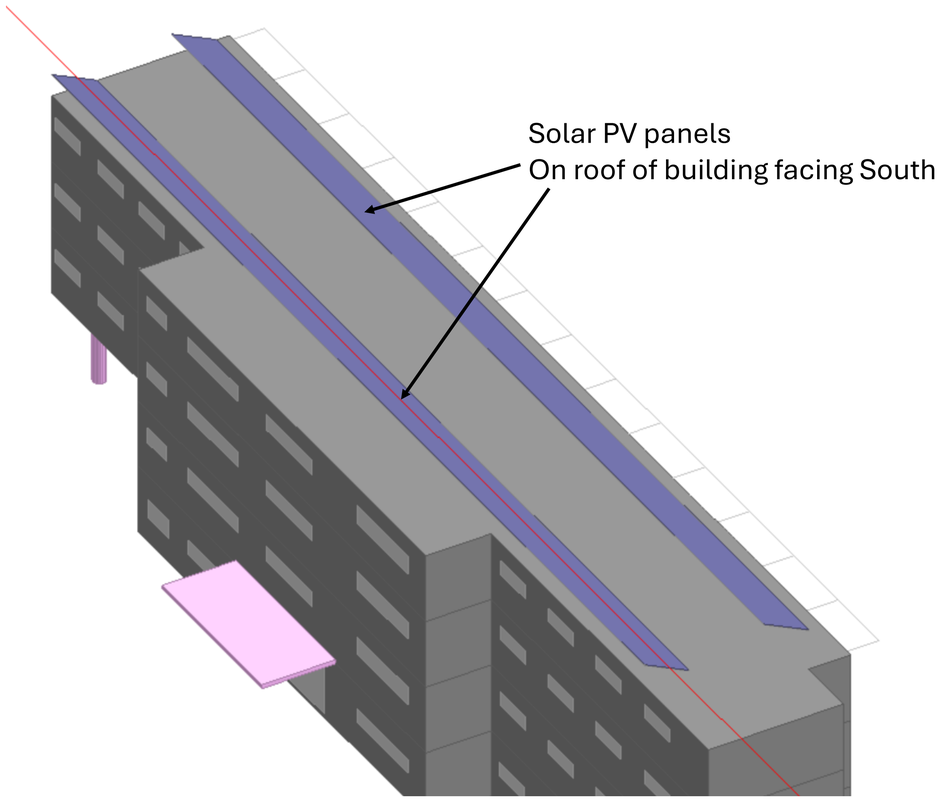


Figure 100: Solar PV on the roof

On-site renewable energy reduces the building's total reliance on the grid.

Examples common in Seattle are; Solar photovoltaics (PV), Solar Collectors for hot water, Building Integrated Photovoltaics(BIPV) and ground source heat pump.

Solar panels were placed on the roof to generate electricity for the project.

A total area of 226.45m2 of solar PV panels will generate 43096.89kWh of electricity for the building.

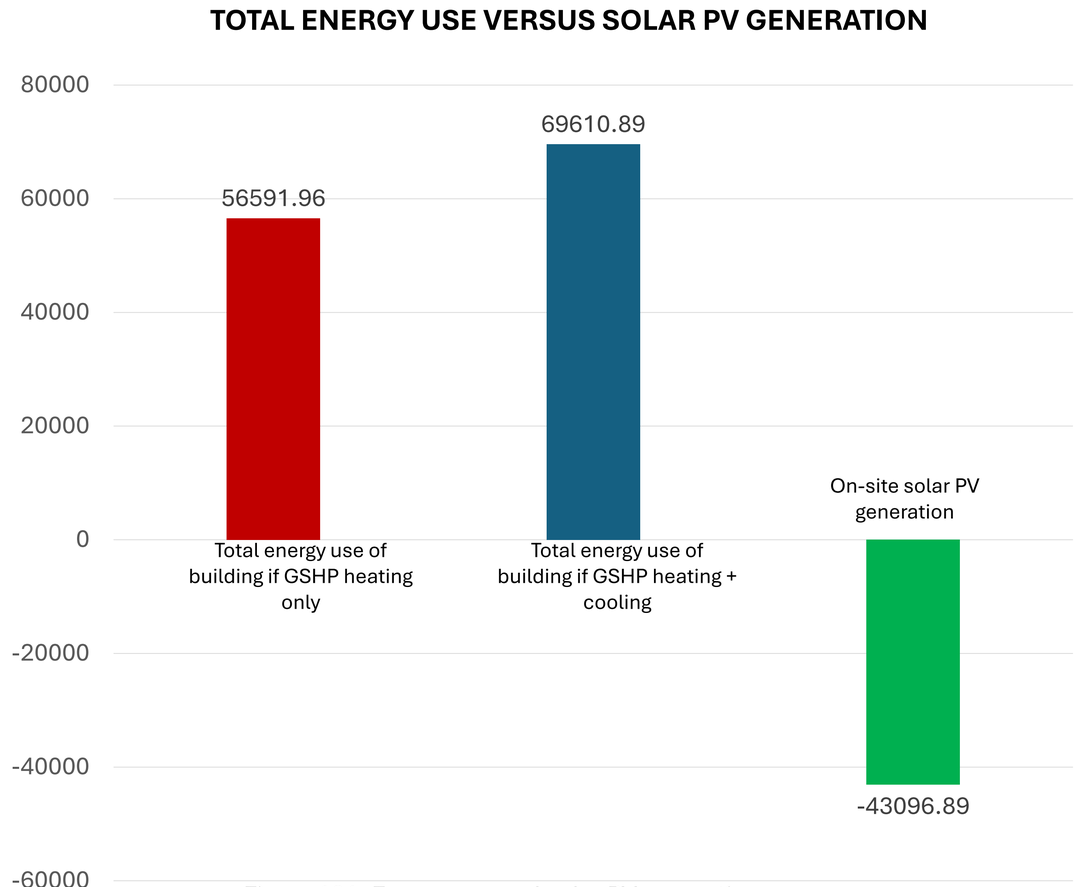


Figure 101: Energy use and solar PV generation

Conclusion: The solar PV system can offset 76.2% of the total energy use of the building if the GSHP provides heating only.

With both heating and cooling, the system will cover 61.9% of the total energy use of the building.

Opting for heating only during the winter and natural ventilation during summer combined with the solar PV will help the project become energy efficient and reduce its reliance on the grid.

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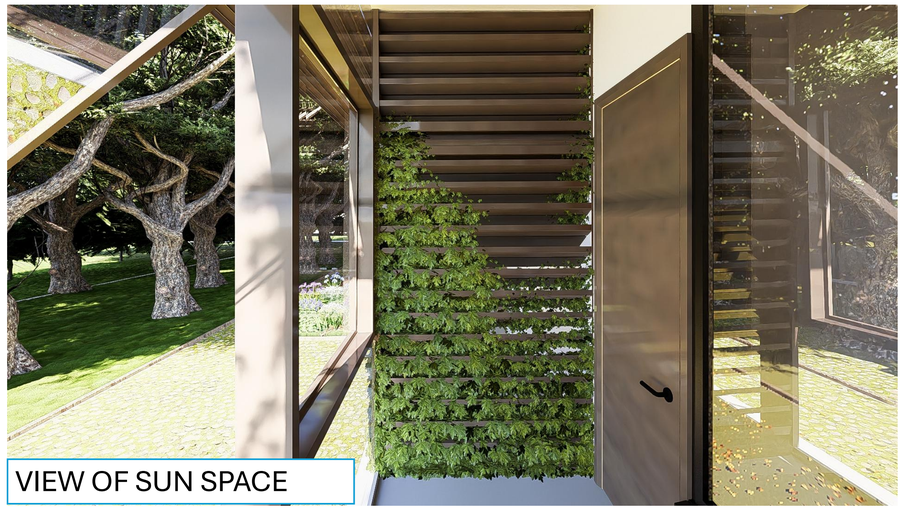
END OF PRESENTATION

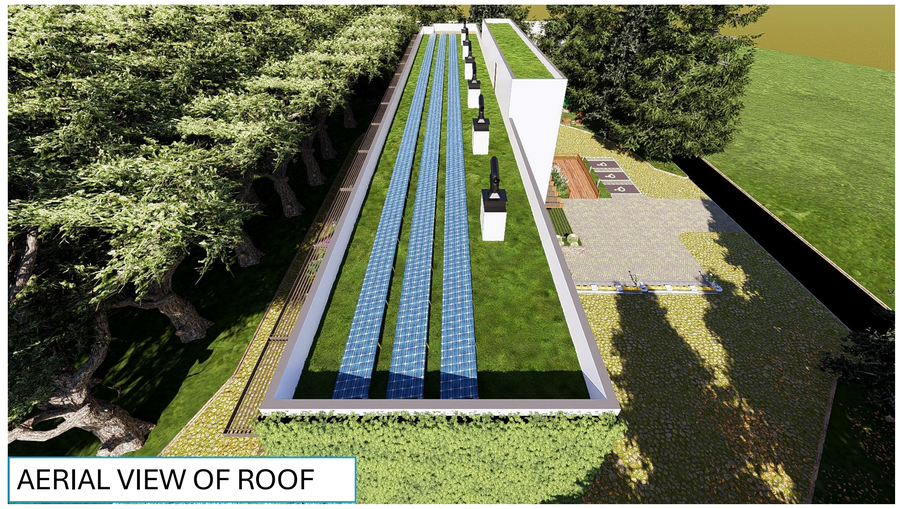
APPENDIX B : ALL THREE-DIMENSIONAL IMPRESSIONS OF PROJECT

SOUTHEAST VIEW

SOUTH VIEW



VIEW OF SUN SPACE



AERIAL VIEW OF ROOF

SOUTHWEST VIEW

NORTHWEST VIEW

Table X : Mechanical Systems

	Total energy of building (kWh) if HVAC system heating only	Energy per total building area (kWh/m2)	**Total energy of building (kWh) if HVAC system heating and cooling**	Energy per total building area (kWh/m2)
1. Variable Refrigerant Flow (VRF) (air cooled) with heat recovery, Dedicated Outside Air System	69096.29	56.41	91550.22	74.74
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TOTAL ANNUAL ENERGY END USES

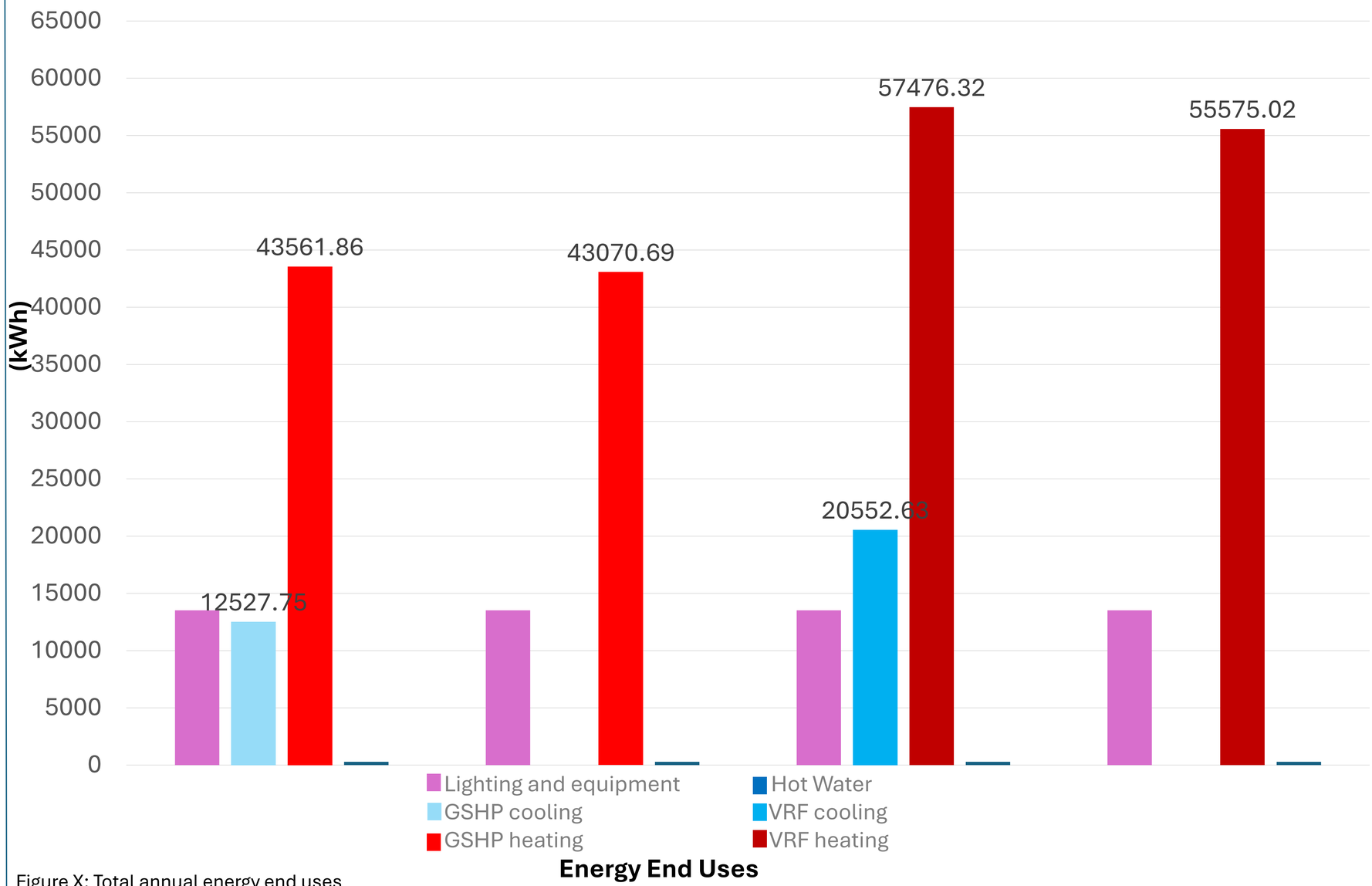


Figure X: Total annual energy end uses

CONCLUSION : MECHANICAL SYSTEMS

The project's heating demand to improve the comfort temperatures (from iteration 5) after applying all passive strategies will be met with a mechanical system that will provide heating during the hours below comfort (Figure X).

Two mechanical systems have been chosen for comparison to choose the most efficient system for the project (Table X).

In heating only mode, the GSHP uses 22.5% less energy compared to the VRF (ie. 43,070.69kWh versus 55,575.02kWh)

When heating and cooling is combined, the GSHP energy use is 24.2% lesser than the VRF.
In cooling mode, the GSHP outperforms the VRF because it consumes 64% less energy than the VRF (ie. 12,527.75kWh versus 20,552.63 kWh)

In comparing both systems, the GSHP presents better efficiency and more energy savings for the Students' Accommodation Project considering all scenarios presented and therefore is the recommended mechanical system for the project.

Sustainable Design of Students' Accommodation Block

RENEWABLE ENERGY

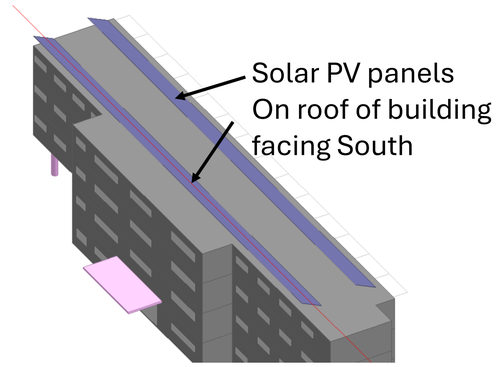


Figure X: Model with solar PV on the roof

Solar panels were placed on the building to receive optimum sun exposure to generate electricity for the project. A total area of 226.45 metre square of solar PV panels will generate 43096.89kWh of electricity for the building.

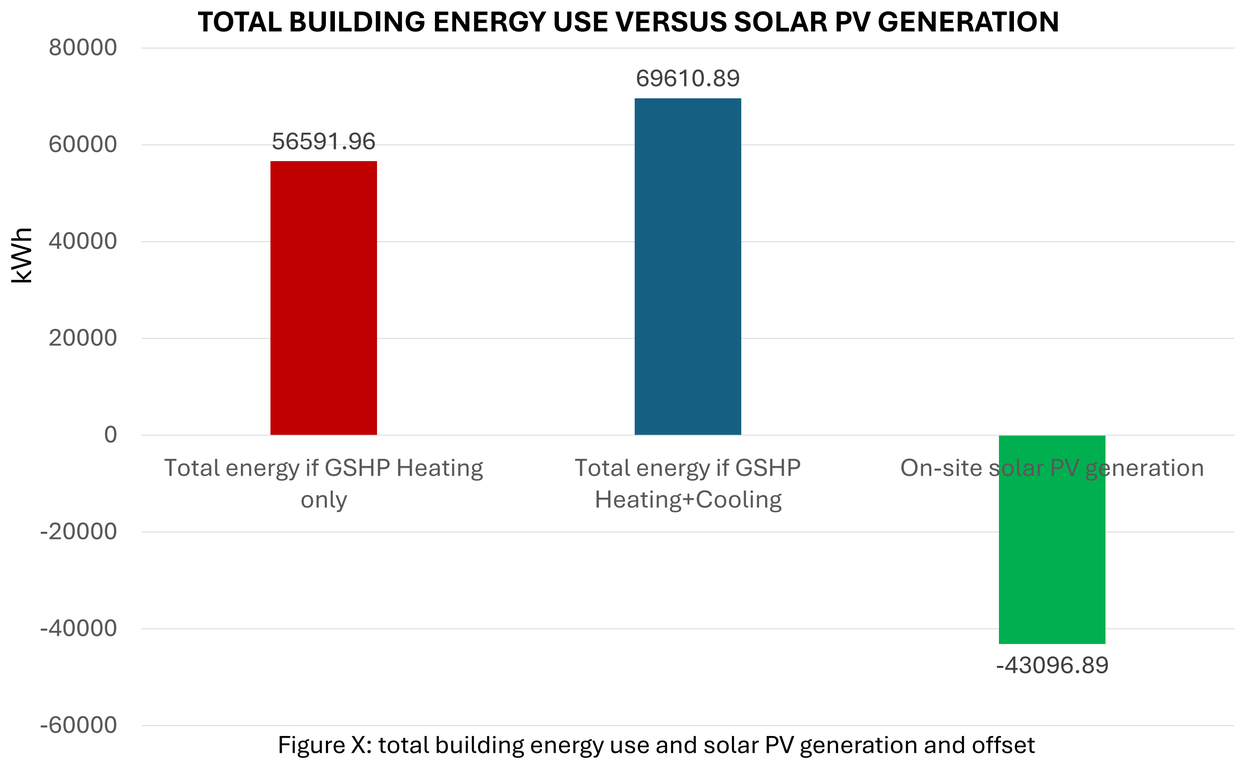


Figure X: total building energy use and solar PV generation and offset

The solar PV system can cover 76.2% of the total energy use of the building if the GSHP provides heating only. When both heating and cooling are considered, the system will cover 61.9% of the total energy use of the building.
This indicates that, opting for heating only during the winter and natural ventilation during summer combined with the solar PV energy generation will help the project become more energy efficient and reduce its reliance on the grid.

CONCLUSION

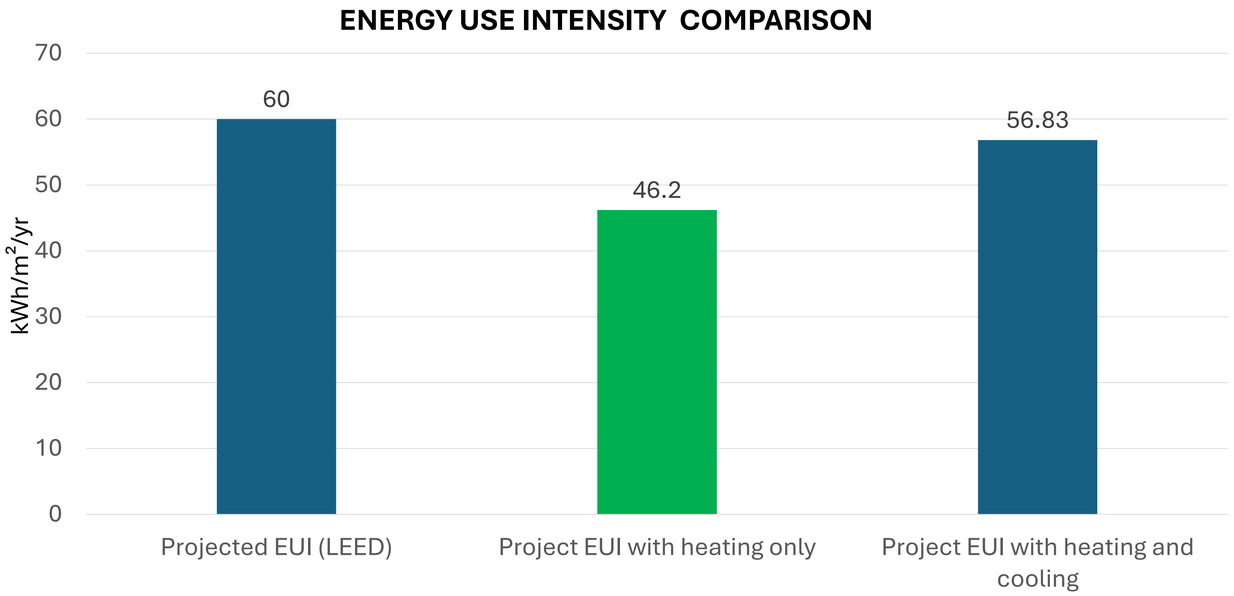


Figure X: Projected EUI versus actual for project

Environmental and passive strategies

Project had a compact linear form with longest side facing south for passive solar heating. East and West facades had no openings.
Building aligned to prevailing wind directions for natural ventilation.
South facing sun space for passive solar heat gain
Increased glazing area for passive solar heat gain
Windows for natural cross ventilation and wind cowls for wind driven ventilation.
Horizontal brise soliel for shading at south façade
Thermal mass incorporated for passive heat gain

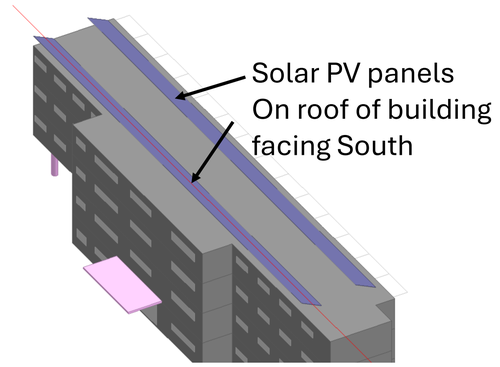
Design iteration and optimisation

From the base case to final iteration 5, increased glazing area, insulation and air tightness of the envelope was improved to reduce cold hours by 26.7% and indoor thermal comfort improved by 44.6%.

Through passive design strategies and improving the performance of the building through iterations, the students' accommodation project in Seattle achieved all performance targets as listed in Table X and achieved energy efficiency with the use of a ground source heat pump with underfloor heating.
The building responds effectively to the climate resulting in a high performance and acceptable CO2 levels.

Sustainable Design of Students' Accommodation Block

RENEWABLE ENERGY



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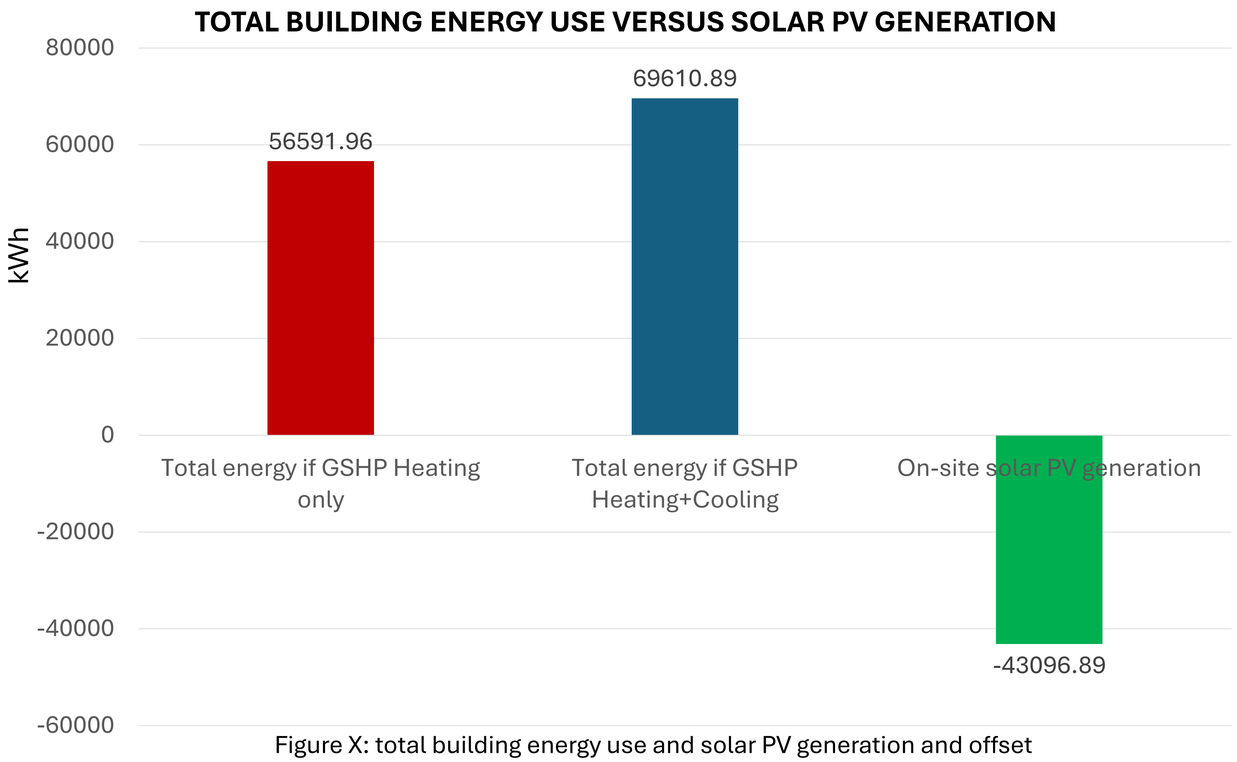

Figure X: total building energy use and solar PV generation and offset

Figure X compares the energy use of the building and the electricity generated by the solar PV. The PV system can cover 76.2% of the total energy use of the building if the GSHP provides heating only. When both heating and cooling are considered, the system will cover 61.9% of the total energy use of the building.
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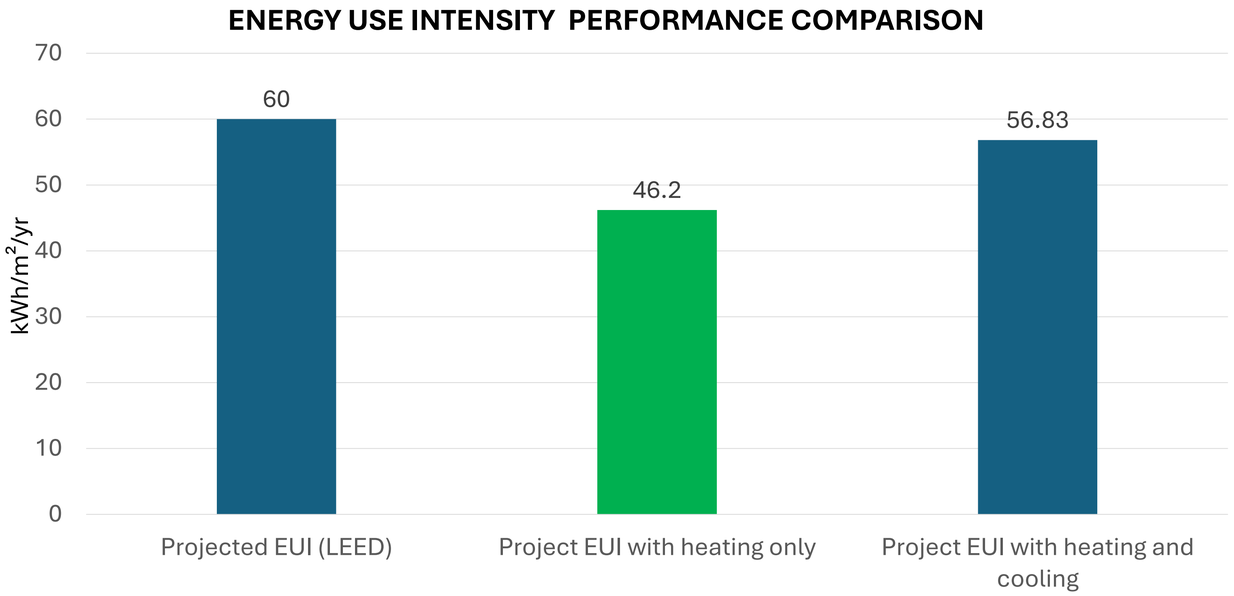
CONCLUSION



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Through passive design strategies and improving the performance of the building through iterations, the students' accommodation project in Seattle achieved all performance targets as listed in Table X and achieved energy efficiency with the use of a ground source heat pump with underfloor heating.
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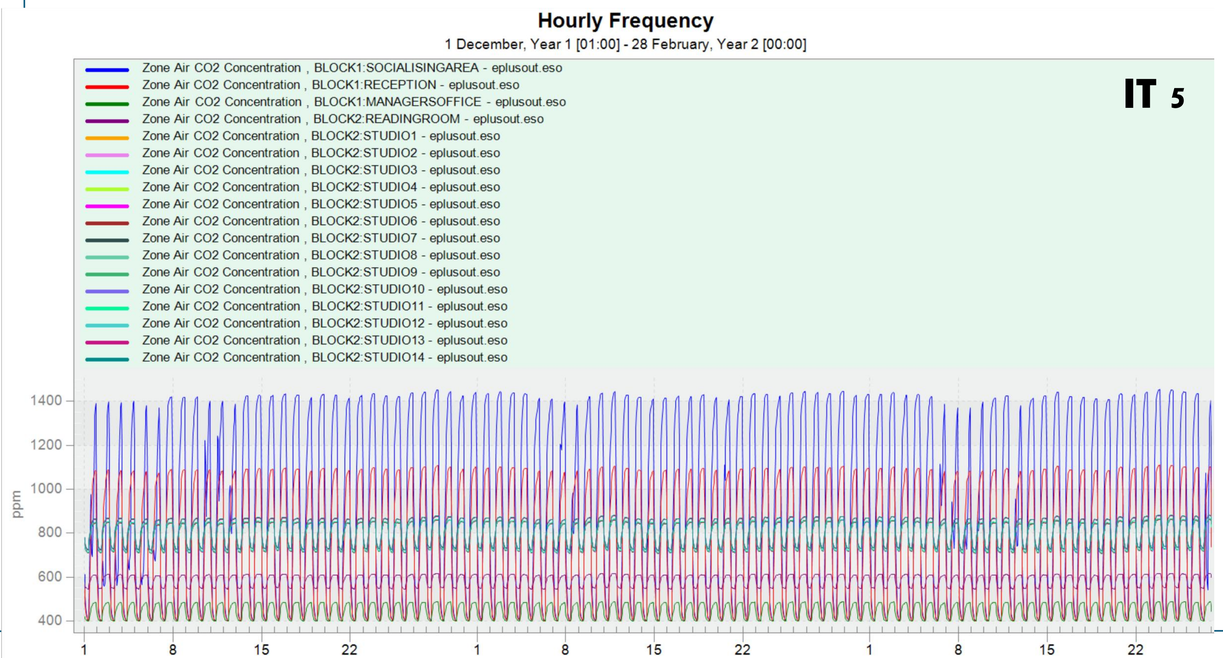
END OF PRESENTATION C_{02} LEVELS OF ZONES



END OF PRESENTATION C_{02} LEVELS OF ZONES

IT 1 – DECEMBER, JAN FEB ONLY

Date/Time	Air Temperature (°C)	Radiant Temperature (°C)	Operative Temperature (°C)	Outside Dry-Bulb Temperature (°C)	Relative Humidity (%)
01/10/2024	17.94758	18.34034	18.14396	11.42456	56.23373
01/11/2024	14.22802	14.44465	14.33633	8.069271	61.34907
01/12/2024	11.38943	11.53806	11.46374	5.416532	60.18064
01/01/2025	10.51825	10.66408	10.59117	4.496606	57.92255
01/02/2025	12.25416	12.49049	12.37232	5.822173	55.70998
01/03/2025	14.93549	15.22827	15.08188	8.320313	51.6541

IT 2 –

Analysis | Summary | Parametric | Optimisation + UA/SA | Data Visualisation

Date/Time	Air Temperature (°C)	Radiant Temperature (°C)	Operative Temperature (°C)	Outside Dry-Bulb Temperature (°C)	Relative Humidity (%)
01/10/2024	17.69244	18.31928	18.00586	11.42456	56.09901
01/11/2024	14.17599	14.55982	14.36791	8.069271	61.09952
01/12/2024	11.79692	12.18935	11.99313	5.416532	59.2726
01/01/2025	10.88434	11.22432	11.05433	4.496606	57.06856
01/02/2025	12.60894	13.08563	12.84728	5.822173	54.43206
01/03/2025	14.71485	15.14102	14.92794	8.320313	51.91319

IT 3 –

Analysis | Summary | Parametric | Optimisation + UA/SA | Data Visualisation

Date/Time	Air Temperature (°C)	Radiant Temperature (°C)	Operative Temperature (°C)	Outside Dry-Bulb Temperature (°C)	Relative Humidity (%)
01/10/2024	18.24313	18.94396	18.59355	11.42456	54.5832
01/11/2024	14.95273	15.38892	15.17082	8.069271	59.14407
01/12/2024	12.60714	13.06018	12.83366	5.416532	57.0877
01/01/2025	11.78156	12.17708	11.97932	4.496606	54.81379
01/02/2025	13.49222	14.04114	13.76668	5.822173	52.34678
01/03/2025	15.57819	16.06421	15.8212	8.320313	50.21038

IT 4 –

Analysis | Summary | Parametric | Optimisation + UA/SA | Data Visualisation

Date/Time	Air Temperature (°C)	Radiant Temperature (°C)	Operative Temperature (°C)	Outside Dry-Bulb Temperature (°C)	Relative Humidity (%)
01/10/2024	18.54948	19.17919	18.86433	11.42456	53.81321
01/11/2024	15.24075	15.65501	15.44788	8.069271	58.15695
01/12/2024	12.79286	13.20097	12.99691	5.416532	56.30856
01/01/2025	11.90798	12.27116	12.08957	4.496606	54.33294
01/02/2025	13.49533	13.96173	13.72853	5.822173	52.50263
01/03/2025	15.74484	16.18938	15.96711	8.320313	49.73003

IT 5 –

Date/Time	Air Temperature (°C)	Radiant Temperature (°C)	Operative Temperature (°C)	Outside Dry-Bulb Temperature (°C)	Relative Humidity (%)
01/10/2024	19.22201	19.81751	19.51976	11.42456	52.44425
01/11/2024	16.28148	16.67759	16.47954	8.069271	56.21189
01/12/2024	13.88673	14.27886	14.08279	5.416532	54.10555
01/01/2025	12.90171	13.22804	13.06488	4.496606	52.67909
01/02/2025	14.33927	14.74673	14.543	5.822173	51.59845
01/03/2025	16.49698	16.86449	16.68073	8.320313	49.10918

Sustainable Design of Students' Accommodation Block

PROJECT BRIEF – BUILDING USE AND OCCUPANCY

Figure 67: Building type (Google, 2025 with modifications by Author

TARGET : GRADUATE STUDENTS

42

STAFF

8

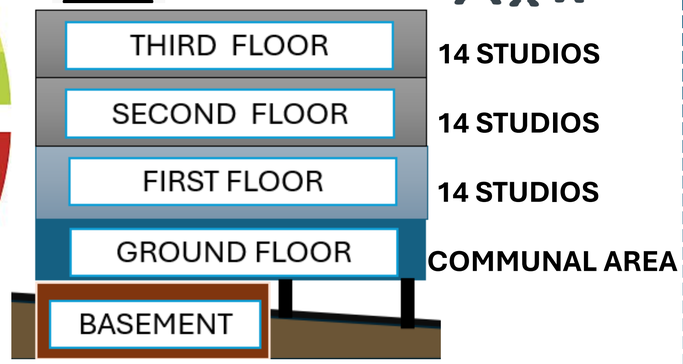


Figure 66: Activity Schedule Source(Author)

DETAILS OF SPACES AND ACTIVITY SCHEDULE

Ground floor - Communal Spaces
Reception
Manager's office
Socialising/play area
Kitchenette
Washroom - 2
Study area
Mailboxes zone
Bicycle storage
Circulation: Lift and stair

1ST, 2ND AND 3RD FLOOR - Studios
sleeping area
study
Kitchenette
toilet and shower
Laundry + utility store
Terrace garden

Basement – Services

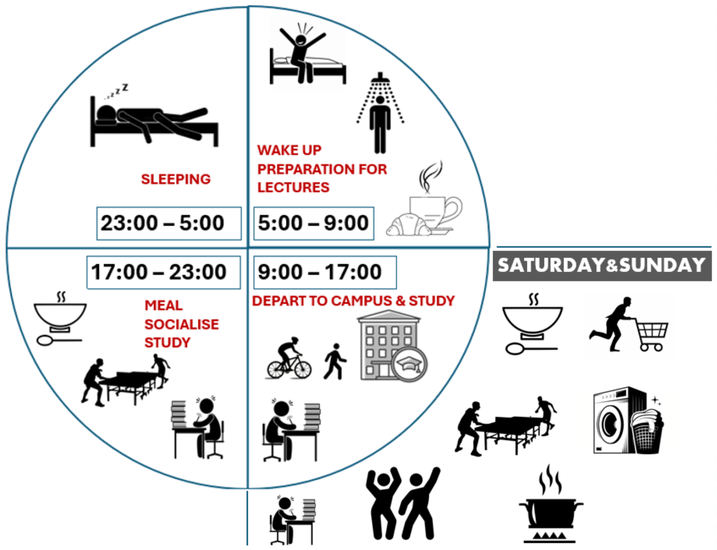


Figure 68: Activity Schedule
Source: (Google 2025 with modifications by Author)

A BUILDING THAT WILL LAST MORE THAN 100 YEARS

HEAVY TIMBER, CONCRETE AND STEEL STRUCTURE

HIGH PERFORMANCE ENVELOPE

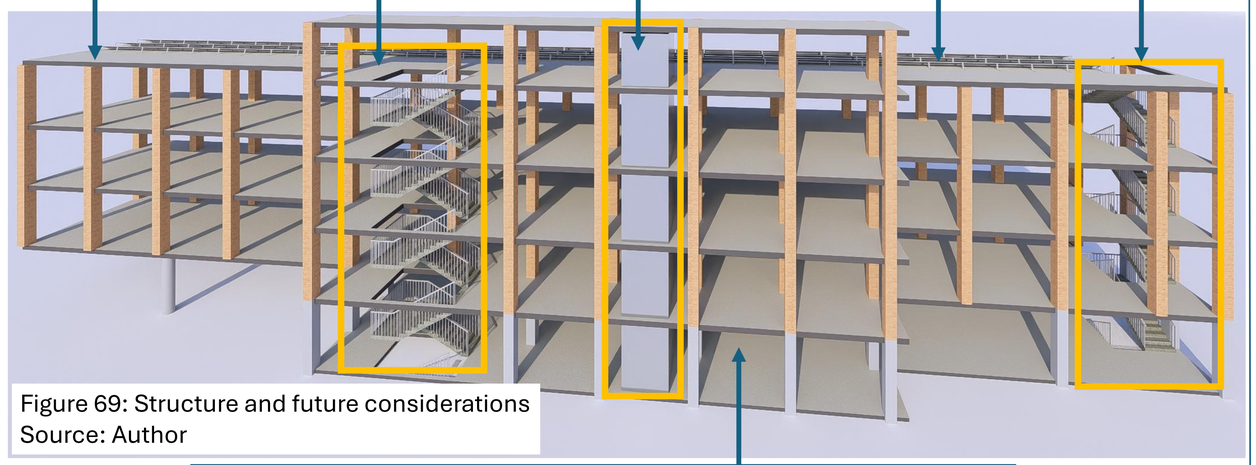


Figure 69: Structure and future considerations
Source: Author

BASEMENT: FOR SERVICES NOW AND THE FUTURE

RAINWATER HARVESTING AND TREATMENT TANKS

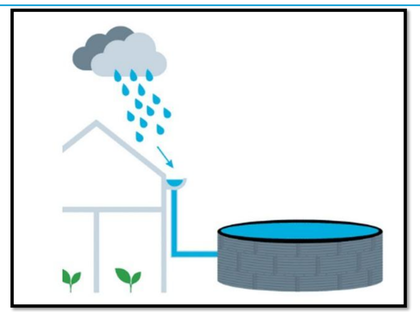


Figure 70: Rainwater harvesting Source(Google, 2025)

GREY WATER TREATMENT

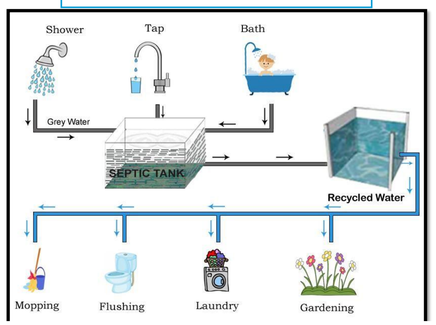


Figure 71: Rainwater harvesting Source(Google, 2025)

OTHER RENEWABLE ENERGY SYSTEMS

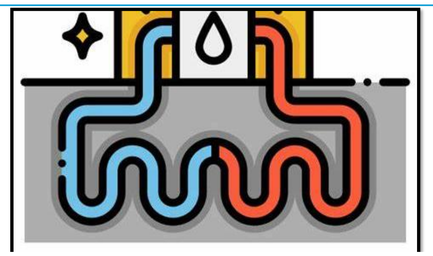


Figure 72: Rainwater harvesting Source(Google, 2025)

WASTE COMPOST

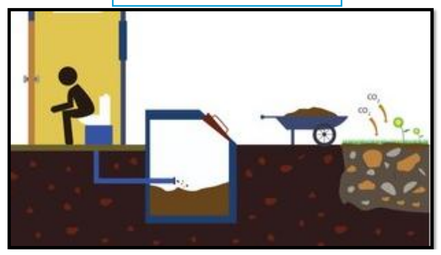
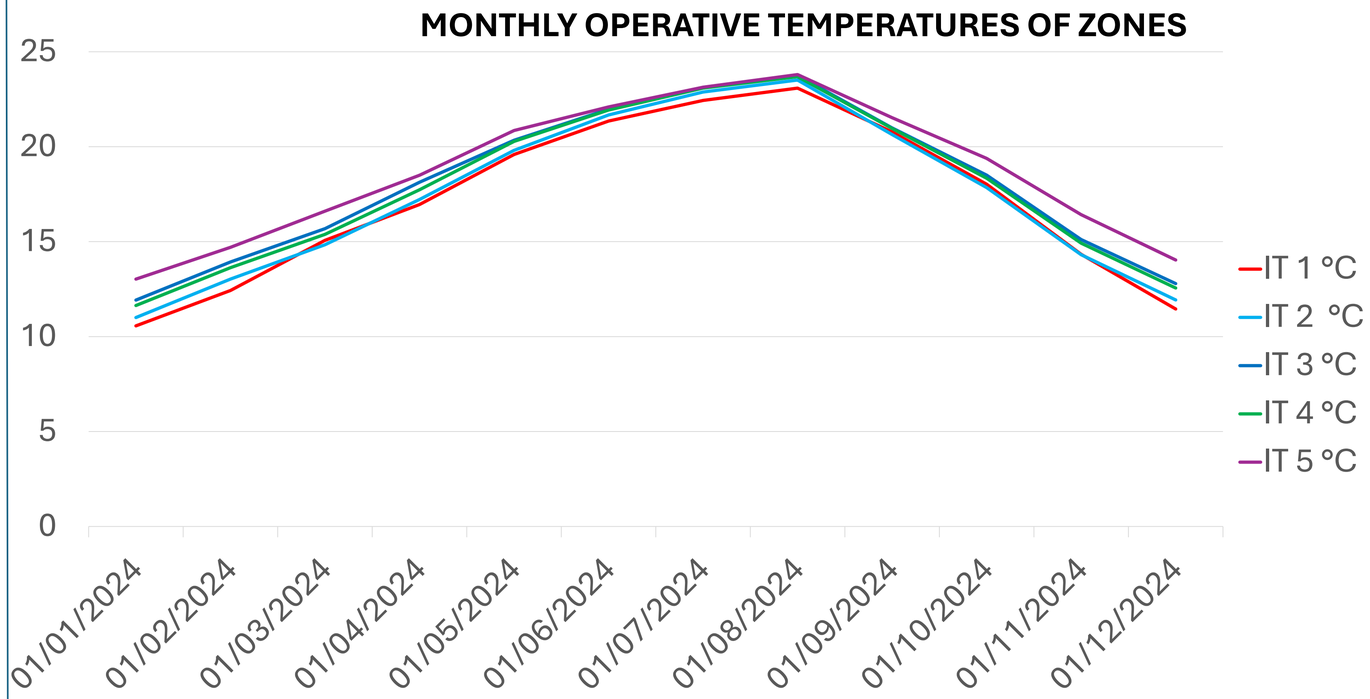
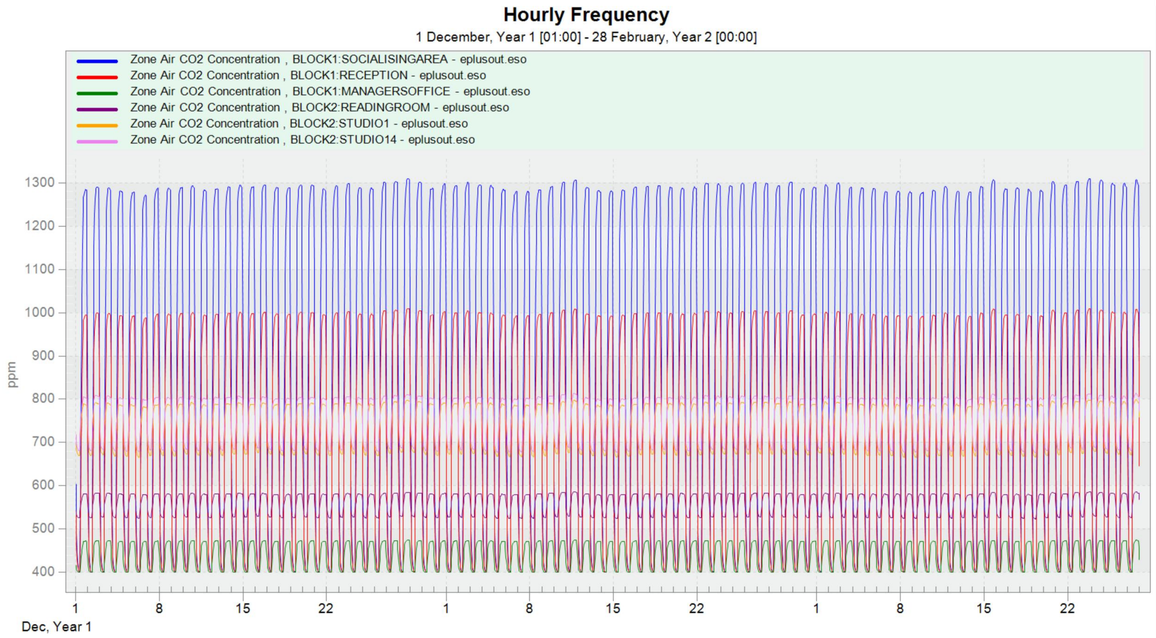


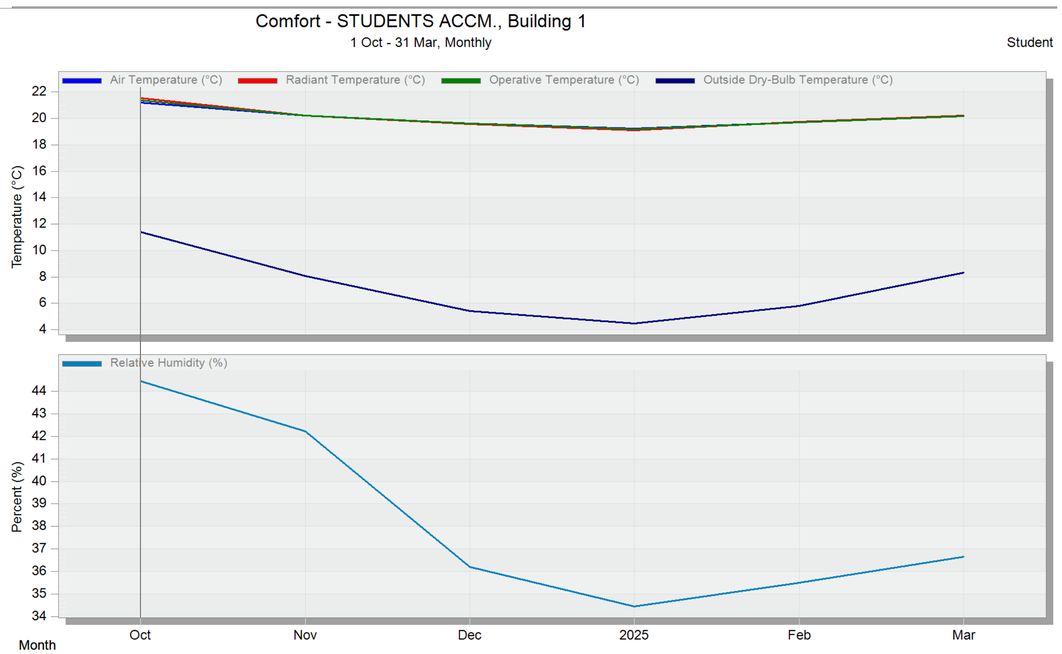
Figure 73: Rainwater harvesting Source(Google, 2025)

TEMPERATURE RANGE COMPARISON OF ITERATIONS





MONTHLY OPERATIVE TEMPERATURES OF ZONES IN WINTER



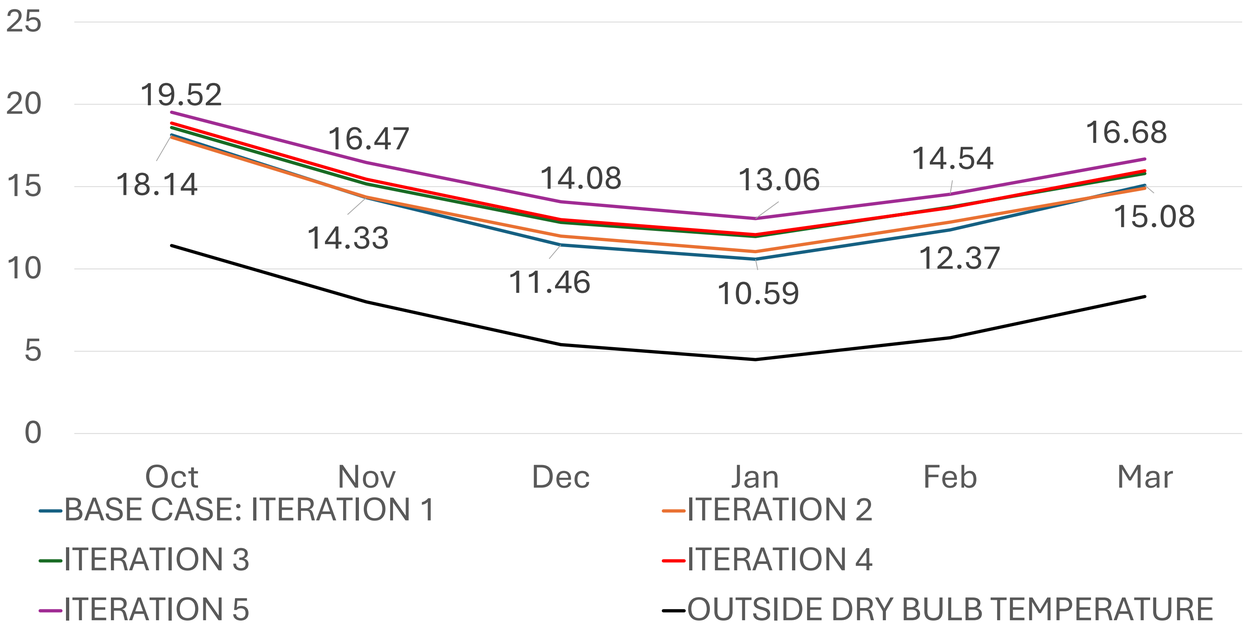
EnergyPlus Output — Student

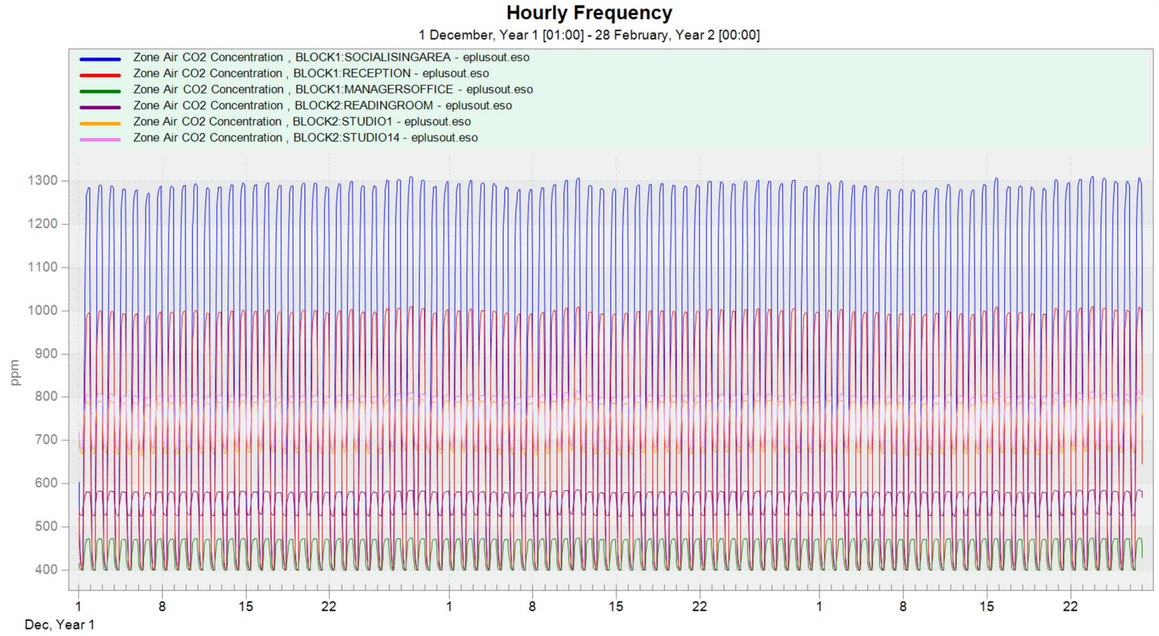
	Oct	Nov	Dec	2025	Feb	Mar
Air Temperature (°C)	21.18	20.22	19.62	19.25	19.70	20.17
Radiant Temperature (°C)	21.52	20.21	19.58	19.12	19.73	20.19
Operative Temperature (°C)	21.35	20.22	19.60	19.18	19.71	20.18
Outside Dry-Bulb Temperature (°C)	11.42	8.07	5.42	4.50	5.82	8.32
Relative Humidity (%)	44.44	42.22	36.22	34.46	35.49	36.64

HAVE SIMULATIONS AND SHOW HOW THE OPERATIVE TEMPERATURES HAVE IMPROVED IN ALL ITERATIONS OVER TIME...IN WINTER ONLY.
SHOW ALSO THE CO2 LEVELS DURING THE WINTER TIME ONLY.
LASTLY SHOW THE MONTH OPERATIVE TEMPEATURES AFTER THE MECH SYSTEM.

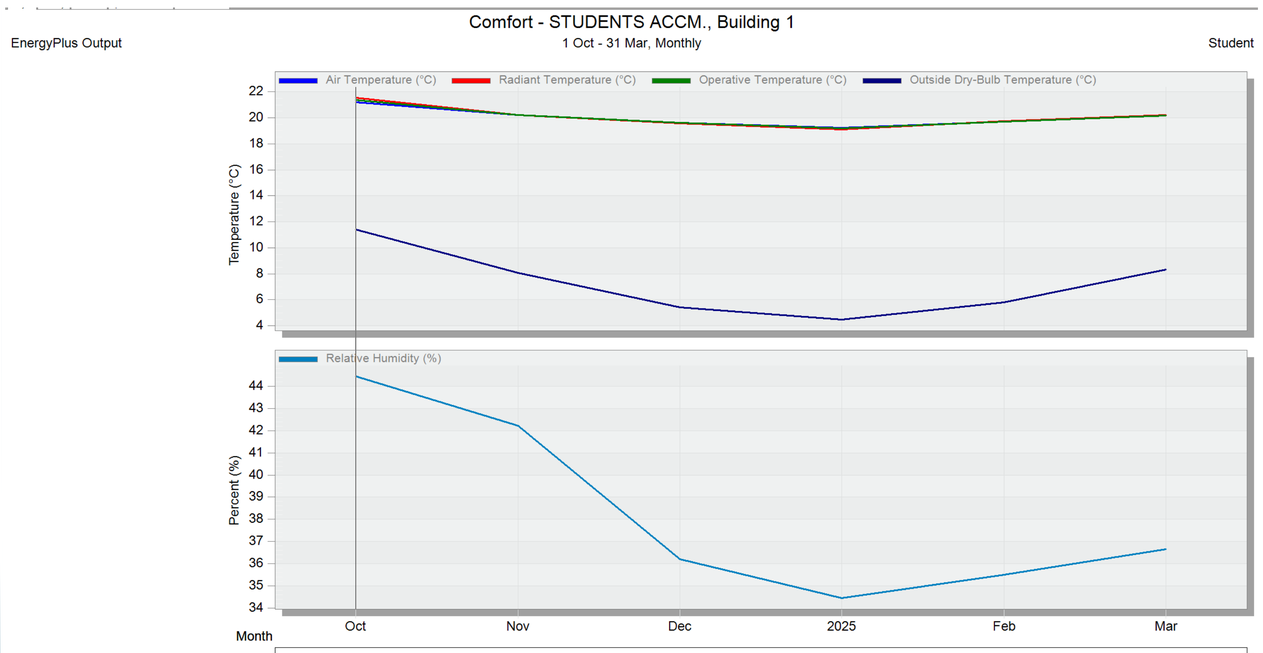
TEMPERATURE RANGE COMPARISON OF ITERATIONS

MONTHLY OPERATIVE TEMPERATURES OF ZONES





MONTHLY OPERATIVE TEMPERATURES OF ZONES IN WINTER



	Oct	Nov	Dec	2025	Feb	Mar
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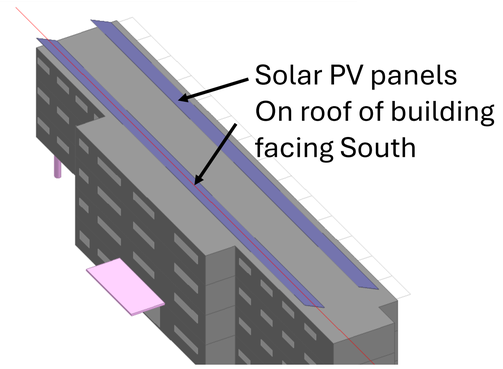
SHOW ALSO THE CO2 LEVELS DURING THE WINTER TIME ONLY.
LASTLY SHOW THE MONTH OPERATIVE TEMPEATURES AFTER THE MECH SYSTEM.

Sustainable Design of Students' Accommodation Block

RENEWABLE ENERGY



On-site renewable energy reduces the building's total reliance on the grid. Very common choices in Seattle are Solar photovoltaics (PV), Solar Collectors for hot water, Building Integrated Photovoltaics(BIPV) and ground source heat pump for heating.

Solar panels were placed on the building to receive optimum sun exposure to generate electricity for the project. A total area of 226.45 metre square of solar PV panels will generate 43096.89kWh of electricity for the building.

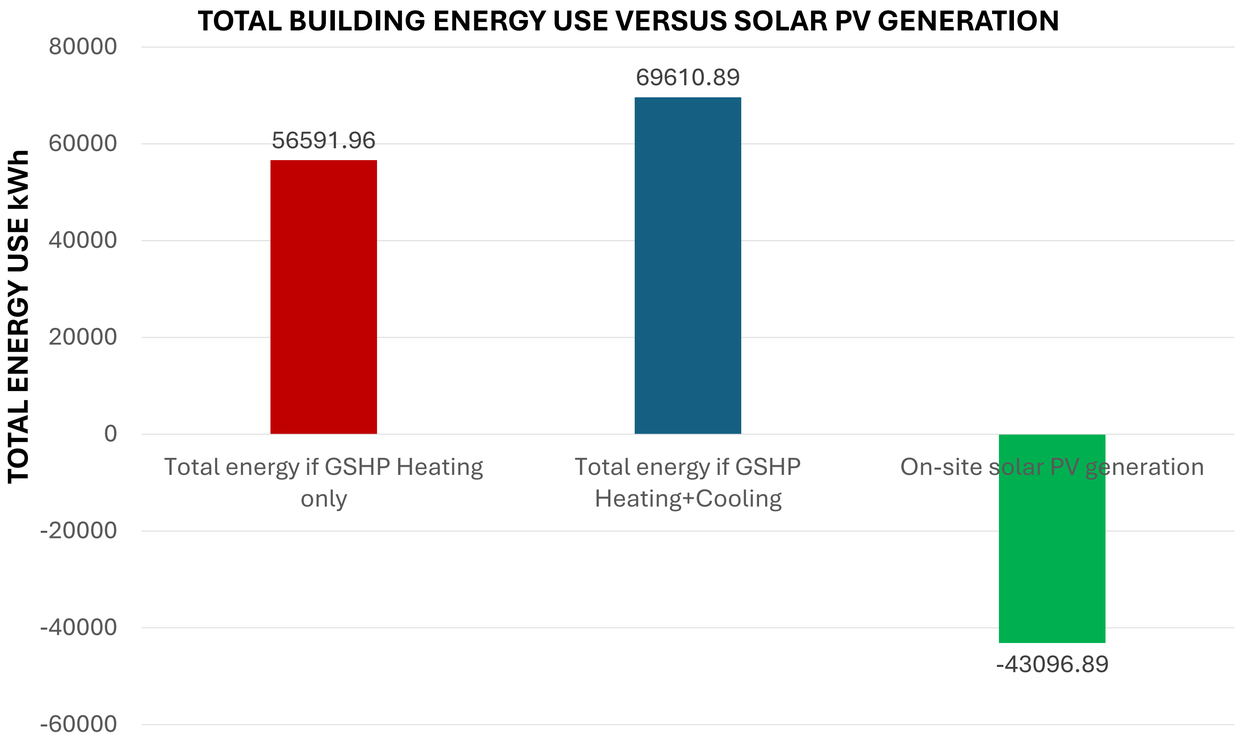


Figure X compares the energy use of the building and the electricity generated by the solar PV. The PV system can cover 76.2% of the total energy use of the building if the GSHP provides heating only. When both heating and cooling are considered, the system will cover 61.9% of the total energy use of the building.
This indicates that, opting for heating only during the winter and natural ventilation during summer combined with the solar PV energy generation will help the project become more energy efficient and reduce its reliance on the grid.

CONCLUSION

The design of the students' accommodation block set out with a brief to targets that were derived from standards to be achieved.
1.Passive strategies employed improved the operative temperatures of all zones to reach an acceptable temperature range.

> if my building current energy use is 56591.96kWh for the whole year as against a predicted value of 133,500kWh, what IS THE PERFORMANCE IMPROVEMENT IN PERCENTAGE?

2. Plug in Your Numbers

- **Actual Energy Use** = 56,591.96 kWh
- **Predicted Energy Use** = 133,500 kWh

$$\text{Improvement \%} = \left(1 - \frac{56{,}591.96}{133{,}500}\right) \times 100 = (1 - 0.424) \times 100 = **57.6\%**$$

3. Interpretation

- Your building used **57.6% less energy** than predicted, indicating **exceptional efficiency**.
- **Possible reasons**:
  - High-performance HVAC (e.g., heat pumps).
  - Superior insulation/tight envelope.
  - Energy-efficient lighting/appliances.

ITERATION 5 – ALL SPACES PASS CIBSE TM59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	0.99	N/A	Pass
SOCIALISINGAREA	0.10	N/A	Pass
READINGROOM	0.00	0.00	Pass
STUDIO1	0.00	0.00	Pass
STUDIO14	0.00	0.00	Pass

ITERATION 5 ACHIEVES ASE AND SDA

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	14.0	7.36	52.52	14.02	100.0
STUDIO 1	10.2	5.79	56.84	9.64	94.7
STUDIO 14	8.7	4.40	50.62	8.49	97.5
RECEPTION	103.2	83.46	80.88	101.19	98.1
MANAGERS OFFICE	4.7	2.62	56.10	4.68	100.0
SOCIALISING AREA	85.6	58.38	68.22	80.94	94.6
	226.3	**162.02**	**71.58**	**218.95**	**96.7**

A **Ground Source Heat Pump (GSHP) with Water-to-Water Configuration and Underfloor Heating** is a highly efficient HVAC system that leverages geothermal energy for space heating (and optionally cooling). Here's a detailed breakdown:

---

1. Definition & Key Components

- **Ground Source Heat Pump (GSHP):**
  A system that extracts heat from the ground (via buried pipes/loops) and transfers it to a building.
- **Water-to-Water Heat Pump:**
  A type of GSHP that uses **water-based heat exchange** (instead of air) for both:
  - **Heat source/sink:** Ground loops circulate water/antifreeze to absorb/release heat.
  - **Heat distribution:** Heated/chilled water is sent to the building's underfloor system.
- **Underfloor Heating (Radiant Floor):**
  Pipes embedded in floors circulate warm water for even, comfortable heat distribution.

MECHANICAL SYSTEMS

Table X : Mechanical Systems

Mechanical System	Total energy (kWh) Heating only	Energy per total building area (kWh/m2)	Total energy (kWh) Heating and cooling	Energy per total building area (kWh/m2)
Variable Refrigerant Flow (air cooled) with heat recovery, Dedicated Outside Air System	69096.29	56.41	91550.22	74.74
Ground Source Heat Pump; water to water heat pump and heated floor system	56591.96	46.20	69610.89	56.83

The total energy includes all floors and circulation zones the excluding lift

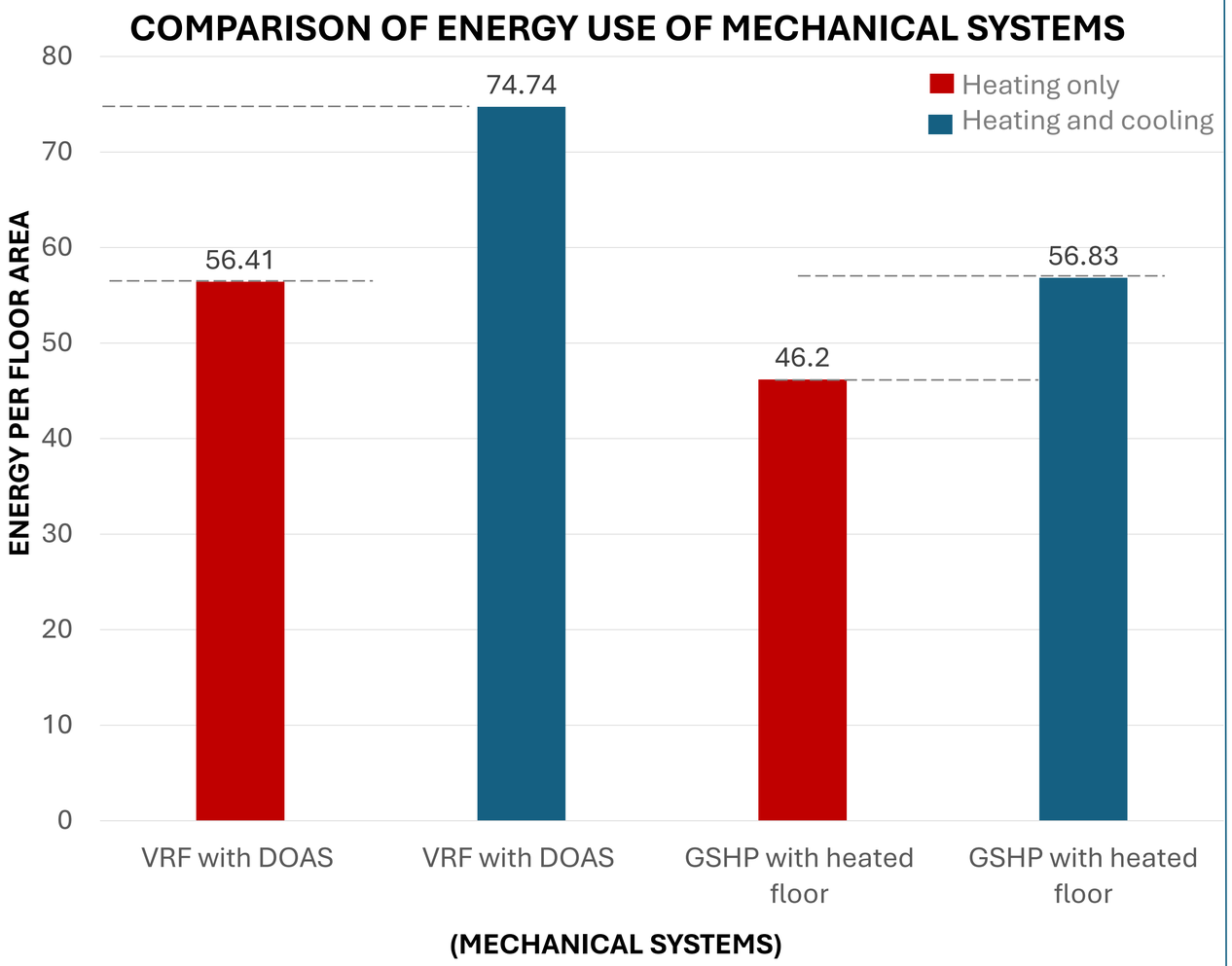


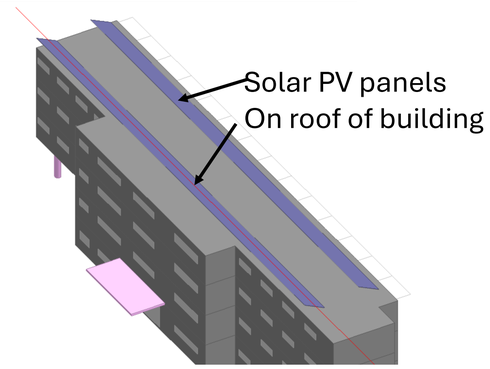
Table X : Conclusion of systems comparison

System	Heating and cooling
VRF	An energy efficient system for heating because of low energy use
	Higher energy use for cooling
GSHP	More energy efficient for heating because it has lower energy use of 18% less than VRF
	Uses less energy for cooling (42% less than the VRF)

The GSHP is the best system for the building due to its energy efficiency as seen on Table X.
The students accommodation block requires heating only during the winter only as the summer periods will employ natural ventilation.
The project will save 18.7% of energy if cooling is excluded.

Sustainable Design of Students' Accommodation Block

RENEWABLE ENERGY



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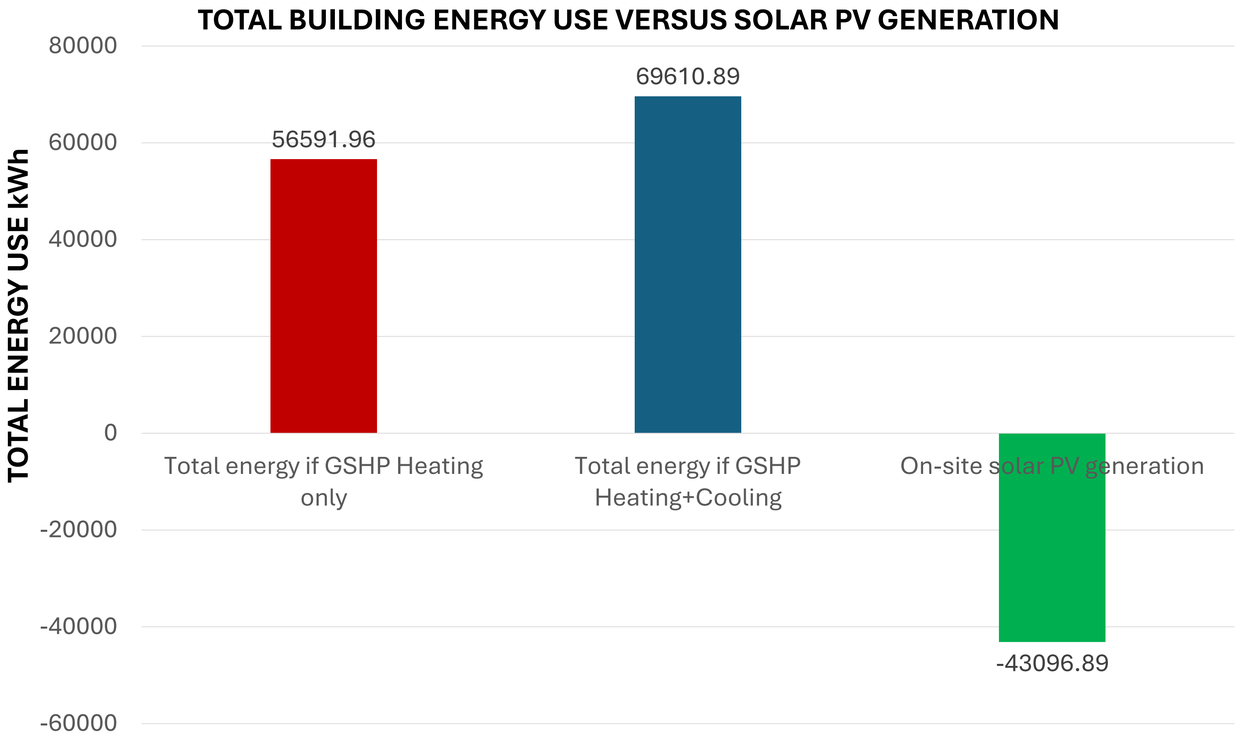

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This indicates that, opting for heating only during the winter and natural ventilation during summer combined with the solar PV energy generation will help the project become more energy efficient and reduce its reliance on the grid.

CONCLUSION

Site and Source Energy

	Total Energy [kWh]	Energy Per Total Building Area [kWh/m2]	Energy Per Conditioned Building Area [kWh/m2]
Total Site Energy	69629.46	56.85	56.85
Net Site Energy	28687.41	23.42	23.42
Total Source Energy	213794.45	174.55	174.55
Net Source Energy	84130.99	68.69	68.69

CONCLUDE AND WRITE HOW THE PROJECT HAS BEEN ABLE TO ADRESS DAYLIGHTING, OVERHEATING AND INDOOR AIR QUALITY
WRITE THIS CONCLUSION AFTER ALL ITERATIONS BEFORE MOVING ON TO CO2 AND JUSTIFY THE INCORPORATION OF AN HVAC SYSTEM TO SOLVE THE WINTER CHALLENGE AND REDUCE CO2.

Electric Loads Satisfied

	Electricity [kWh]	Percent Electricity [%]
Fuel-Fired Power Generation	0.000	0.00
High Temperature Geothermal*	0.000	0.00
Photovoltaic Power	43096.890	318.73
Wind Power	0.000	0.00
Power Conversion	-2154.84	-15.9
Net Decrease in On-Site Storage	0.000	0.00
Total On-Site Electric Sources	40942.046	302.80
Electricity Coming From Utility	11102.320	82.11
Surplus Electricity Going To Utility	38523.089	284.91
Net Electricity From Utility	-27420.77	-202.8
Total On-Site and Utility Electric Sources	13521.277	100.00
Total Electricity End Uses	13521.277	100.00

GSHP HEATING + COOLING BREAKDOWN

End Uses

	Electricity [kWh]	Natural Gas [kWh]	Gasoline [kWh]	Diesel [kWh]	Coal [kWh]	Fuel Oil No 1 [kWh]	Fuel Oil No 2 [kWh]	Propane [kWh]	Other Fuel 1 [kWh]	Other Fuel 2 [kWh]	District Cooling [kWh]	District Heating [kWh]	Water [m3]
Heating	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25085.47	0.00
Cooling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12527.75	0.00	0.00
Interior Lighting	13139.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Exterior Lighting	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Interior Equipment	382.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Exterior Equipment	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fans	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pumps	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Rejection	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Humidification	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Recovery	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Water Systems	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18476.40	289.32
Refrigeration	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Generators	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total End Uses	13521.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	12527.75	43561.86	289.32

Note: District heat appears to be the principal heating source based on energy usage.

GSHP HEATING ONLY

End Uses

	Electricity [kWh]	Natural Gas [kWh]	Gasoline [kWh]	Diesel [kWh]	Coal [kWh]	Fuel Oil No 1 [kWh]	Fuel Oil No 2 [kWh]	Propane [kWh]	Other Fuel 1 [kWh]	Other Fuel 2 [kWh]	District Cooling [kWh]	District Heating [kWh]	Water [m3]
Heating	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	24594.29	0.00
Cooling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Interior Lighting	13139.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Exterior Lighting	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Interior Equipment	382.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Exterior Equipment	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fans	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pumps	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Rejection	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Humidification	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Recovery	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Water Systems	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18476.40	289.32
Refrigeration	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Generators	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total End Uses	13521.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43070.69	289.32

Note: District heat appears to be the principal heating source based on energy usage.

GSHP HEATING + COOLING BREAKDOWN

GSHP HEATING ONLY IN WINTER ONLY.

Site and Source Energy

	Total Energy [kWh]	Energy Per Total Building Area [kWh/m2]	Energy Per Conditioned Building Area [kWh/m2]
Total Site Energy	39716.50	32.43	32.43
Net Site Energy	39716.50	32.43	32.43
Total Source Energy	140516.27	114.72	114.72
Net Source Energy	140516.27	114.72	114.72

End Uses

	Electricity [kWh]	Natural Gas [kWh]	Gasoline [kWh]	Diesel [kWh]	Coal [kWh]	Fuel Oil No 1 [kWh]	Fuel Oil No 2 [kWh]	Propane [kWh]	Other Fuel 1 [kWh]	Other Fuel 2 [kWh]	District Cooling [kWh]	District Heating [kWh]	Water [m3]
Heating	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	23821.96	0.00
Cooling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Interior Lighting	6515.82	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Exterior Lighting	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Interior Equipment	189.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Exterior Equipment	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fans	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pumps	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Rejection	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Humidification	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Recovery	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Water Systems	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9189.47	143.90
Refrigeration	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Generators	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total End Uses	6705.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	33011.43	143.90

Note: District heat appears to be the principal heating source based on energy usage.

VRF+DOAS HEATING + COOLING BREAKDOWN

End Uses

	Electricity [kWh]	Natural Gas [kWh]	Gasoline [kWh]	Diesel [kWh]	Coal [kWh]	Fuel Oil No 1 [kWh]	Fuel Oil No 2 [kWh]	Propane [kWh]	Other Fuel 1 [kWh]	Other Fuel 2 [kWh]	District Cooling [kWh]	District Heating [kWh]	Water [m3]
Heating	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38999.92	0.00
Cooling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20552.63	0.00	0.00
Interior Lighting	13139.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Exterior Lighting	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Interior Equipment	382.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Exterior Equipment	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fans	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pumps	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Rejection	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Humidification	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Recovery	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Water Systems	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18476.40	289.32
Refrigeration	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Generators	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total End Uses	13521.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20552.63	57476.32	289.32

Note: District heat appears to be the principal heating source based on energy usage.

VRF HEATING ONLY

End Uses

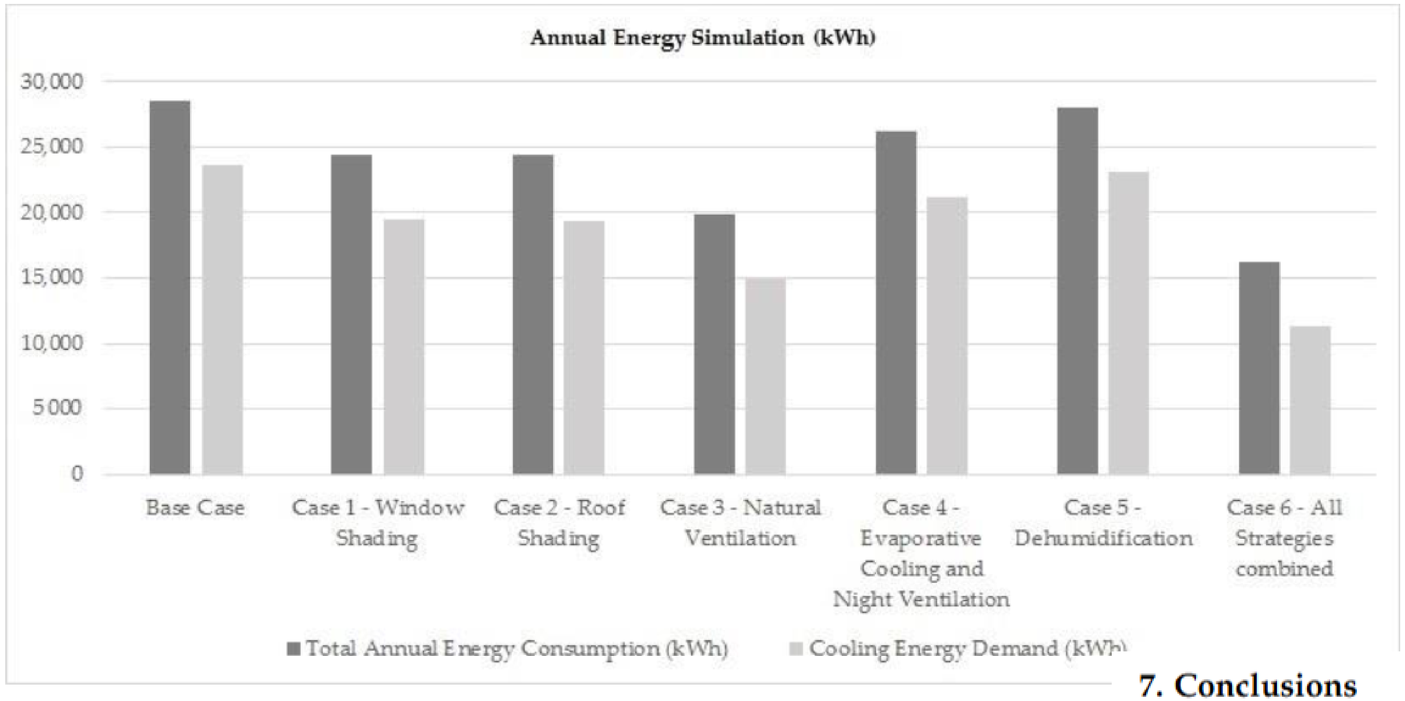
	Electricity [kWh]	Natural Gas [kWh]	Gasoline [kWh]	Diesel [kWh]	Coal [kWh]	Fuel Oil No 1 [kWh]	Fuel Oil No 2 [kWh]	Propane [kWh]	Other Fuel 1 [kWh]	Other Fuel 2 [kWh]	District Cooling [kWh]	District Heating [kWh]	Water [m3]
Heating	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37098.62	0.00
Cooling	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Interior Lighting	13139.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Exterior Lighting	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Interior Equipment	382.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Exterior Equipment	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fans	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pumps	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Rejection	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Humidification	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Heat Recovery	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Water Systems	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18476.40	289.32
Refrigeration	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Generators	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total End Uses	13521.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	55575.02	289.32

Note: District heat appears to be the principal heating source based on energy usage.

TEMPERATURE RANGE COMPARISON OF ITERATIONS

It appeared that the base model had a total of 1985 h of discomfort, representing an average of 70% indoor temperature discomfort hours a year.

8784 hours



7. Conclusions

In this study, different scenarios representing each passive cooling technique have been simulated and compared to evaluate the thermal comfort and the energy consumption technique of a conventional office building in Ouagadougou using EnergyPlus to conduct the simulations. The following conclusions may be taken from the findings and analysis:

i. The simulations showed massive potential for reducing the discomfort hours with the use of roof shading and natural ventilation, which are the most effective passive techniques, reducing the annual discomfort hours by up to 40%.
ii. Window shading has a reverse effect on energy consumption and discomfort hours.
iii. A 53% reduction in the wall U-value reduces the annual discomfort hours by 5% and the annual energy consumption by 8%.
iv. Using a single passive technique cannot guarantee optimal thermal comfort and energy reduction throughout a full year.
v. Overall, the combined use of the strategies decreases the discomfort hours by 69% and the annual energy consumption by 43%.

TEMPERATURE RANGE COMPARISON OF ITERATIONS

For example, CIBSE provides recommended benchmark summertime peak temperatures and overheating criteria as follows:

Offices, schools, dwellings (living areas) should have no more than 1% of the annual occupied hours over operative temperature of 28°C.

Bedrooms in dwellings should have no more than 1% of the annual occupied hours over operative temperature of 26°C.

Sustainable Design of Students' Accommodation Block

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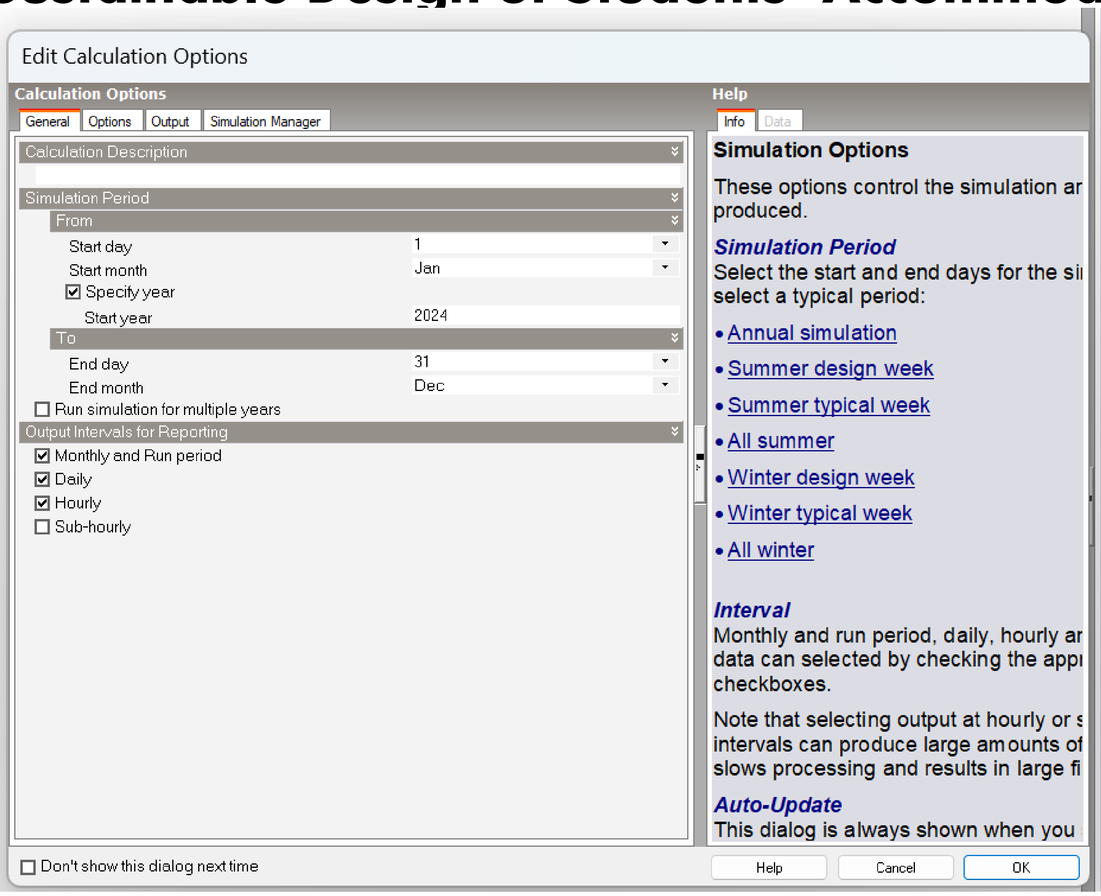

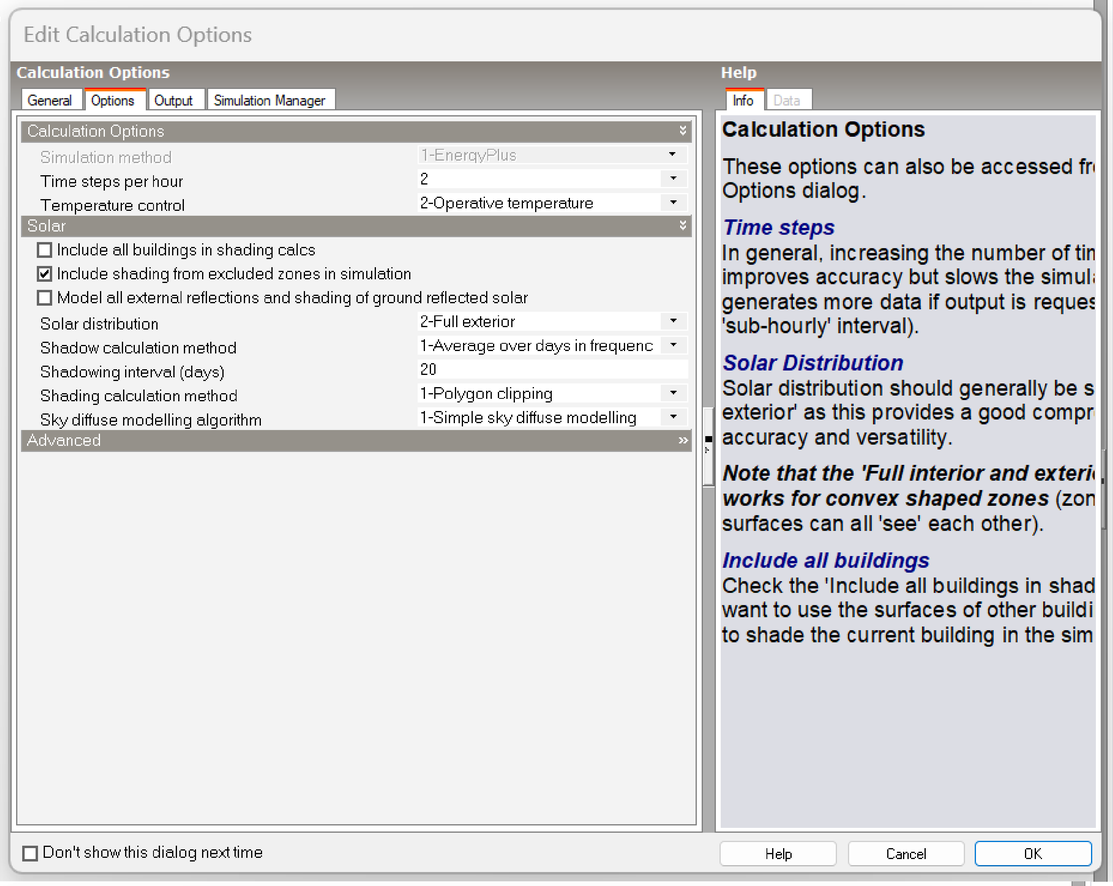

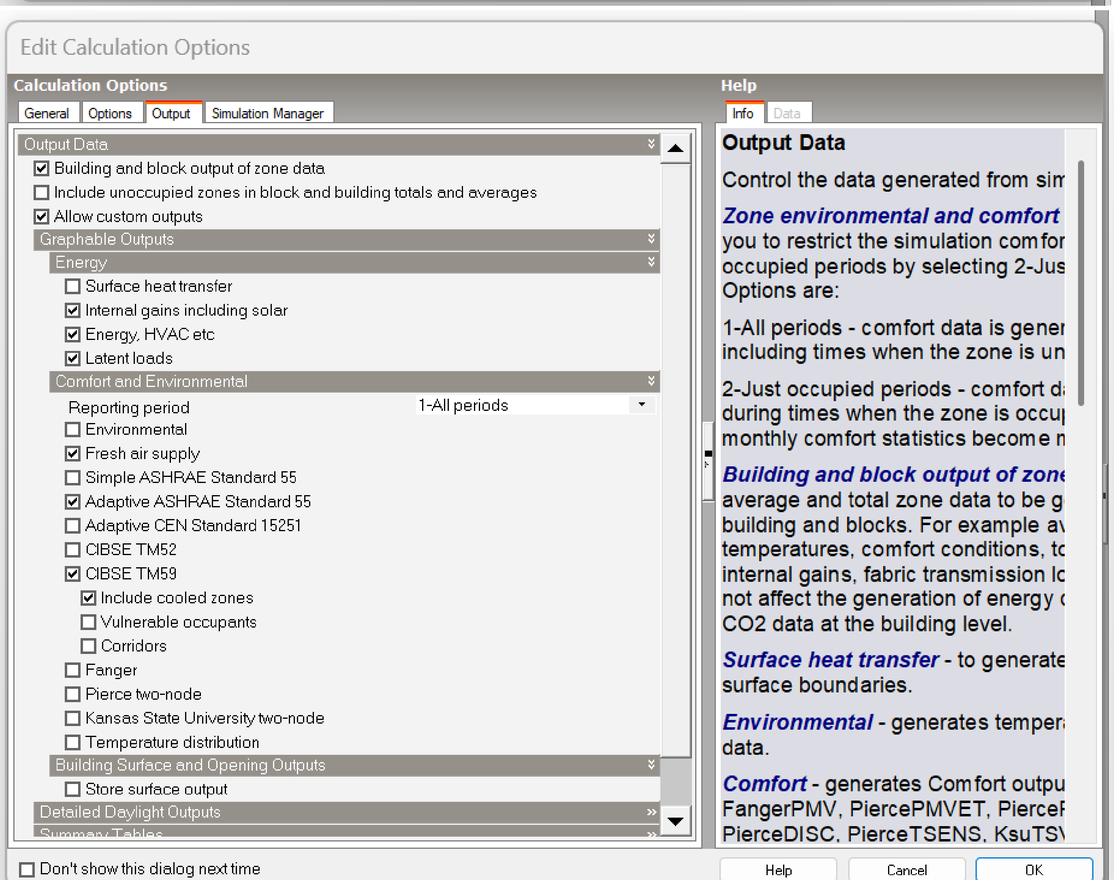

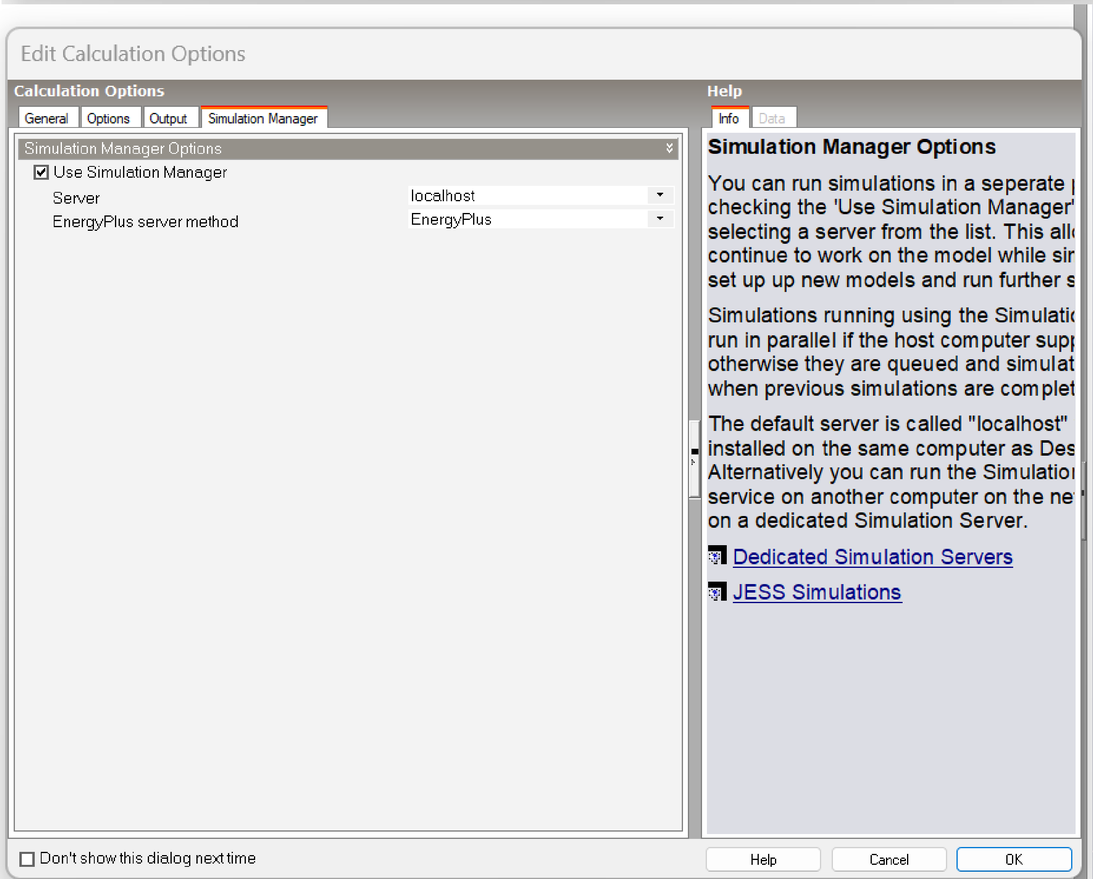

Sustainable Design of Students' Accommodation Block

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Display Options

General | Detailed

Data

Data	3-Comfort
Interval	6-Distribution
Show as	3-Graph and tab
Days per page	365

☐ Normalise by area

Y-Axis

Appearance

Script Manager

Scripts

General

General

☑ Enable scripts

Script — Building Average Zone Air Te

Help

Info | Data

Script

- Script
  - CS-Script
  - EMS
    - Building Average Zone Air Temperature
    - Change Heating and Cooling Setpoints Daily
    - CIBSE TM52 Reports
    - CIBSE TM59 Reports
    - CIBSE TM59 Reports Include Cooled zones
    - CIBSE TM59 Ventilation temperature control enhancement
    - CO2 control (On/Off)
    - CO2 control (Proportional)
    - DIN 4108-2 2013-02 DEFAULT INT BLIND - AUTOMATIC

Data Report (Not Editable)

General

Building Average Zone Air Temperature

Calculate building average zone temperature for the occupied zones

Category — EMS

Script

Enable program — **No**

```
! report average air temperature of all occupied zones in building

! for all occupied zones create sensors and internal variables for temperature
and zone air volume respectively
<ForAllOccupiedZones>,
EnergyManagementSystem:Sensor,
  Temp_<LoopZoneVariableName>,
  <LoopZoneIDFName> ,
  Zone Mean Air Temperature ;

EnergyManagementSystem:InternalVariable,
  Vol_<LoopZoneVariableName>,
```

Help | Cancel | OK

Summary Tables

Summary output units (SI): 1-kWh

Summary Annual Reports

- ☐ All Summary
- ☐ LEED Summary
- ☑ Annual Building Utility Performance Summary (ABUPS)
- ☑ Demand End Use Components Summary
- ☑ Sensible heat gain summary
- ☑ Input Verification and Results Summary
- ☐ Source Energy End Use Components Summary
- ☑ Adaptive Comfort Summary
- ☑ Zone Component Load Summary
- ☑ Standard 62.1 Summary
- ☐ Energy Meters
- ☑ Climatic Data Summary
- ☑ Equipment Summary
- ☑ Envelope Summary
- ☐ Initialization Summary
- ☐ Surface Shadowing Summary
- ☐ Shading Summary
- ☑ Lighting Summary
- ☑ HVAC Sizing Summary
- ☑ System Summary
- ☑ Component Sizing Summary
- ☑ Outdoor Air Summary
- ☑ Object Count Summary
- ☐ Component Cost Economics Summary

Summary Monthly Reports

Miscellaneous Outputs

Summary Monthly Reports

- ☐ All Monthly
- ☑ Zone Cooling Summary Monthly
- ☑ Zone Heating Summary Monthly
- ☐ Zone Electric Summary Monthly
- ☐ Space Gains Monthly
- ☐ Peak Space Gains Monthly
- ☐ Space Gain Components At Cooling Peak Monthly
- ☐ Energy Consumption Electricity Natural Gas Monthly
- ☐ Energy Consumption Electricity Generated Propane Monthly
- ☐ Energy Consumption Diesel Fuel Oil Monthly
- ☐ Energy Consumption District Heating Cooling Monthly
- ☐ Energy Consumption Coal Gasoline Monthly
- ☐ End Use Energy Consumption Electricity Monthly
- ☐ End Use Energy Consumption Natural Gas Monthly
- ☐ End Use Energy Consumption Diesel Monthly
- ☐ End Use Energy Consumption FuelOil Monthly
- ☐ End Use Energy Consumption Coal Monthly
- ☐ End Use Energy Consumption Propane Monthly
- ☐ End Use Energy Consumption Gasoline Monthly
- ☑ End Use Energy Consumption Other Fuels Monthly
- ☐ Peak Energy End Use Electricity Part1 Monthly
- ☐ Peak Energy End Use Electricity Part2 Monthly
- ☐ Electric Components Of Peak Demand Monthly
- ☐ Peak Energy End Use Natural Gas Monthly
- ☐ Peak Energy End Use Diesel Monthly
- ☐ Peak Energy End Use Fuel Oil Monthly
- ☐ Peak Energy End Use Coal Monthly
- ☐ Peak Energy End Use Propane Monthly

- ☐ Peak Energy End Use Propane Monthly
- ☐ Peak Energy End Use Gasoline Monthly
- ☑ Peak Energy End Use Other Fuels Monthly
- ☐ Setpoints Not Met With Temperatures Monthly
- ☐ Comfort Report Simple 55 Monthly
- ☐ Unglazed Transpired Solar Collector Summary Monthly
- ☐ Occupant Comfort Data Summary Monthly
- ☐ Chiller Report Monthly
- ☐ Tower Report Monthly
- ☐ Boiler Report Monthly
- ☐ DX Report Monthly
- ☐ Window Report Monthly
- ☐ Window Energy Report Monthly
- ☐ Window Zone Summary Monthly
- ☐ Window Energy Zone Summary Monthly
- ☐ Average Outdoor Conditions Monthly
- ☐ Outdoor Conditions Maximum Dry-Bulb Monthly
- ☐ Outdoor Conditions Minimum Dry-Bulb Monthly
- ☐ Outdoor Conditions Maximum Wet-Bulb Monthly
- ☐ Outdoor Conditions Maximum Dew Point Monthly
- ☐ Outdoor Ground Conditions Monthly
- ☐ Window AC Report Monthly
- ☐ Water Heater Report Monthly
- ☐ Generator Report Monthly
- ☐ Daylighting Report Monthly
- ☐ Coil Report Monthly
- ☐ Plant Loop Demand Report Monthly
- ☐ Fan Report Monthly
- ☐ Pump Report Monthly

Sustainable Design of Students' Accommodation Block

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Model Options - Building and Block

Model Options

Data | Advanced | Heating Design | Cooling Design | Simulation | Display | Drawing tools | Block | Project details | Carbon

Construction and Glazing Data

Gains Data

Gains data

Lumped — Early — Detailed

Early gains

Internal gains are separated into various categories (e.g. occupancy, lighting, computing etc.)

Gain definition level	1-Globally in model options
Occupancy method	1-Occupancy density
Occupancy latent gains method	1-Dynamic calculation
Equipment gain units	1-Power density
Lighting gain units	2-Normalised power density

Timing

HVAC

HVAC

Simple — Detailed

Simple HVAC

HVAC systems are modelled using Ideal Loads, fuel consumption is calculated from loads using seasonal efficiencies

HVAC sizing	3-Autosize
Simple HVAC autosize method	1-EnergyPlus
☑ Specify Simple/Design HVAC details	
Auxiliary energy calculations	2-Separate fans and pumps
Mechanical ventilation method	2-Ideal loads

Natural Ventilation and Infiltration

Natural ventilation

Scheduled — Calculated

Scheduled ventilation

Ventilation is defined as an air-change rate modified by an operation schedule and controlled using a set-point temperature.

Infiltration units	1-ac/h

Imported Surfaces

Model Options - Building and Block

Model Options

Data | Advanced | Heating Design | Cooling Design | Simulation | Display | Drawing tools | Block | Project details | Carbon

Simplification

- ☐ Merge zones of same activity
- ☐ Merge zones connected by holes
- ☐ Merge zones by selection
- ☑ Lump similar windows on surface
- ☐ Lump similar cracks on surface

Adjacency Settings

Adjacency separation tolerance (m)	0.010
Adjacency angular tolerance (°)	5.0

☐ Standard component block adjacencies

Natural Ventilation

Lighting

Filters

Component Block

Model Options - Building and Block

Model Options

Data | Advanced | Heating Design | Cooling Design | Simulation | Display | Drawing tools | Block | Project details | Carbon

Winter design day

Calculation Options

Simulation method	1-EnergyPlus
Temperature control	1-Air temperature

System Sizing

Design margin	1.25

Output

Advanced

Model Options - Building and Block

Model Options

Data | Advanced | Heating Design | Cooling Design | Simulation | Display | Drawing tools | Block | Project details | Carbon

Calculation Options

Simulation method	1-EnergyPlus
Temperature control	1-Air temperature

- ☐ Exclude all zone natural ventilation (infiltration is always included)
- ☐ Exclude all zone mechanical ventilation
- ☑ Exclude heat recovery

Summer Design Day

Day	15
Start month	Jun
End month	Sep
Day of week	9-SummerDesignDay

System Sizing

Design margin	1.15
Sizing method	1-ASHRAE

Solar

☐

☑

☐

Sc…terior

Out…

Adv…

Model Options - Building and Block

Model Options

Data | Advanced | Heating Design | Cooling Design | Simulation | Display | Drawing tools | Block | Project details | Carbon

Simulation Options

From

Start day	1
Start month	Jan

To

End day	31
End month	Dec

Calculation Options

Simulation method	1-EnergyPlus
Time steps per hour	2
Temperature control	2-Operative temperature

Solar

- ☐ Include all buildings in shading calcs
- ☑ Include shading from excluded zones in simulation
- ☐ Model all external reflections and shading of ground reflected solar

Solar distribution	2-Full exterior
Shadow calculation method	1-Average over days in frequency
Shadowing interval (days)	20
Shading calculation method	1-Polygon clipping
Sky diffuse modelling algorithm	1-Simple sky diffuse modelling

Advanced

Output

- ☑ Building and block output of zone data
- ☐ Include unoccupied zones in block and building totals and averages
- ☑ Allow custom outputs

Graphable Outputs

Detailed Daylight Outputs

Summary Tables

Miscellaneous Outputs

Time Setpoints not Met Tolerances

Sustainable Design of Students' Accommodation Block

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Model Options - Building and Block

Model Options

Data | Advanced | Heating Design | Cooling Design | Simulation | Display | Drawing tools | Block | Project details | Carbon

Show Data for Calculations

Analysis type: 1-EnergyPlus

Edit Screen

Dynamics: 1-Rendered

- [x] Show zone labels

Zone label size: 2-Medium

- [ ] Display zone inner surfaces
- [x] Display filled external walls
- [ ] Display zone filled walls
- [ ] Display zone inner volume floors
- [x] Display zone inner floor perimeters
- [ ] Fit includes imported floor plan
- [x] Show imported 2-D drawing at zone and surface levels
- [x] Show shading devices
- [x] Show openings and sub-surfaces at building level
- [ ] Surrounding block display
- [ ] Surrounding block zone display

Zone floor shading at block level: 1-Activity

- [x] Automatic HVAC network breaks

Visualisation Display

Dynamics: 1-Rendered

- [ ] Show shadows
- [x] Show North arrow
- [x] Show ground plane

View (°): 50

- [ ] Antialiasing

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- Activity Template
  - Template: TM59_1-BedLivingKitchen
  - Sector: Residential spaces
  - Zone multiplier: 1
  - [x] Include zone in thermal calculations
  - [x] Include zone in Radiance daylighting calculations
- Floor Areas and Volumes
- Occupancy
  - [x] Occupied?
  - **Occupancy density (people/m2)**: 0.1000
  - Schedule: TM59_Living/Kitchen_Occ
  - Metabolic
  - Clothing
  - Comfort Radiant Temperature Weighting
  - Air Velocity
- Contaminant Generation and Removal
- Holidays
- DHW
- Environmental Control
  - Heating Setpoint Temperatures
    - Heating (°C): 21.0
    - Heating set back (°C): 12.0
  - Cooling Setpoint Temperatures
    - Cooling (°C): 25.0
    - Cooling set back (°C): 28.0
  - Humidity Control
  - Ventilation Setpoint Temperatures
  - Minimum Fresh Air
  - Lighting
- Computers
  - [x] On
  - Power density (W/m2): 0.00
  - Schedule: Dwell_DomCommonAreas_Equip
  - Radiant fraction: 0.200
- Office Equipment
  - [x] On
  - Power density (W/m2): 0.00
  - Schedule: TM59_Kitchen_Equip
  - Radiant fraction: 0.000
- Miscellaneous
  - [x] On
  - Power density (W/m2): 0.00
  - Schedule: TM59_Living/Kitchen_Equip
  - Fuel: 1-Electricity from grid
  - Fraction lost: 0.000000
  - Latent fraction: 0.000000
  - Radiant fraction: 0.200000
- Catering
- Process

- Construction Template
  - Template: Project construction template
- Construction
  - External walls: Project wall
  - Below grade walls: Project below grade wall
  - Flat roof: Project flat roof
  - Pitched roof (occupied): Project pitched roof
  - Pitched roof (unoccupied): Project unoccupied pitched roof
  - Internal partitions: Project partition
  - Semi-Exposed
    - Semi-exposed walls: Project semi-exposed wall
    - Semi-exposed ceiling: Project semi-exposed ceiling
    - Semi-exposed floor: Project semi-exposed floor
  - Floors
    - Ground floor: Project ground floor
    - External floor: Project external floor
    - Internal floor: Project internal floor
  - Sub-Surfaces
  - Internal Thermal Mass
  - Component Block
  - Geometry, Areas and Volumes
  - Surface Convection
  - Linear Thermal Bridging at Junctions
- Airtightness
  - [x] Model infiltration
  - Constant rate (ac/h): 0.700
  - Schedule: On 24/7
  - Delta T and Wind Speed Coefficients
- Cost

- Glazing Template
  - Template: Project glazing template
- External Windows
  - Glazing type: Project external glazing
  - Layout: Preferred height 1.5m, 30% glazed
  - Dimensions
    - Type: 3-Preferred height
    - Window to wall %: 30.00
    - Window height (m): 1.50
    - Window spacing (m): 5.00
    - Sill height (m): 0.80
    - Outside reveal depth (m): 0.000
  - Frame and Dividers
  - Shading
  - Airflow Control Windows
  - Free Aperture
- Internal Windows
- Sloped Roof Windows/Skylights
- Doors
- Vents

- Lighting Template
  - Template: IECC-1998
- General Lighting
  - [x] On
  - Normalised power density (W/m2-100 lux): 5.0000
  - Schedule: TM59_Default_Light
  - Luminaire type: 1-Suspended
  - Radiant fraction: 0.420
  - Visible fraction: 0.180
  - Convective fraction: 0.400
- Lighting Control
  - [ ] On
- Task and Display Lighting
  - [ ] On
- Exterior Lighting
  - [ ] On
- Cost

- HVAC Template
  - Template: Natural ventilation - No Heating/Cooling
- Mechanical Ventilation
  - [ ] On
- Auxiliary Energy
  - Pump etc energy (W/m2): 0.0000
  - Schedule: TM59_Living/Kitchen_Occ
- Heating
  - [ ] Heated
- Cooling
  - [ ] Cooled
- Humidity Control
- DHW
  - [x] On
  - **DHW Template**: Project DHW
  - Type: 4-Instantaneous hot water only
  - DHW CoP: 0.8500
  - Fuel: 1-Electricity from grid
  - Water Temperatures
    - Delivery temperature (°C): 65.00
    - Mains supply temperature (°C): 10.00
  - Operation
    - Schedule: Dwell_DomCommonAreas_Occ
- Natural Ventilation
  - [x] On
  - Outside air definition method: 1-By zone
  - Outside air (ac/h): 5.000
  - Operation
    - Schedule: TM59_Living/Kitchen_Occ
  - Outdoor Temperature Limits
  - Delta T Limits
  - Delta T and Wind Speed Coefficients
  - Mixed Mode Zone Equipment
    - [ ] Mixed mode on
- Earth Tube
- Air Temperature Distribution
- Cost

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

END OF SUBMISSION

BT

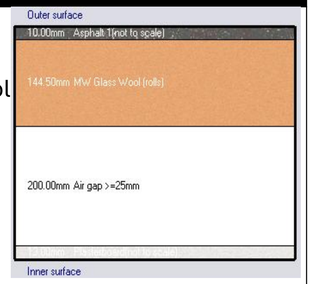
Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

BASE CASE

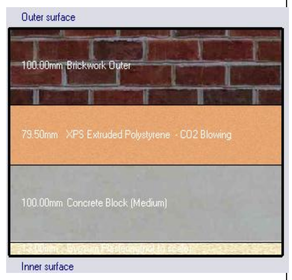
Roof (367.5mm)
10mm asphalt
144.5mm MW glass wool
200mm air gap
13mm plasterboard

U-value(0.25)

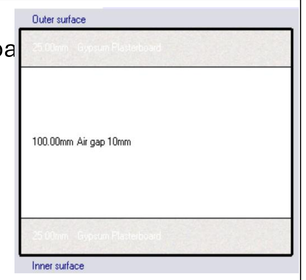


External wall (292mm)
100mm brickwork
79.5mm XPS insulation
100mm concrete block
13mm gypsum plastering

U-value(0.350)



Internal wall (450mm)
25mm gypsum plasterboa
100mm air gap
25mm gypsum plasterboard
U-value(1.64)



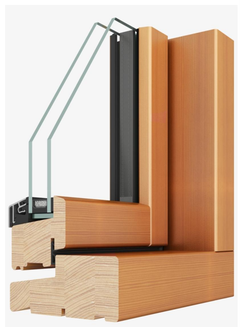
Ground floor slab (300mm)
30mm timber flooring
70mm floor screed
100mm cast concrete
132.7mm UREA fomaldehyde foam

U-value(0.25)

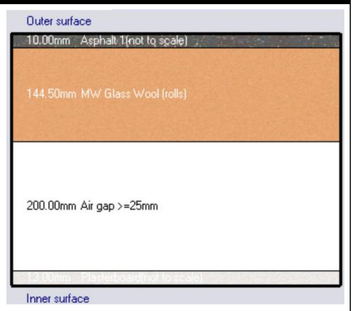
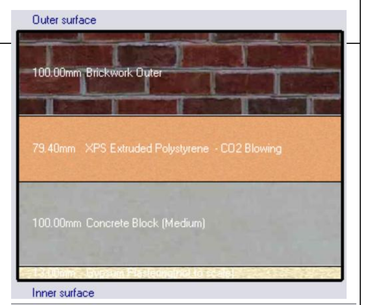
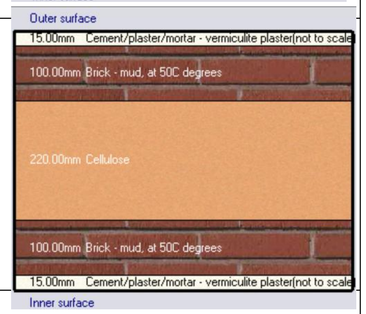
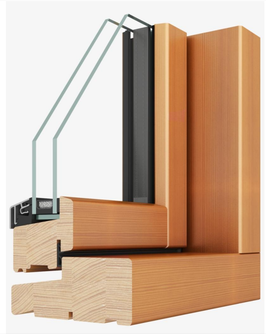
Window:Double Clear 3mm/13mmAir
Frame:painted wood

G-value: 0.691
VT: 0.744

U-value(1.96)



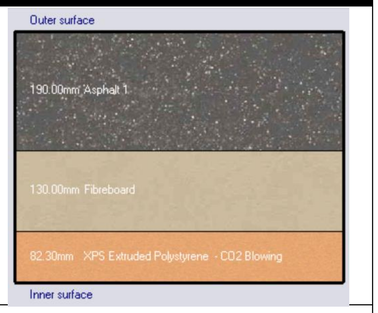
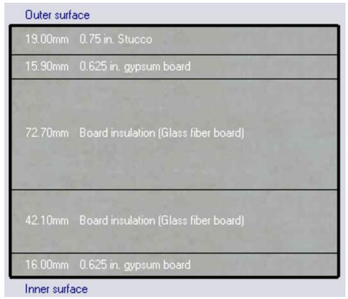
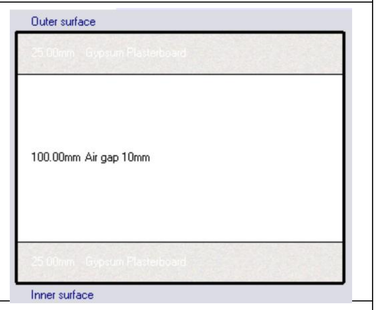
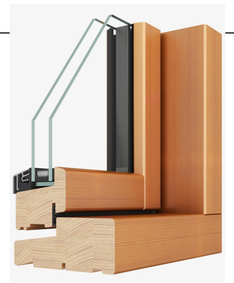
IECC-2000

ELEMENT(mm)	CROSS-SECTION
Roof (367.5mm) 10mm asphalt 144.5mm glass wool 200mm air gap 13mm plasterboard **U-value(0.25)**	
External wall (292mm) 100mm brickwork 79.4mm XPS extruded polystyrene insulation 100mm concrete block (medium) 13mm gypsum plastering **U-value(0.351)**	
Internal wall (450mm) 15mm plaster 100mm brick-mud 220mm cellulose 100mm brick-mud 15mmplaster **U-value(1.64)**	
Ground floor slab (300mm) 100mm UREA fomaldehyde foam 100mm cast concrete 70mm floor screed 30mm timber flooring **U-value(0.314)**	
Window:Double Clear 3mm/13mmAir **Frame:painted wood** G-value: 0.764 VT: 0.812 **U-value(2.7)**	

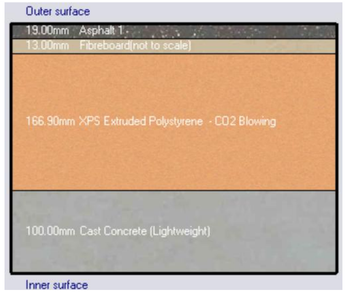
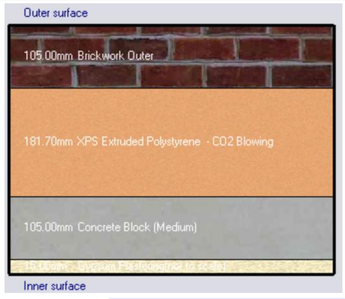
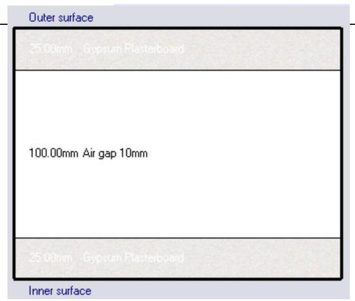
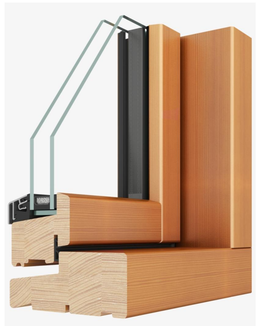
Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

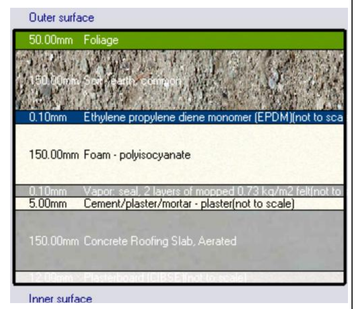
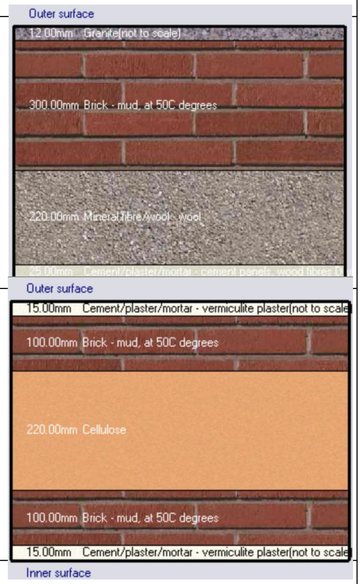
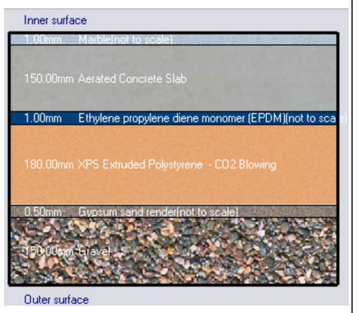
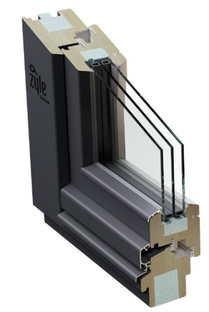
SEATTLE CODE

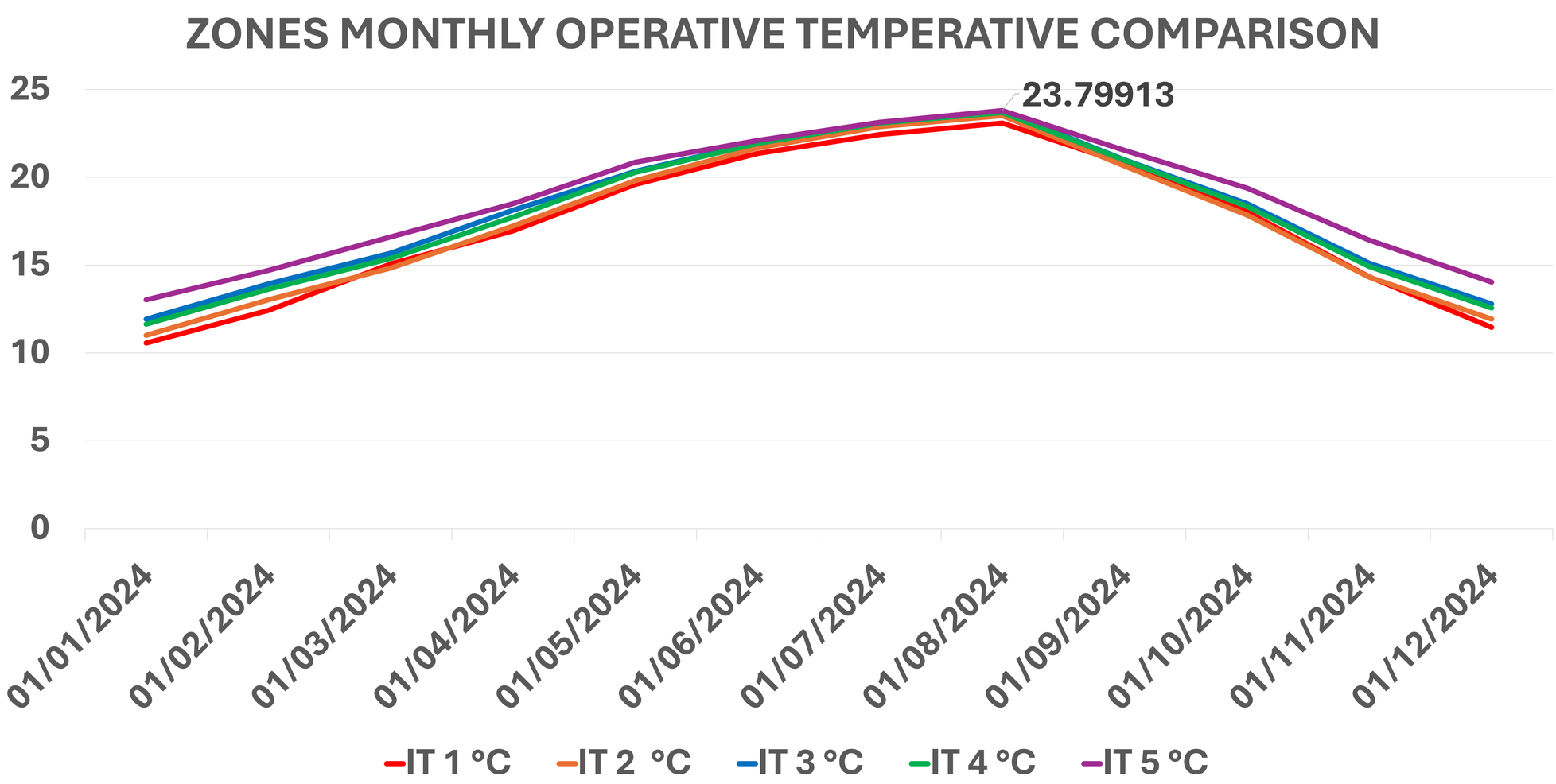
ELEMENT(mm)	CROSS-SECTION
Roof (402mm) 190mm asphalt 130mm fibreboard 82.3mm XPS **U-value(0.20)**	
External wall (166mm) 19mm Stucco 16mm Gypsumboard 72.7mm glass fibre insulation 42.1mm glass fibre insulation 16mm gypsumboard **U-value(0.28)**	
Internal wall (150mm) 25mmplasterboard 100mm air gap 25mmplasterboard **U-value(1.64)**	
Ground floor slab (358mm) 50mm floor screed 182.4mm EPS insul. 100mm concrete 25mm brick slips Clay underfloor **U-value(0.20)**	
Window:LoE Clr 3mm double – 13mm Air G-value: 0.65 VT: 0.76 **U-value(1.78)**	

LEED

ELEMENT(mm)	CROSS-SECTION
Roof (300mm) 19mm asphalt 13mm fibreboard 167mm XPS 100mm concrete **U-value(0.18)**	
External wall (407mm) 105mm brickwork 182mm XPS insulation 105mm concrete 15mm gypsum plaster **U-value(0.17)**	
Internal wall (150mm) 25mmplasterboard 100mm air gap 25mmplasterboard **U-value(1.639)**	
Ground floor slab (440mm) 30mm timber floor 70mm floor screed 100mm concrete 240mm fomaldehyde foam **U-value(0.15)**	
Window:LoE Elec Abs bleached 6mm double Argon G-value: 0.48 VT: 0.657 **U-value(1.3)**	

PASSIVHAUS

ELEMENT(mm)	CROSS-SECTION
Roof (532mm) 50mm vegetation 120mm soil Waterproofing 150mm polysiocyanate insulation Vapour seal 50mm screed 150mm concrete 12mm plasterboard **U-value(0.148)**	
External wall (550mm) glass-fiber-reinforced concrete panels 300mm brick mud 220mm mineral fibre wool 12mm white granite **U-value(0.15)**	
Internal wall (450mm) 15mm plaster 100mm brick-mud 220mm cellulose 100mm brick-mud 15mmplaster **U-value(1.64)**	
Ground floor slab (540mm) 10mm marble 150mm aerated concrete 1.2mm damp proof 180mm extruded polystrene 50mm blinding 150mm hardcore **U-value(0.149)**	
Window:Triple LowE clear 3mm/13mmAr Frame:wood;ext.aluminum G-value: 0.47 VT: 0.66 **U-value(0.78)**	


ZONES MONTHLY OPERATIVE TEMPERATIVE COMPARISON
23.79913
25
20
15
10
5
0
01/01/2024
01/02/2024
01/03/2024
01/04/2024
01/05/2024
01/06/2024
01/07/2024
01/08/2024
01/09/2024
01/10/2024
01/11/2024
01/12/2024
IT 1 °C
IT 2 °C
IT 3 °C
IT 4 °C
IT 5 °C

Sustainable Design of Students' Accommodation Block

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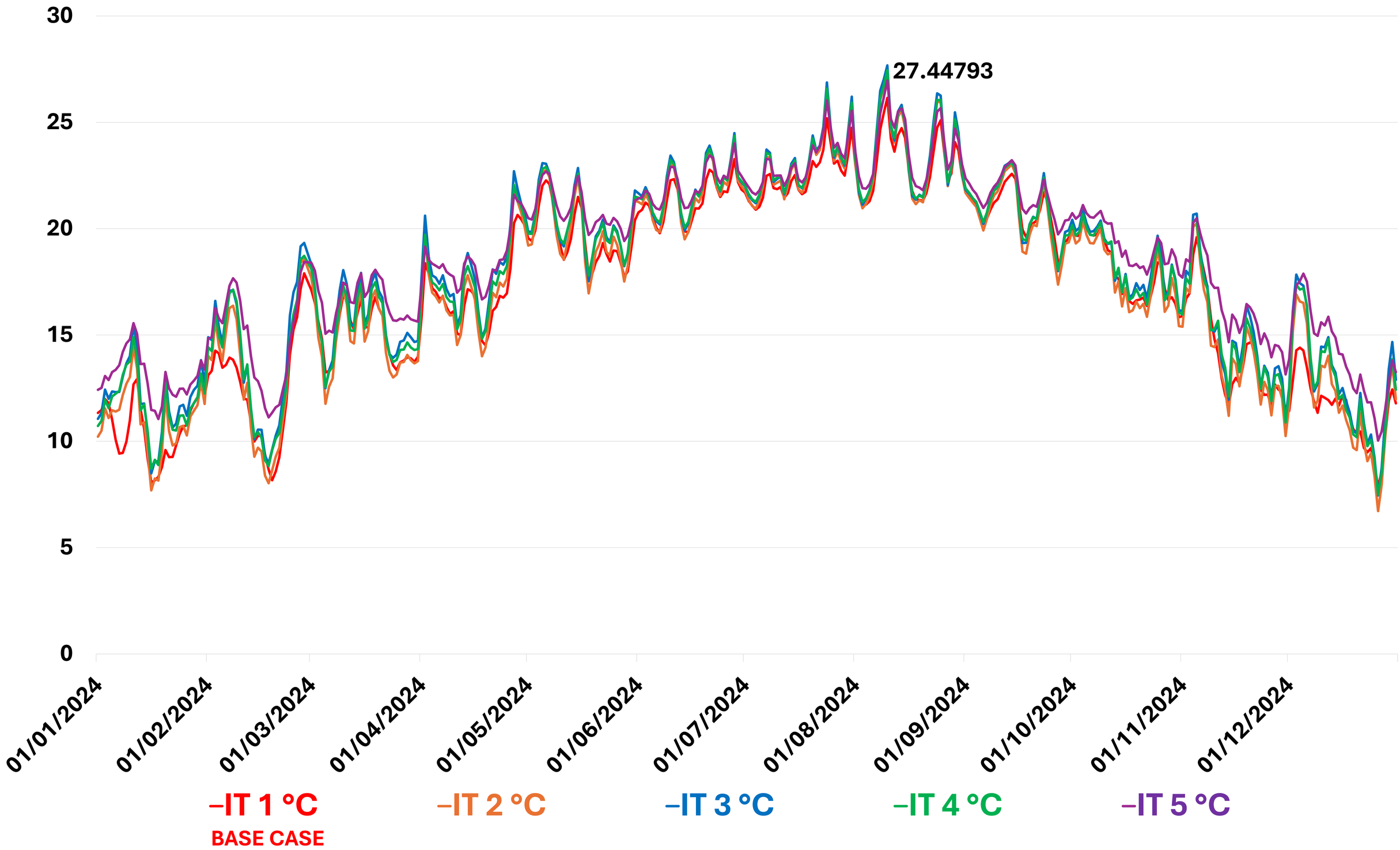
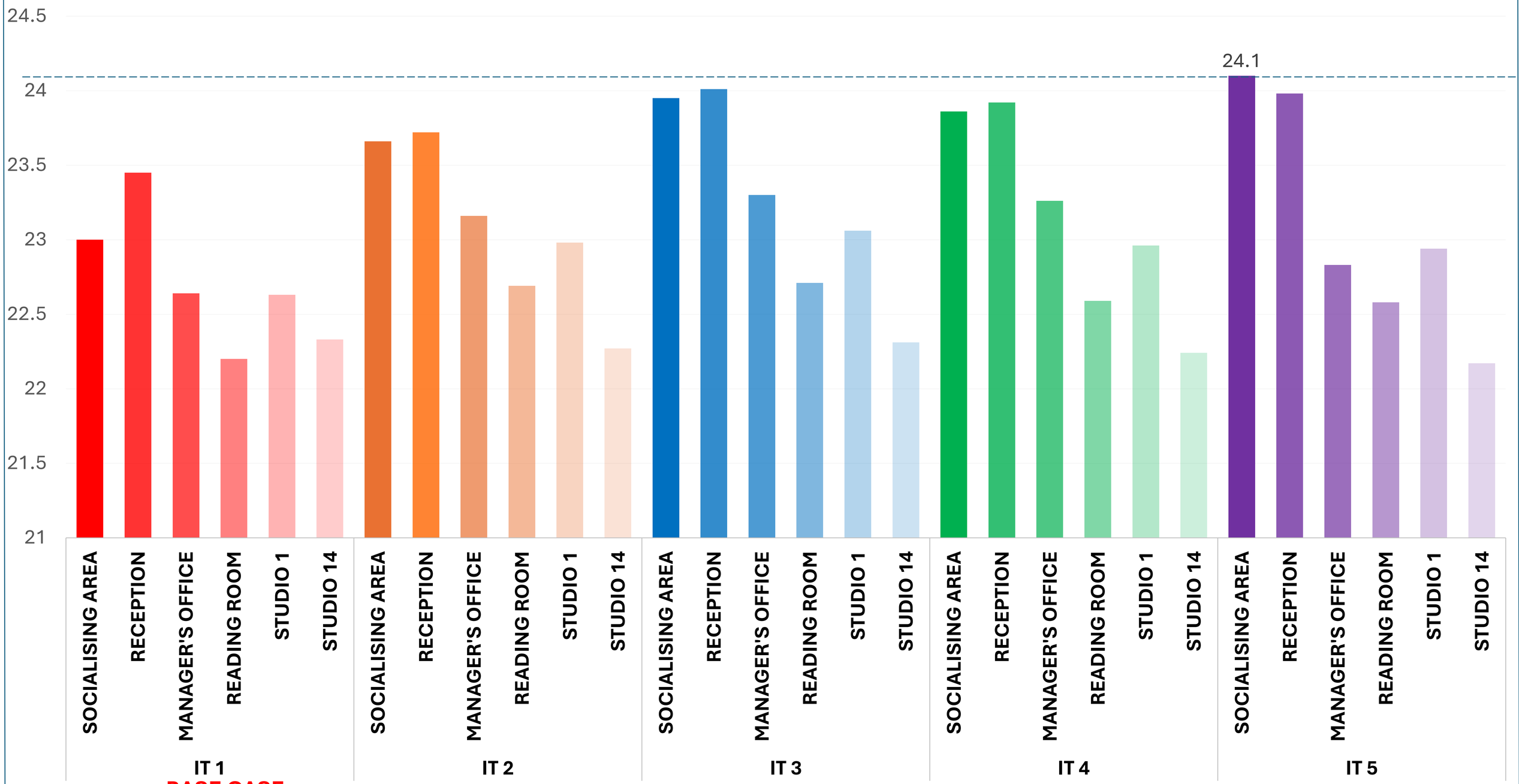
Afua Ampomah-Amoako (201838915)

ZONES DAILY OPERATIVE TEMPERATURE COMPARISON



ZONES ONLY

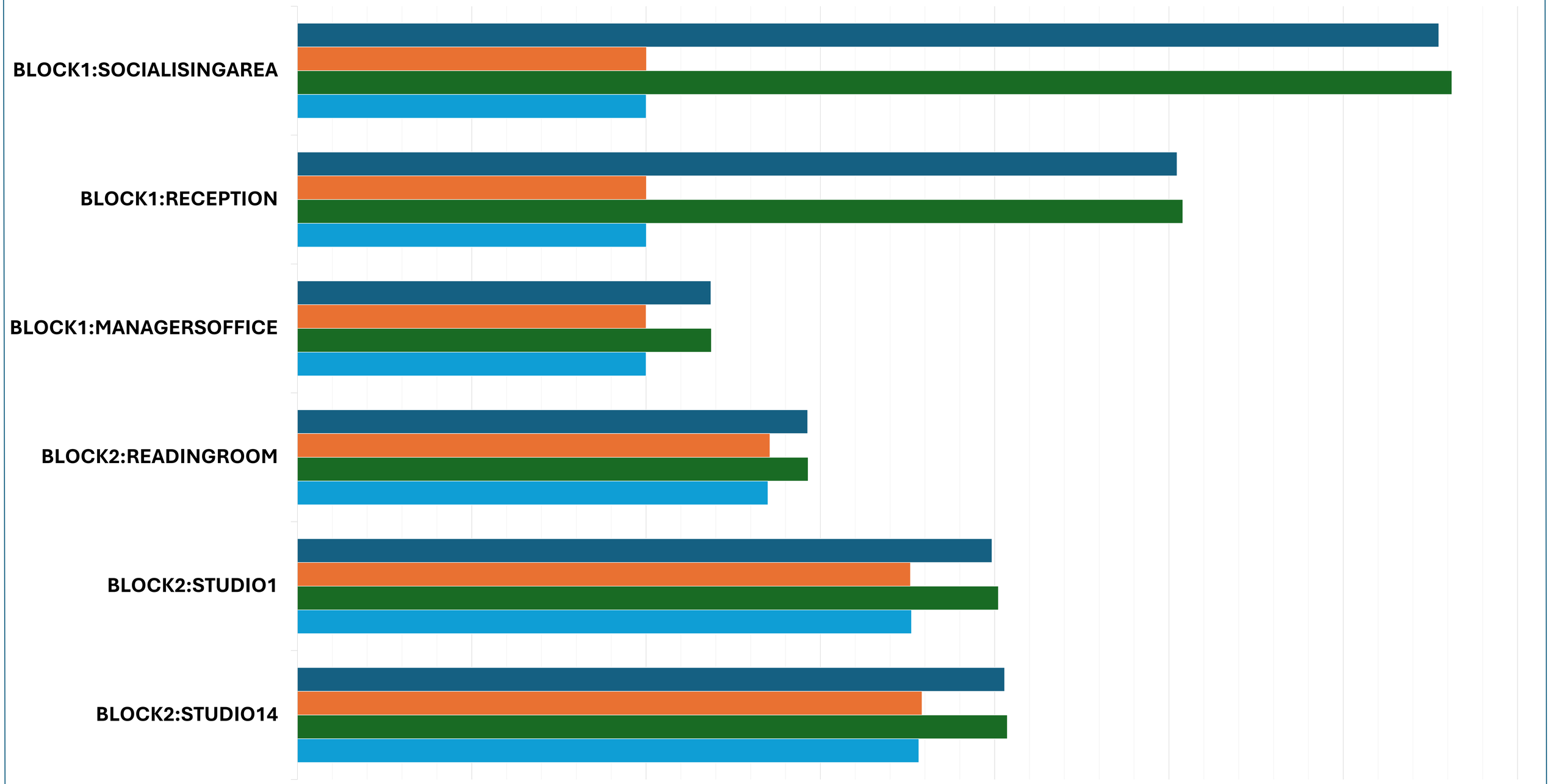
MONTHLY HIGHEST OPERATIVE TEMPERATURE OF ZONES IN AUGUST

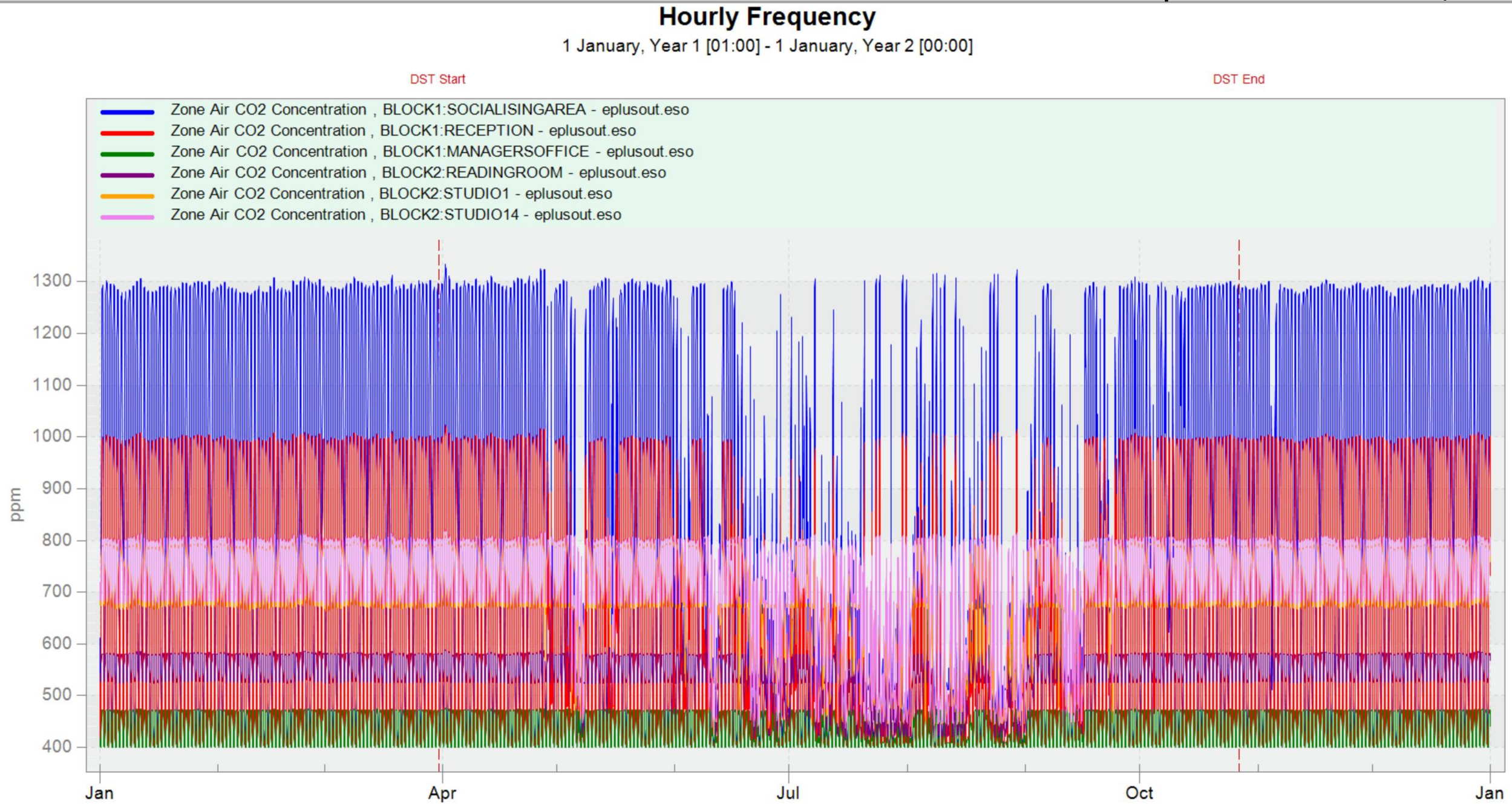


Sustainable Design of Students' Accommodation Block

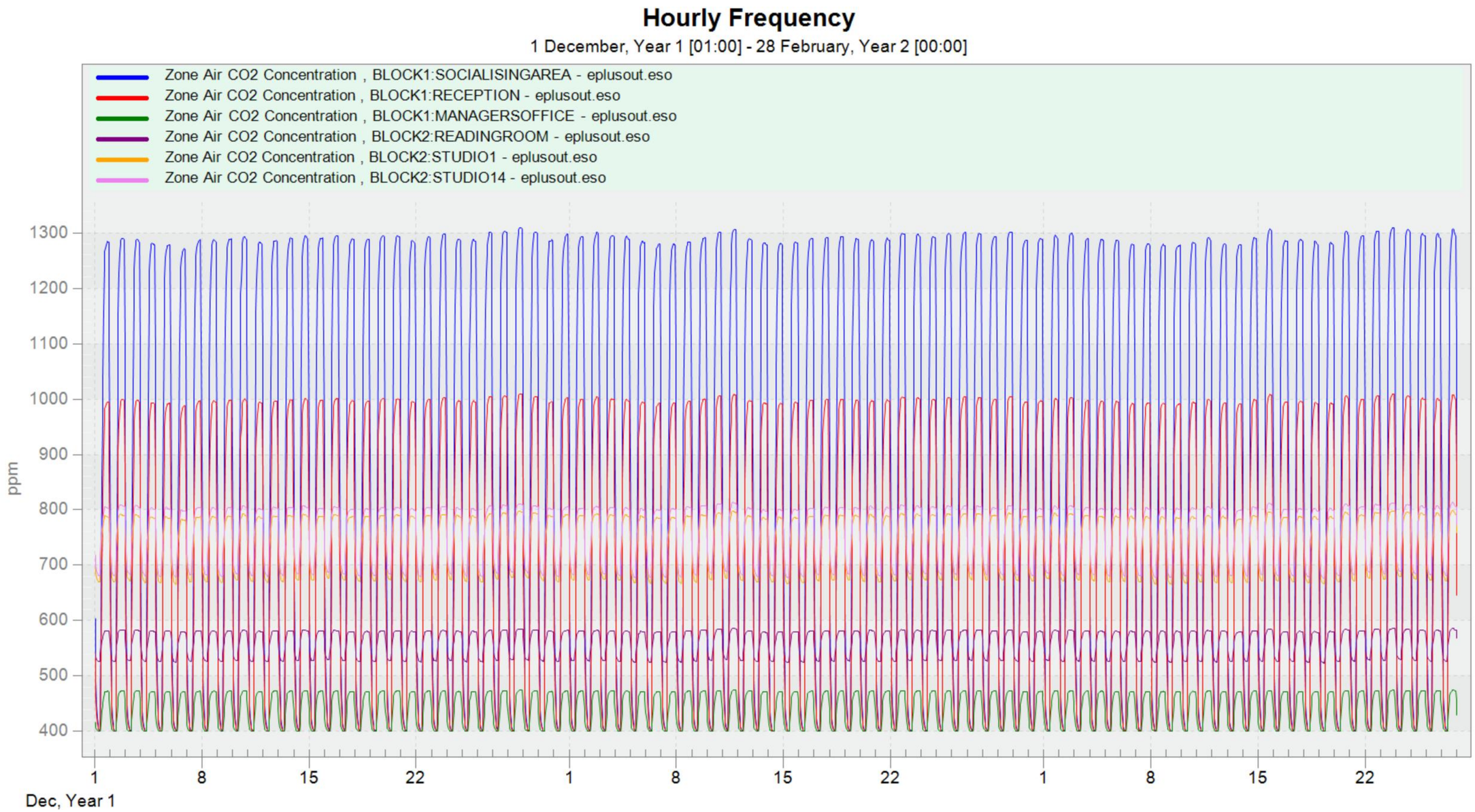
Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

IT 1 WINTER AND SUMMER CO2 LEVELS COMPARISON *Not true representation of data*

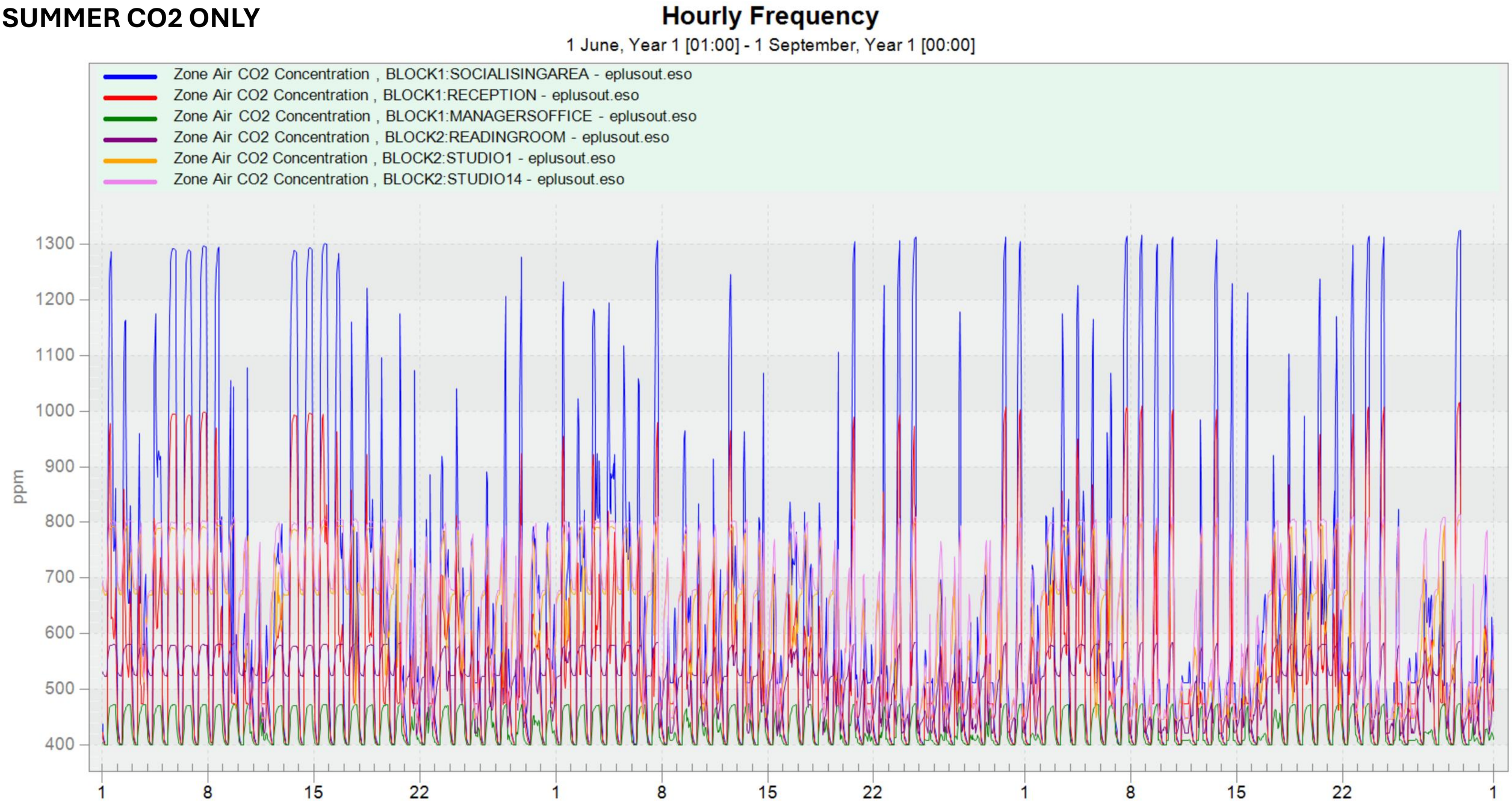



Hourly Frequency
1 January, Year 1 [01:00] - 1 January, Year 2 [00:00]
DST Start
DST End
Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso
ppm
1300
1200
1100
1000
900
800
700
600
500
400
Jan
Apr
Jul
Oct
Jan

WINTER CO2 ONLY



SUMMER CO2 ONLY



BT

Outer surface
50.00mm Foliage
120.00mm Soil, earth / common
0.10mm Ethylene propylene diene monomer (EPDM)(not to sca
247.00mm Foam - polyisocyanate
0.10mm Vapor: seal, 2 layers of mopped 0.73 kg/m2 felt(not to
50.00mm Cement/plaster/mortar - plaster
150.00mm Concrete Roofing Slab, Aerated
Inner surface

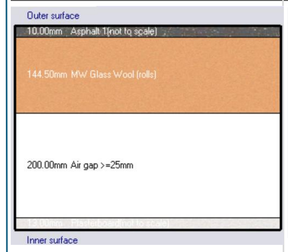
DESIGNBUILDER

BASE CASE IT 1.

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

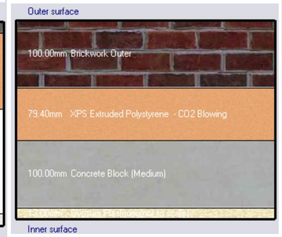
BASE CASE SUMMARY



Roof (367.5mm)

10mm asphalt
144.5mm glass wool
200mm air gap
13mm plasterboard

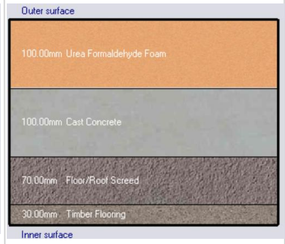
U-value(0.25)



External wall (292mm)

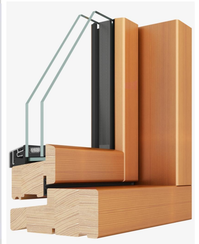
100mm brickwork
79.4mm XPS extruded polystyrene insulation
100mm concrete block (medium)
13mm gypsum plastering

U-value(0.351)



Ground floor slab (300mm)
100mm UREA fomaldehyde foam
100mm cast concrete
70mm floor screed
30mm timber flooring

U-value(0.314)



Window:Double Clear 3mm/13mmAir
Frame:painted wood

G-value: 0.764
VT: 0.812

U-value(2.7)

WINDOWS (NO SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1m x 0.7m	2	North
STUDIO 1	1m x 0.9m	1	South
STUDIO 14	1m x 0.9m	1	South
RECEPTION	(South)1.5mx1m ; 1.5x1(East)	(South)4 ; (East)1	South & East
MANAGER'S OFFICE	1m x 0.7m	1	North
SOCIALISING AREA	1.5m	5	South

RESULT: EXTREMELY LOW SDA VALUES FOR ALL SPACES EXCEPT THE RECEPTION

ACHIEVE DAYLIGHTING TARGET BY INCREASING WINDOW SIZES AND CHANGING GLASS TYPE

BASE CASE PASS CIBSE TM59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	0.28	N/A	Pass
SOCIALISINGAREA	0.00	N/A	Pass
READINGROOM	0.00	0.00	Pass
STUDIO1	0.00	0.00	Pass
STUDIO14	0.00	0.00	Pass

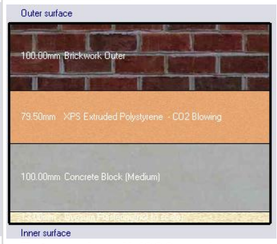
BASE CASE PASS ASE BUT NOT SDA

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	17.7	0.00	0.00	17.66	100.0
STUDIO 1	12.5	0.00	0.00	12.55	100.0
STUDIO 14	11.5	0.11	0.95	11.51	100.0
RECEPTION	113.7	56.42	49.63	113.49	99.8
MANAGERS OFFICE	6.6	0.00	0.00	6.65	100.0
SOCIALISING AREA	95.8	8.76	9.14	95.74	99.9
	257.9	**65.28**	**25.32**	**257.59**	**99.9**

ITERATION 2 SUMMARY

Roof (367.5mm)
10mm asphalt
144.5mm MW glass wool
200mm air gap
13mm plasterboard

U-value(0.25)

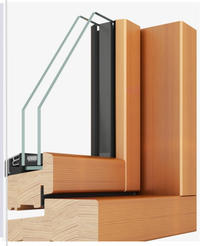


External wall (292mm)
100mm brickwork
79.5mm XPS insulation
100mm concrete block
13mm gypsum plastering

U-value(0.350)

Ground floor slab (300mm)
30mm timber flooring
70mm floor screed
100mm cast concrete
132.7mm UREA fomaldehyde
Foam

U-value(0.25)



Window:Double Clear
3mm/13mmAir
Frame:painted wood
G-value: 0.691
VT: 0.744

U-value(1.96)

WINDOWS (1m SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.78m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	1.7m x 0.9m	1	North
SOCIALISING AREA	1.9mx2.1m	10	South

RESULT: OVERHEATING OF SPACES AFTER INCREASING WINDOW SIZES. IMPROVED SDA BUT ASE SHOULD BE IMPROVED

ADDRESS OVERHEATING AND IMPROVE ASE. INCREASE U-VALUE OF BUILDING FABRIC.
ADDRESS OVERHEATING, INCREASE WINDOW SHADING AND CHNAGE GLASS TYPE TO IMPROVE ASE.

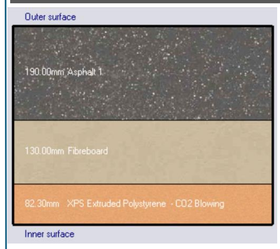
ITERATION 2 – TWO SPACES FAIL CIBSE TM59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.18	N/A	Pass
RECEPTION	5.25	N/A	Fail
SOCIALISINGAREA	5.58	N/A	Fail
READINGROOM	0.00	0.00	Pass
STUDIO1	0.06	5.00	Pass
STUDIO14	0.00	0.00	Pass

ITERAION 2 PASS BOTH ASE AND SDA

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	17.7	14.37	81.38	17.66	100.0
STUDIO 1	12.5	9.10	72.50	11.08	88.3
STUDIO 14	11.5	7.68	66.67	10.09	87.6
RECEPTION	113.7	113.21	99.59	101.29	89.1
MANAGERS OFFICE	6.6	4.77	71.83	6.65	100.0
SOCIALISING AREA	95.8	95.17	99.30	40.73	42.5
	257.9	**244.30**	**94.73**	**187.51**	**72.7**

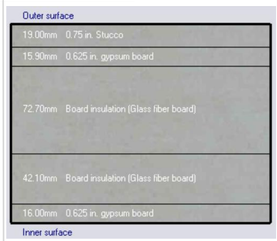
ITERATION 3 SUMMARY



Roof (402mm)

190mm asphalt
130mm fibreboard
82.3mm XPS

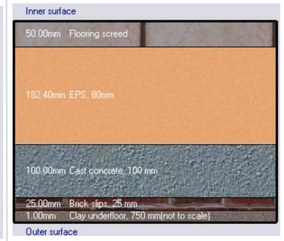
U-value(0.20)



External wall (166mm)

19mm Stucco
16mm Gypsumboard
72.7mm glass fibre insulation
42.1mm glass fibre insulation
16mm gypsumboard

U-value(0.28)



Ground floor slab (358mm)

50mm floor screed
182.4mm EPS insul.
100mm concrete
25mm brick slips
Clay underfloor

U-value(0.20)

Window: LoE Clr
3mm double – 13mm Air

G-value: 0.65
VT: 0.76

U-value(1.78)

WINDOWS (1.5m SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.78m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	1.7m x 0.9m	1	North
SOCIALISING AREA	1.9mx2.1m	10	South

RESULT: SDA ACHIEVED BUT ASE IN SOME SPACES HAS TO BE IMPROVED AND OVERHEATING REMEDIED FOR RECEPTION

IMPROVE U-VALUE OF FABRIC AND CHANGE GLASS TYPE

ITERATION 3 – ONE SPACE FAIL CIBSE TM59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.41	N/A	Pass
RECEPTION	3.34	N/A	Fail
SOCIALISINGAREA	2.95	N/A	Pass
READINGROOM	0.00	1.00	Pass
STUDIO1	0.00	6.50	Pass
STUDIO14	0.00	2.50	Pass

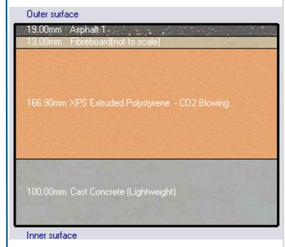
ITERAION 3 PASES BOTH ASE AND SDA

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	18.8	15.73	83.51	18.84	100.0
STUDIO 1	12.5	8.05	64.17	11.08	88.3
STUDIO 14	12.1	6.87	56.67	10.70	88.3
RECEPTION	117.2	112.85	96.27	99.07	84.5
MANAGERS OFFICE	7.3	4.12	56.34	7.32	100.0
SOCIALISING AREA	98.6	95.77	97.13	36.74	37.3
	266.7	**243.39**	**91.28**	**183.75**	**68.9**

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

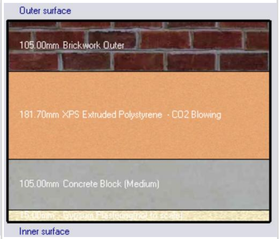
ITERATION 4 SUMMARY



Roof (300mm)

19mm asphalt
13mm fibreboard
167mm XPS
100mm concrete

U-value(0.18)



External wall (407mm)

105mm brickwork
182mm XPS insulation
105mm concrete
15mm gypsum plaster

U-value(0.17)

Ground floor slab (440mm)

30mm timber floor
70mm floor screed
100mm concrete
240mm fomaldehyde foam

U-value(0.15)

Window:LoE Elec Abs bleached 6mm double Argon

G-value: 0.48
VT: 0.657

U-value(1.3)

WINDOWS (1.5m SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.78m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	1.7m x 0.9m	1	North
SOCIALISING AREA	1.9mx2.1m	10	South

RESULT: OVERHEATING SOLVED BUT THE SDA FOR MANAGER'S OFFICE AND ASE FOR SOCIALISING AREA SHOULD BE IMPROVED

IMPROVE U-VALUE OF FABRIC TO IMPROVE CIIBSE VALUES FOR RECEPTION, SOCIALISING AREA AND STUDIO 1.
CHANGE WINDOW SIZES FOR MAGER'S OFFICE AND SOCIALISING AREA.
CHANGE GLASS TYPE TO IMPROVE SDA AND ASE

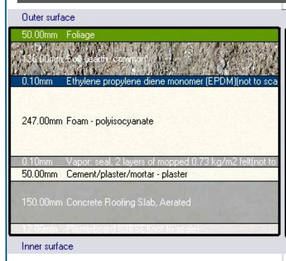
ITERATION 4 – ALL SPACES PASS CIBSE TM59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	1.50	N/A	Pass
SOCIALISINGAREA	1.48	N/A	Pass
READINGROOM	0.00	0.00	Pass
STUDIO1	0.00	1.50	Pass
STUDIO14	0.00	0.00	Pass

ITERAION 4 PASES BOTH ASE AND SDA – IMPROVEMENT REQUIRED

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	16.6	9.95	59.88	16.62	100.0
STUDIO 1	12.5	6.69	53.33	11.40	90.8
STUDIO 14	11.0	5.54	50.48	10.34	94.3
RECEPTION	110.5	104.16	94.26	107.37	97.2
MANAGERS OFFICE	6.1	2.98	49.09	6.07	100.0
SOCIALISING AREA	93.3	87.86	94.14	48.01	51.4
	250.0	**217.18**	**86.86**	**199.81**	**79.9**

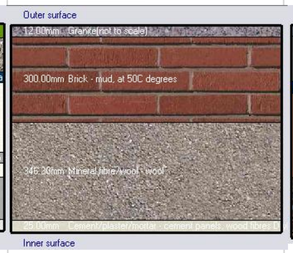
ITERATION 5 SUMMARY



Roof (629mm)

50mm vegetation
120mm soil
Waterproofing
247mm polysiocyanate insulation
Vapour seal
50mm screed
150mm concrete
12mm plasterboard

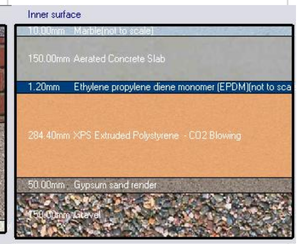
U-value(0.1)



External wall (683mm)

glass-fiber-reinforced concrete panels
300mm brick mud
346.3mm mineral fibre wool
12mm white granite

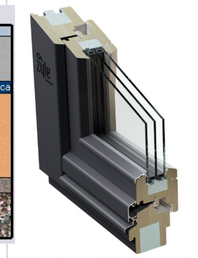
U-value(0.1)



Ground floor slab (540mm)

10mm marble
150mm aerated concrete
1.2mm damp proof
284.4mm extruded polystrene
50mm blinding
150mm hardcore

U-value(0.149)



Window: Triple lowE clear 3mm/13mm Argon
Frame: Wood

G-value: 0.47
VT: 0.66

U-value(0.78)

WINDOWS (1.5m SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.65m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	2m x 0.9m	1	North
SOCIALISING AREA	1.7mx1.65m	10	South

RESULT: MANAGER'S OFFICE SDA AND SOCIALISING ASE ACHIEVED
RESULT: CIBSE TM 59 VALUES IMPROVED, SDA AND ASE TARGETS ACHIEVED.

ITERATION 5 – ALL SPACES PASS CIBSE TM59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	0.99	N/A	Pass
SOCIALISINGAREA	0.10	N/A	Pass
READINGROOM	0.00	0.00	Pass
STUDIO1	0.00	0.00	Pass
STUDIO14	0.00	0.00	Pass

ITERATION 5 ACHIEVES ASE AND SDA

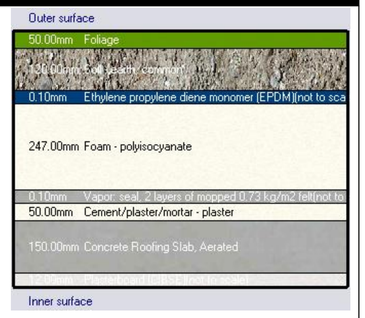
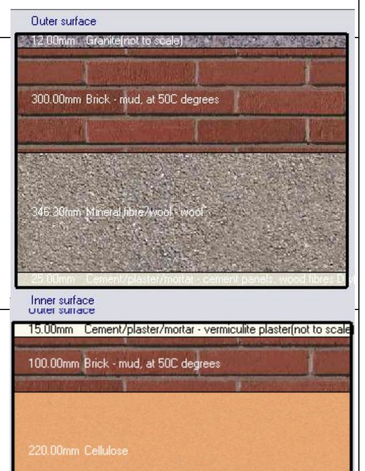
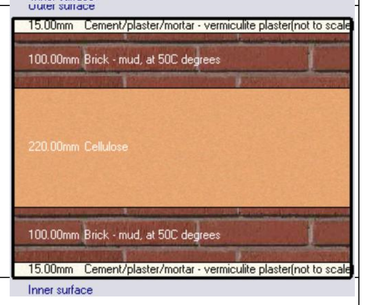
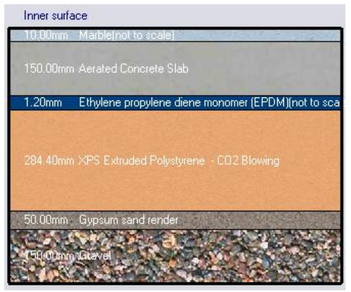
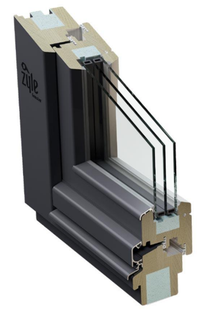
Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	14.0	7.36	52.52	14.02	100.0
STUDIO 1	10.2	5.79	56.84	9.64	94.7
STUDIO 14	8.7	4.40	50.62	8.49	97.5
RECEPTION	103.2	83.46	80.88	101.19	98.1
MANAGERS OFFICE	4.7	2.62	56.10	4.68	100.0
SOCIALISING AREA	85.6	58.38	68.22	80.94	94.6
	226.3	**162.02**	**71.58**	**218.95**	**96.7**

CONCLUSION

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

ITERATION 5 – MATERIALS

ELEMENT(mm)	CROSS-SECTION
Roof (629mm) 50mm vegetation 120mm soil Waterproofing 247mm polysiocyanate insulation Vapour seal 50mm screed 150mm concrete 12mm plasterboard **U-value(0.1)**	
External wall (683mm) glass-fiber-reinforced concrete panels 300mm brick mud 346.3mm mineral fibre wool 12mm white granite **U-value(0.1)**	
Internal wall (450mm) 15mm plaster 100mm brick-mud 220mm cellulose 100mm brick-mud 15mmplaster **U-value(1.64)**	
Ground floor slab (540mm) 10mm marble 150mm aerated concrete 1.2mm damp proof 284.4mm extruded polystrene 50mm blinding 150mm hardcore **U-value(0.149)**	
Window: Triple lowE clear 3mm/13mm Argon **Frame: Wood;ext.aluminum** G-value: 0.47 VT: 0.66 **U-value(0.78)**	

ITERATION 2 - OVERHEATING CRITERIA – CIBSE TM 59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	0.99	N/A	Pass
SOCIALISINGAREA	0.10	N/A	Pass
READINGROOM	0.00	0.00	Pass
STUDIO1	0.00	0.00	Pass
STUDIO14	0.00	0.00	Pass

ITERATION 2 - SPATIAL DAYLIGHT AUTONOMY AND ANNUAL SUNLIGHT EXPOSURE

Eligible zones for daylighting

Block	Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
Block 2	READING ROOM	14.0	7.36	52.52	14.02	100.0
Block 2	STUDIO 1	10.2	5.79	56.84	9.64	94.7
Block 2	STUDIO 14	8.7	4.40	50.62	8.49	97.5
Block 1	RECEPTION	103.2	83.46	80.88	101.19	98.1
Block 1	MANAGERS OFFICE	4.7	2.62	56.10	4.68	100.0
Block 1	SOCIALISING AREA	85.6	58.38	68.22	80.94	94.6
Total		**226.3**	**162.02**	**71.58**	**218.95**	**96.7**

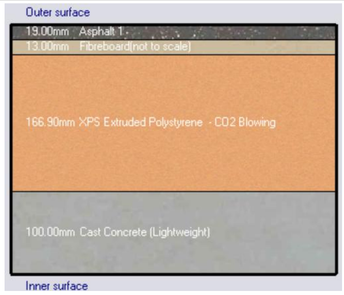
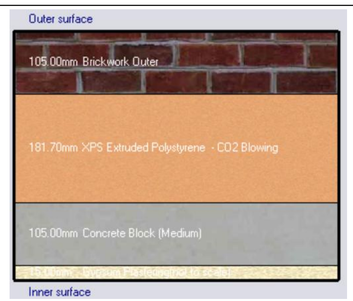
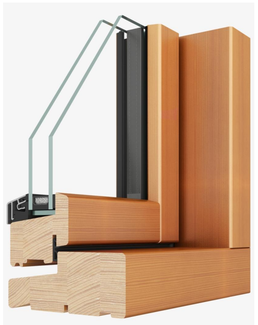
ALL SOUTH WINDOWS SHADED – 1M OVERHANG
MANAGER'S OFFICE WINDOW LENGTH CHANGED FROM 1.7m to 2m
SUN SPACE GLASS TYPE WAS CHANGED TO DOUBLE
HEIGHT OF SOCIALISING AREA WINDOW CHANGED FROM 1.7m to 1.65m

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.65m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	2m x 0.9m	1	North
SOCIALISING AREA	1.7mx1.65m	10	South

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

IT 4

ELEMENT(mm)	CROSS-SECTION
Roof (300mm) 19mm asphalt 13mm fibreboard 167mm XPS 100mm concrete **U-value(0.18)**	
External wall (407mm) 105mm brickwork 182mm XPS insulation 105mm concrete 15mm gypsum plaster **U-value(0.17)**	
Ground floor slab (440mm) 30mm timber floor 70mm floor screed 100mm concrete 240mm fomaldehyde foam **U-value(0.15)**	
Window:LoE Elec Abs bleached 6mm double Argon G-value: 0.48 VT: 0.657 **U-value(1.3)**	

BASE CASE - OVERHEATING CRITERIA – CIBSE TM 59

Naturally ventilated

Criteria for predominantly naturally ventilated homes

Block	Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
BLOCK1	MANAGERSOFFICE	0.00	N/A	Pass
BLOCK1	RECEPTION	1.50	N/A	Pass
BLOCK1	SOCIALISINGAREA	1.48	N/A	Pass
BLOCK2	READINGROOM	0.00	0.00	Pass
BLOCK2	STUDIO1	0.00	1.50	Pass
BLOCK2	STUDIO14	0.00	0.00	Pass

Points credited	3

* This construction must identify how it will address glare

Eligible zones for daylighting

Block	Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
Block 2	READING ROOM	16.6	9.95	59.88	16.62	100.0
Block 2	STUDIO 1	12.5	6.69	53.33	11.40	90.8
Block 2	STUDIO 14	11.0	5.54	50.48	10.34	94.3
Block 1	RECEPTION	110.5	104.16	94.26	107.37	97.2
Block 1	MANAGERS OFFICE	6.1	2.98	49.09	6.07	100.0
Block 1	SOCIALISING AREA	93.3	87.86	94.14	48.01	51.4
Total		**250.0**	**217.18**	**86.86**	**199.81**	**79.9**

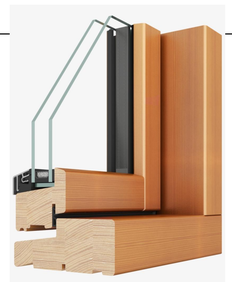
ALL SOUTH WINDOWS SHADED – 1M OVERHANG

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1m x 0.7m	2	North
STUDIO 1	1m x 0.9m	1	South
STUDIO 14	1m x 0.9m	1	South
RECEPTION	(S)1.5mx1m ; 1.5x1(E)	(S)4 ; (E)1	South & East
MANAGER'S OFFICE	1m x 0.7m	1	North
SOCIALISING AREA	1.5m	5	South

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

IT 3

ELEMENT(mm)	CROSS-SECTION
Roof (402mm) 190mm asphalt 130mm fibreboard 82.3mm XPS **U-value(0.20)**	
External wall (166mm) 19mm Stucco 16mm Gypsumboard 72.7mm glass fibre insulation 42.1mm glass fibre insulation 16mm gypsumboard **U-value(0.28)**	
Ground floor slab (358mm) 50mm floor screed 182.4mm EPS insul. 100mm concrete 25mm brick slips Clay underfloor **U-value(0.20)**	
Window:LoE Clr 3mm double – 13mm Air G-value: 0.65 VT: 0.76 **U-value(1.78)**	

BASE CASE - OVERHEATING CRITERIA – CIBSE TM 59

Naturally ventilated

Criteria for predominantly naturally ventilated homes

Block	Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
BLOCK1	MANAGERSOFFICE	0.41	N/A	Pass
BLOCK1	RECEPTION	3.34	N/A	Fail
BLOCK1	SOCIALISINGAREA	2.95	N/A	Pass
BLOCK2	READINGROOM	0.00	1.00	Pass
BLOCK2	STUDIO1	0.00	6.50	Pass
BLOCK2	STUDIO14	0.00	2.50	Pass

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	18.8	15.73	83.51	18.84	100.0
STUDIO 1	12.5	8.05	64.17	11.08	88.3
STUDIO 14	12.1	6.87	56.67	10.70	88.3
RECEPTION	117.2	112.85	96.27	99.07	84.5
MANAGERS OFFICE	7.3	4.12	56.34	7.32	100.0
SOCIALISING AREA	98.6	95.77	97.13	36.74	37.3
	266.7	**243.39**	**91.28**	**183.75**	**68.9**

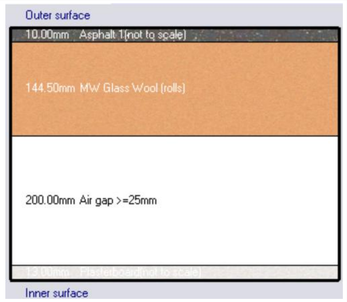
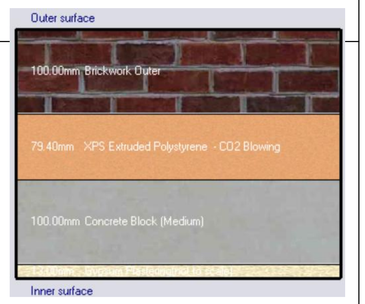
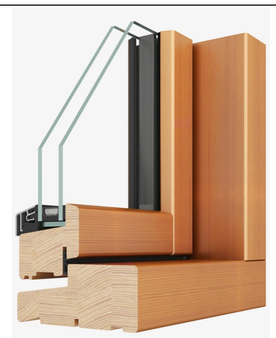
ALL SOUTH WINDOWS SHADED – 1M OVERHANG

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1m x 0.7m	2	North
STUDIO 1	1m x 0.9m	1	South
STUDIO 14	1m x 0.9m	1	South
RECEPTION	(S)1.5mx1m ; 1.5x1(E)	(S)4 ; (E)1	South & East
MANAGER'S OFFICE	1m x 0.7m	1	North
SOCIALISING AREA	1.5m	5	South

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

ITERATION 2 SUMMARY

ELEMENT(mm)	CROSS-SECTION
Roof (367.5mm) 10mm asphalt 144.5mm glass wool 200mm air gap 13mm plasterboard **U-value(0.25)**	
External wall (292mm) 100mm brickwork 79.4mm XPS extruded polystyrene insulation 100mm concrete block (medium) 13mm gypsum plastering **U-value(0.351)**	
Ground floor slab (300mm) 100mm UREA fomaldehyde foam 100mm cast concrete 70mm floor screed 30mm timber flooring **U-value(0.314)**	
Window:Double Clear 3mm/13mmAir **Frame:painted wood** G-value: 0.764 VT: 0.812 **U-value(2.7)**	

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.18	N/A	Pass
RECEPTION	5.25	N/A	Fail
SOCIALISINGAREA	5.58	N/A	Fail
READINGROOM	0.00	0.00	Pass
STUDIO1	0.06	5.00	Pass
STUDIO14	0.00	0.00	Pass

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	17.7	14.37	81.38	17.66	100.0
STUDIO 1	12.5	9.10	72.50	11.08	88.3
STUDIO 14	11.5	7.68	66.67	10.09	87.6
RECEPTION	113.7	113.21	99.59	101.29	89.1
MANAGERS OFFICE	6.6	4.77	71.83	6.65	100.0
SOCIALISING AREA	95.8	95.17	99.30	40.73	42.5
	257.9	**244.30**	**94.73**	**187.51**	**72.7**

ALL SOUTH WINDOWS SHADED – 1M OVERHANG

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.78m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	1.7m x 0.9m	1	North
SOCIALISING AREA	1.9mx2.1m	10	South

Sustainable Design of Students' Accommodation Block

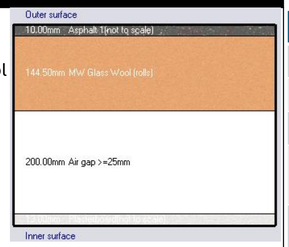
Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

BASE CASE SUMMARY

ELEMENT	CROSS-SECTION

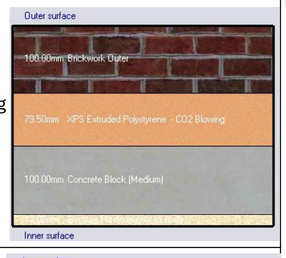
Roof (367.5mm)
10mm asphalt
144.5mm MW glass wool
200mm air gap
13mm plasterboard

U-value(0.25)



External wall (292mm)
100mm brickwork
79.5mm XPS insulation
100mm concrete block
13mm gypsum plastering

U-value(0.350)



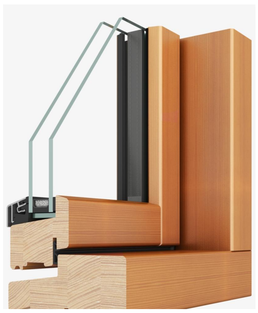
Ground floor slab (300mm)
30mm timber flooring
70mm floor screed
100mm cast concrete
132.7mm UREA fomaldehyde foam

U-value(0.25)

Window:Double Clear 3mm/13mmAir
Frame:painted wood

G-value: 0.691
VT: 0.744

U-value(1.96)



WINDOWS (NO SHADING)

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1m x 0.7m	2	North
STUDIO 1	1m x 0.9m	1	South
STUDIO 14	1m x 0.9m	1	South
RECEPTION	(South)1.5mx1m ; 1.5x1(East)	(South)4 ; (East)1	South & East
MANAGER'S OFFICE	1m x 0.7m	1	North
SOCIALISING AREA	1.5m	5	South

BASE CASE PASSES CIBSE TM59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	0.28	N/A	Pass
SOCIALISINGAREA	0.00	N/A	Pass
READINGROOM	0.00	0.00	Pass
STUDIO1	0.00	0.00	Pass
STUDIO14	0.00	0.00	Pass

BASE CASE PASSES ASE BUT NOT SDA

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	17.7	0.00	0.00	17.66	100.0
STUDIO 1	12.5	0.00	0.00	12.55	100.0
STUDIO 14	11.5	0.11	0.95	11.51	100.0
RECEPTION	113.7	56.42	49.63	113.49	99.8
MANAGERS OFFICE	6.6	0.00	0.00	6.65	100.0
SOCIALISING AREA	95.8	8.76	9.14	95.74	99.9
	257.9	**65.28**	**25.32**	**257.59**	**99.9**

ITERATION 2 SUMMARY

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., STUDENTS ACCOMMODATION

EnergyPlus Output | 1 Jan - 31 Dec, Run period | Student

Comfort - STUDENTS ACCM., STUDENTS ACCOMMODATION

EnergyPlus Output | 1 Jan - 31 Dec, Run period

Air Temperature (°C)	19.42
Radiant Temperature (°C)	20.36
Operative Temperature (°C)	19.89
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	47.36

Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., STUDENTS ACCOMMODATION

EnergyPlus Output | 1 Jan - 31 Dec, Run period | Student

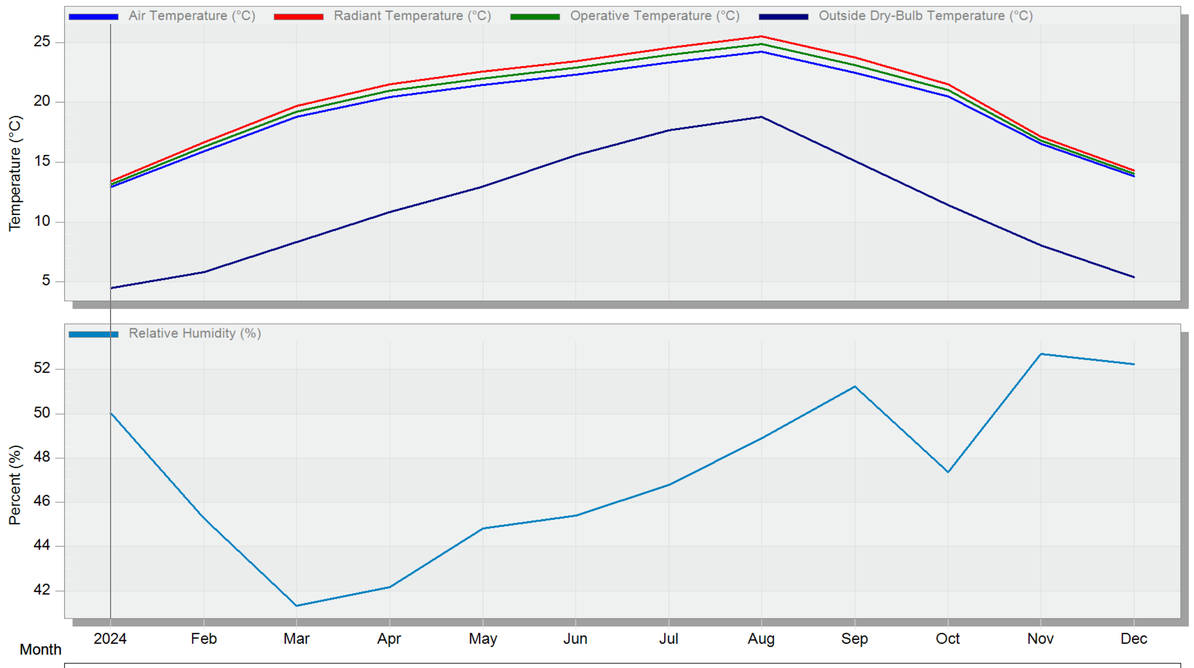
	Year
Lighting (kWh)	12363.78
DHW (Electricity) (kWh)	12304.89
Air Temperature (°C)	19.42
Radiant Temperature (°C)	20.36
Operative Temperature (°C)	19.89
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-67927.30
External Vent. (kWh)	-87640.59
General Lighting (kWh)	12363.78
Occupancy (kWh)	31761.24
Solar Gains Interior Windows (kWh)	15786.38
Solar Gains Exterior Windows (kWh)	223384.05
Mech Vent + Nat Vent + Infiltration (ac/h)	1.75

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Comfort - STUDENTS ACCM., STUDENTS ACCOMMODATION

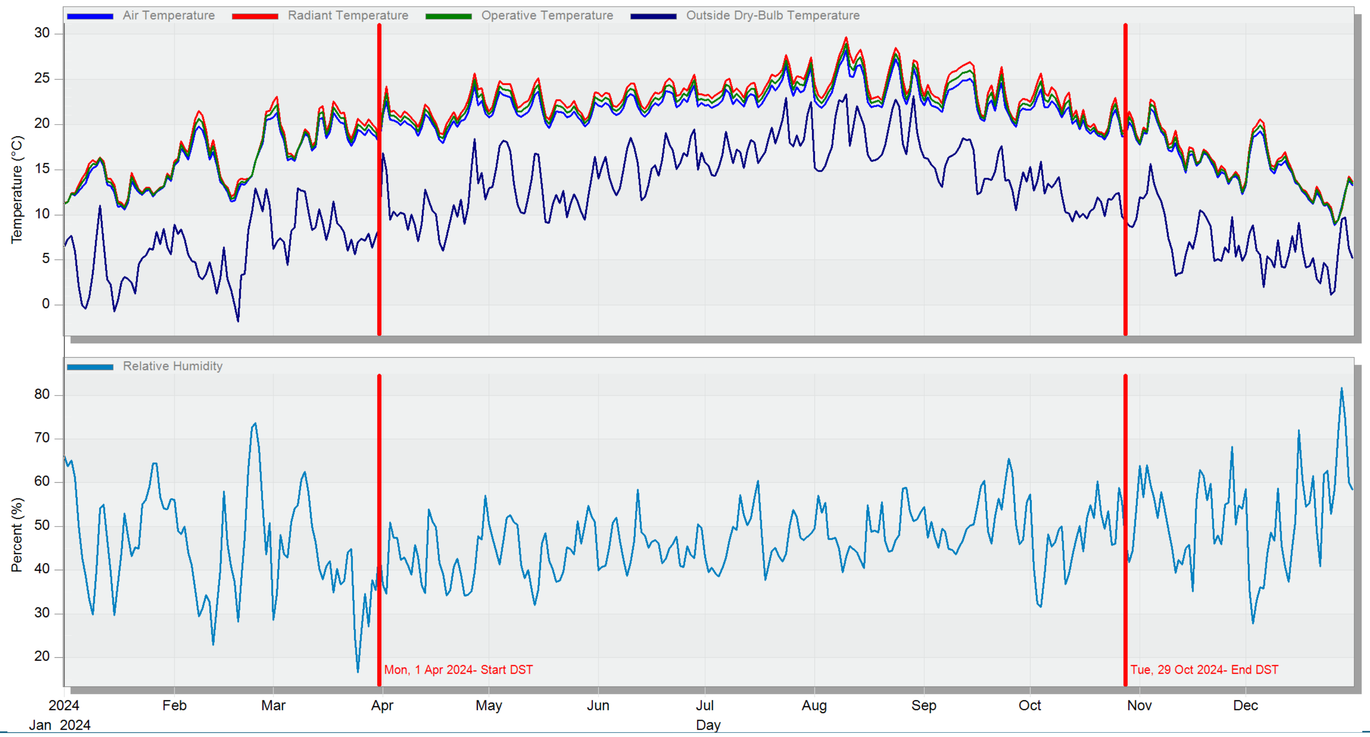
EnergyPlus Output | 1 Jan - 31 Dec, Monthly | Student



	2024	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Air Temperature (°C)	12.93	15.93	18.78	20.47	21.48	22.31	23.34	24.22	22.50	20.51	16.56	13.81
Radiant Temperature (°C)	13.40	16.66	19.69	21.51	22.57	23.46	24.58	25.51	23.78	21.51	17.13	14.31
Operative Temperature (°C)	13.16	16.30	19.24	20.99	22.02	22.88	23.96	24.87	23.14	21.01	16.85	14.06
Outside Dry-Bulb Temperature (°C)	4.50	5.82	8.33	10.85	12.97	15.59	17.66	18.80	15.09	11.43	8.07	5.42
Relative Humidity (%)	50.04	45.28	41.33	42.18	44.81	45.39	46.79	48.88	51.24	47.36	52.70	52.25

Comfort - STUDENTS ACCM., STUDENTS ACCOMMODATION

EnergyPlus Output | 1 Jan - 31 Dec, Daily | Student

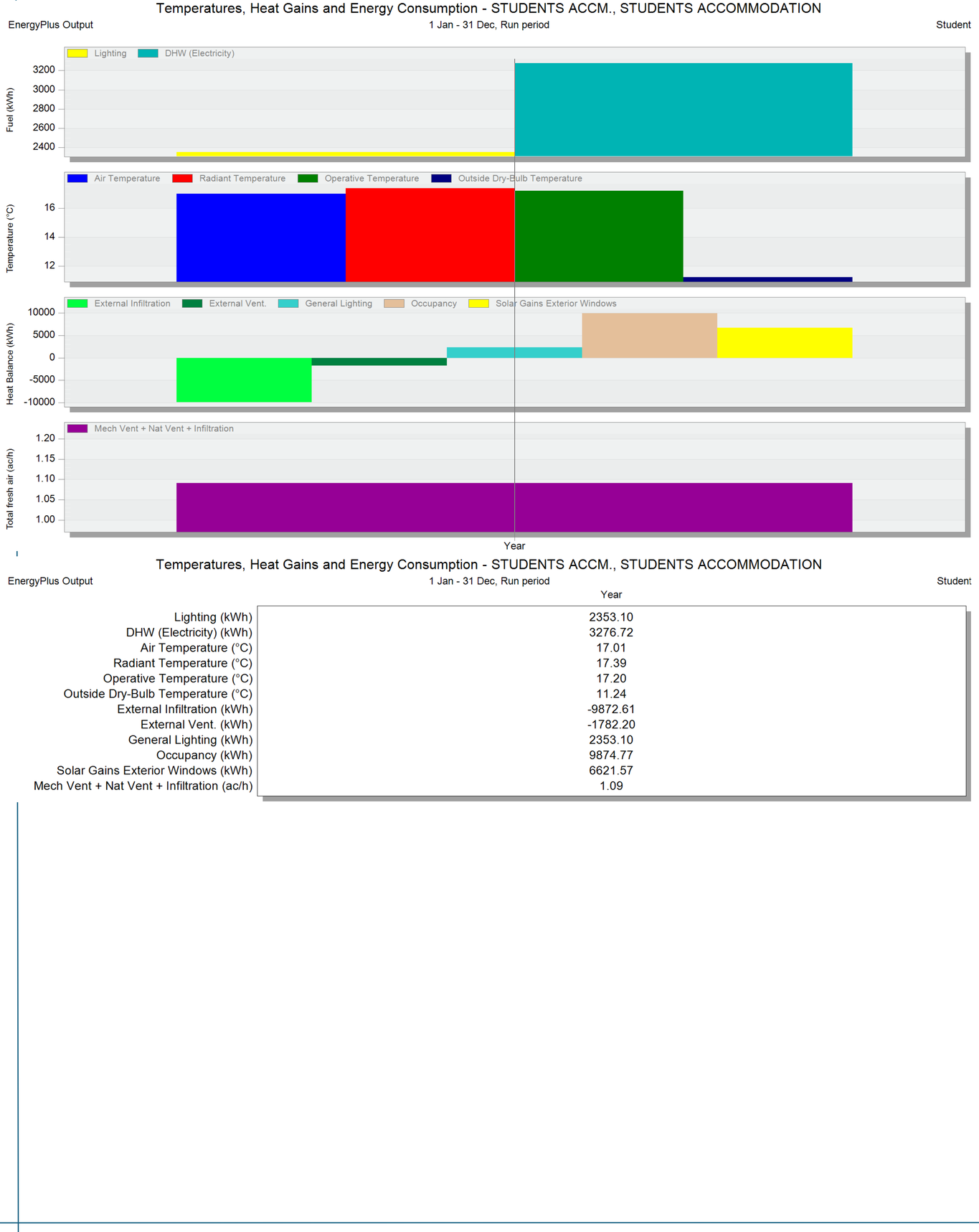


10th AUGUST 28.1 OPERATIVE TEMPERATURE

CALCULATIONS FOR ZONES ONLY

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

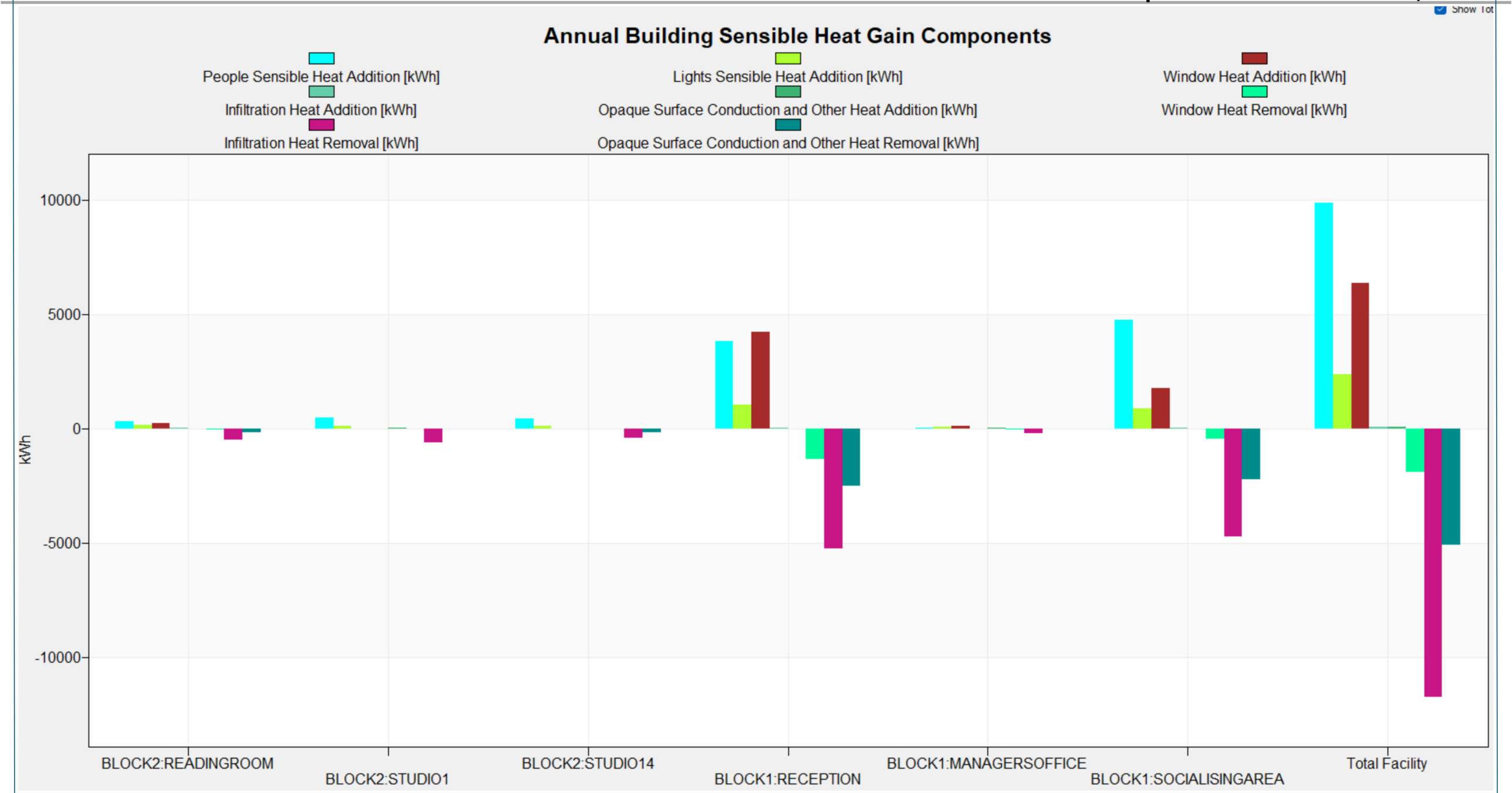


	Year
Lighting (kWh)	2353.10
DHW (Electricity) (kWh)	3276.72
Air Temperature (°C)	17.01
Radiant Temperature (°C)	17.39
Operative Temperature (°C)	17.20
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-9872.61
External Vent. (kWh)	-1782.20
General Lighting (kWh)	2353.10
Occupancy (kWh)	9874.77
Solar Gains Exterior Windows (kWh)	6621.57
Mech Vent + Nat Vent + Infiltration (ac/h)	1.09

Air Temperature (°C)	17.01
Radiant Temperature (°C)	17.39
Operative Temperature (°C)	17.20
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	55.21

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)



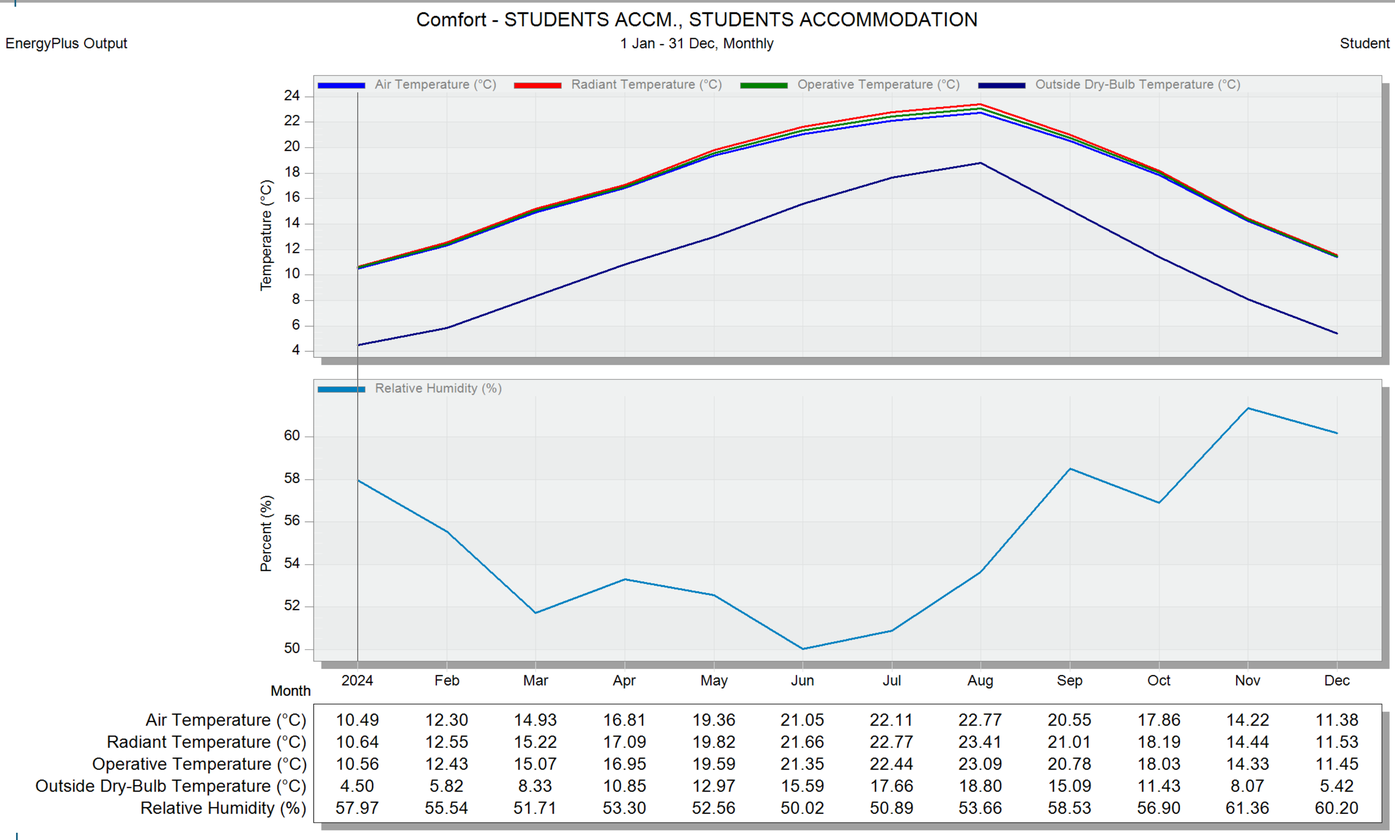
Time Not Comfortable Based on Simple ASHRAE 55-2004

	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	6587.00	8661.00	6535.50
BLOCK2:STUDIO1	5755.50	8598.00	5647.00
BLOCK2:STUDIO14	6431.50	8594.50	6317.50
BLOCK1:RECEPTION	3078.50	4467.50	2887.50
BLOCK1:MANAGERSOFFICE	3561.50	4621.50	3488.00
BLOCK1:SOCIALISINGAREA	3014.50	4598.50	2918.00
Facility	6882.00	8724.00	6695.50

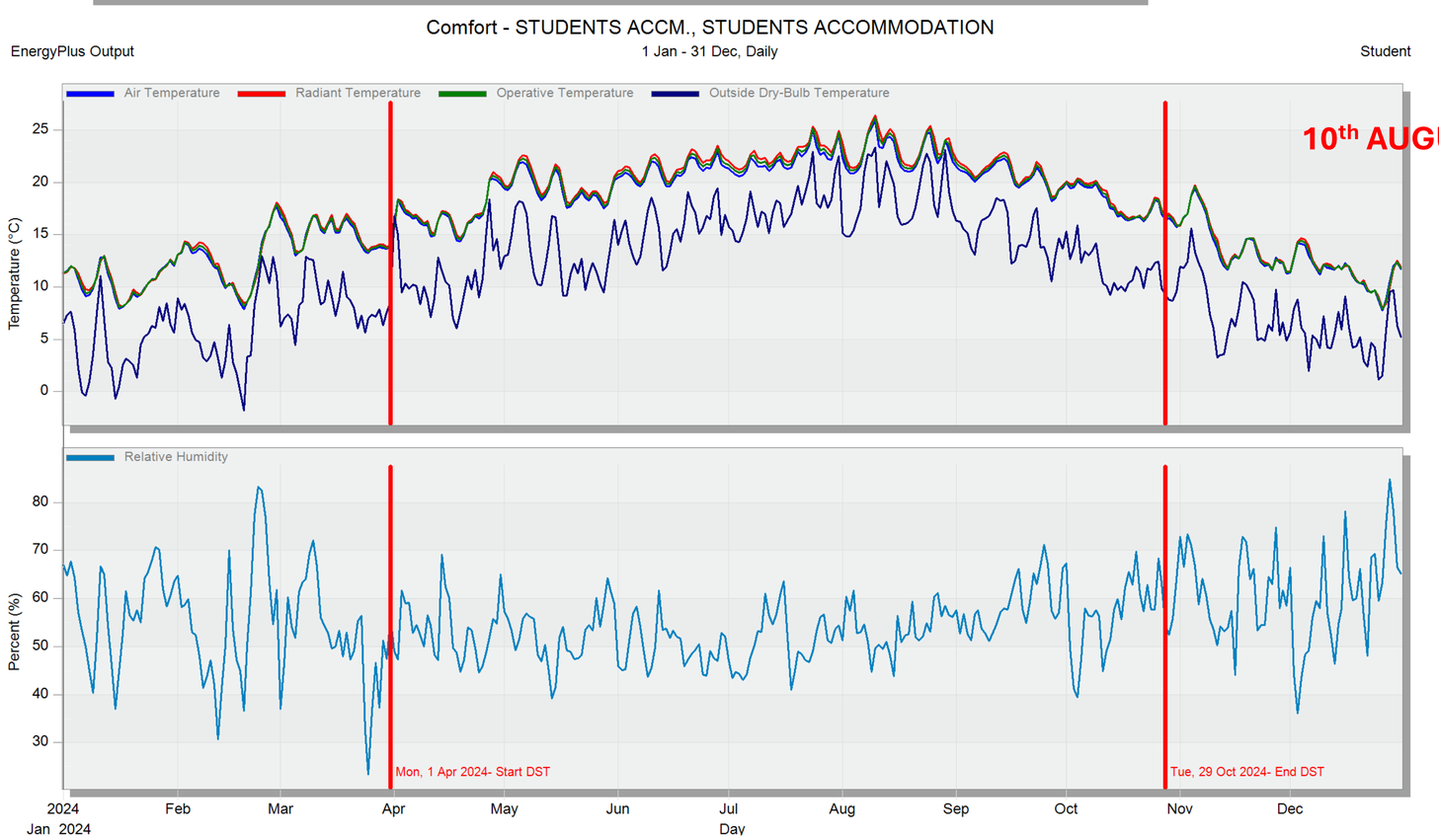
Aggregated over the RunPeriods for Weather

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)



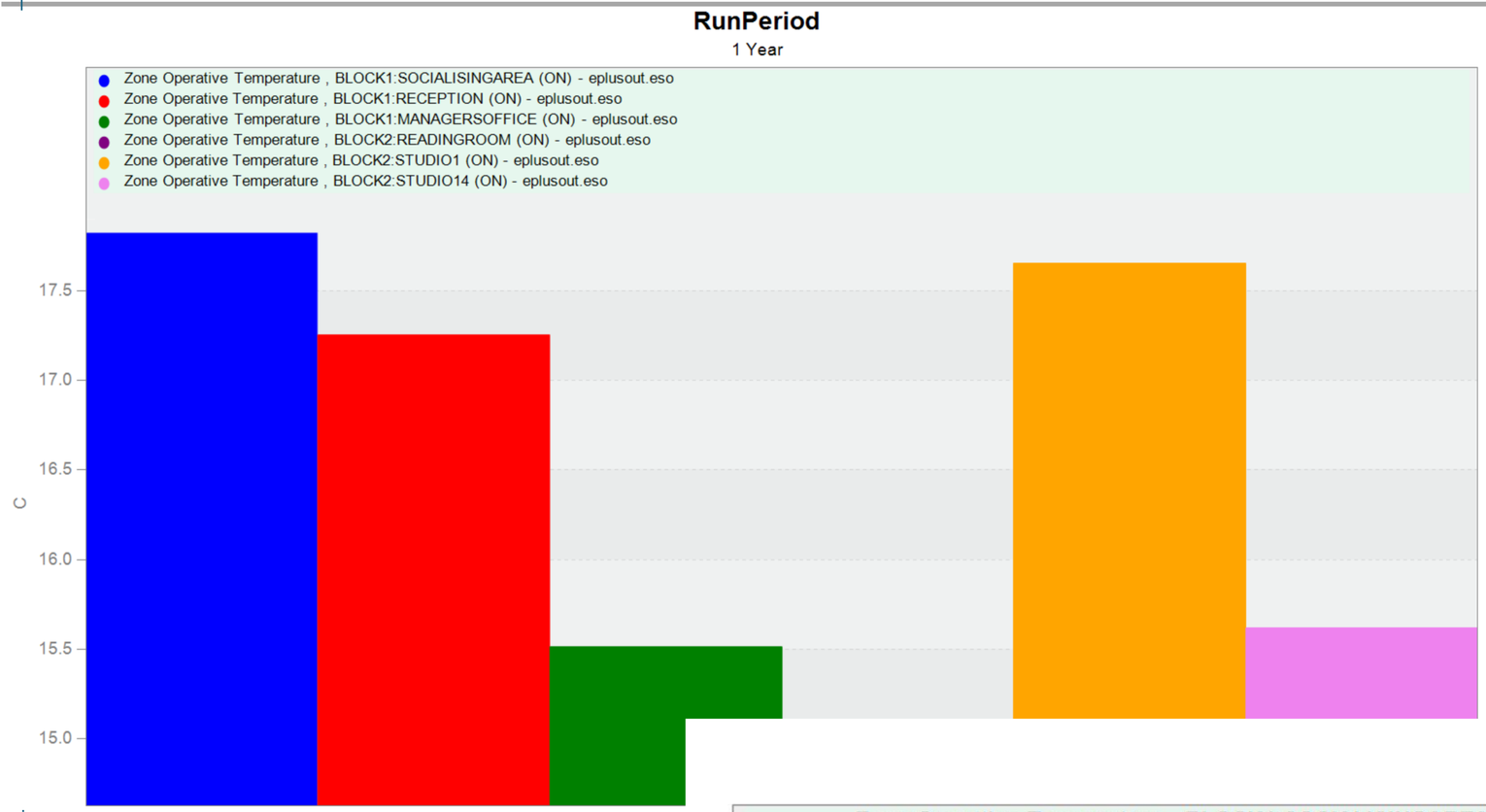
	2024	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Air Temperature (°C)	10.49	12.30	14.93	16.81	19.36	21.05	22.11	22.77	20.55	17.86	14.22	11.38
Radiant Temperature (°C)	10.64	12.55	15.22	17.09	19.82	21.66	22.77	23.41	21.01	18.19	14.44	11.53
Operative Temperature (°C)	10.56	12.43	15.07	16.95	19.59	21.35	22.44	23.09	20.78	18.03	14.33	11.45
Outside Dry-Bulb Temperature (°C)	4.50	5.82	8.33	10.85	12.97	15.59	17.66	18.80	15.09	11.43	8.07	5.42
Relative Humidity (%)	57.97	55.54	51.71	53.30	52.56	50.02	50.89	53.66	58.53	56.90	61.36	60.20

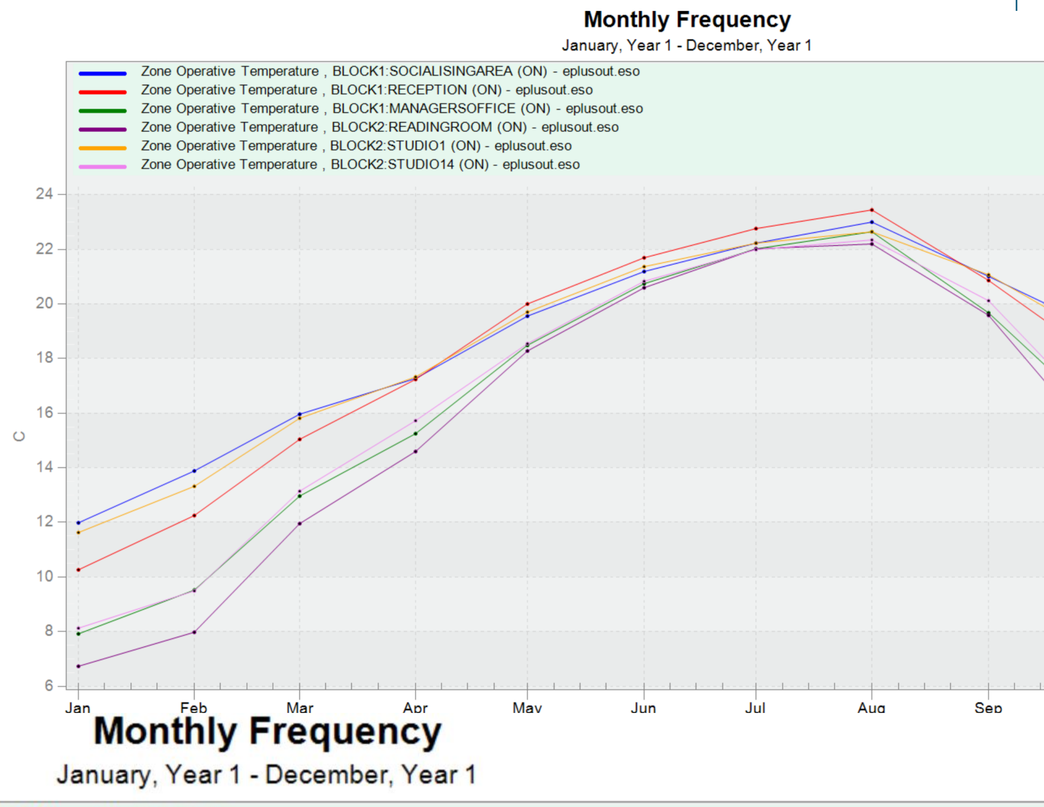


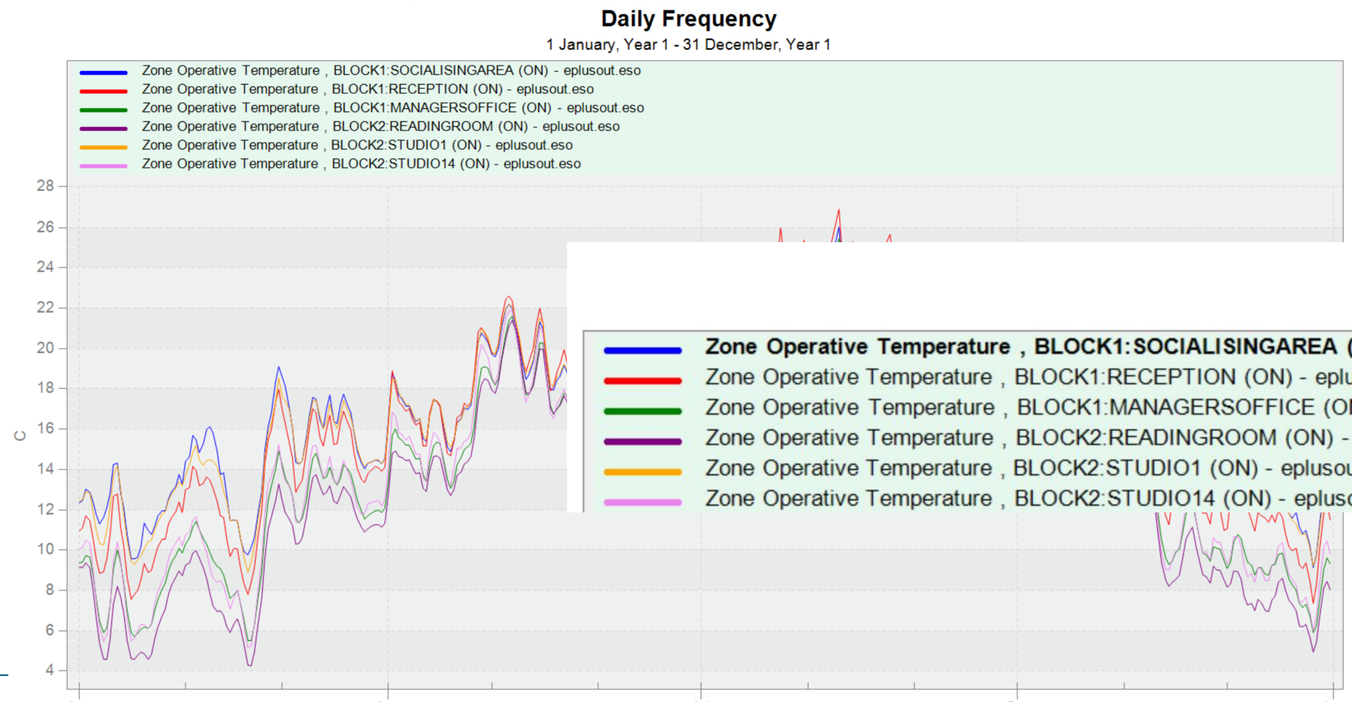
10th AUGUST 26.1 OPERATIVE TEMPERATURE

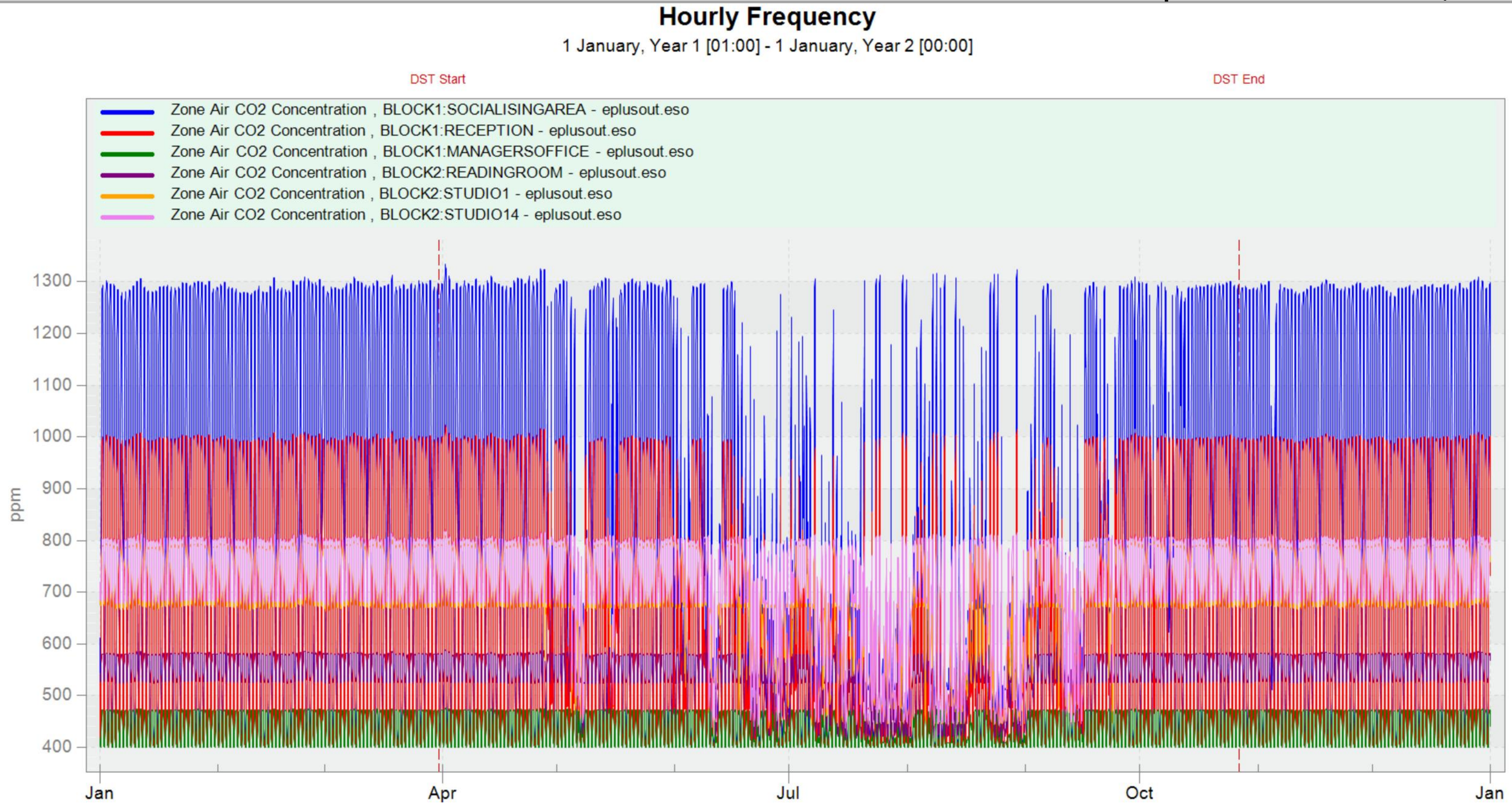
Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

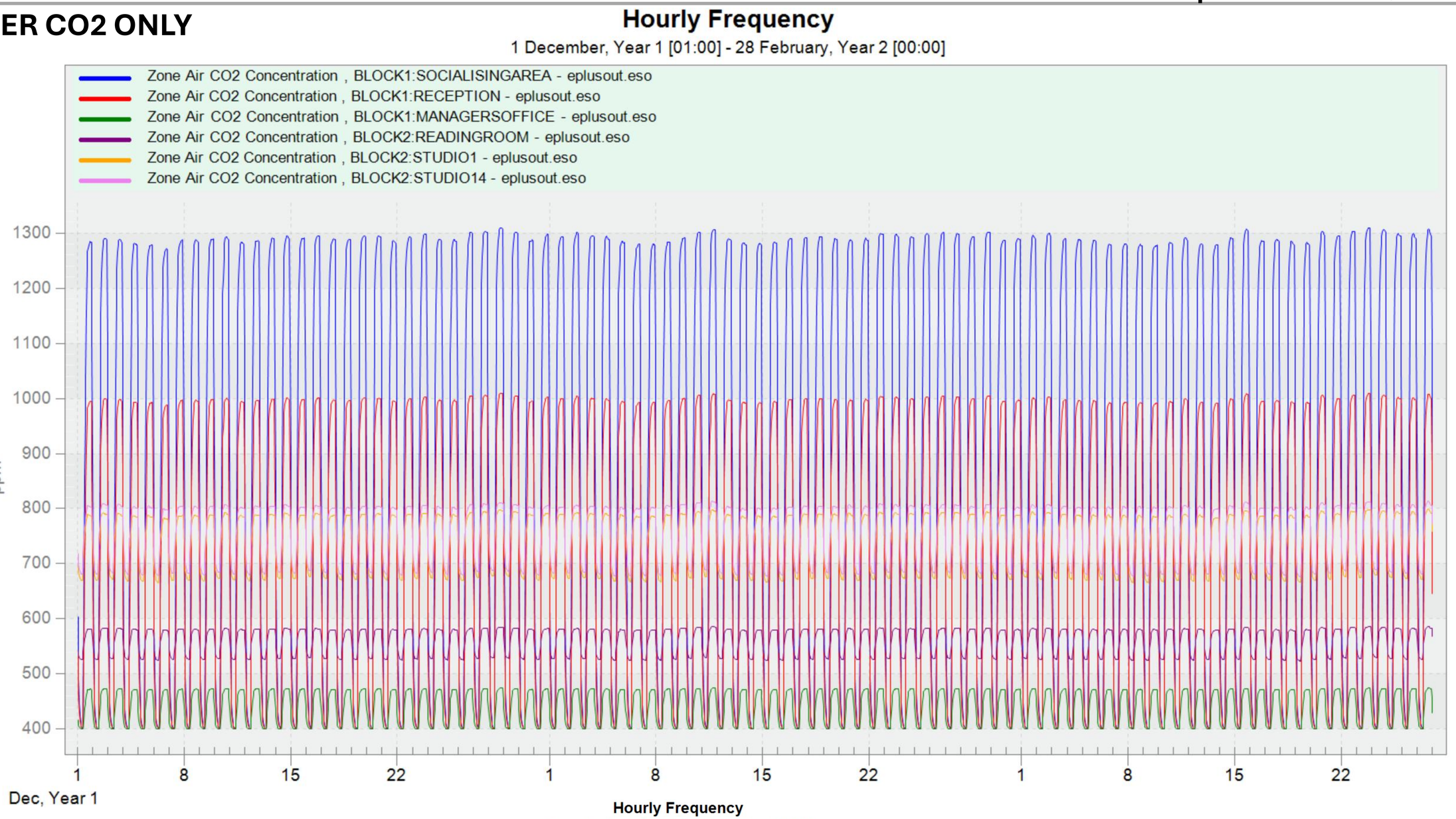







Hourly Frequency
1 January, Year 1 [01:00] - 1 January, Year 2 [00:00]
DST Start
DST End
Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso
ppm
1300
1200
1100
1000
900
800
700
600
500
400
Jan
Apr
Jul
Oct
Jan

WINTER CO2 ONLY



Hourly Frequency

1 December, Year 1 [01:00] - 28 February, Year 2 [00:00]

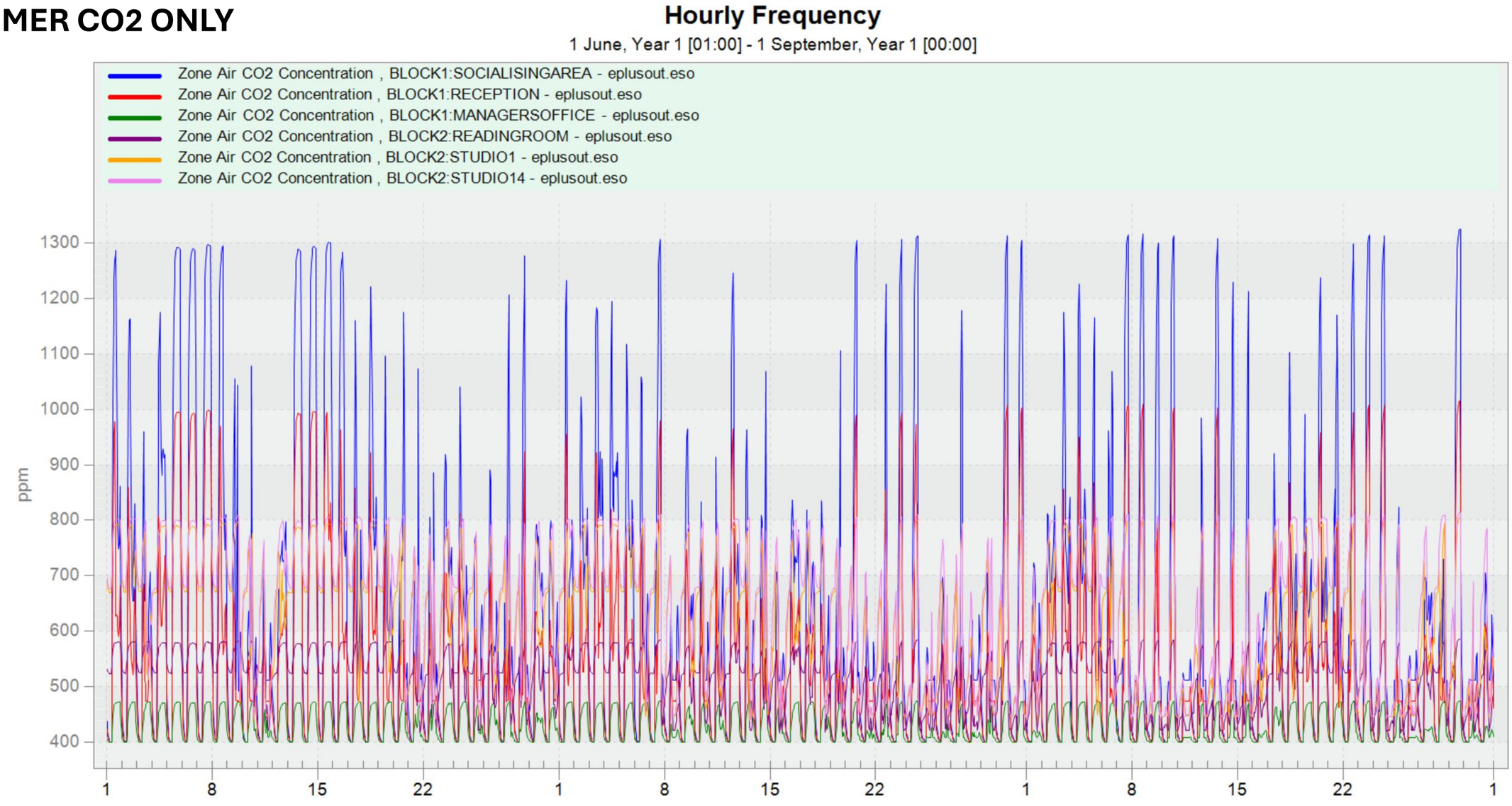
Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso	1306.154479 ppm
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso	1007.220119 ppm
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso	473.854177 ppm
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso	584.942109 ppm
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso	797.309257 ppm
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso	810.971836 ppm

Time Not Comfortable Based on Simple ASHRAE 55-2004

	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	2136.00	2136.00	2136.00
BLOCK2:STUDIO1	2136.00	2136.00	2136.00
BLOCK2:STUDIO14	2136.00	2136.00	2136.00
BLOCK1:RECEPTION	1157.00	1157.00	1157.00
BLOCK1:MANAGERSOFFICE	1157.00	1157.00	1157.00
BLOCK1:SOCIALISINGAREA	1149.50	1157.00	1149.50
Facility	2136.00	2136.00	2136.00

Aggregated over the RunPeriods for Weather

SUMMER CO2 ONLY



Hourly Frequency

1 June, Year 1 [01:00] - 1 September, Year 1 [00:00]

Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso	1185.848731 ppm
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso	722.492324 ppm
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso	470.388370 ppm
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso	581.562586 ppm
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso	744.256342 ppm
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso	805.765203 ppm

Time Not Comfortable Based on Simple ASHRAE 55-2004

	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	394.50	2108.50	344.00
BLOCK2:STUDIO1	244.50	2044.50	135.00
BLOCK2:STUDIO14	424.50	2048.00	314.00
BLOCK1:RECEPTION	270.00	942.50	85.00
BLOCK1:MANAGERSOFFICE	232.00	1069.50	160.50
BLOCK1:SOCIALISINGAREA	221.00	1052.00	126.50
Facility	672.00	2172.00	490.50

Sustainable Design of Students' Accommodation Block

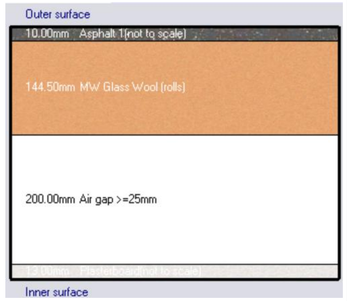
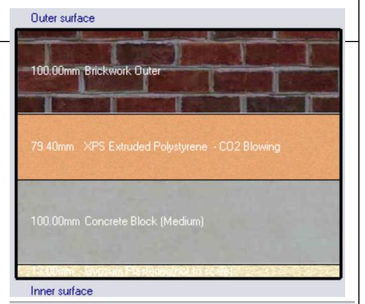
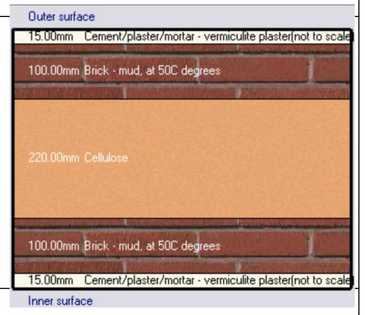
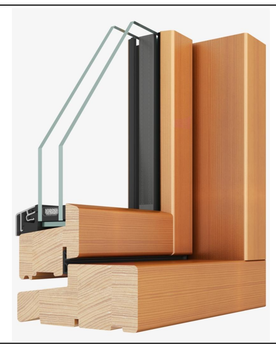
Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

ITERATION 2

Template	IECC-2000, Medium weight

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

ELEMENT(mm)	CROSS-SECTION
Roof (367.5mm) 10mm asphalt 144.5mm glass wool 200mm air gap 13mm plasterboard **U-value(0.25)**	
External wall (292mm) 100mm brickwork 79.4mm XPS extruded polystyrene insulation 100mm concrete block (medium) 13mm gypsum plastering **U-value(0.351)**	
Internal wall (450mm) 15mm plaster 100mm brick-mud 220mm cellulose 100mm brick-mud 15mmplaster **U-value(1.64)**	
Ground floor slab (300mm) 100mm UREA fomaldehyde foam 100mm cast concrete 70mm floor screed 30mm timber flooring **U-value(0.314)**	
Window:Double Clear 3mm/13mmAir **Frame:painted wood** G-value: 0.764 VT: 0.812 **U-value(2.7)**	

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.18	N/A	Pass
RECEPTION	5.25	N/A	Fail
SOCIALISINGAREA	5.58	N/A	Fail
READINGROOM	0.00	0.00	Pass
STUDIO1	0.06	5.00	Pass
STUDIO14	0.00	0.00	Pass

Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
READING ROOM	17.7	14.37	81.38	17.66	100.0
STUDIO 1	12.5	9.10	72.50	11.08	88.3
STUDIO 14	11.5	7.68	66.67	10.09	87.6
RECEPTION	113.7	113.21	99.59	101.29	89.1
MANAGERS OFFICE	6.6	4.77	71.83	6.65	100.0
SOCIALISING AREA	95.8	95.17	99.30	40.73	42.5
	257.9	**244.30**	**94.73**	**187.51**	**72.7**

ALL SOUTH WINDOWS SHADED – 1M OVERHANG

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.78m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	1.7m x 0.9m	1	North
SOCIALISING AREA	1.9mx2.1m	10	South

CALCULATIONS FOR ALL FLOORS

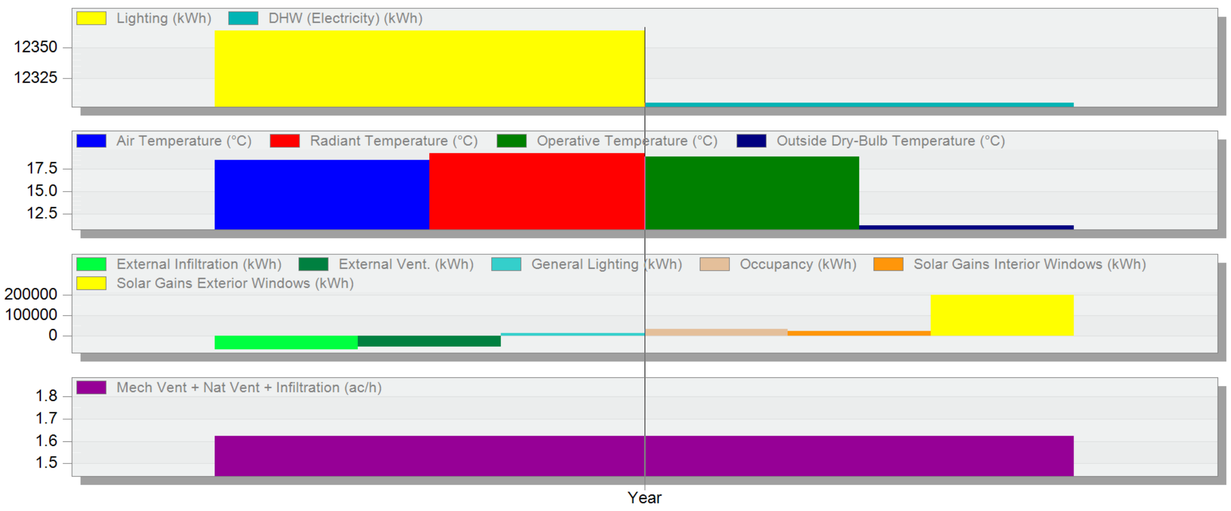
Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

CALCULATIONS FOR ZONES ONLY

Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Run period | Student

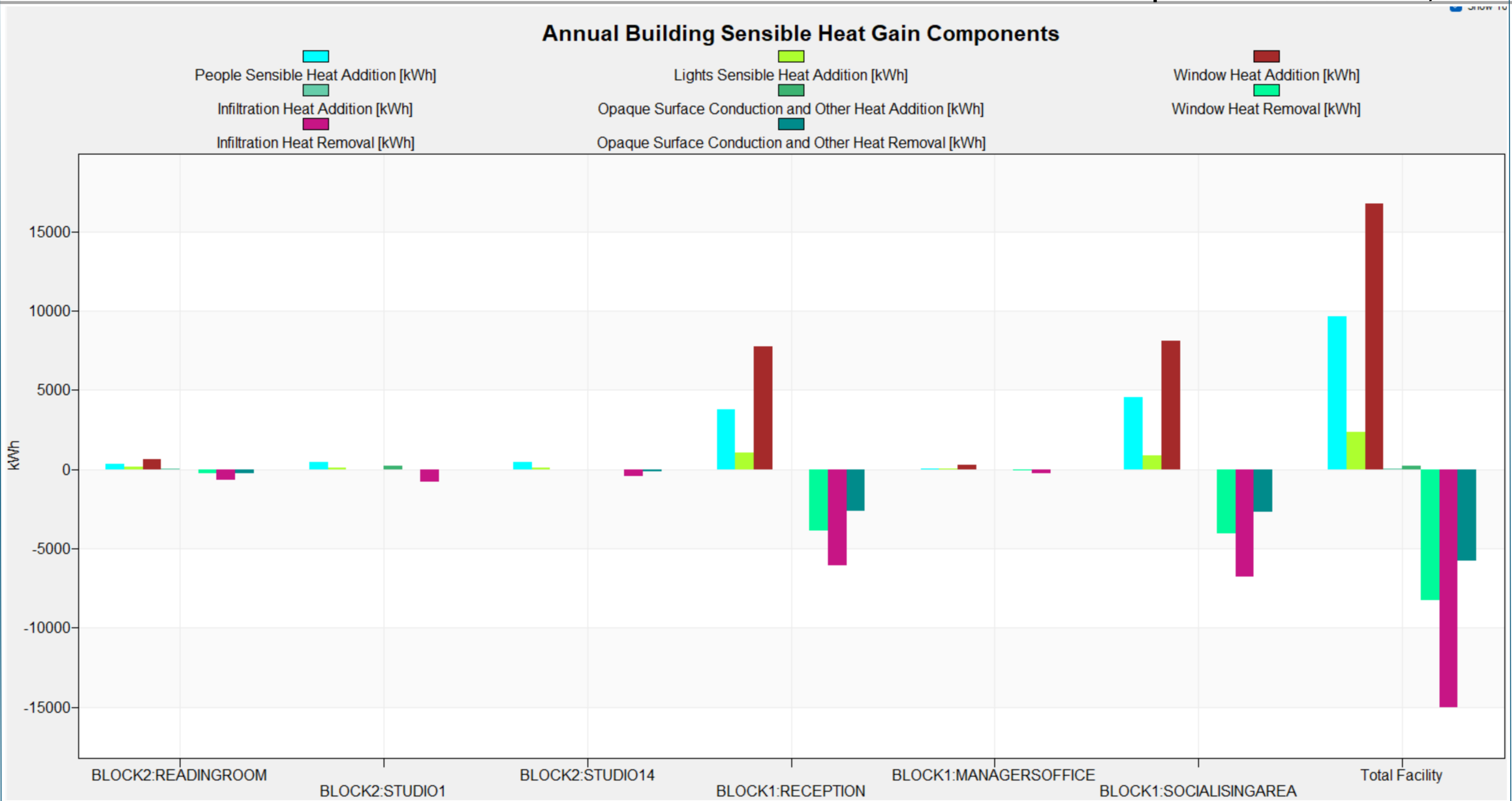


Lighting (kWh)	12364.09
DHW (Electricity) (kWh)	12305.15
Air Temperature (°C)	18.51
Radiant Temperature (°C)	19.27
Operative Temperature (°C)	18.89
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-67380.65
External Vent. (kWh)	-51176.36
General Lighting (kWh)	12364.09
Occupancy (kWh)	34127.27
Solar Gains Interior Windows (kWh)	22365.87
Solar Gains Exterior Windows (kWh)	200407.59
Mech Vent + Nat Vent + Infiltration (ac/h)	1.62

Comfort - STUDENTS ACCM., Building 1

1 Jan - 31 Dec, Run period | Student

Air Temperature (°C)	18.59
Radiant Temperature (°C)	19.34
Operative Temperature (°C)	18.96
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	49.41


Annual Building Sensible Heat Gain Components
People Sensible Heat Addition [kWh]
Lights Sensible Heat Addition [kWh]
Window Heat Addition [kWh]
Infiltration Heat Addition [kWh]
Opaque Surface Conduction and Other Heat Addition [kWh]
Window Heat Removal [kWh]
Infiltration Heat Removal [kWh]
Opaque Surface Conduction and Other Heat Removal [kWh]
kWh
15000
10000
5000
0
-5000
-10000
-15000
BLOCK2:READINGROOM
BLOCK2:STUDIO1
BLOCK2:STUDIO14
BLOCK1:RECEPTION
BLOCK1:MANAGERSOFFICE
BLOCK1:SOCIALISINGAREA
Total Facility

Time Not Meeting the Adaptive Comfort Models during Occupied Hours

	ASHRAE55 90% Acceptability Limits [Hours]	ASHRAE55 80% Acceptability Limits [Hours]	CEN15251 Category I Acceptability Limits [Hours]	CEN15251 Category II Acceptability Limits [Hours]	CEN15251 Category III Acceptability Limits [Hours]
PEOPLE BLOCK2:READINGROOM	2054.00	1596.00	4354.00	3126.50	2317.00
PEOPLE BLOCK2:STUDIO1	1304.50	711.00	4190.00	2450.00	1501.50
PEOPLE BLOCK2:STUDIO14	2430.50	1878.00	4709.00	3872.50	2656.50
PEOPLE BLOCK1:RECEPTION	945.00	623.50	1677.50	1108.00	789.50
PEOPLE BLOCK1:MANAGERSOFFICE	1137.00	880.50	2195.50	1563.50	1247.00
PEOPLE BLOCK1:SOCIALISINGAREA	738.00	420.00	1507.00	901.00	600.50

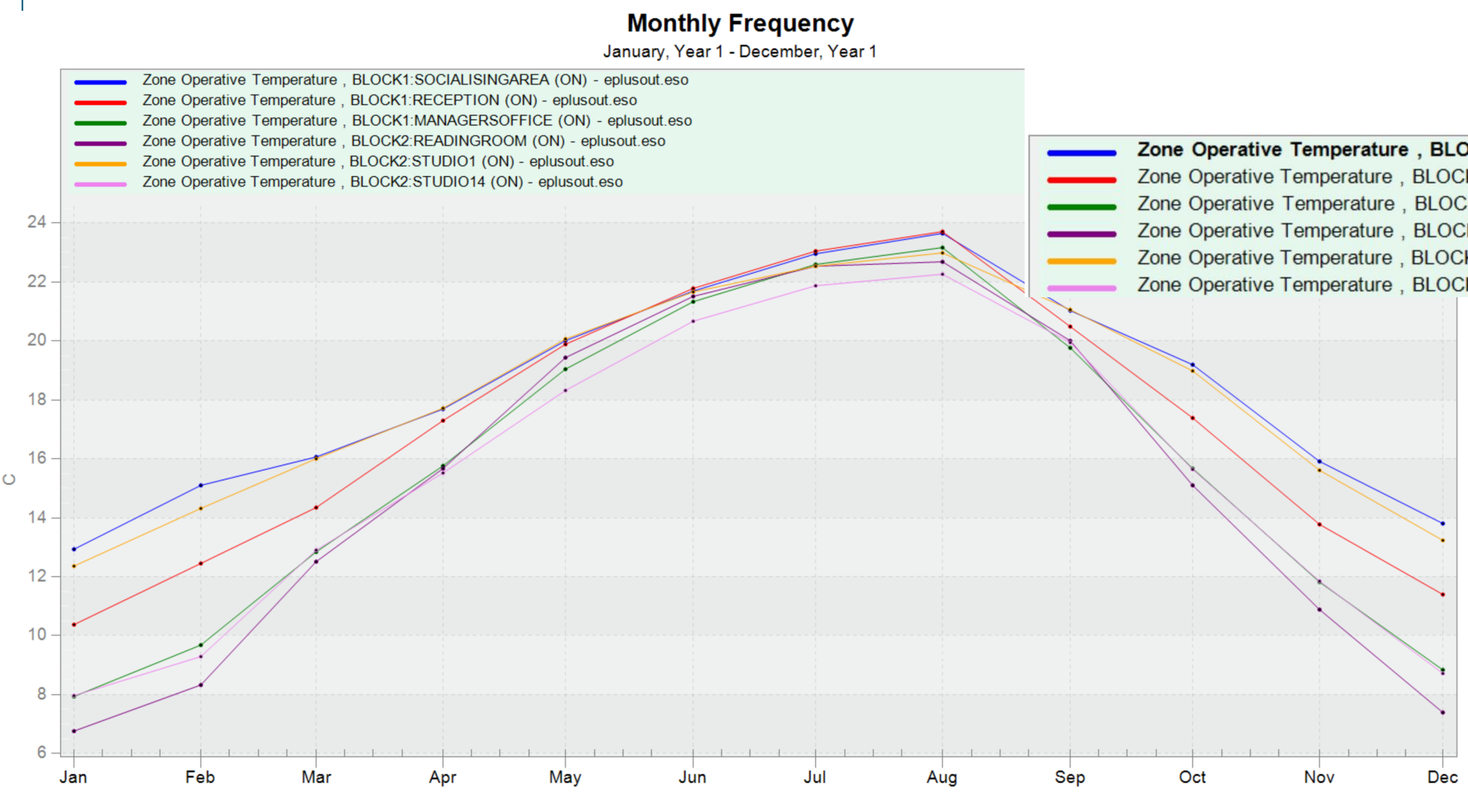
Time Not Comfortable Based on Simple ASHRAE 55-2004

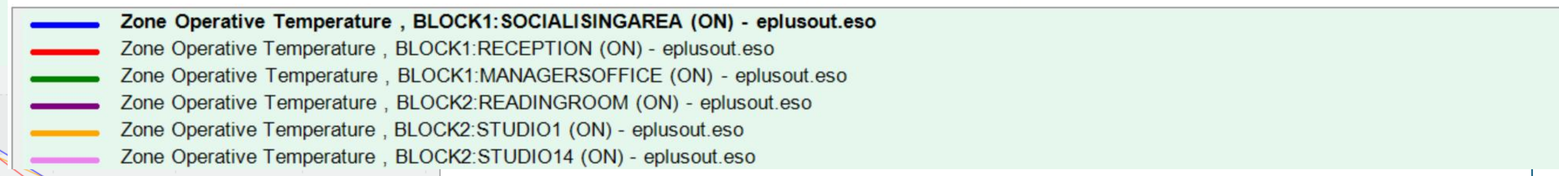
	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	6280.50	8531.00	6125.00
BLOCK2:STUDIO1	5695.00	8492.00	5506.00
BLOCK2:STUDIO14	6529.50	8595.00	6416.00
BLOCK1:RECEPTION	3229.50	4344.00	2933.50
BLOCK1:MANAGERSOFFICE	3488.00	4519.00	3328.50
BLOCK1:SOCIALISINGAREA	2793.50	4285.50	2464.50
Facility	6960.00	8712.50	6589.00

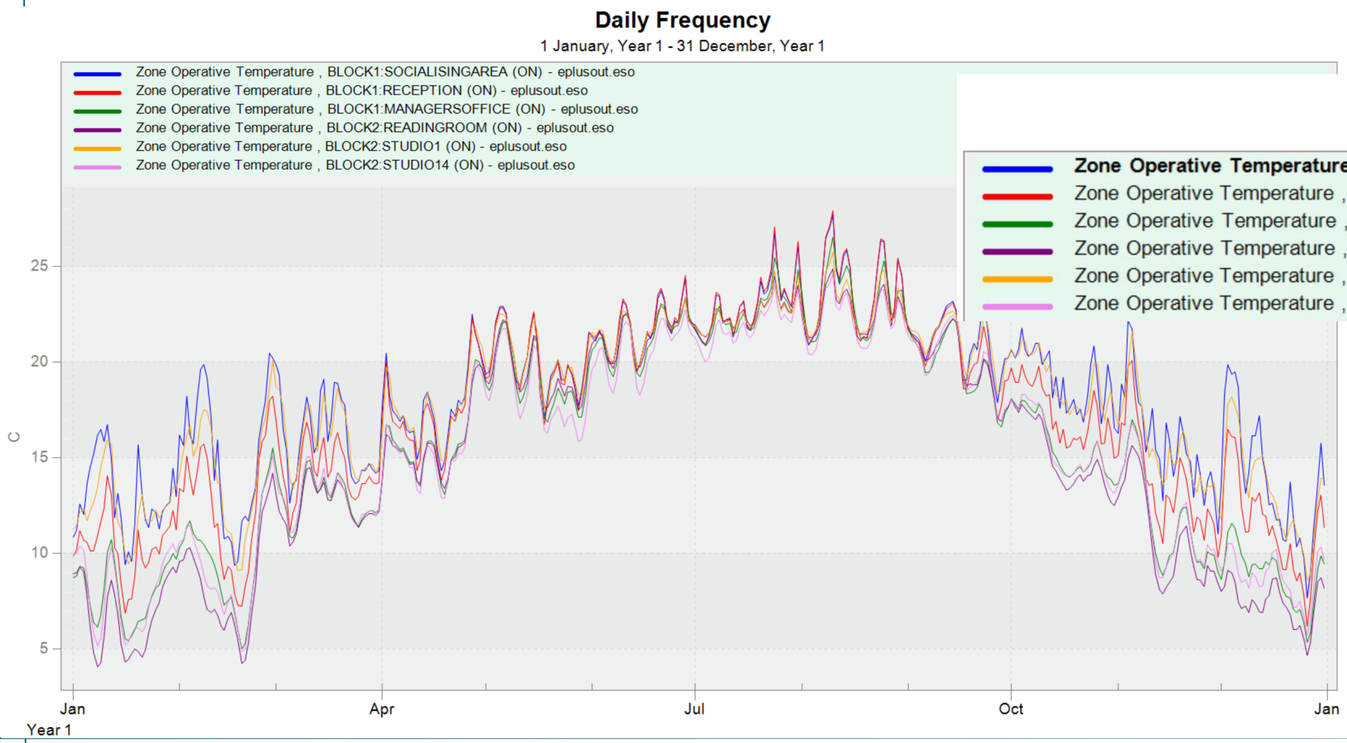
Aggregated over the RunPeriods for Weather

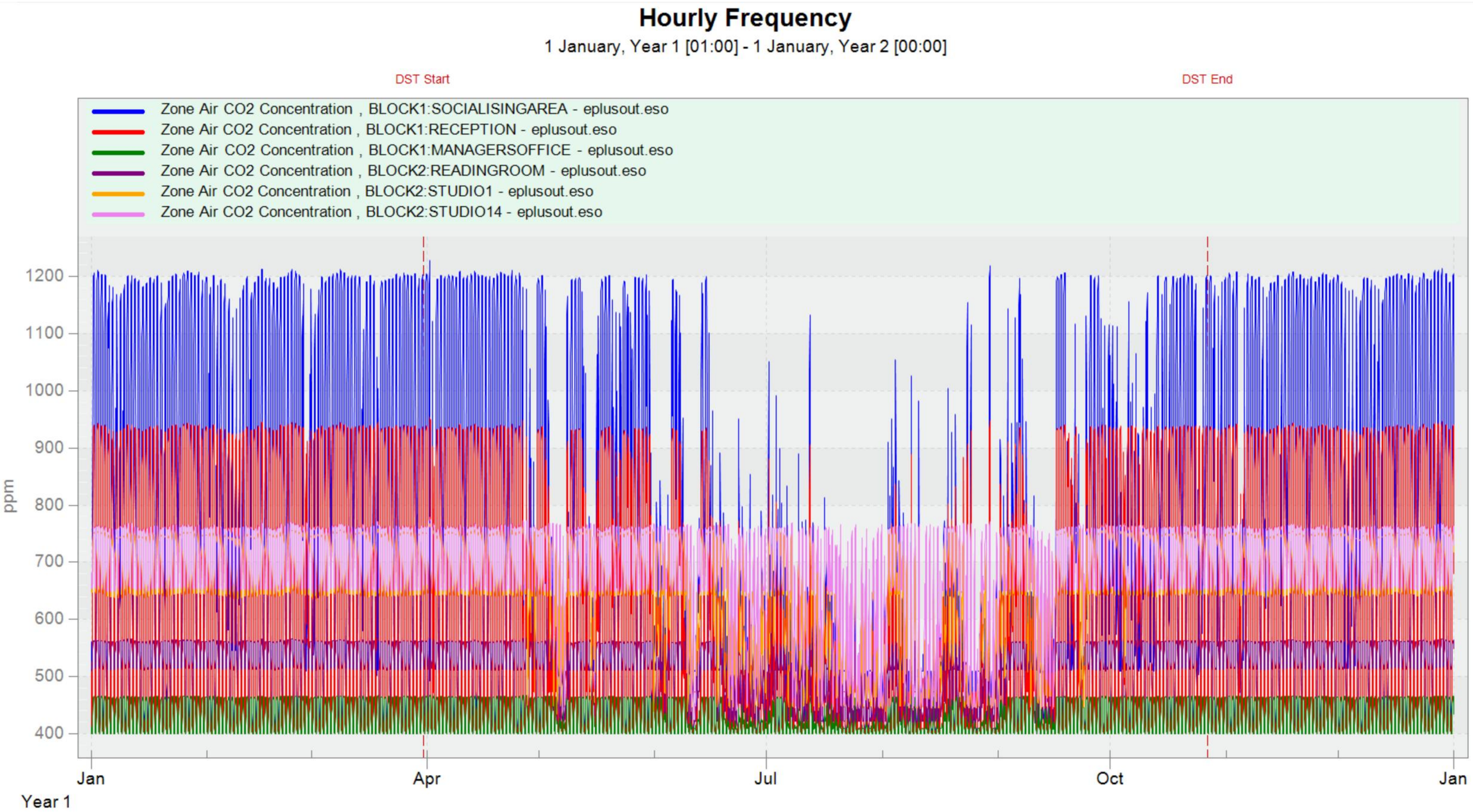
Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

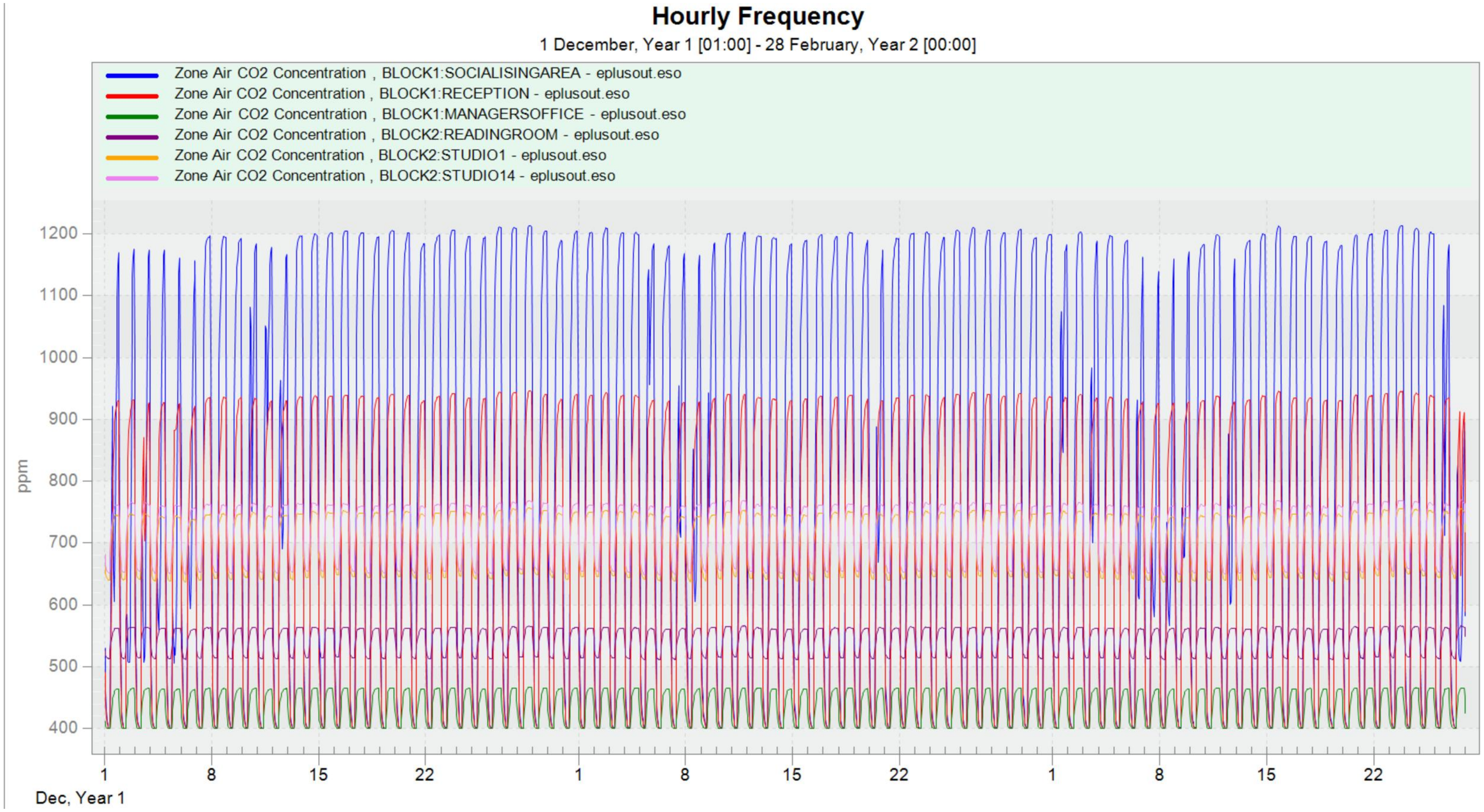




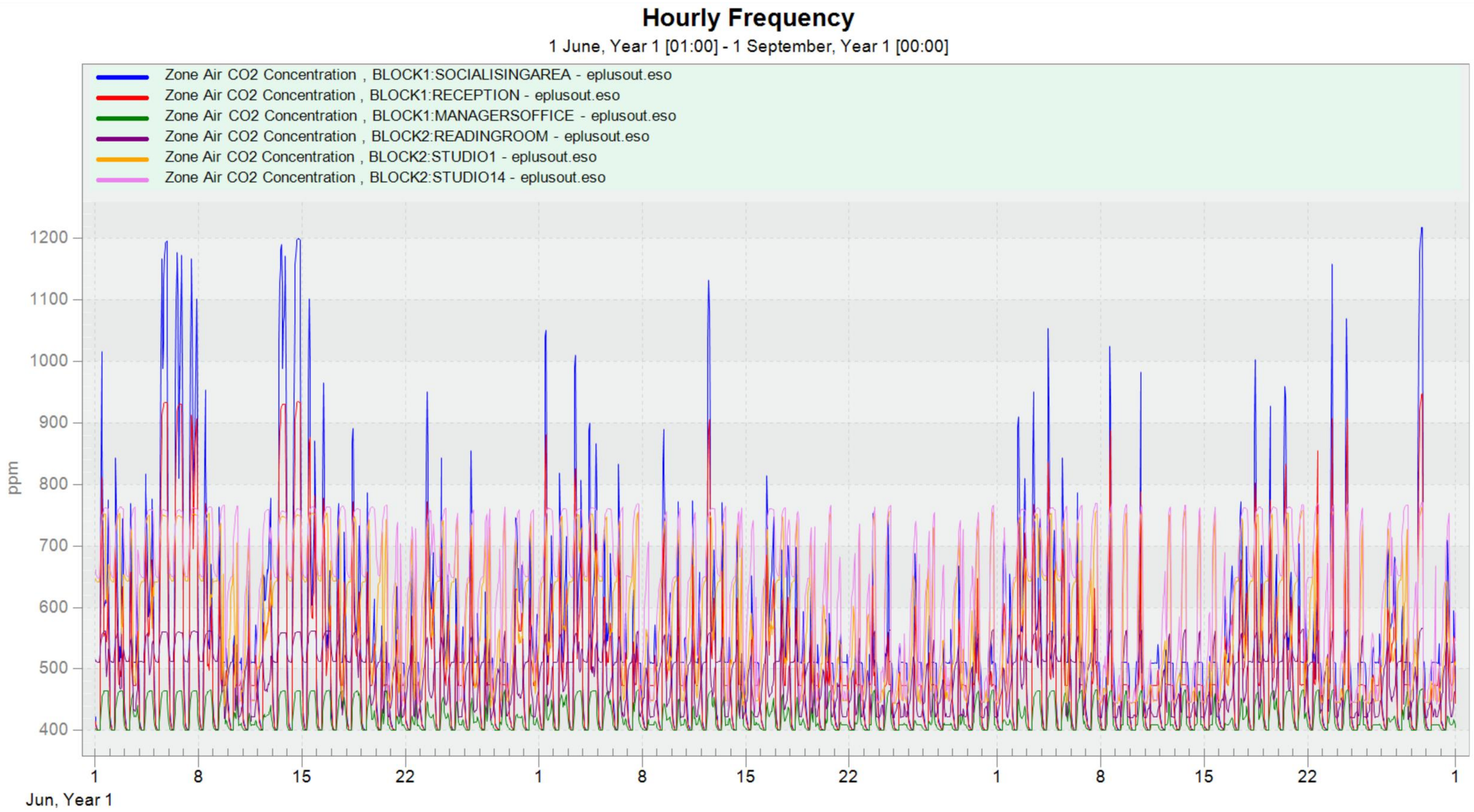



Hourly Frequency
1 January, Year 1 [01:00] - 1 January, Year 2 [00:00]
DST Start
DST End
Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso
ppm
1200
1100
1000
900
800
700
600
500
400
Jan
Apr
Jul
Oct
Jan
Year 1

WINTER CO2 ONLY



SUMMER CO2 ONLY



Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

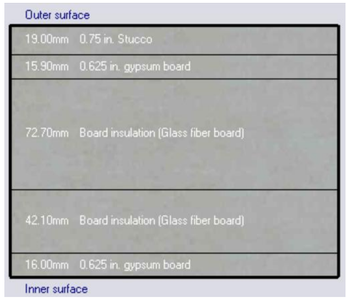
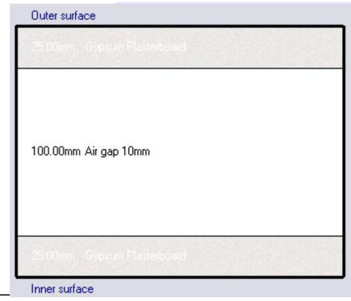
ITERATION 3

SEATTLE CODE.

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

BASE CASE – MATERIALS

ELEMENT(mm)	CROSS-SECTION
Roof (402mm) 190mm asphalt 130mm fibreboard 82.3mm XPS **U-value(0.20)**	
External wall (166mm) 19mm Stucco 16mm Gypsumboard 72.7mm glass fibre insulation 42.1mm glass fibre insulation 16mm gypsumboard **U-value(0.28)**	
Internal wall (150mm) 25mmplasterboard 100mm air gap 25mmplasterboard **U-value(1.64)**	
Ground floor slab (358mm) 50mm floor screed 182.4mm EPS insul. 100mm concrete 25mm brick slips Clay underfloor **U-value(0.20)**	
Window:LoE Clr 3mm double – 13mm Air G-value: 0.65 VT: 0.76 **U-value(1.78)**	

BASE CASE - OVERHEATING CRITERIA – CIBSE TM 59

Naturally ventilated

Criteria for predominantly naturally ventilated homes

Block	Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
BLOCK1	MANAGERSOFFICE	0.41	N/A	Pass
BLOCK1	RECEPTION	3.34	N/A	Fail
BLOCK1	SOCIALISINGAREA	2.95	N/A	Pass
BLOCK2	READINGROOM	0.00	1.00	Pass
BLOCK2	STUDIO1	0.00	6.50	Pass
BLOCK2	STUDIO14	0.00	2.50	Pass

Points credited	Exemplary performance

* 1 additional point awarded because each zone achieved at least 55% coverage
* This construction must identify how it will address glare

Eligible zones for daylighting

Block	Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
Block 2	READING ROOM	18.8	15.73	83.51	18.84	100.0
Block 2	STUDIO 1	12.5	8.05	64.17	11.08	88.3
Block 2	STUDIO 14	12.1	6.87	56.67	10.70	88.3
Block 1	RECEPTION	117.2	112.85	96.27	99.07	84.5
Block 1	MANAGERS OFFICE	7.3	4.12	56.34	7.32	100.0
Block 1	SOCIALISING AREA	98.6	95.77	97.13	36.74	37.3
Total		**266.7**	**243.39**	**91.28**	**183.75**	**68.9**

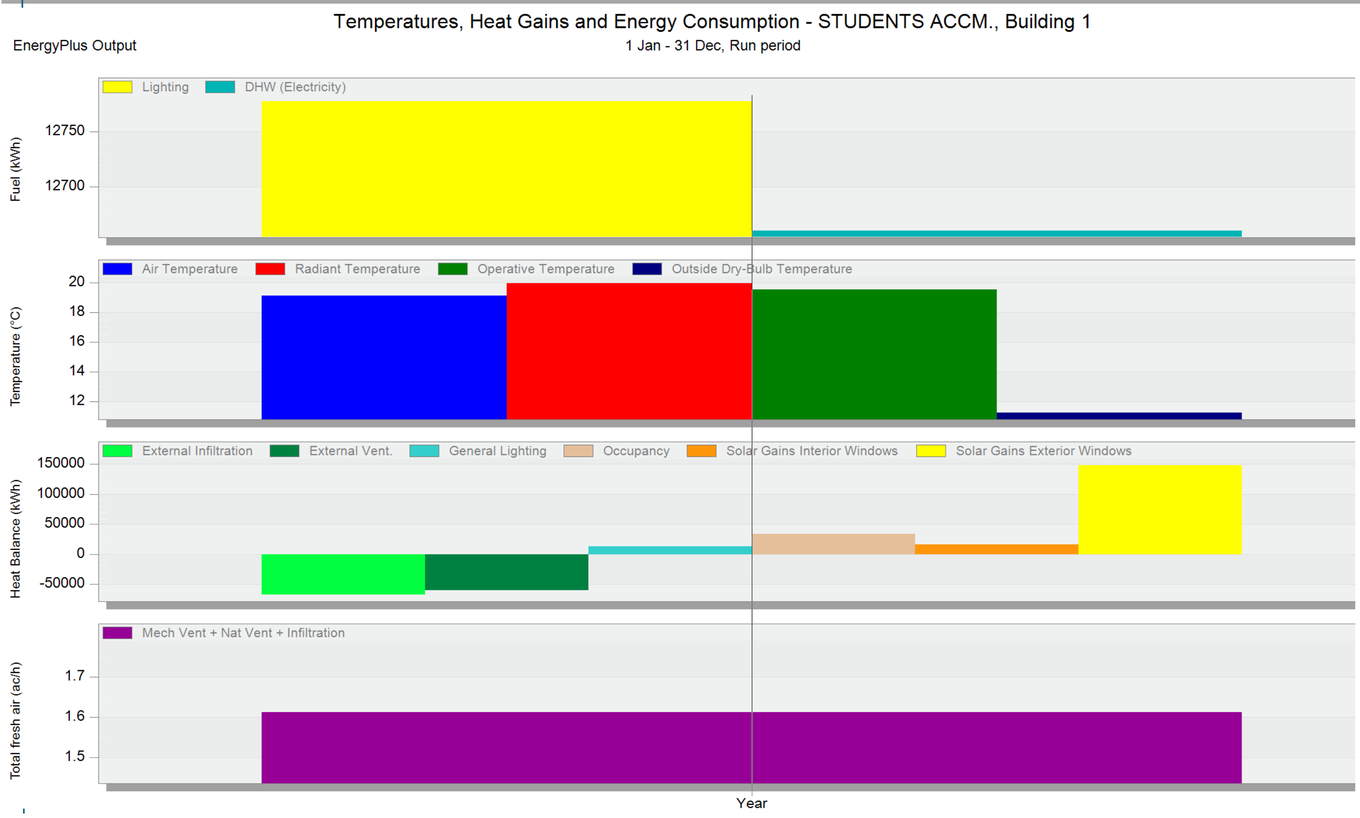
ALL SOUTH WINDOWS SHADED – 1M OVERHANG

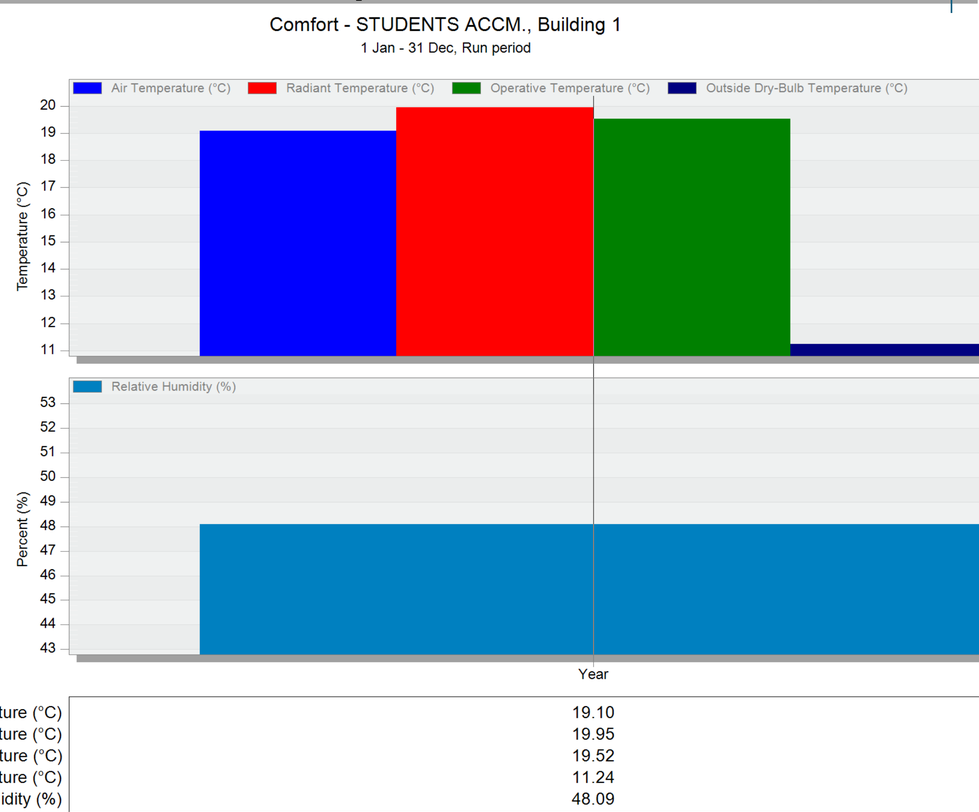
ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1m x 0.7m	2	North
STUDIO 1	1m x 0.9m	1	South
STUDIO 14	1m x 0.9m	1	South
RECEPTION	(S)1.5mx1m ; 1.5x1(E)	(S)4 ; (E)1	South & East
MANAGER'S OFFICE	1m x 0.7m	1	North
SOCIALISING AREA	1.5m	5	South

CALCULATIONS FOR ALL FLOORS

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)





Air Temperature (°C)	19.10
Radiant Temperature (°C)	19.95
Operative Temperature (°C)	19.52
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	48.09

Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., Building 1

EnergyPlus Output 1 Jan - 31 Dec, Run period Student

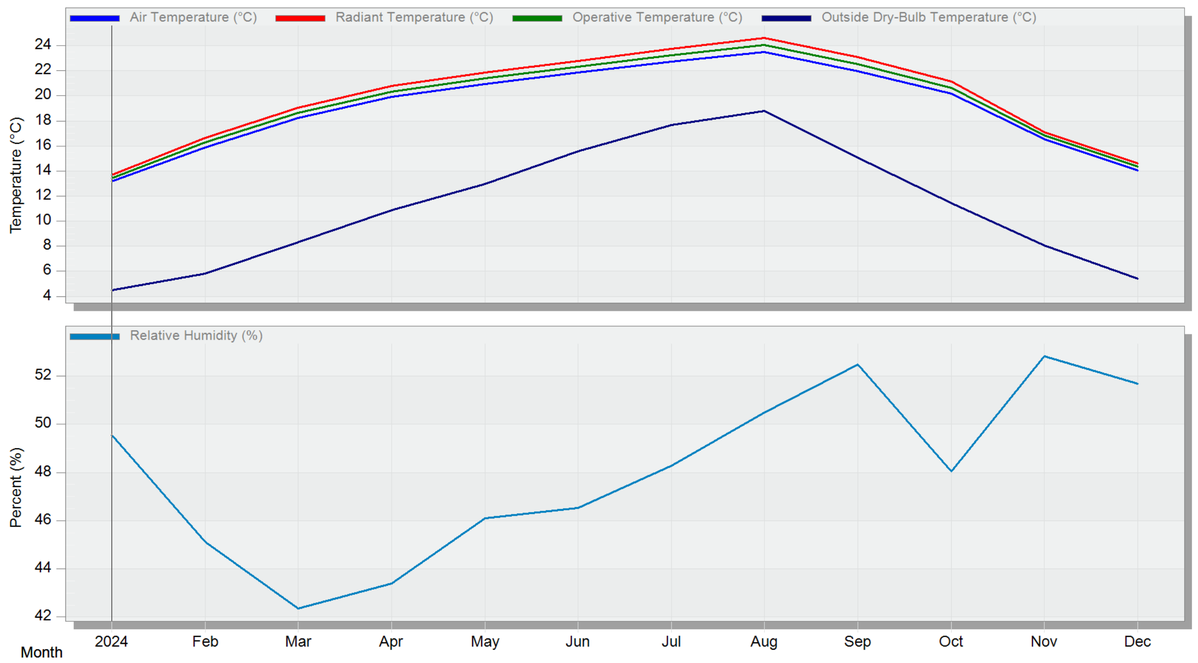
	Year
Lighting (kWh)	12777.18
DHW (Electricity) (kWh)	12660.68
Air Temperature (°C)	19.10
Radiant Temperature (°C)	19.95
Operative Temperature (°C)	19.52
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-67293.71
External Vent. (kWh)	-60141.98
General Lighting (kWh)	12777.18
Occupancy (kWh)	34121.28
Solar Gains Interior Windows (kWh)	16252.99
Solar Gains Exterior Windows (kWh)	148264.63
Mech Vent + Nat Vent + Infiltration (ac/h)	1.61

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Comfort - STUDENTS ACCM., Building 1

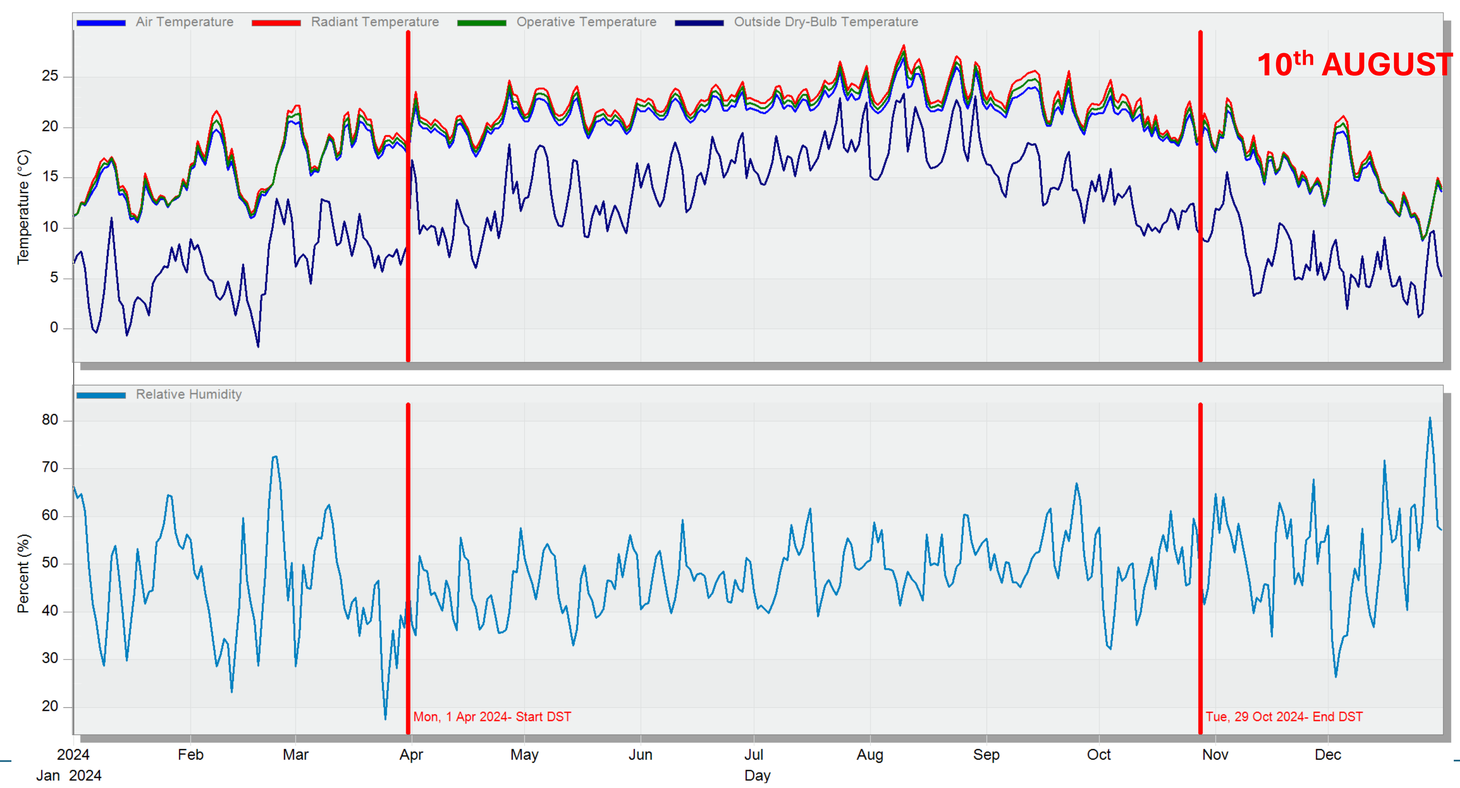
EnergyPlus Output — 1 Jan - 31 Dec, Monthly — Student



	2024	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Air Temperature (°C)	13.20	15.90	18.24	19.90	20.94	21.86	22.72	23.52	21.96	20.17	16.54	14.06
Radiant Temperature (°C)	13.70	16.67	19.06	20.77	21.84	22.81	23.75	24.62	23.08	21.13	17.13	14.61
Operative Temperature (°C)	13.45	16.28	18.65	20.34	21.39	22.33	23.24	24.07	22.52	20.65	16.84	14.34
Outside Dry-Bulb Temperature (°C)	4.50	5.82	8.33	10.85	12.97	15.59	17.66	18.80	15.09	11.43	8.07	5.42
Relative Humidity (%)	49.55	45.13	42.36	43.39	46.11	46.52	48.28	50.50	52.48	48.04	52.83	51.70

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output — 1 Jan - 31 Dec, Daily — Student

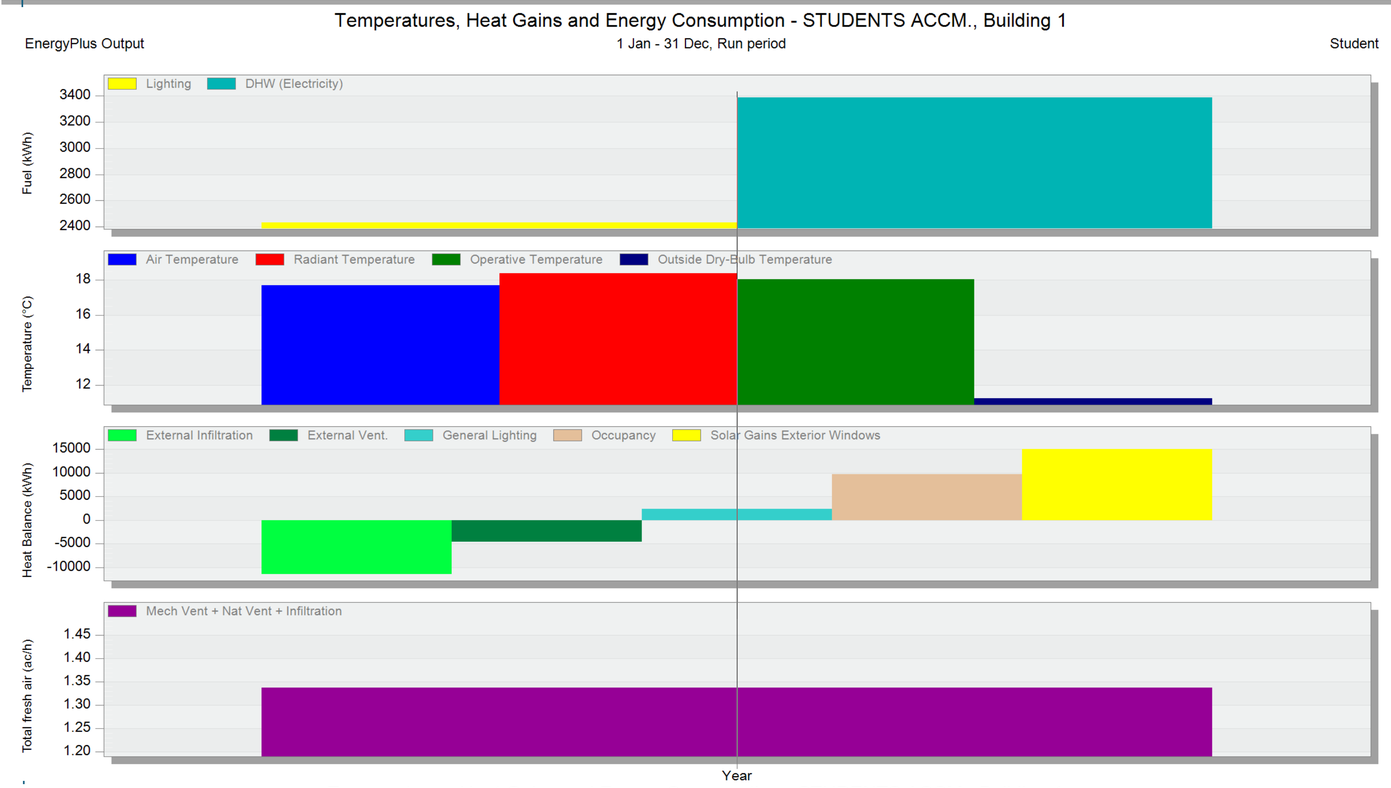


10th AUGUST 27.6 OPERATIVE TEMPERATURE

CALCULATIONS FOR ZONES ONLY

Sustainable Design of Students' Accommodation Block

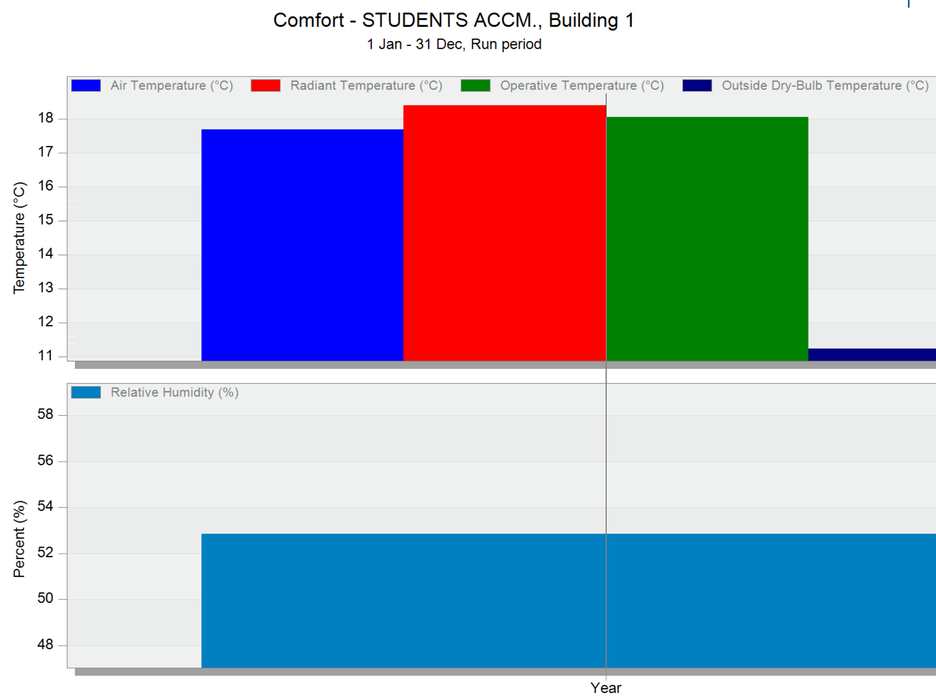
Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)



Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., Building 1
EnergyPlus Output
1 Jan - 31 Dec, Run period
Student

Lighting (kWh)	2433.20
DHW (Electricity) (kWh)	3386.98
Air Temperature (°C)	17.69
Radiant Temperature (°C)	18.39
Operative Temperature (°C)	18.04
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-11407.77
External Vent. (kWh)	-4588.70
General Lighting (kWh)	2433.20
Occupancy (kWh)	9754.29
Solar Gains Exterior Windows (kWh)	15044.91
Mech Vent + Nat Vent + Infiltration (ac/h)	1.34

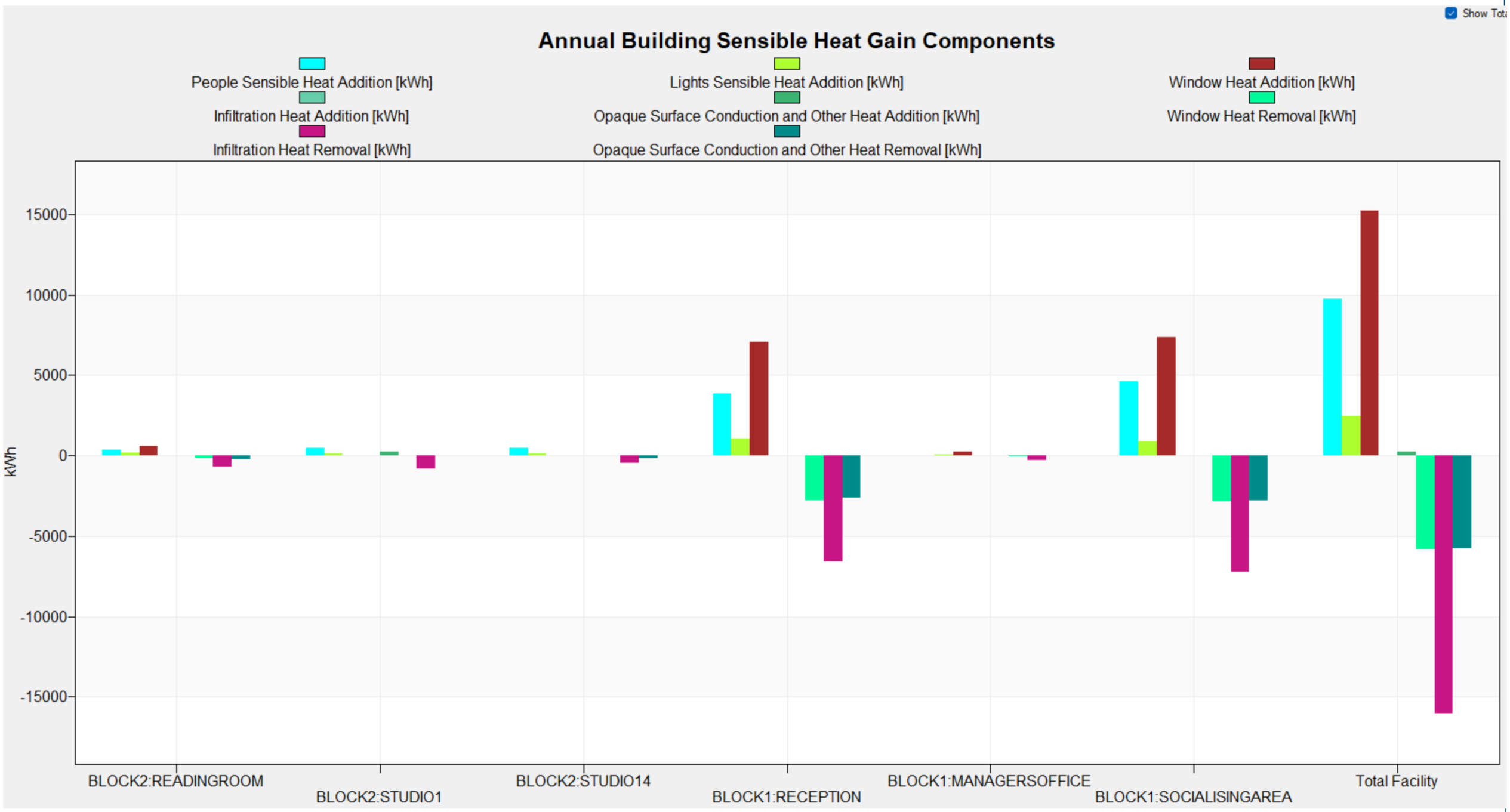
EnergyPlus Output



Air Temperature (°C)	17.69
Radiant Temperature (°C)	18.39
Operative Temperature (°C)	18.04
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	52.85

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)



Time Not Comfortable Based on Simple ASHRAE 55-2004

	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	6139.50	8505.50	5967.00
BLOCK2:STUDIO1	5301.00	8509.50	5123.00
BLOCK2:STUDIO14	6473.00	8567.00	6330.50
BLOCK1:RECEPTION	3083.50	4311.50	2757.00
BLOCK1:MANAGERSOFFICE	3508.00	4487.00	3331.00
BLOCK1:SOCIALISINGAREA	2592.00	4247.50	2234.50
Facility	6892.50	8697.00	6497.00

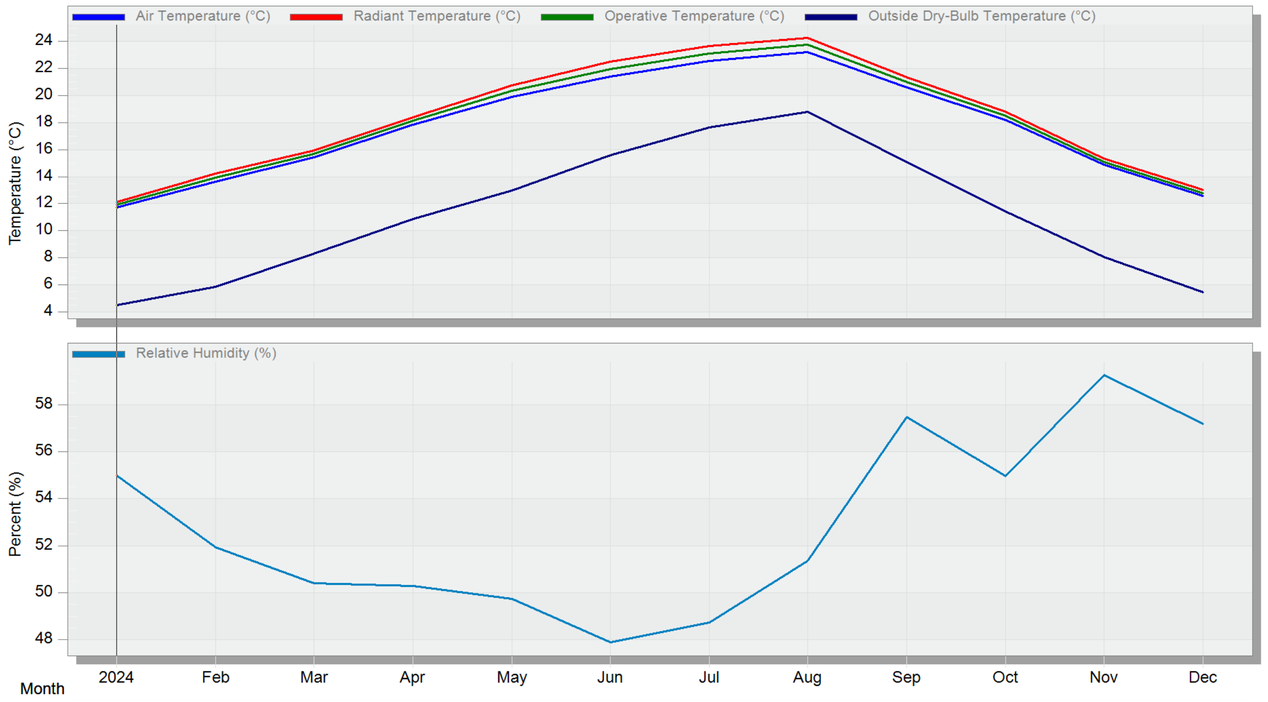
Aggregated over the RunPeriods for Weather

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Comfort - STUDENTS ACCM., Building 1

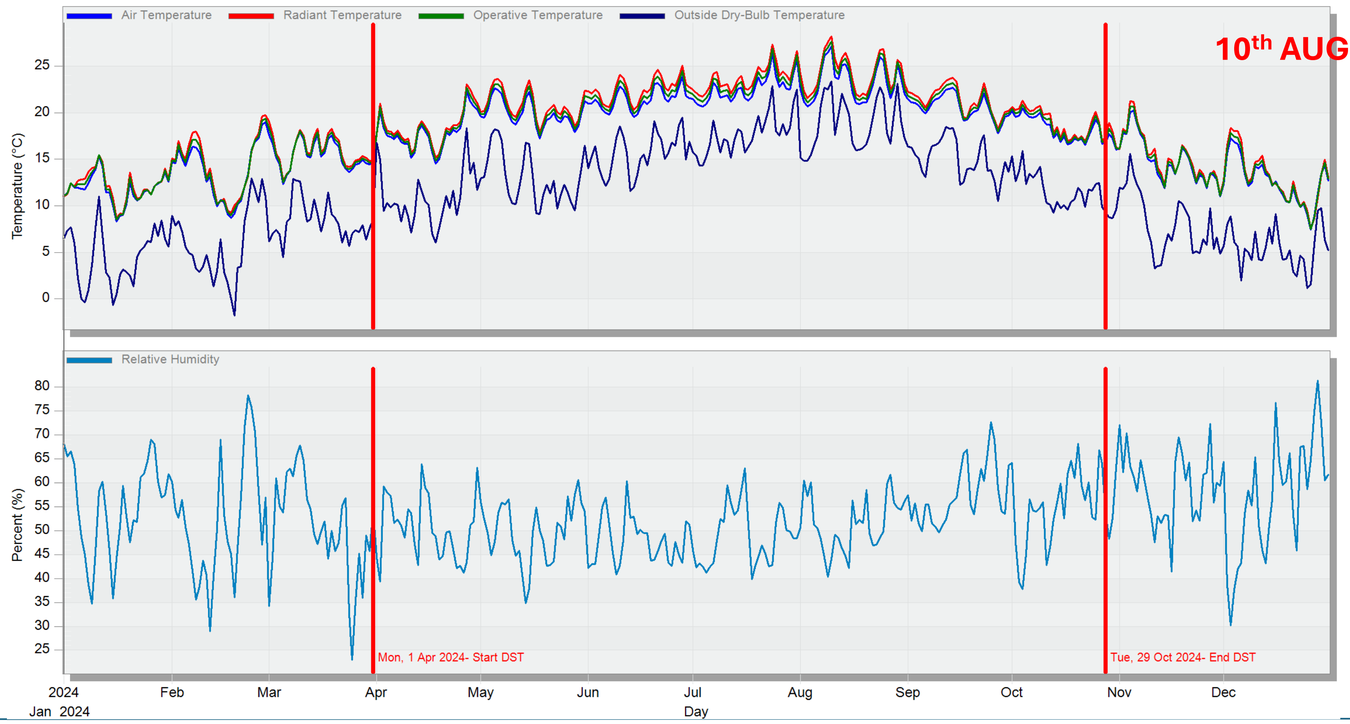
EnergyPlus Output — 1 Jan - 31 Dec, Monthly — Student



Month	2024	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Air Temperature (°C)	11.73	13.64	15.45	17.84	19.91	21.42	22.55	23.24	20.61	18.19	14.89	12.57
Radiant Temperature (°C)	12.12	14.22	15.92	18.42	20.77	22.52	23.68	24.27	21.37	18.81	15.32	13.01
Operative Temperature (°C)	11.92	13.93	15.68	18.13	20.34	21.97	23.12	23.75	20.99	18.50	15.11	12.79
Outside Dry-Bulb Temperature (°C)	4.50	5.82	8.33	10.85	12.97	15.59	17.66	18.80	15.09	11.43	8.07	5.42
Relative Humidity (%)	54.99	51.92	50.40	50.28	49.74	47.90	48.74	51.36	57.47	54.97	59.25	57.18

Comfort - STUDENTS ACCM., Building 1

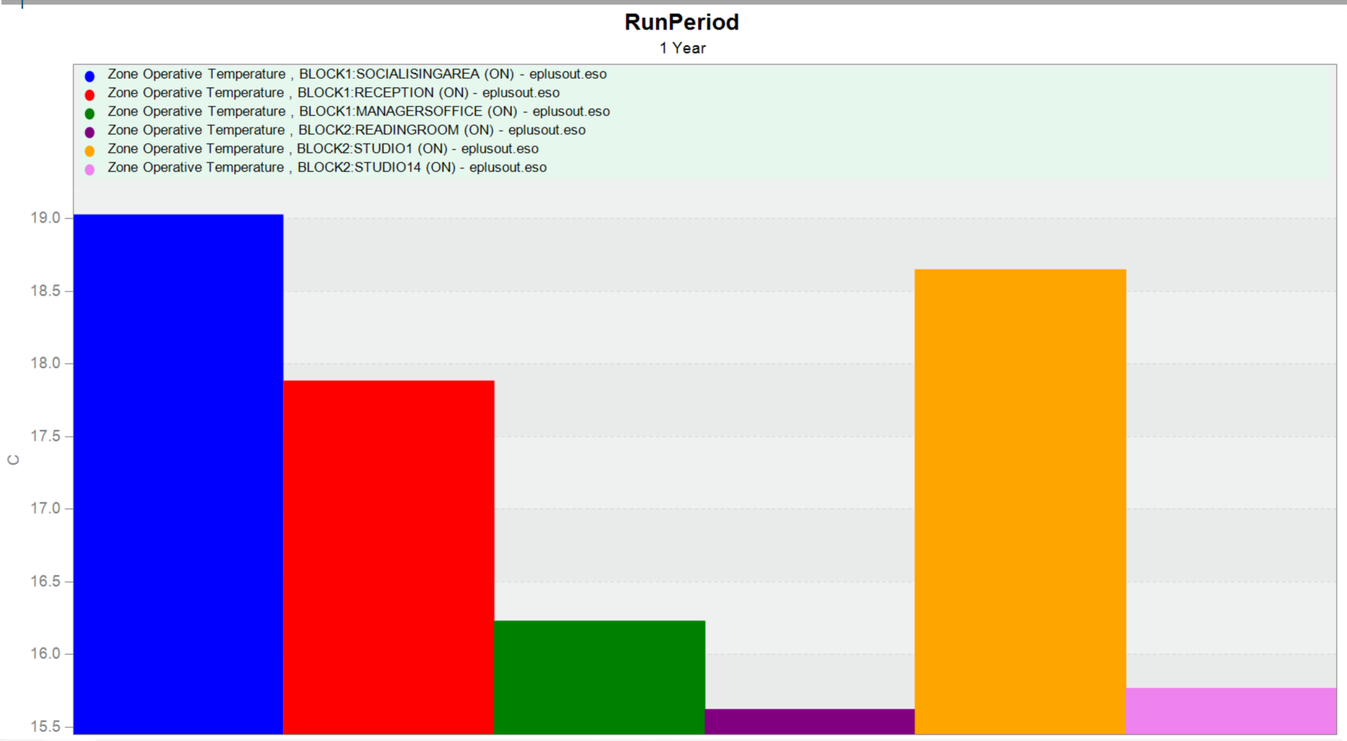
EnergyPlus Output — 1 Jan - 31 Dec, Daily — Student

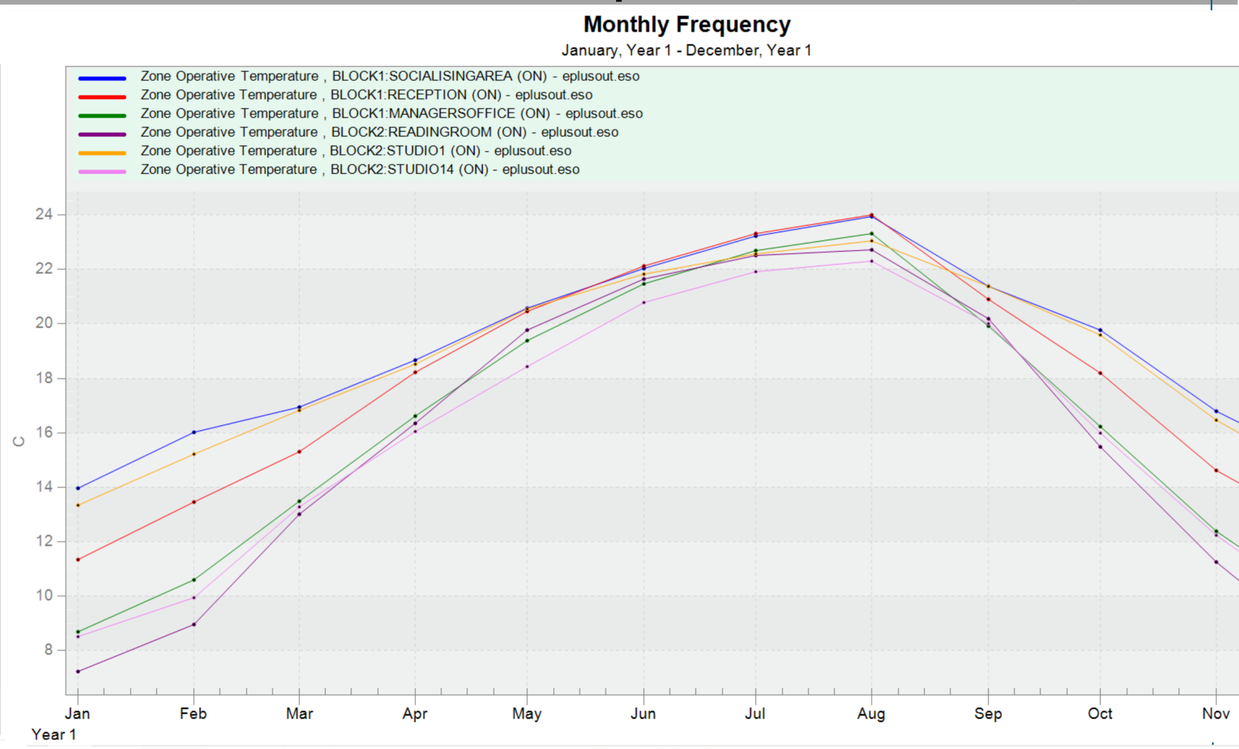


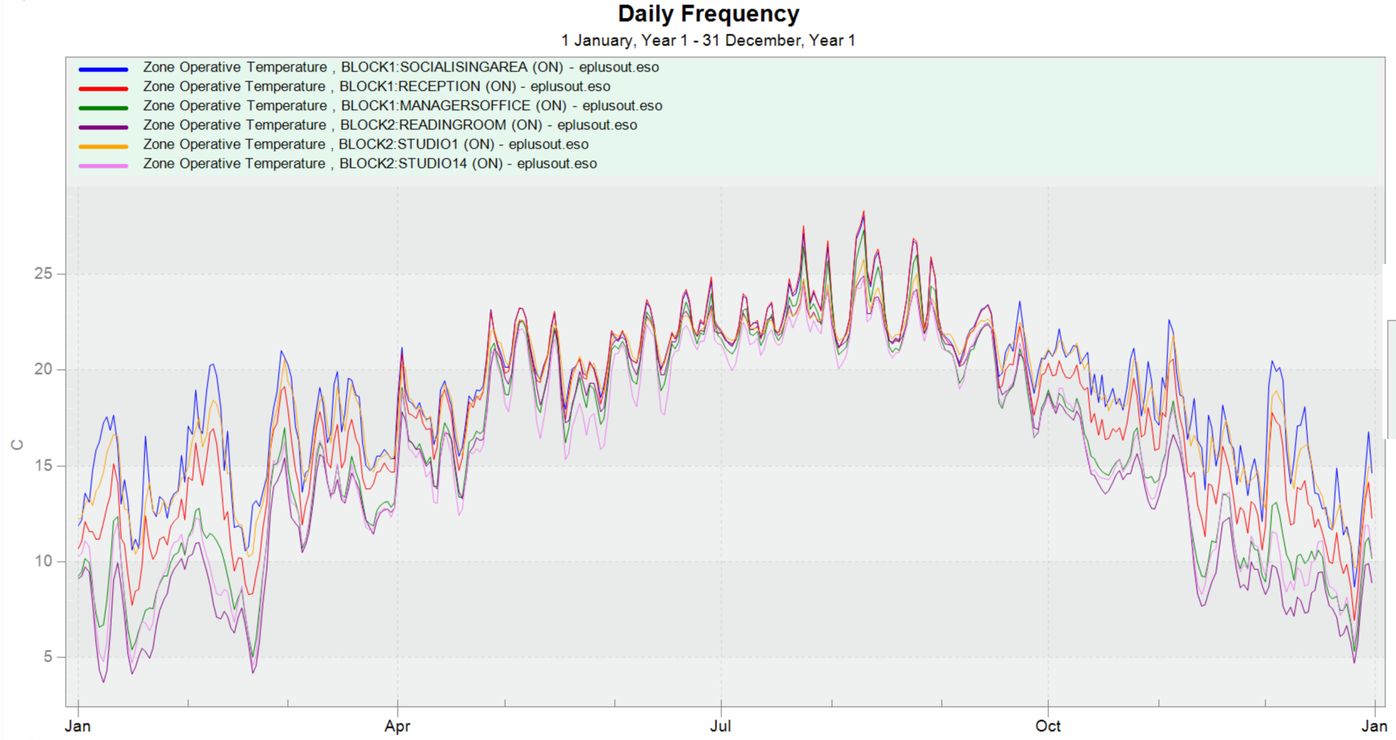
10th AUGUST 27.6 OPERATIVE TEMPERATURE

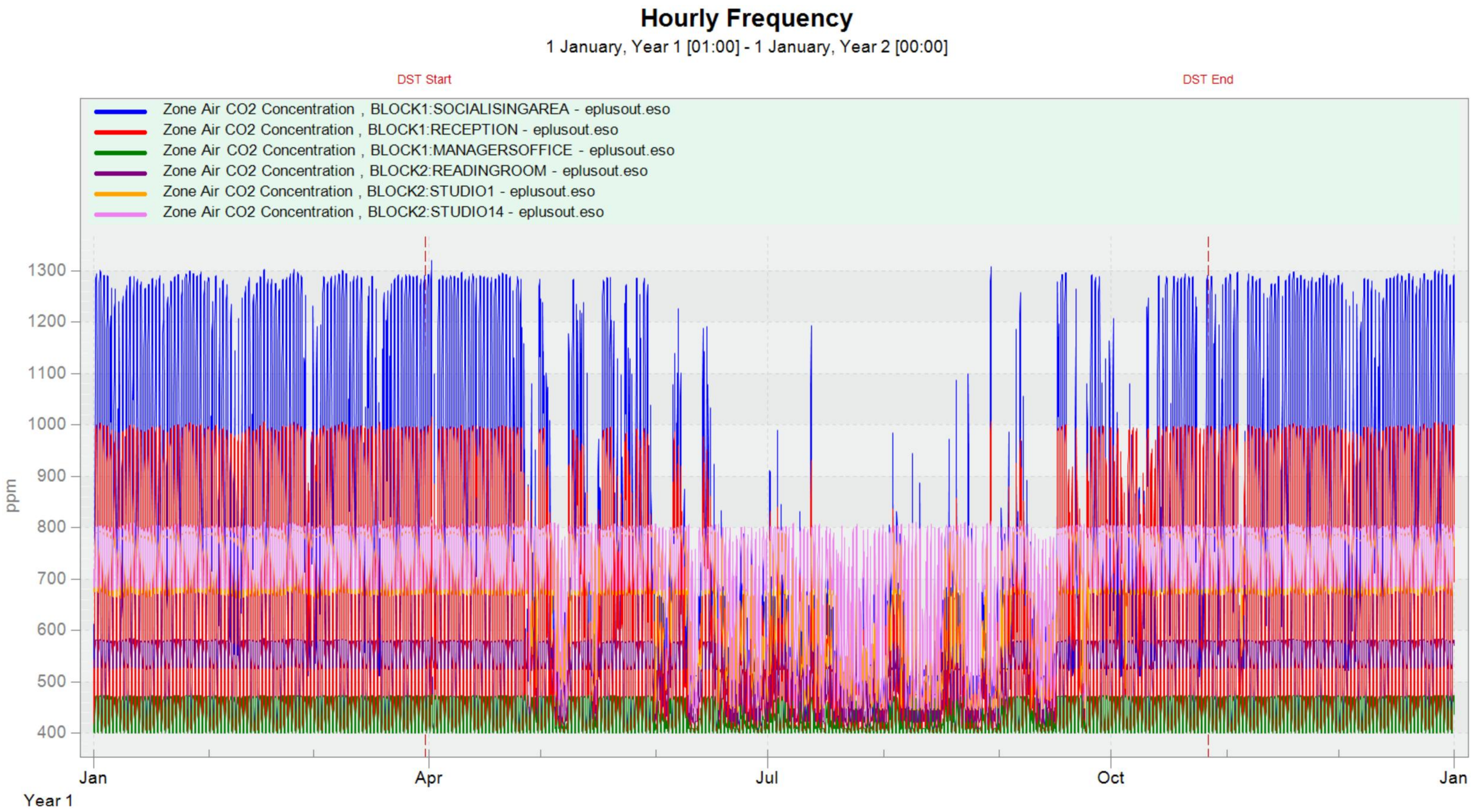
Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

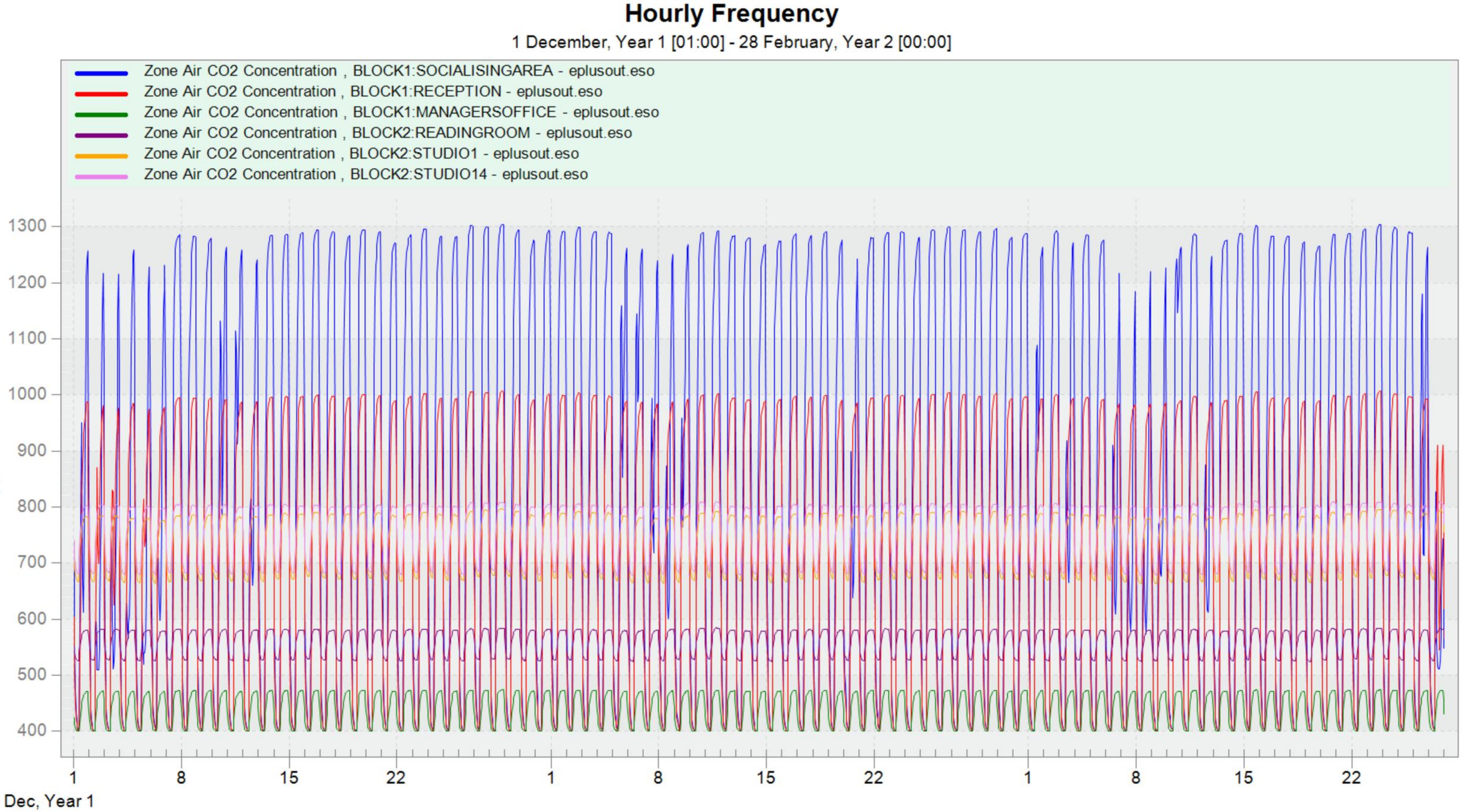







Hourly Frequency
1 January, Year 1 [01:00] - 1 January, Year 2 [00:00]
DST Start
DST End
Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso
ppm
1300
1200
1100
1000
900
800
700
600
500
400
Jan
Apr
Jul
Oct
Jan
Year 1

WINTER CO2 ONLY

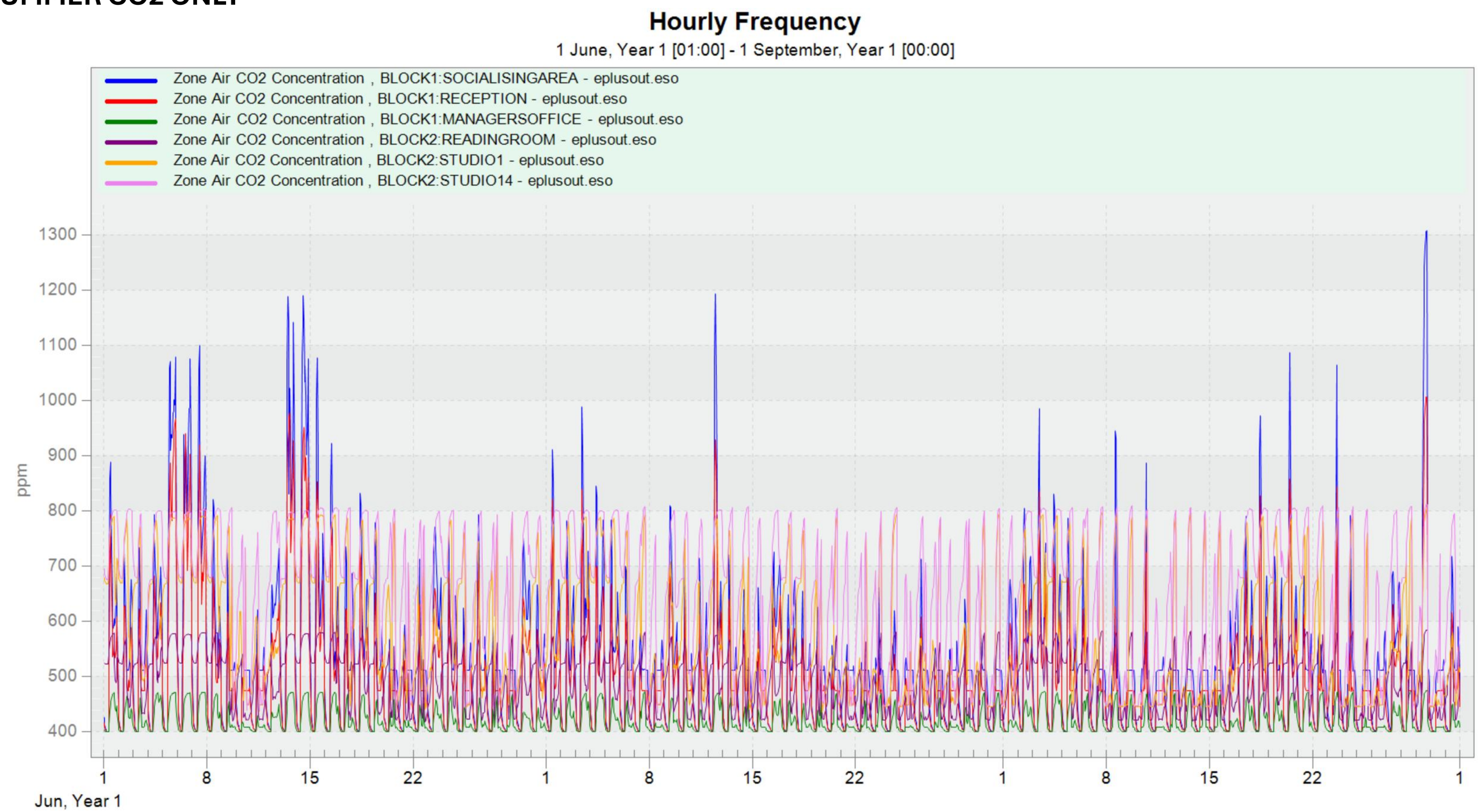


Time Not Comfortable Based on Simple ASHRAE 55-2004

	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	2136.00	2136.00	2136.00
BLOCK2:STUDIO1	2130.00	2136.00	2130.00
BLOCK2:STUDIO14	2136.00	2136.00	2136.00
BLOCK1:RECEPTION	1127.00	1157.00	1127.00
BLOCK1:MANAGERSOFFICE	1157.00	1157.00	1157.00
BLOCK1:SOCIALISINGAREA	961.50	1135.50	953.00
Facility	2136.00	2136.00	2136.00

Aggregated over the RunPeriods for Weather

SUMMER CO2 ONLY



Time Not Comfortable Based on Simple ASHRAE 55-2004

	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	299.00	1961.00	127.50
BLOCK2:STUDIO1	233.50	1957.50	55.00
BLOCK2:STUDIO14	521.50	2019.00	378.00
BLOCK1:RECEPTION	416.00	866.50	158.50
BLOCK1:MANAGERSOFFICE	327.50	945.50	148.00
BLOCK1:SOCIALISINGAREA	416.00	863.00	154.50
Facility	819.00	2145.50	506.00

Aggregated over the RunPeriods for Weather

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

ITERATION 4

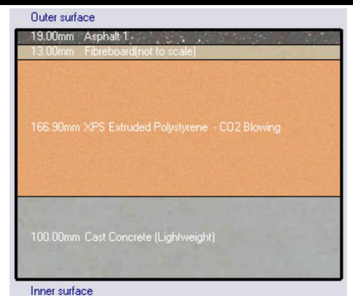
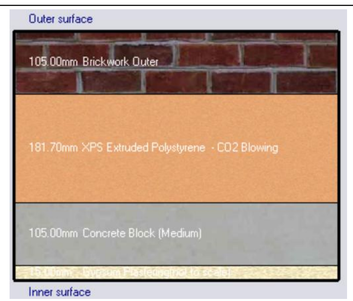
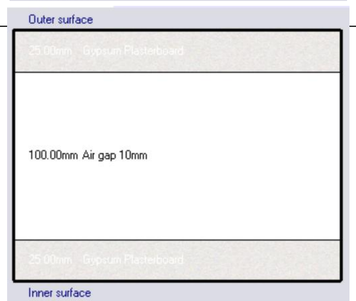
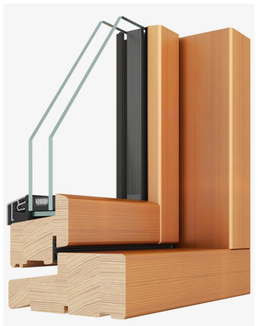
LEED PLAT.

SECOND ITERATION

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

BASE CASE – MATERIALS

ELEMENT(mm)	CROSS-SECTION
Roof (300mm) 19mm asphalt 13mm fibreboard 167mm XPS 100mm concrete **U-value(0.18)**	
External wall (407mm) 105mm brickwork 182mm XPS insulation 105mm concrete 15mm gypsum plaster **U-value(0.17)**	
Internal wall (150mm) 25mmplasterboard 100mm air gap 25mmplasterboard **U-value(1.639)**	
Ground floor slab (440mm) 30mm timber floor 70mm floor screed 100mm concrete 240mm fomaldehyde foam **U-value(0.15)**	
Window:LoE Elec Abs bleached 6mm double Argon G-value: 0.48 VT: 0.657 **U-value(1.3)**	

BASE CASE - OVERHEATING CRITERIA – CIBSE TM 59

Naturally ventilated

Criteria for predominantly naturally ventilated homes

Block	Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
BLOCK1	MANAGERSOFFICE	0.00	N/A	Pass
BLOCK1	RECEPTION	1.50	N/A	Pass
BLOCK1	SOCIALISINGAREA	1.48	N/A	Pass
BLOCK2	READINGROOM	0.00	0.00	Pass
BLOCK2	STUDIO1	0.00	1.50	Pass
BLOCK2	STUDIO14	0.00	0.00	Pass

Points credited	3

* This construction must identify how it will address glare

Eligible zones for daylighting

Block	Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
Block 2	READING ROOM	16.6	9.95	59.88	16.62	100.0
Block 2	STUDIO 1	12.5	6.69	53.33	11.40	90.8
Block 2	STUDIO 14	11.0	5.54	50.48	10.34	94.3
Block 1	RECEPTION	110.5	104.16	94.26	107.37	97.2
Block 1	MANAGERS OFFICE	6.1	2.98	49.09	6.07	100.0
Block 1	SOCIALISING AREA	93.3	87.86	94.14	48.01	51.4
Total		**250.0**	**217.18**	**86.86**	**199.81**	**79.9**

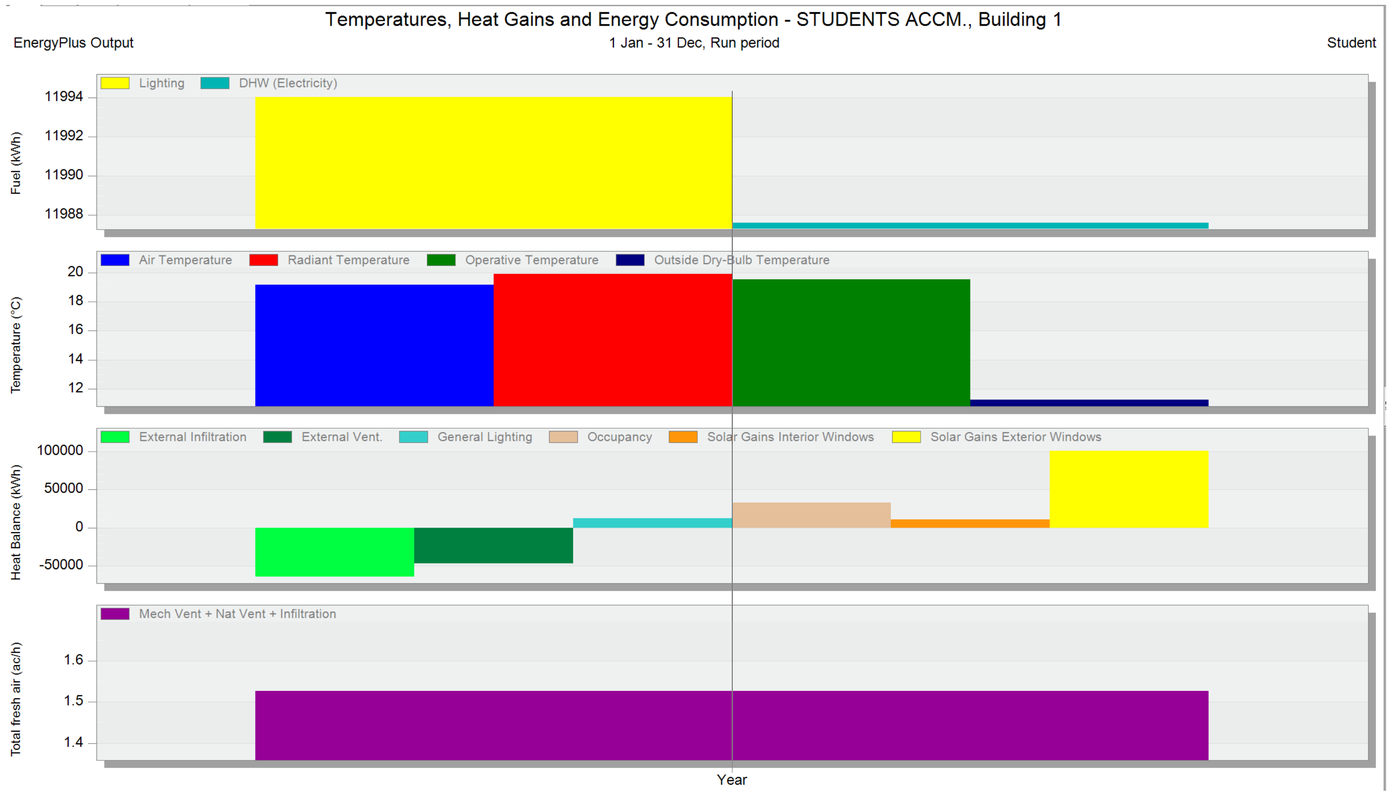
ALL SOUTH WINDOWS SHADED – 1M OVERHANG

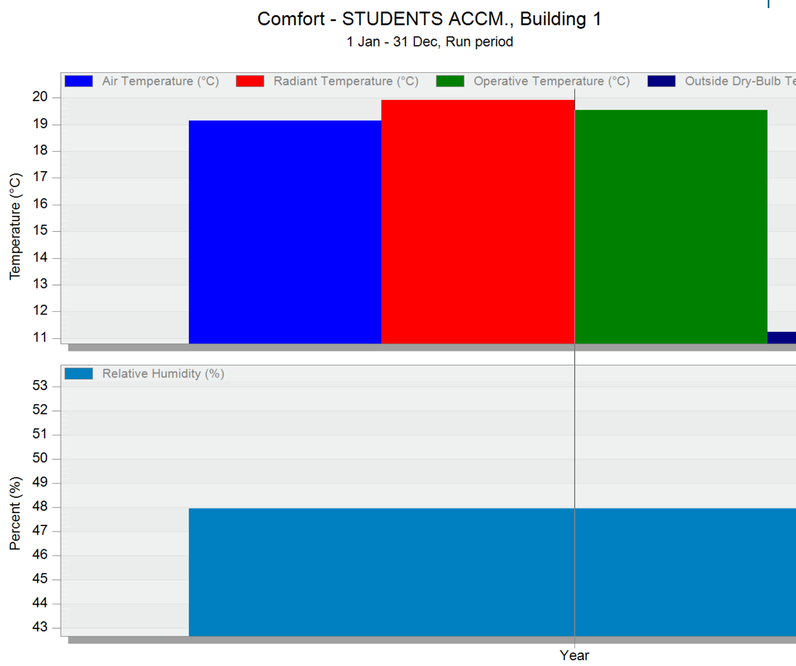
ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1m x 0.7m	2	North
STUDIO 1	1m x 0.9m	1	South
STUDIO 14	1m x 0.9m	1	South
RECEPTION	(S)1.5mx1m ; 1.5x1(E)	(S)4 ; (E)1	South & East
MANAGER'S OFFICE	1m x 0.7m	1	North
SOCIALISING AREA	1.5m	5	South

CALCULATIONS FOR ALL FLOORS

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)





Air Temperature (°C)	19.15
Radiant Temperature (°C)	19.91
Operative Temperature (°C)	19.53
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	47.95

Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., Building 1

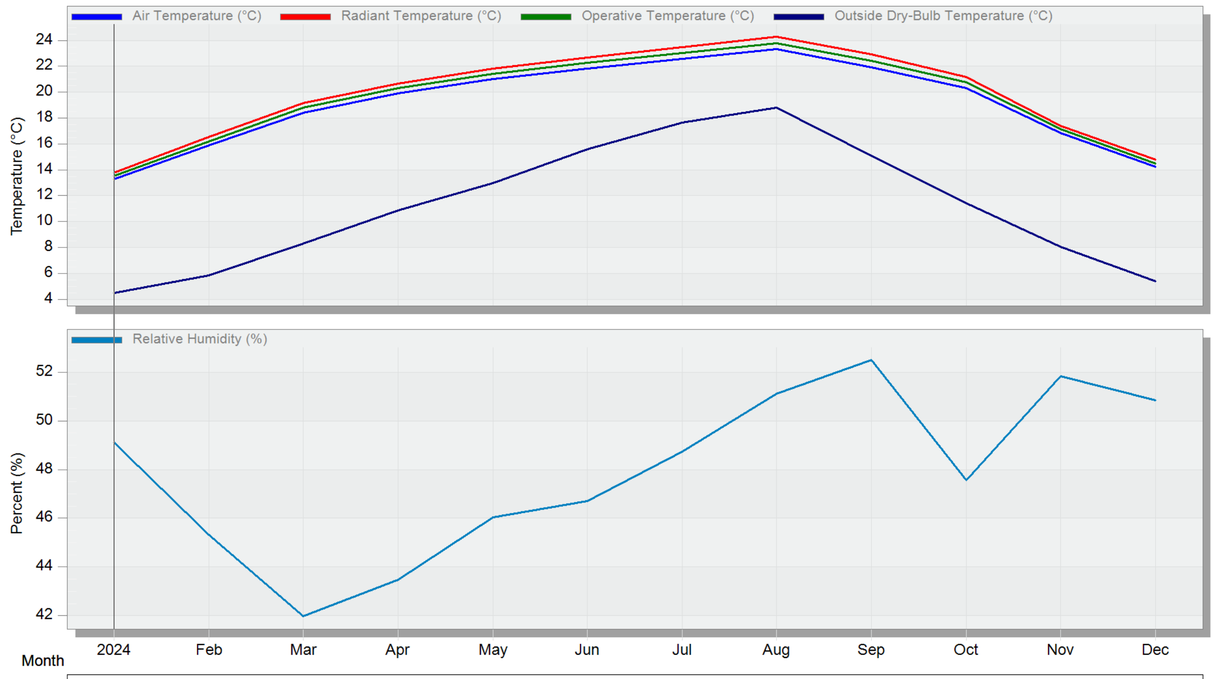
EnergyPlus Output | 1 Jan - 31 Dec, Run period | Student

Year

Lighting (kWh)	11994.03
DHW (Electricity) (kWh)	11987.61
Air Temperature (°C)	19.15
Radiant Temperature (°C)	19.91
Operative Temperature (°C)	19.53
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-64064.70
External Vent. (kWh)	-46700.95
General Lighting (kWh)	11994.03
Occupancy (kWh)	32910.61
Solar Gains Interior Windows (kWh)	10861.64
Solar Gains Exterior Windows (kWh)	100477.27
Mech Vent + Nat Vent + Infiltration (ac/h)	1.53

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Monthly | Student

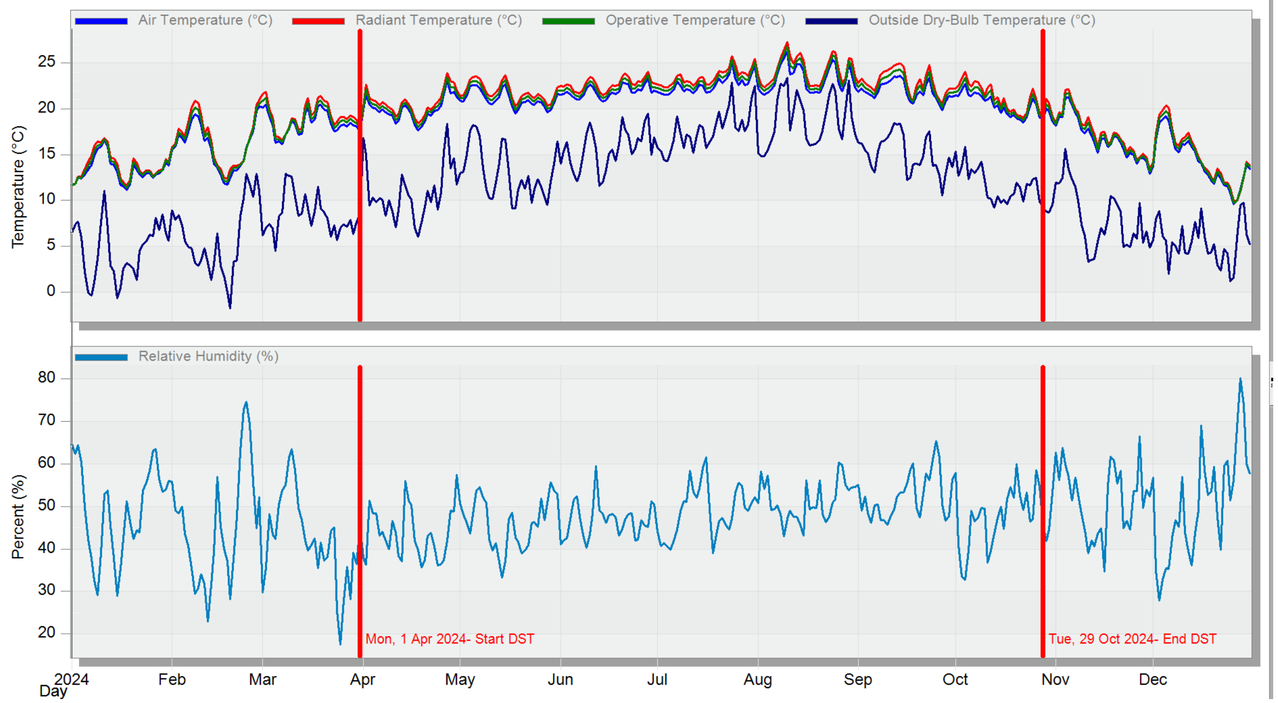


	2024	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Air Temperature (°C)	13.30	15.88	18.43	19.91	21.04	21.85	22.59	23.31	21.92	20.33	16.86	14.25
Radiant Temperature (°C)	13.77	16.55	19.17	20.67	21.83	22.68	23.48	24.28	22.92	21.19	17.42	14.77
Operative Temperature (°C)	13.53	16.22	18.80	20.29	21.43	22.27	23.03	23.79	22.42	20.76	17.14	14.51
Outside Dry-Bulb Temperature (°C)	4.50	5.82	8.33	10.85	12.97	15.59	17.66	18.80	15.09	11.43	8.07	5.42
Relative Humidity (%)	49.10	45.31	41.98	43.48	46.05	46.71	48.73	51.11	52.50	47.56	51.84	50.84

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Daily | Student

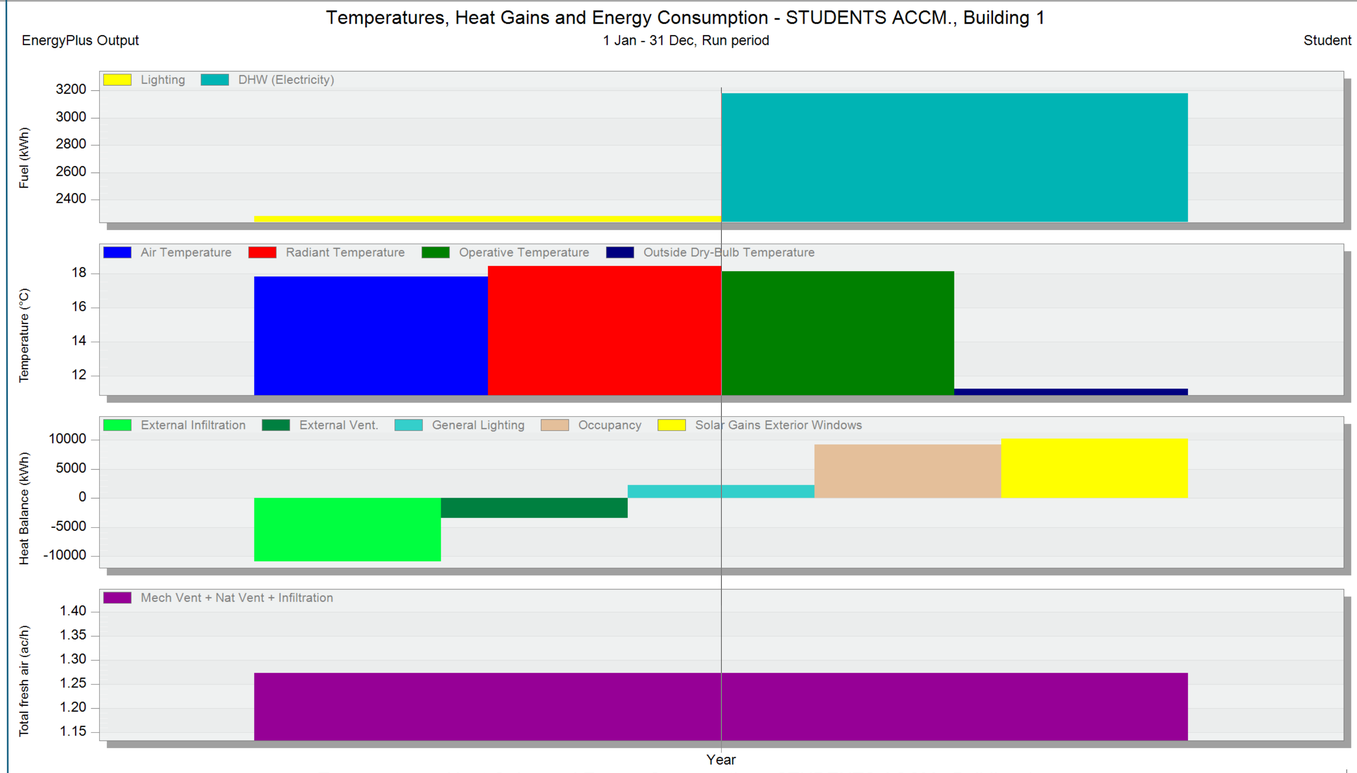
10th AUGUST 27.2 OPERATIVE TEMPERATURE

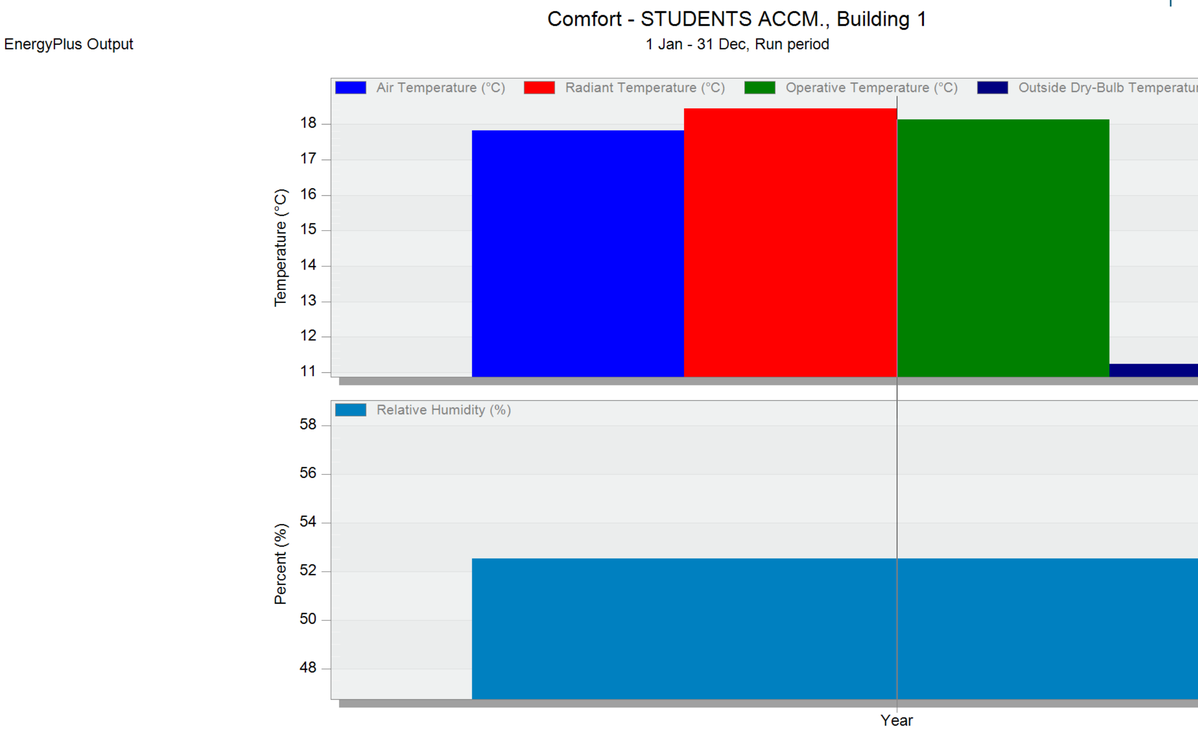


CALCULATIONS FOR ZONES ONLY

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)





Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Run period | Student

	Year
Lighting (kWh)	2281.53
DHW (Electricity) (kWh)	3178.16
Air Temperature (°C)	17.82
Radiant Temperature (°C)	18.44
Operative Temperature (°C)	18.13
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-10913.67
External Vent. (kWh)	-3468.36
General Lighting (kWh)	2281.53
Occupancy (kWh)	9240.26
Solar Gains Exterior Windows (kWh)	10225.31
Mech Vent + Nat Vent + Infiltration (ac/h)	1.27

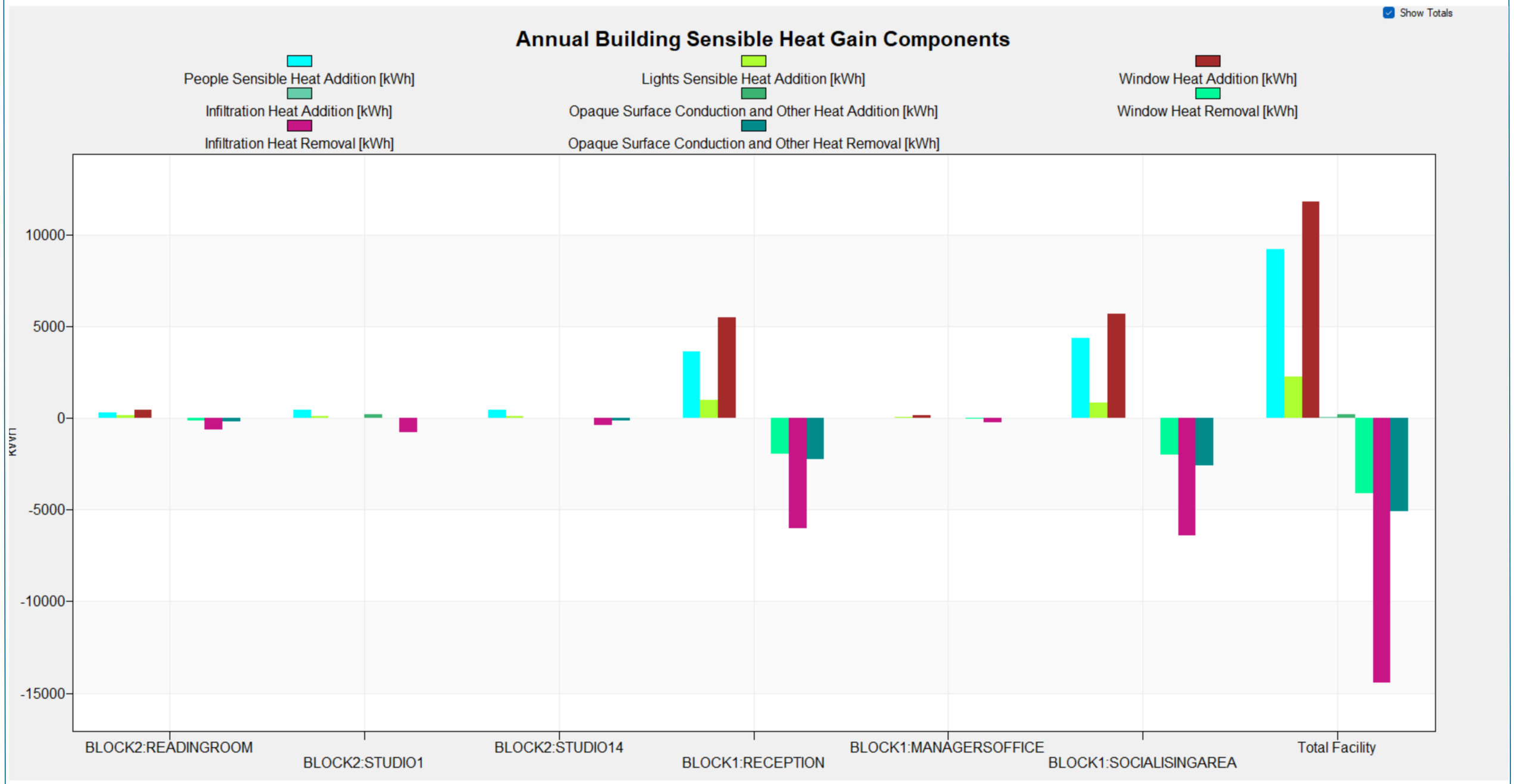
Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Run period

	Year
Air Temperature (°C)	17.82
Radiant Temperature (°C)	18.44
Operative Temperature (°C)	18.13
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	52.52

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

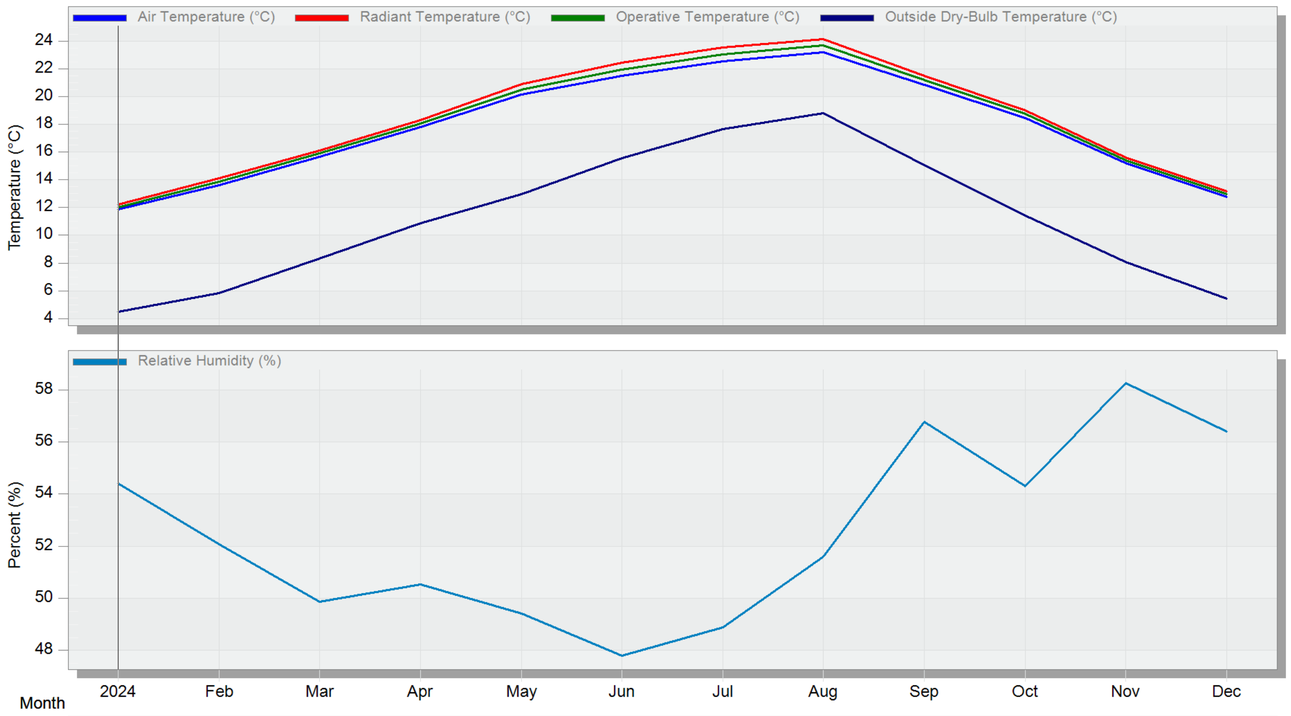


Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output — 1 Jan - 31 Dec, Monthly — Student

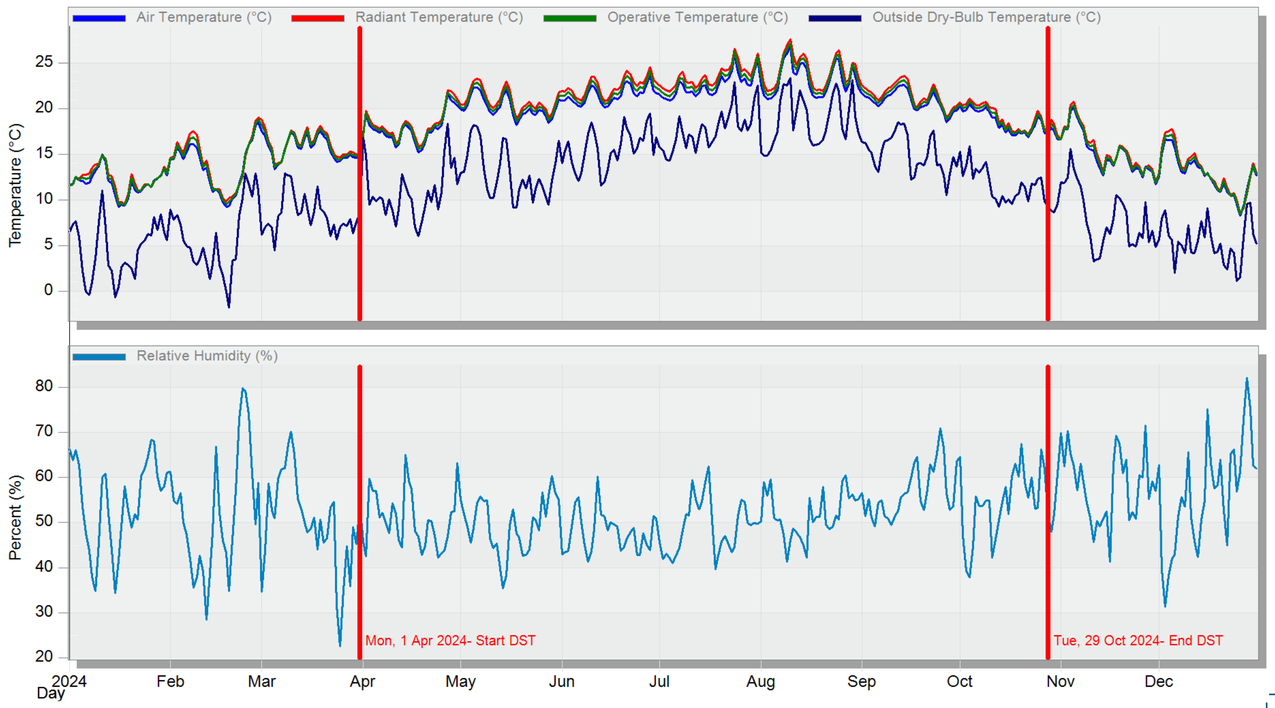


Month	2024	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Air Temperature (°C)	11.86	13.63	15.67	17.81	20.13	21.48	22.53	23.21	20.84	18.45	15.19	12.75
Radiant Temperature (°C)	12.22	14.13	16.11	18.30	20.88	22.44	23.53	24.13	21.52	19.01	15.60	13.14
Operative Temperature (°C)	12.04	13.88	15.89	18.05	20.50	21.96	23.03	23.67	21.18	18.73	15.40	12.95
Outside Dry-Bulb Temperature (°C)	4.50	5.82	8.33	10.85	12.97	15.59	17.66	18.80	15.09	11.43	8.07	5.42
Relative Humidity (%)	54.41	52.07	49.87	50.54	49.43	47.81	48.89	51.60	56.78	54.30	58.25	56.40

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output — 1 Jan - 31 Dec, Daily — Student

10th AUGUST 27.1 OPERATIVE TEMPERATURE



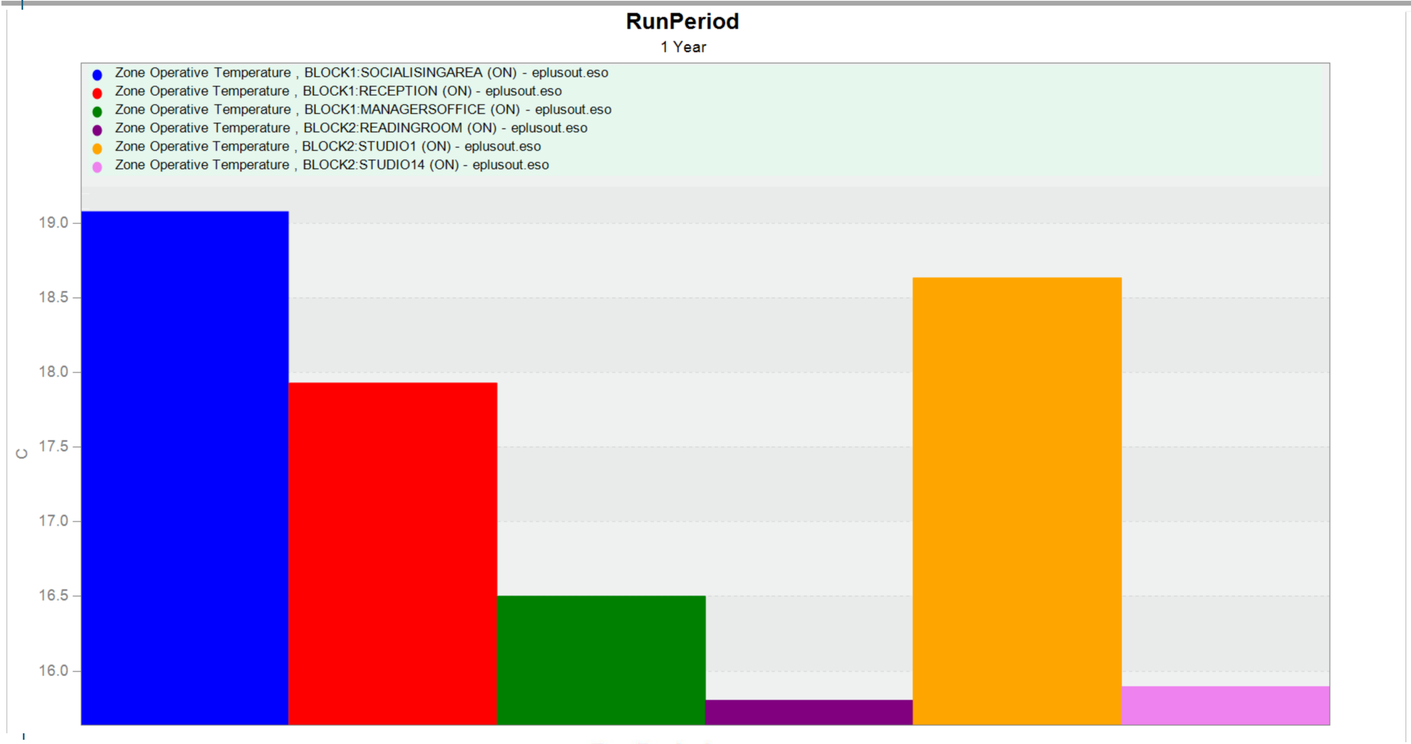
Time Not Comfortable Based on Simple ASHRAE 55-2004

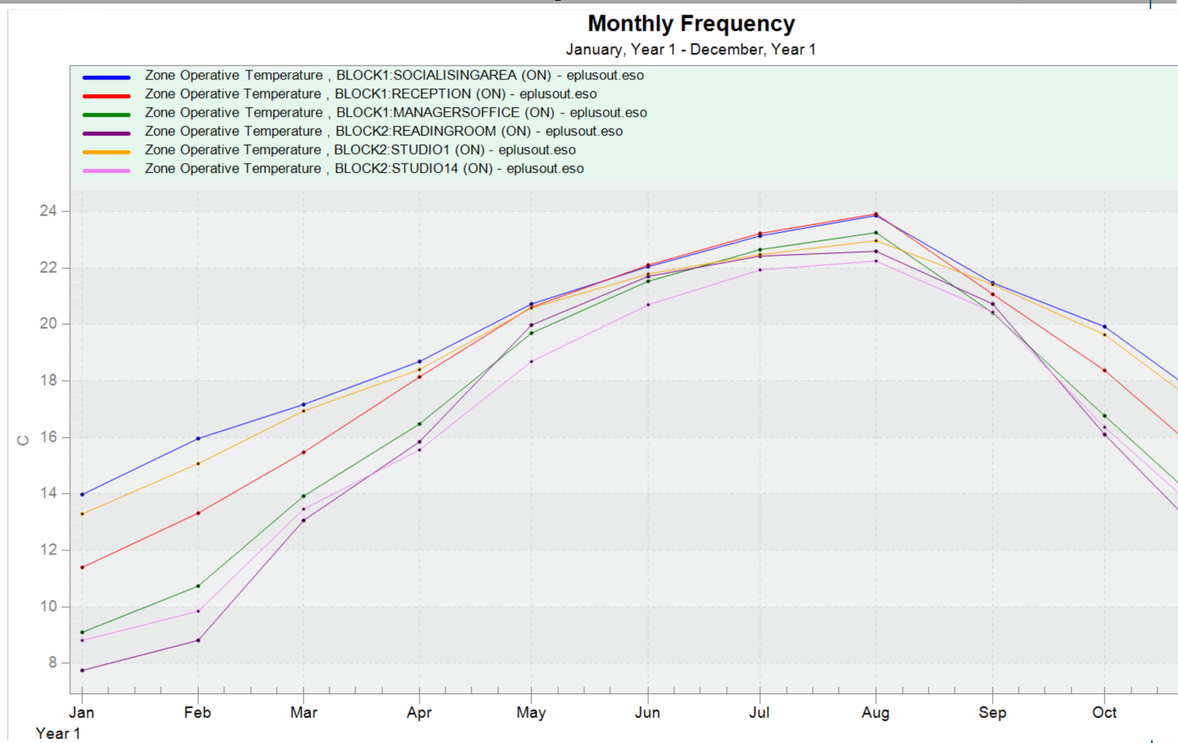
	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	5899.00	8654.50	5849.00
BLOCK2:STUDIO1	5266.50	8551.00	5130.00
BLOCK2:STUDIO14	6321.00	8689.00	6287.00
BLOCK1:RECEPTION	3040.50	4359.00	2760.50
BLOCK1:MANAGERSOFFICE	3347.50	4532.00	3204.50
BLOCK1:SOCIALISINGAREA	2557.50	4332.00	2251.00
Facility	6750.00	8744.50	6421.00

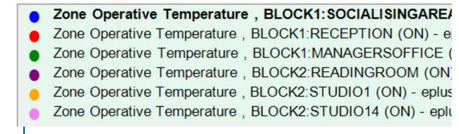
Aggregated over the RunPeriods for Weather

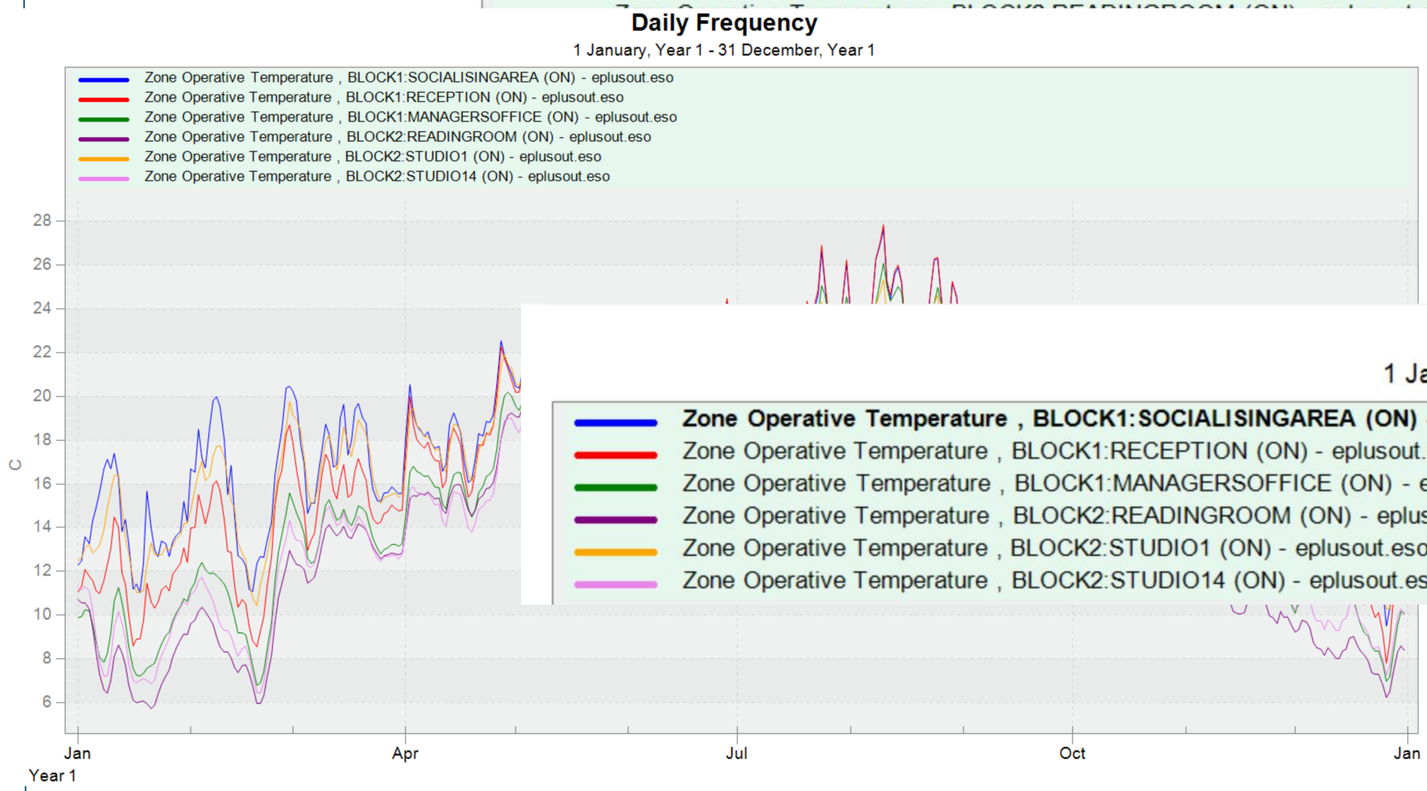
Sustainable Design of Students' Accommodation Block

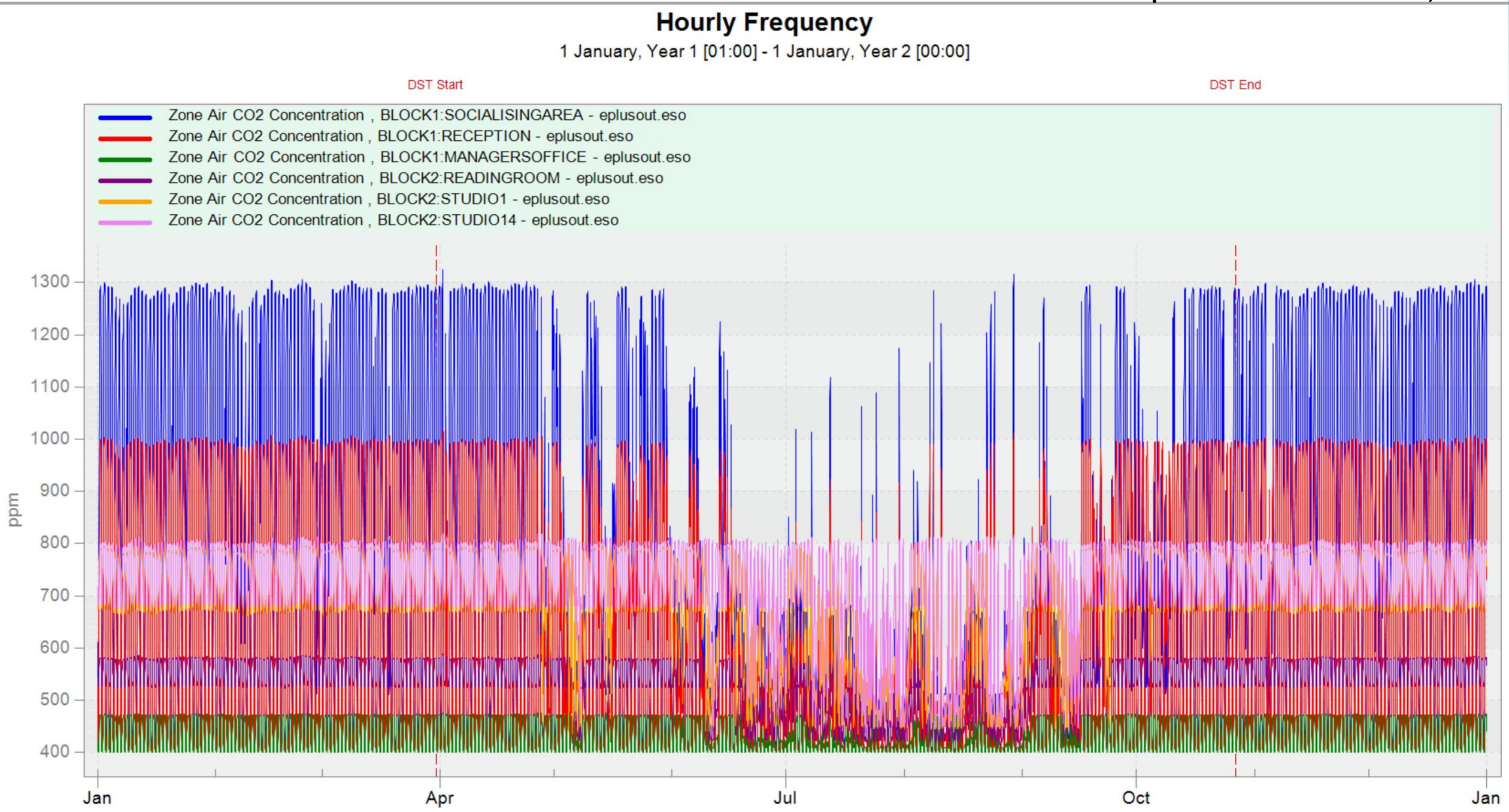
Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)







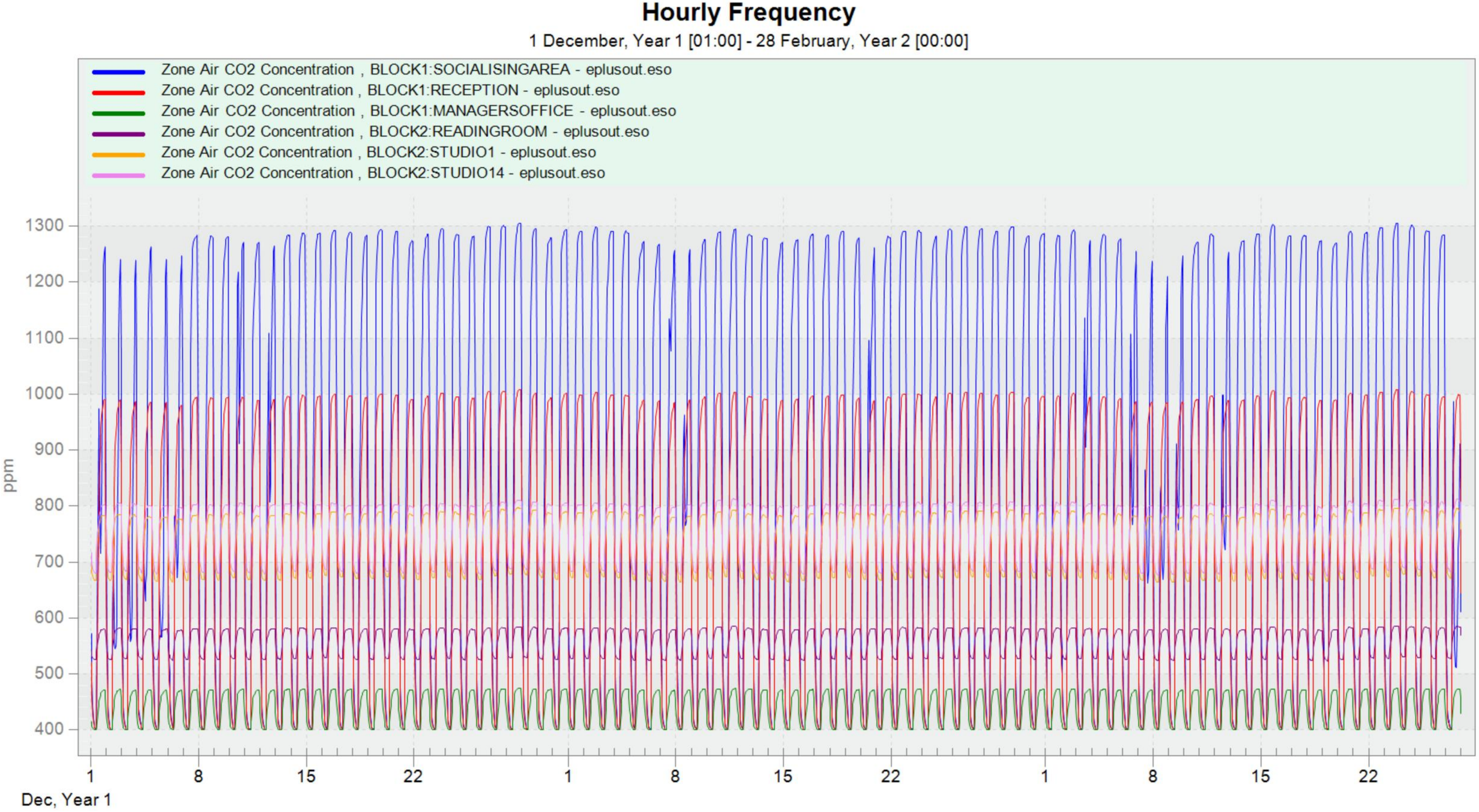




Hourly Frequency
1 January, Year 1 [01:00] - 1 January, Year 2 [00:00]
DST Start
DST End

Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso	1227.886277 ppm
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso	953.604738 ppm
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso	466.755323 ppm
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso	576.391361 ppm
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso	782.730162 ppm
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso	795.216436 ppm

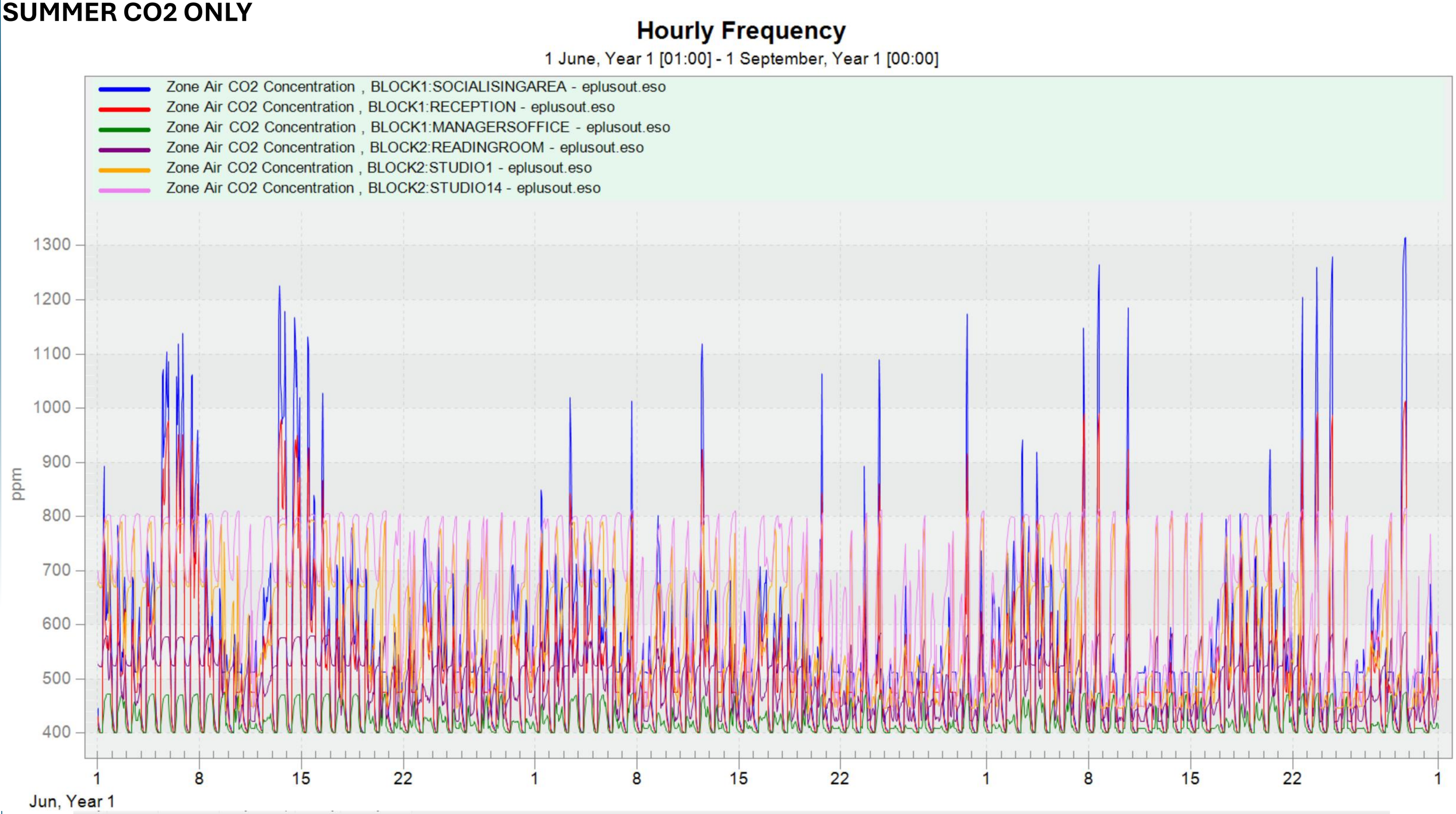
WINTER CO2 ONLY



Hourly Frequency
1 December, Year 1 [01:00] - 28 February, Year 2 [00:00]

Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso	1163.302904 ppm
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso	911.290883 ppm
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso	461.525577 ppm
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso	574.065303 ppm
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso	775.142019 ppm
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso	788.109684 ppm

SUMMER CO2 ONLY



Hourly Frequency
1 June, Year 1 [01:00] - 1 September, Year 1 [00:00]

Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso	1081.421462 ppm
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso	854.853378 ppm
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso	454.634479 ppm
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso	568.797422 ppm
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso	768.063359 ppm
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso	778.328898 ppm

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

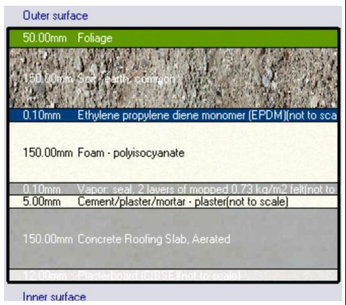
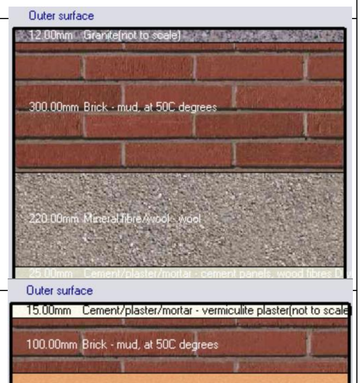
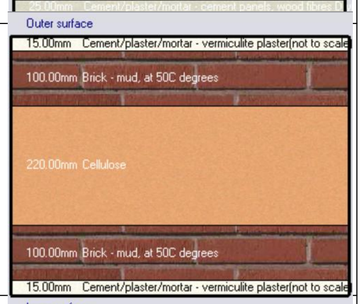
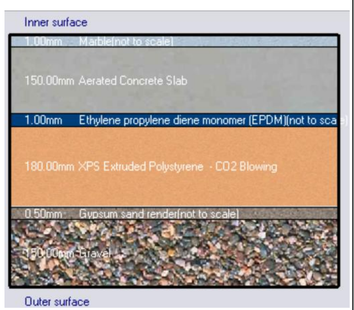
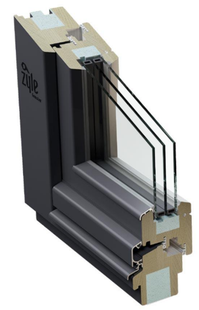
ITERATION 5

PASSIVHAUS

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

ITERATION 2 – MATERIALS

ELEMENT(mm)	CROSS-SECTION
Roof (532mm) 50mm vegetation 120mm soil Waterproofing 150mm polysiocyanate insulation Vapour seal 50mm screed 150mm concrete 12mm plasterboard **U-value(0.148)**	
External wall (550mm) glass-fiber-reinforced concrete panels 300mm brick mud 220mm mineral fibre wool 12mm white granite **U-value(0.15)**	
Internal wall (450mm) 15mm plaster 100mm brick-mud 220mm cellulose 100mm brick-mud 15mmplaster **U-value(1.64)**	
Ground floor slab (540mm) 10mm marble 150mm aerated concrete 1.2mm damp proof 180mm extruded polystrene 50mm blinding 150mm hardcore **U-value(0.149)**	
Window:Triple LowE clear 3mm/13mmAr **Frame:wood;ext.aluminum** G-value: 0.47 VT: 0.66 **U-value(0.78)**	

ITERATION 2 - OVERHEATING CRITERIA – CIBSE TM 59

Naturally ventilated

Criteria for predominantly naturally ventilated homes

Block	Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
BLOCK1	MANAGERSOFFICE	0.00	N/A	Pass
BLOCK1	RECEPTION	0.99	N/A	Pass
BLOCK1	SOCIALISINGAREA	0.66	N/A	Pass
BLOCK2	READINGROOM	0.00	0.00	Pass
BLOCK2	STUDIO1	0.00	0.00	Pass
BLOCK2	STUDIO14	0.00	0.00	Pass

ITERATION 2 - SPATIAL DAYLIGHT AUTONOMY AND ANNUAL SUNLIGHT EXPOSURE

Eligible zones for daylighting SDA AND ASE IN WINTER

Block	Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
Block 2	READING ROOM	14.0	7.16	51.08	14.02	100.0
Block 2	STUDIO 1	10.2	2.36	23.16	10.18	100.0
Block 2	STUDIO 14	8.7	2.15	24.69	8.70	100.0
Block 1	RECEPTION	103.2	78.24	75.83	103.18	100.0
Block 1	MANAGERS OFFICE	4.7	2.05	43.90	4.68	100.0
Block 1	SOCIALISING AREA	85.6	77.50	90.56	42.23	49.3
Total		**226.3**	**169.46**	**74.87**	**182.99**	**80.8**

ALL SOUTH WINDOWS SHADED – 1.5M OVERHANG

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.78m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	1.7m x 0.9m	1	North
SOCIALISING AREA	1.9mx2.1m	10	South

CALCULATIONS FOR ALL FLOORS

Report: **Annual Building Utility Performance Summary**

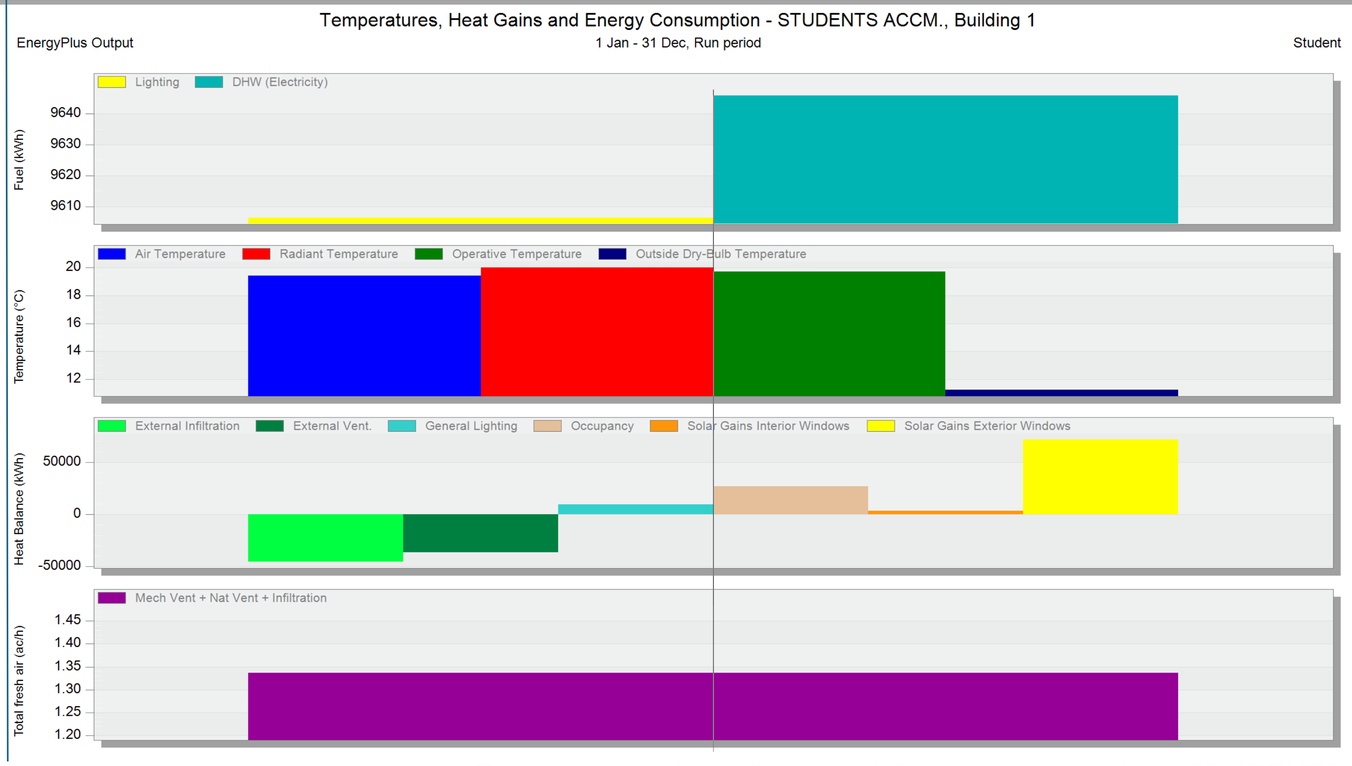
For: **Entire Facility**

Timestamp: **2025-04-06 09:49:13**

Values gathered over 8760.00 hours

Site and Source Energy

	Total Energy [kWh]	Energy Per Total Building Area [kWh/m2]	Energy Per Conditioned Building Area [kWh/m2]
Total Site Energy	17805.34	18.64	
Net Site Energy	17805.34	18.64	
Total Source Energy	60049.02	62.86	
Net Source Energy	60049.02	62.86	

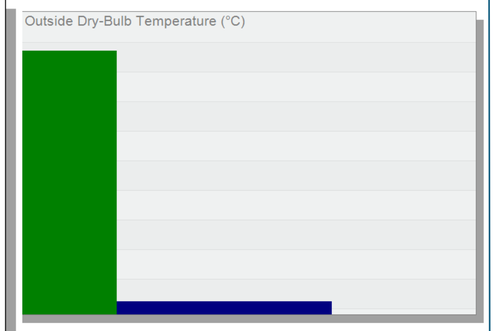


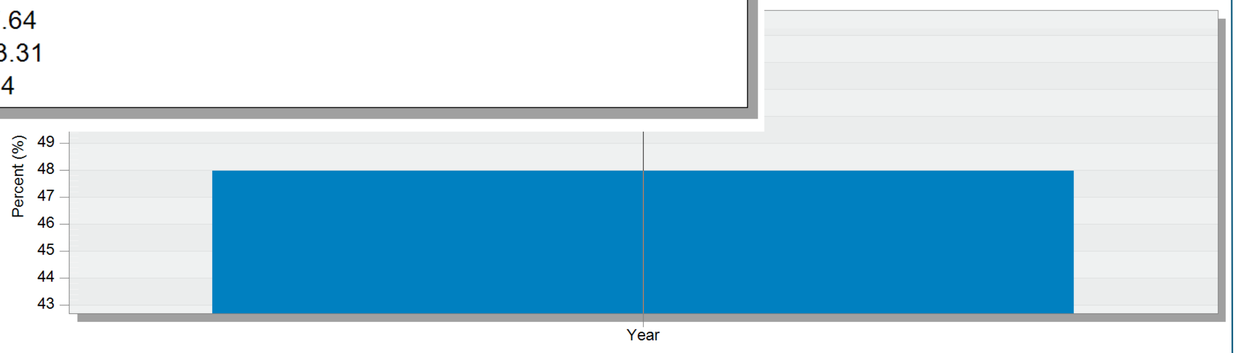
Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., Building 1
EnergyPlus Output
1 Jan - 31 Dec, Run period
Student

Year

Student

Lighting (kWh)	9606.31
DHW (Electricity) (kWh)	9645.92
Air Temperature (°C)	19.44
Radiant Temperature (°C)	20.00
Operative Temperature (°C)	19.72
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-45413.69
External Vent. (kWh)	-36242.67
General Lighting (kWh)	9606.31
Occupancy (kWh)	26731.12
Solar Gains Interior Windows (kWh)	3417.64
Solar Gains Exterior Windows (kWh)	71603.31
Mech Vent + Nat Vent + Infiltration (ac/h)	1.34





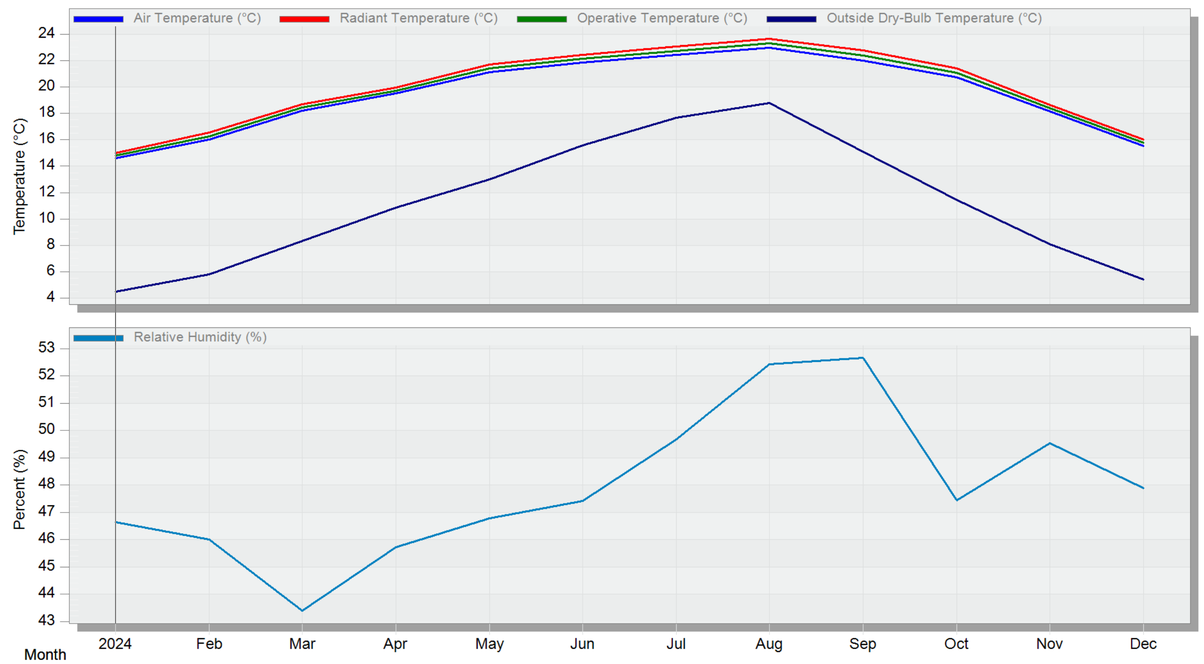
Air Temperature (°C)	19.44
Radiant Temperature (°C)	20.00
Operative Temperature (°C)	19.72
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	47.98

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Comfort - STUDENTS ACCM., Building 1

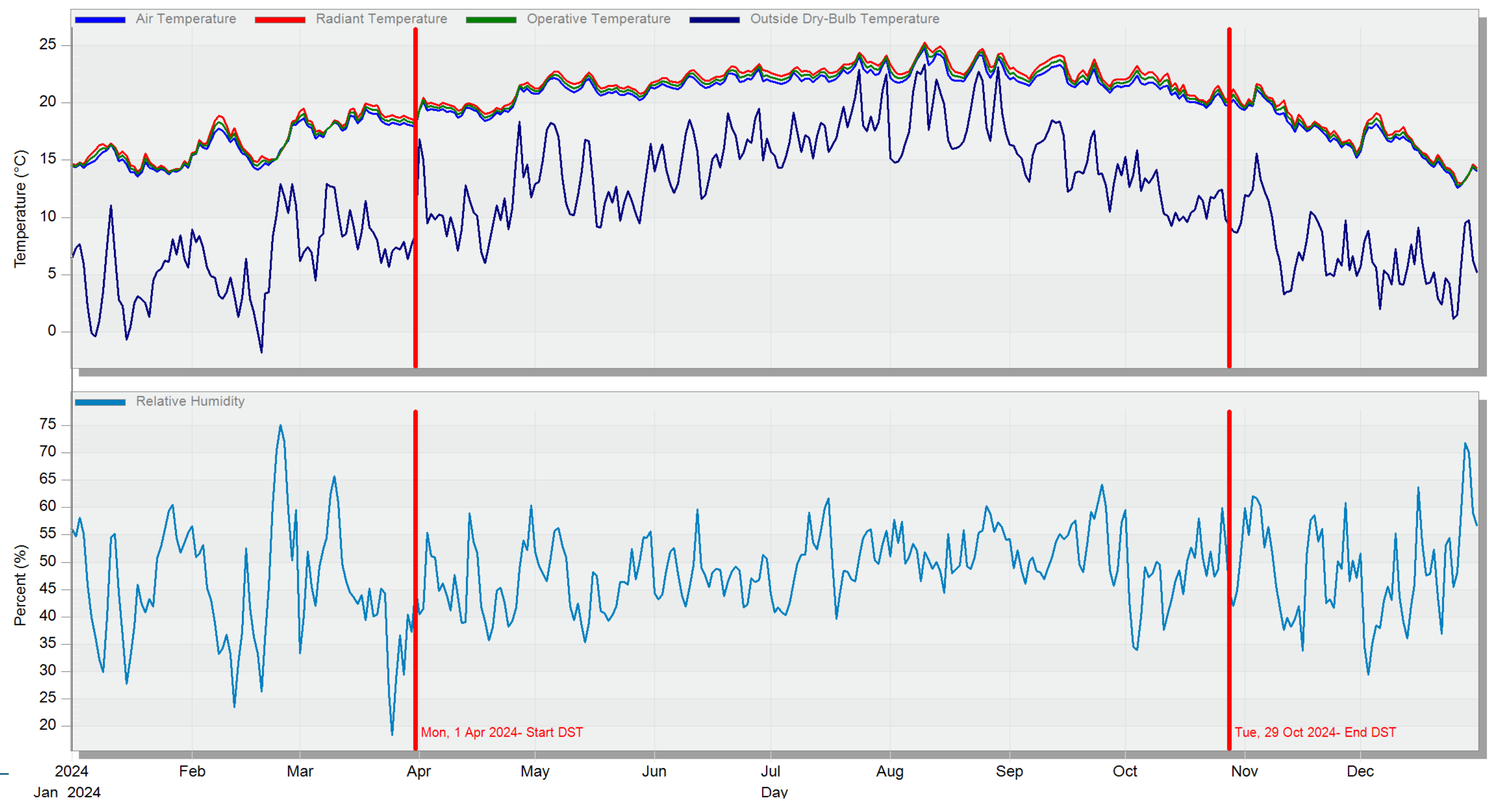
EnergyPlus Output — 1 Jan - 31 Dec, Monthly — Student



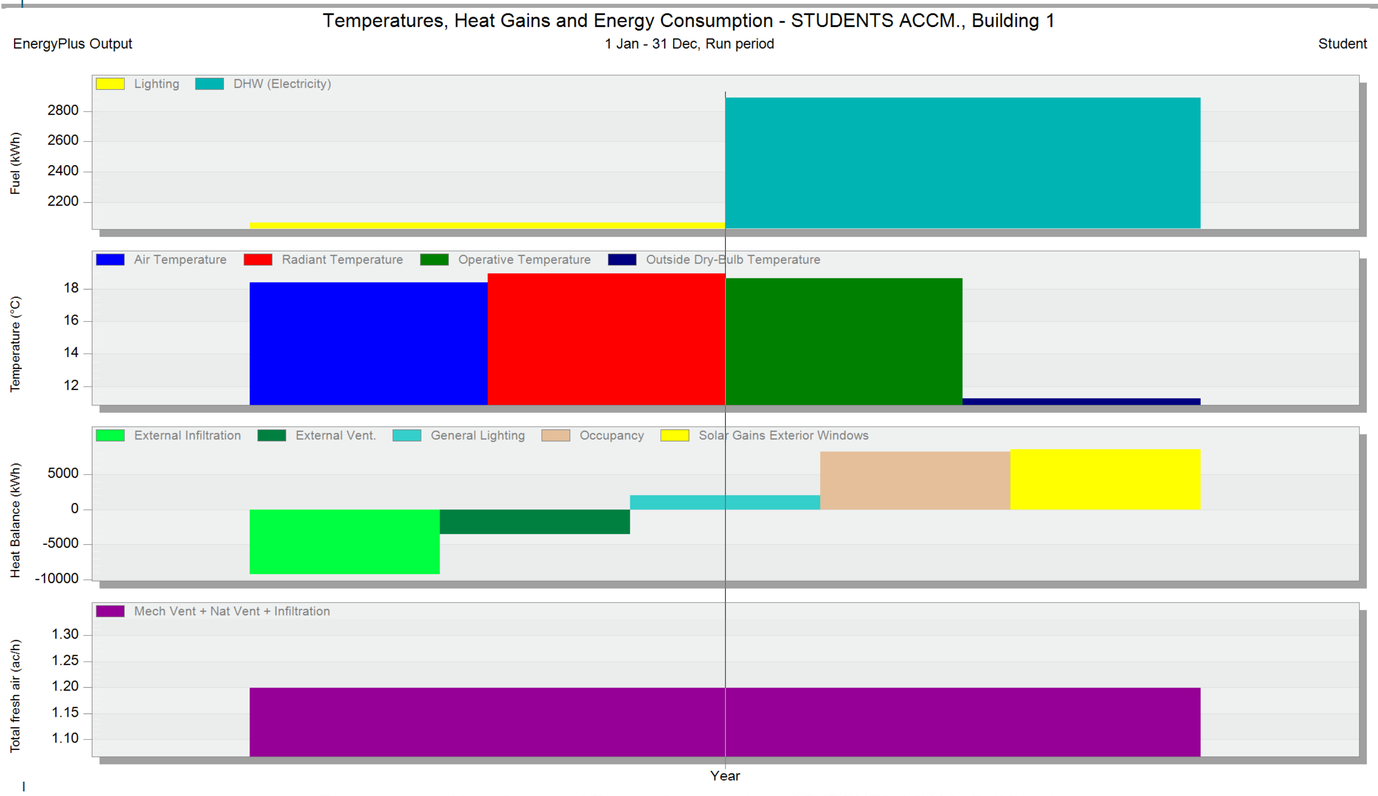
	2024	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Air Temperature (°C)	14.58	16.03	18.18	19.49	21.13	21.83	22.41	22.95	22.02	20.73	18.15	15.54
Radiant Temperature (°C)	15.00	16.53	18.70	19.96	21.68	22.43	23.05	23.64	22.76	21.42	18.66	16.01
Operative Temperature (°C)	14.79	16.28	18.44	19.73	21.40	22.13	22.73	23.30	22.39	21.07	18.41	15.78
Outside Dry-Bulb Temperature (°C)	4.50	5.82	8.33	10.85	12.97	15.59	17.66	18.80	15.09	11.43	8.07	5.42
Relative Humidity (%)	46.66	46.01	43.40	45.72	46.79	47.42	49.69	52.43	52.67	47.44	49.55	47.89

Comfort - STUDENTS ACCM., Building 1 **10th AUGUST 25.2 OPERATIVE TEMPERATURE**

EnergyPlus Output — 1 Jan - 31 Dec, Daily — Student



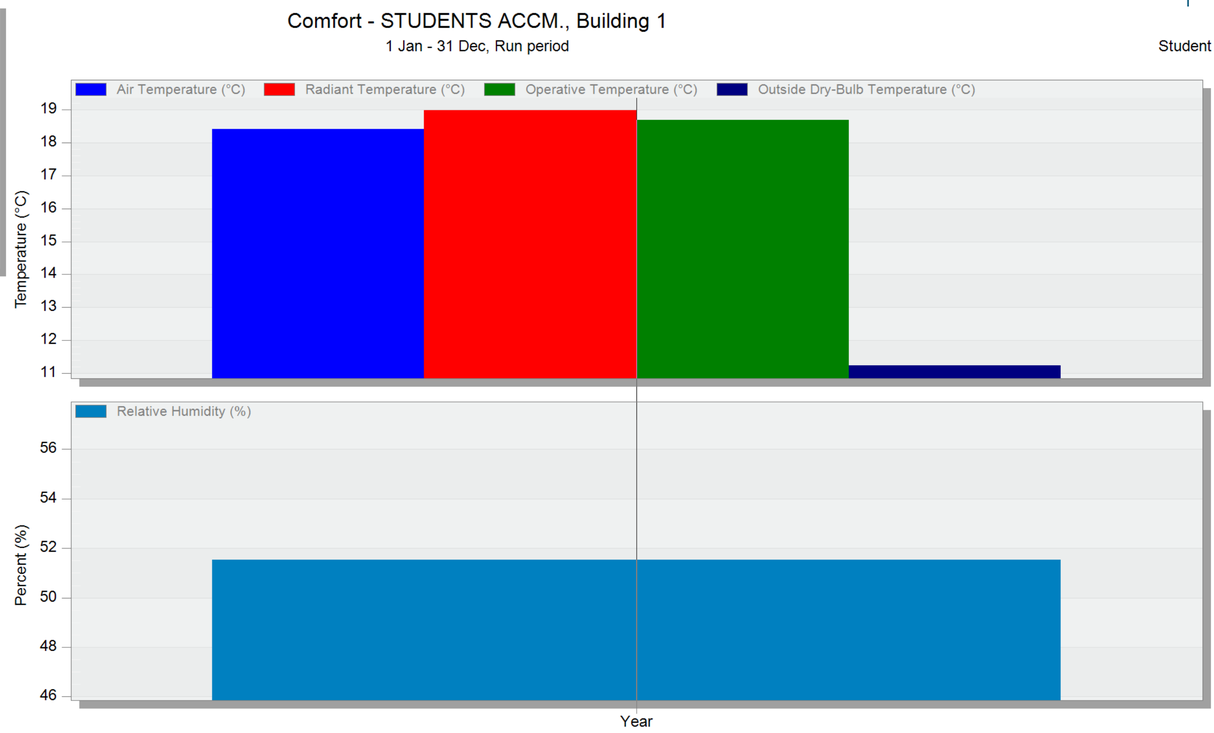
CALCULATIONS FOR ZONES ONLY



Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., Building 1
EnergyPlus Output
1 Jan - 31 Dec, Run period
Student

Year

Lighting (kWh)	2065.32
DHW (Electricity) (kWh)	2889.01
Air Temperature (°C)	18.41
Radiant Temperature (°C)	18.98
Operative Temperature (°C)	18.70
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-9236.00
External Vent. (kWh)	-3521.47
General Lighting (kWh)	2065.32
Occupancy (kWh)	8252.12
Solar Gains Exterior Windows (kWh)	8615.32
Mech Vent + Nat Vent + Infiltration (ac/h)	1.20



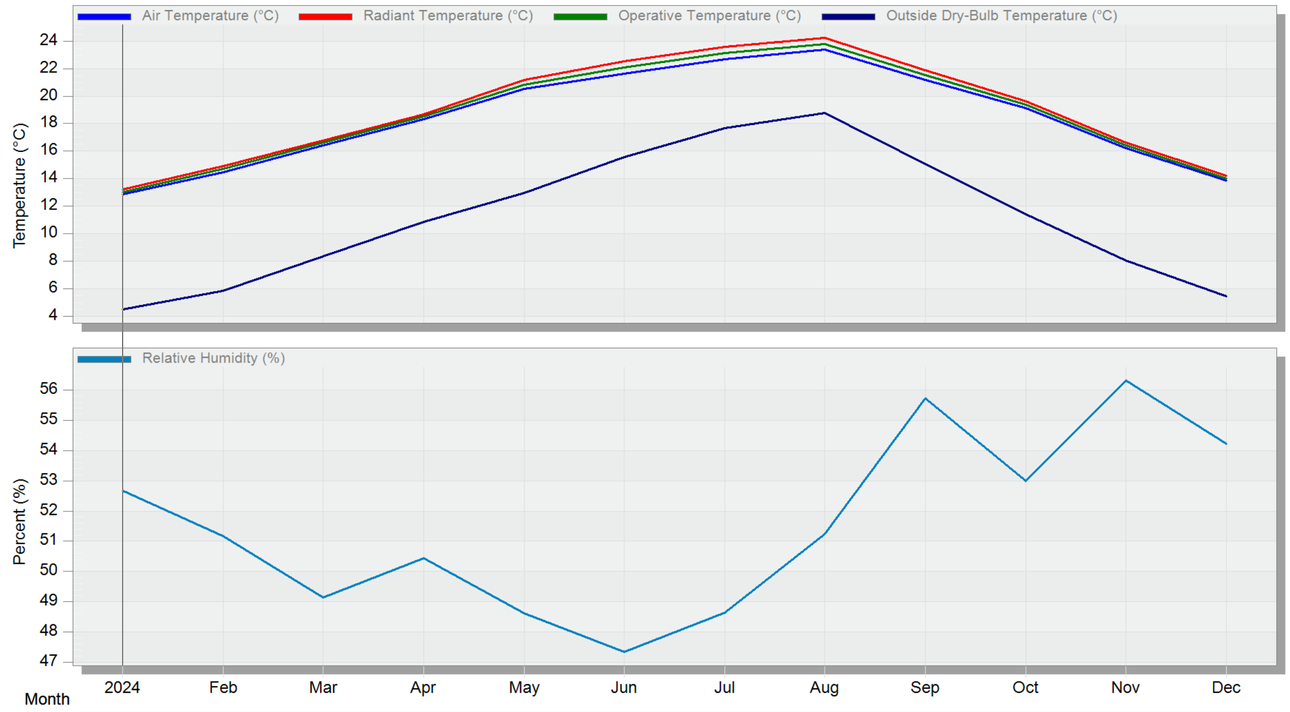
Air Temperature (°C)	18.41
Radiant Temperature (°C)	18.98
Operative Temperature (°C)	18.70
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	51.54

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Monthly | Student

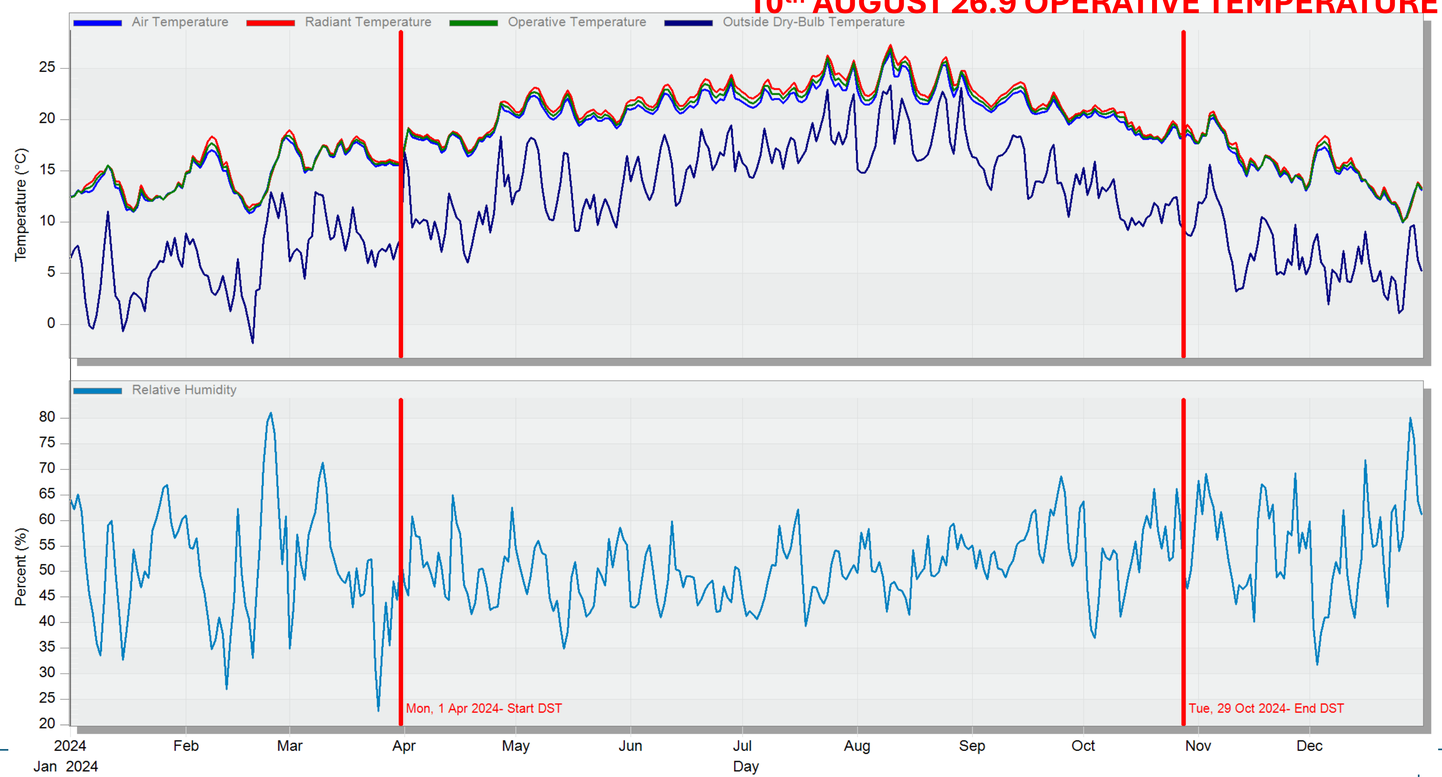


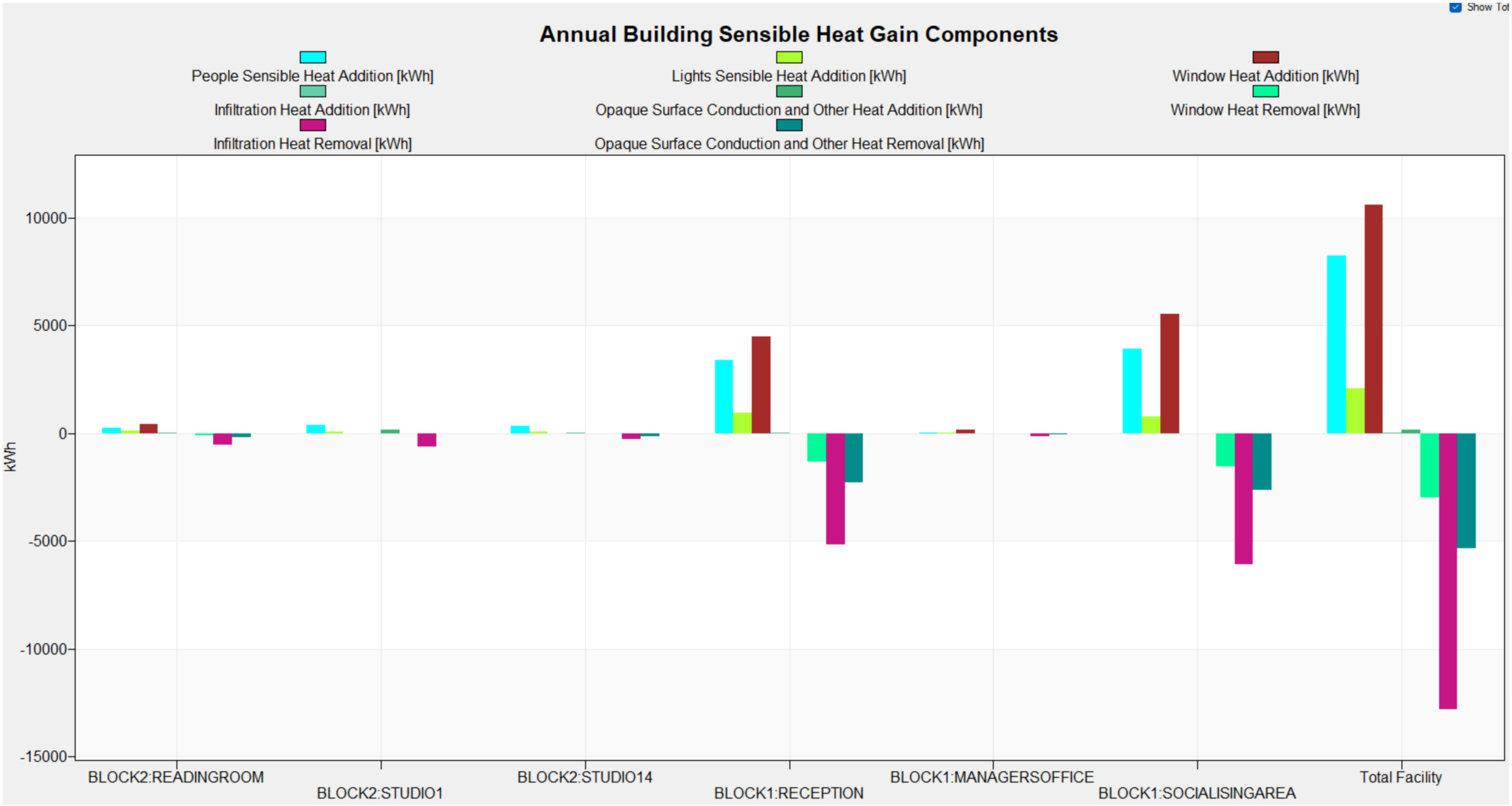
Month	2024	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Air Temperature (°C)	12.87	14.48	16.43	18.31	20.52	21.66	22.67	23.37	21.20	19.12	16.23	13.84
Radiant Temperature (°C)	13.19	14.92	16.79	18.69	21.19	22.53	23.59	24.23	21.86	19.64	16.61	14.22
Operative Temperature (°C)	13.03	14.70	16.61	18.50	20.85	22.09	23.13	23.80	21.53	19.38	16.42	14.03
Outside Dry-Bulb Temperature (°C)	4.50	5.82	8.33	10.85	12.97	15.59	17.66	18.80	15.09	11.43	8.07	5.42
Relative Humidity (%)	52.68	51.17	49.13	50.44	48.61	47.34	48.64	51.24	55.72	52.99	56.32	54.22

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Daily | Student

10th AUGUST 26.9 OPERATIVE TEMPERATURE




Show Tot
Annual Building Sensible Heat Gain Components
People Sensible Heat Addition [kWh]
Lights Sensible Heat Addition [kWh]
Window Heat Addition [kWh]
Infiltration Heat Addition [kWh]
Opaque Surface Conduction and Other Heat Addition [kWh]
Window Heat Removal [kWh]
Infiltration Heat Removal [kWh]
Opaque Surface Conduction and Other Heat Removal [kWh]
10000
5000
0
-5000
-10000
-15000
kWh
BLOCK2:READINGROOM
BLOCK2:STUDIO1
BLOCK2:STUDIO14
BLOCK1:RECEPTION
BLOCK1:MANAGERSOFFICE
BLOCK1:SOCIALISINGAREA
Total Facility

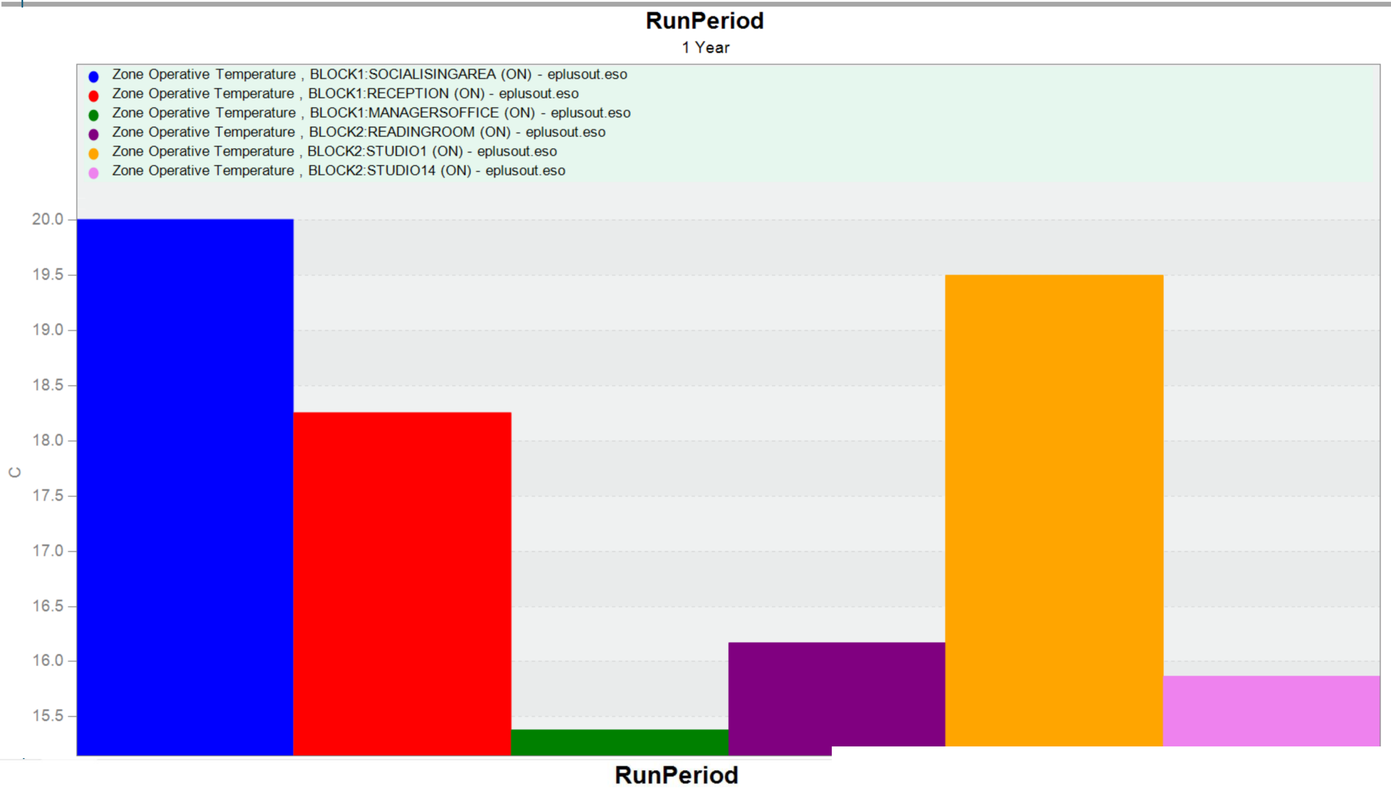
Time Not Comfortable Based on Simple ASHRAE 55-2004

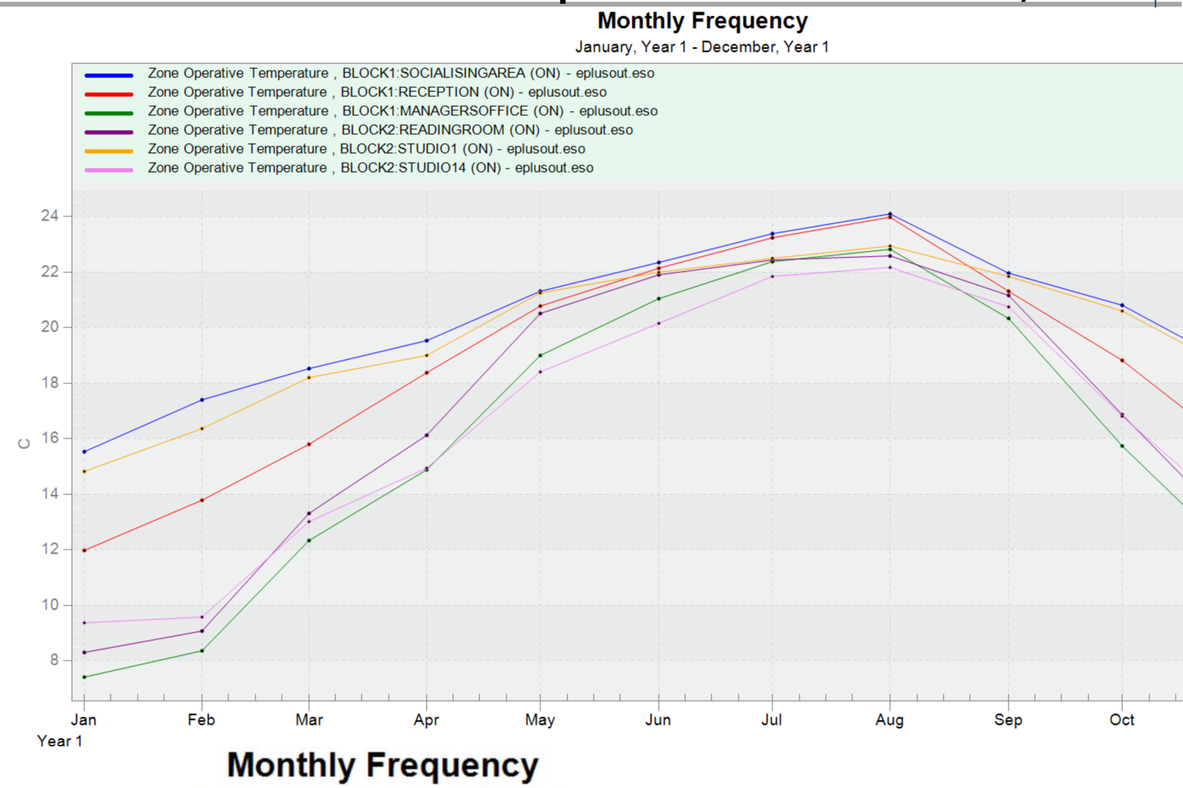
	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	5638.50	8676.00	5607.50
BLOCK2:STUDIO1	4600.00	8615.00	4519.50
BLOCK2:STUDIO14	6342.50	8749.00	6339.50
BLOCK1:RECEPTION	2968.50	4379.50	2703.00
BLOCK1:MANAGERSOFFICE	3390.00	4636.50	3335.50
BLOCK1:SOCIALISINGAREA	2160.50	4327.50	1857.00
Facility	6792.50	8759.50	6503.50

Aggregated over the RunPeriods for Weather

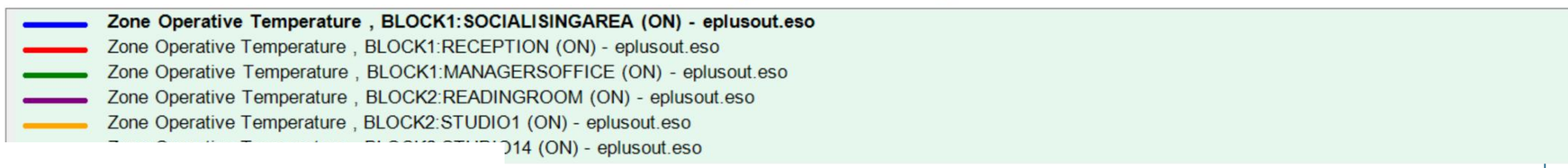
Sustainable Design of Students' Accommodation Block

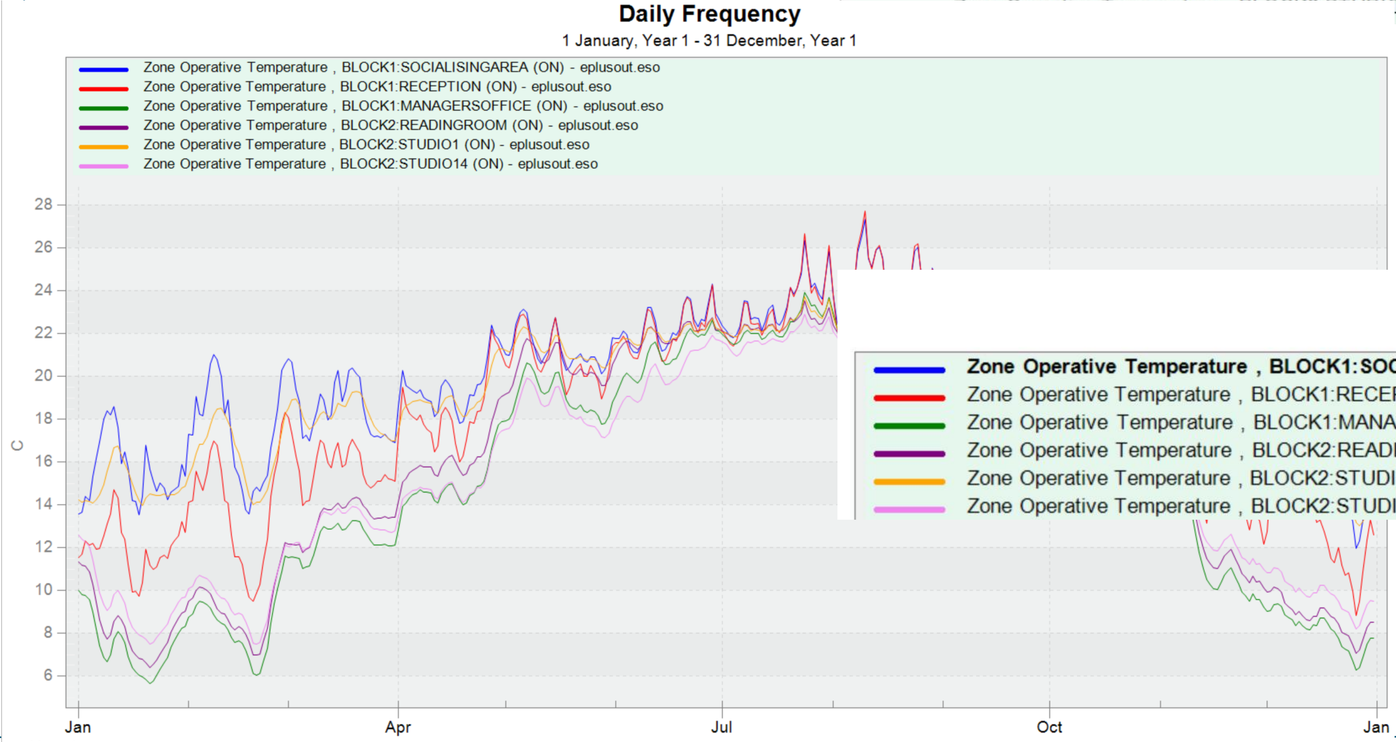
Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

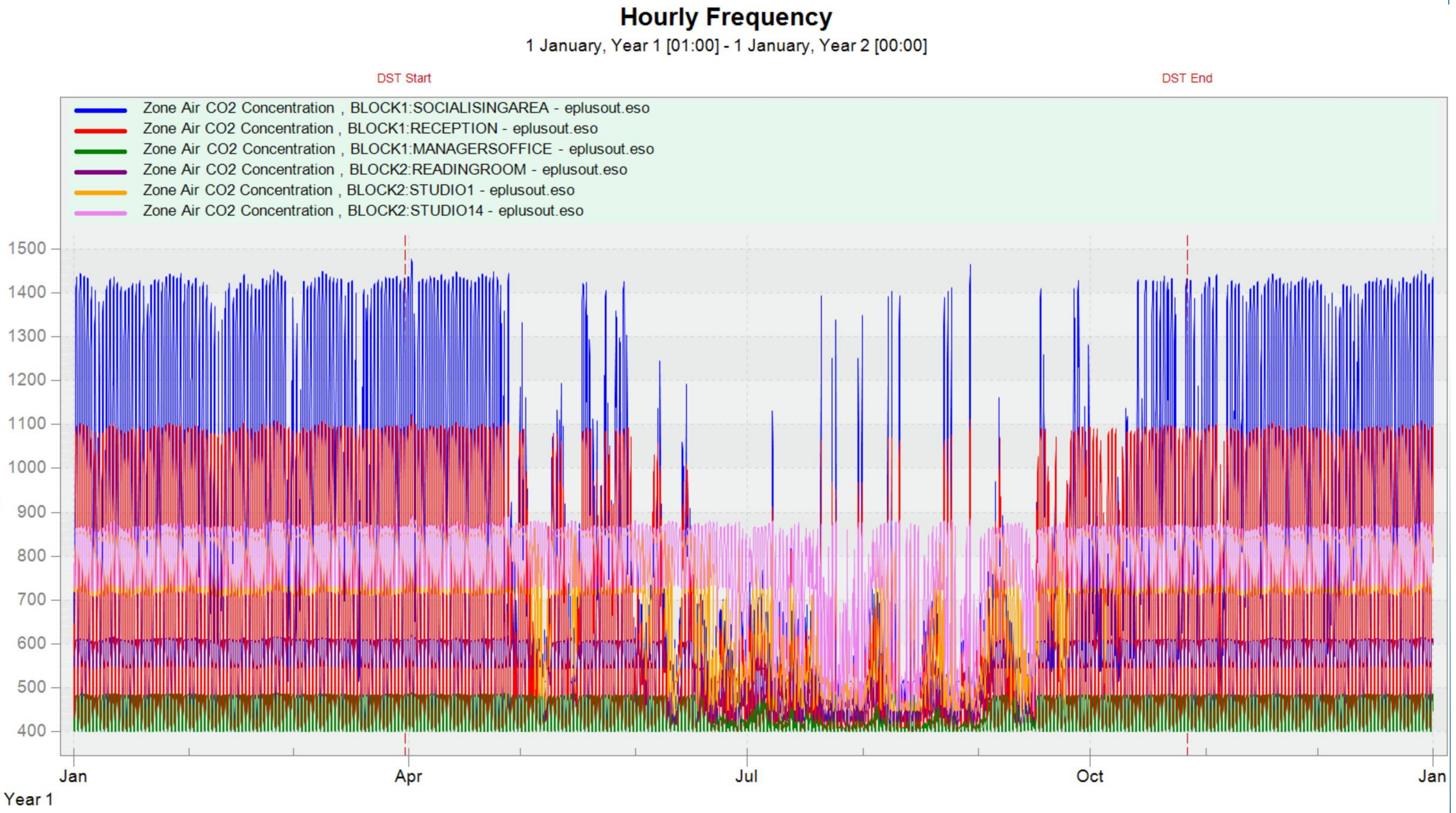




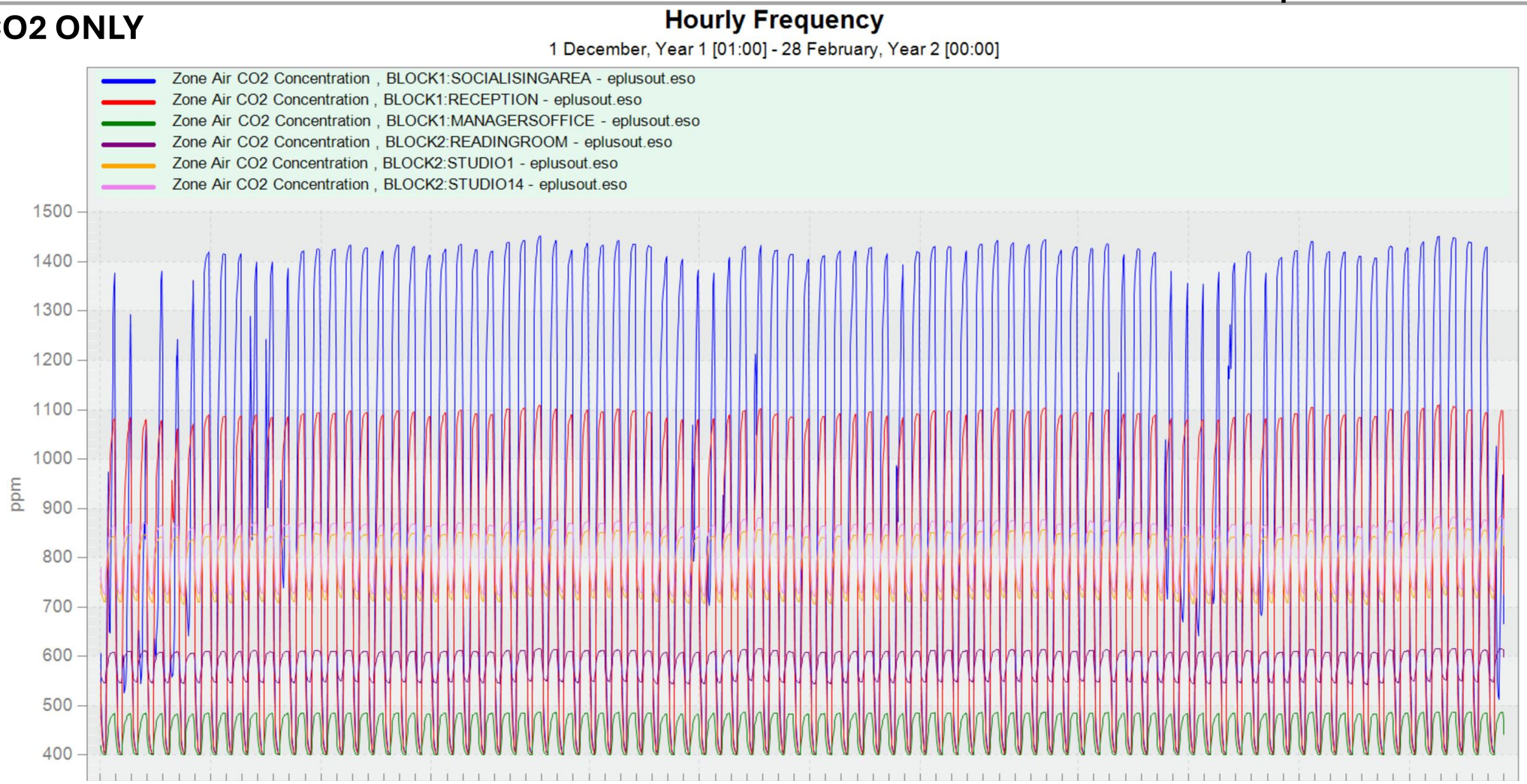
Monthly Frequency
January, Year 1 - December, Year 1






Hourly Frequency
1 January, Year 1 [01:00] - 1 January, Year 2 [00:00]
DST Start
DST End
Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso
1500
1400
1300
1200
1100
1000
900
800
700
600
500
400
ppm
Jan
Apr
Jul
Oct
Jan
Year 1

WINTER CO2 ONLY



Time Not Meeting the Adaptive Comfort Models during Occupied Hours

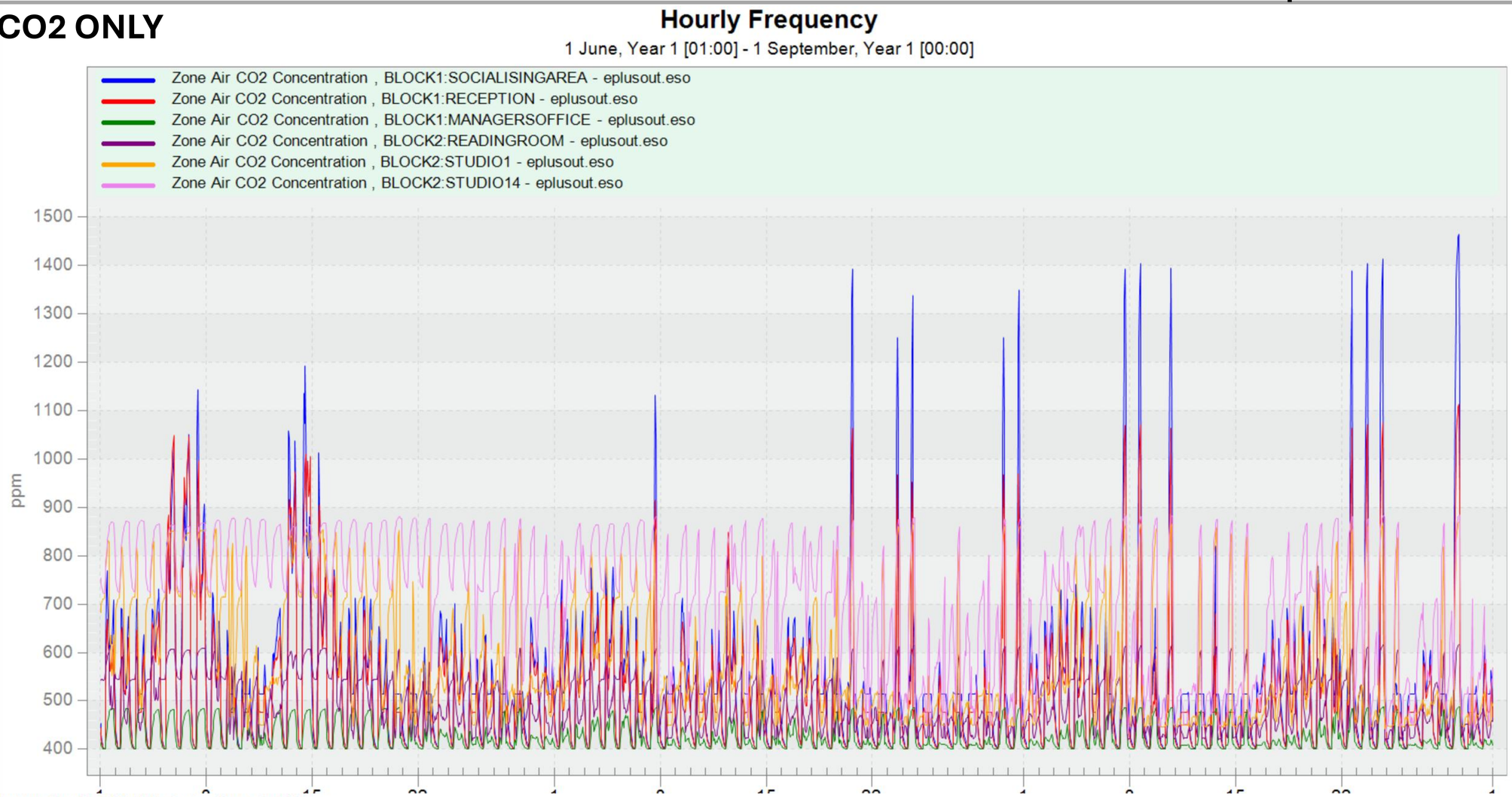
	ASHRAE55 90% Acceptability Limits [Hours]	ASHRAE55 80% Acceptability Limits [Hours]	CEN15251 Category I Acceptability Limits [Hours]	CEN15251 Category II Acceptability Limits [Hours]	CEN15251 Category III Acceptability Limits [Hours]
PEOPLE BLOCK2:READINGROOM	0.00	0.00	96.00	96.00	96.00
PEOPLE BLOCK2:STUDIO1	0.00	0.00	96.00	96.00	96.00
PEOPLE BLOCK2:STUDIO14	0.00	0.00	96.00	96.00	96.00
PEOPLE BLOCK1:RECEPTION	0.00	0.00	52.00	52.00	49.00
PEOPLE BLOCK1:MANAGERSOFFICE	0.00	0.00	52.00	52.00	52.00
PEOPLE BLOCK1:SOCIALISINGAREA	0.00	0.00	45.50	38.50	35.50

Time Not Comfortable Based on Simple ASHRAE 55-2004

	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	2136.00	2136.00	2136.00
BLOCK2:STUDIO1	2136.00	2136.00	2136.00
BLOCK2:STUDIO14	2136.00	2136.00	2136.00
BLOCK1:RECEPTION	1139.00	1157.00	1139.00
BLOCK1:MANAGERSOFFICE	1157.00	1157.00	1157.00
BLOCK1:SOCIALISINGAREA	950.00	1152.50	950.00
Facility	2136.00	2136.00	2136.00

Aggregated over the RunPeriods for Weather

SUMMER CO2 ONLY



Time Not Meeting the Adaptive Comfort Models during Occupied Hours

	ASHRAE55 90% Acceptability Limits [Hours]	ASHRAE55 80% Acceptability Limits [Hours]	CEN15251 Category I Acceptability Limits [Hours]	CEN15251 Category II Acceptability Limits [Hours]	CEN15251 Category III Acceptability Limits [Hours]
PEOPLE BLOCK2:READINGROOM	0.00	0.00	1431.00	245.00	2.50
PEOPLE BLOCK2:STUDIO1	0.00	0.00	1276.00	90.00	0.00
PEOPLE BLOCK2:STUDIO14	3.00	0.00	2102.50	1026.50	110.50
PEOPLE BLOCK1:RECEPTION	206.00	94.00	315.50	28.50	1.00
PEOPLE BLOCK1:MANAGERSOFFICE	0.00	0.00	727.00	82.50	0.00
PEOPLE BLOCK1:SOCIALISINGAREA	209.50			12.00	0.00

Time Not Comfortable Based on Simple ASHRAE 55-2004

	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	33.00	2122.50	0.00
BLOCK2:STUDIO1	94.50	2063.00	13.50
BLOCK2:STUDIO14	197.50	2197.50	194.50
BLOCK1:RECEPTION	327.00	882.50	88.00
BLOCK1:MANAGERSOFFICE	61.00	1089.00	9.00
BLOCK1:SOCIALISINGAREA	347.00	855.50	89.00
Facility	535.50	2207.50	279.00

Aggregated over the RunPeriods for Weather

Sustainable Design of Students' Accommodation Block

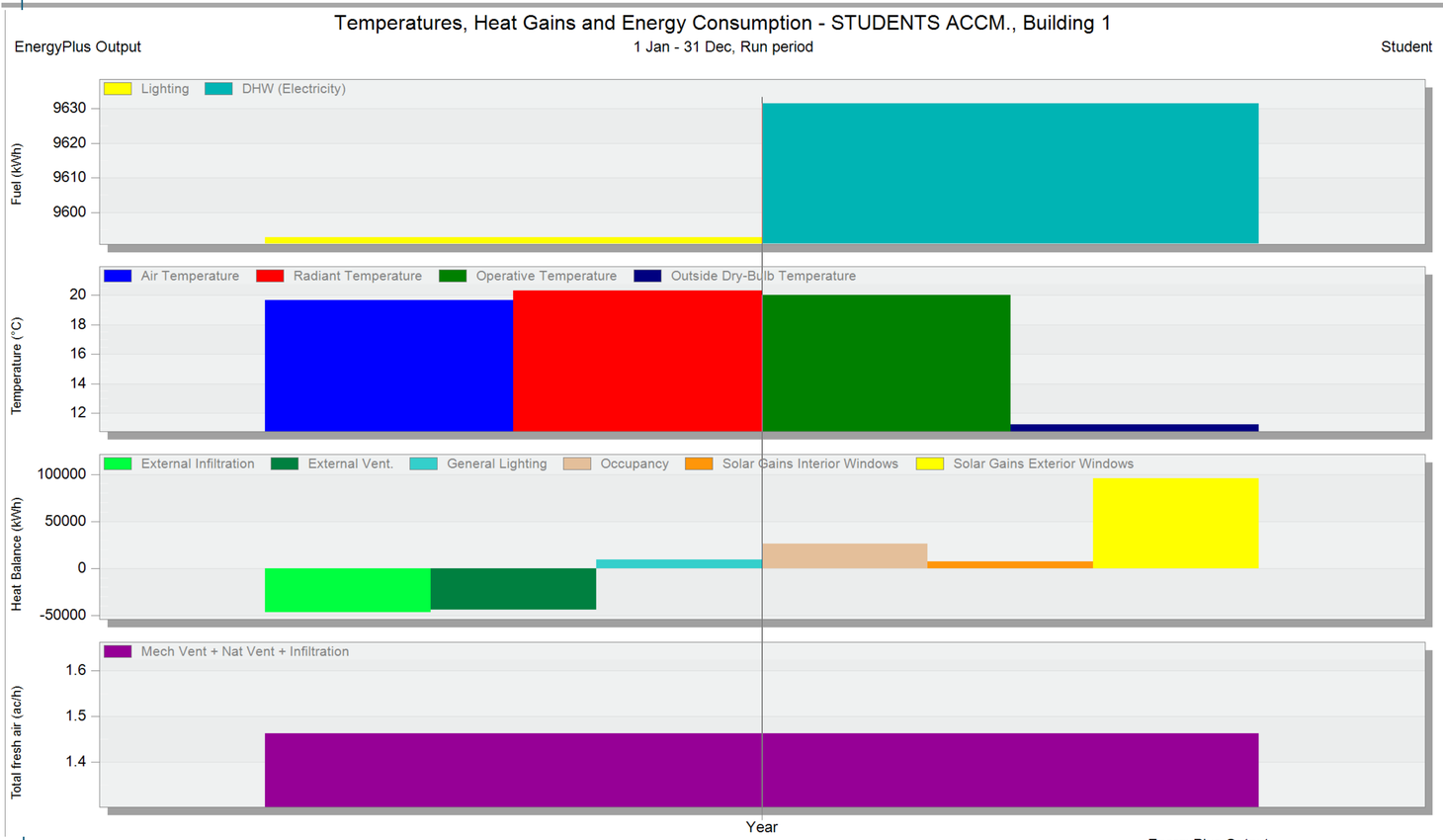
Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

PASSIVHAUS EXT

N/A

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)



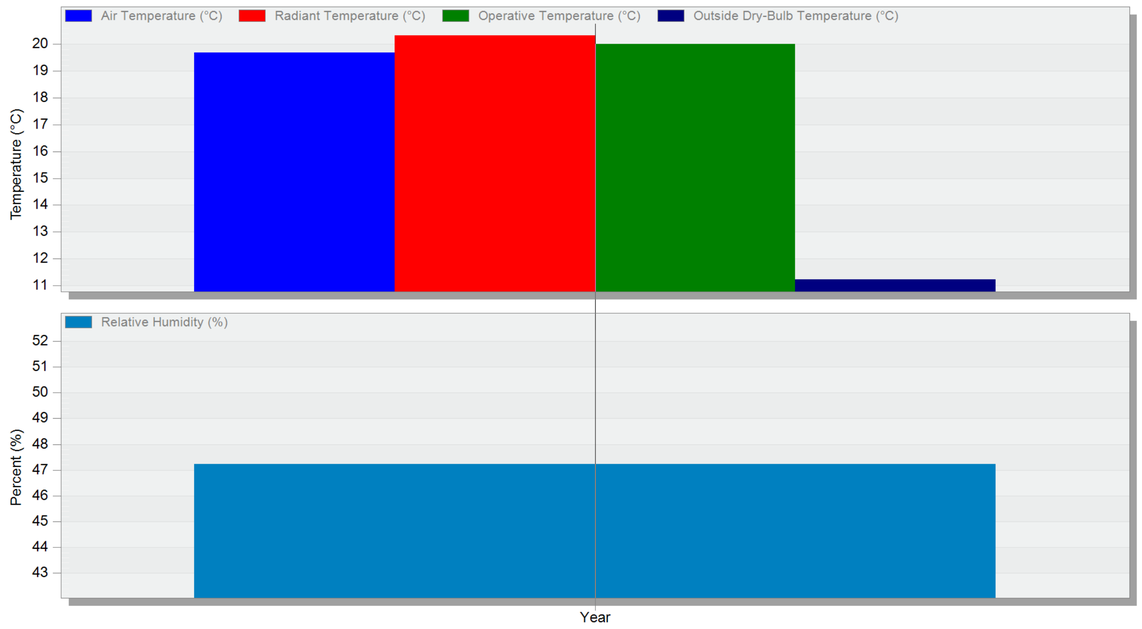
Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Run period | Student

	Year
Lighting (kWh)	9592.79
DHW (Electricity) (kWh)	9631.45
Air Temperature (°C)	19.68
Radiant Temperature (°C)	20.31
Operative Temperature (°C)	20.00
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-46663.69
External Vent. (kWh)	-44327.49
General Lighting (kWh)	9592.79
Occupancy (kWh)	26399.71
Solar Gains Interior Windows (kWh)	7108.59
Solar Gains Exterior Windows (kWh)	95602.91
Mech Vent + Nat Vent + Infiltration (ac/h)	1.46

Comfort - STUDENTS ACCM., Building 1

1 Jan - 31 Dec, Run period | Student



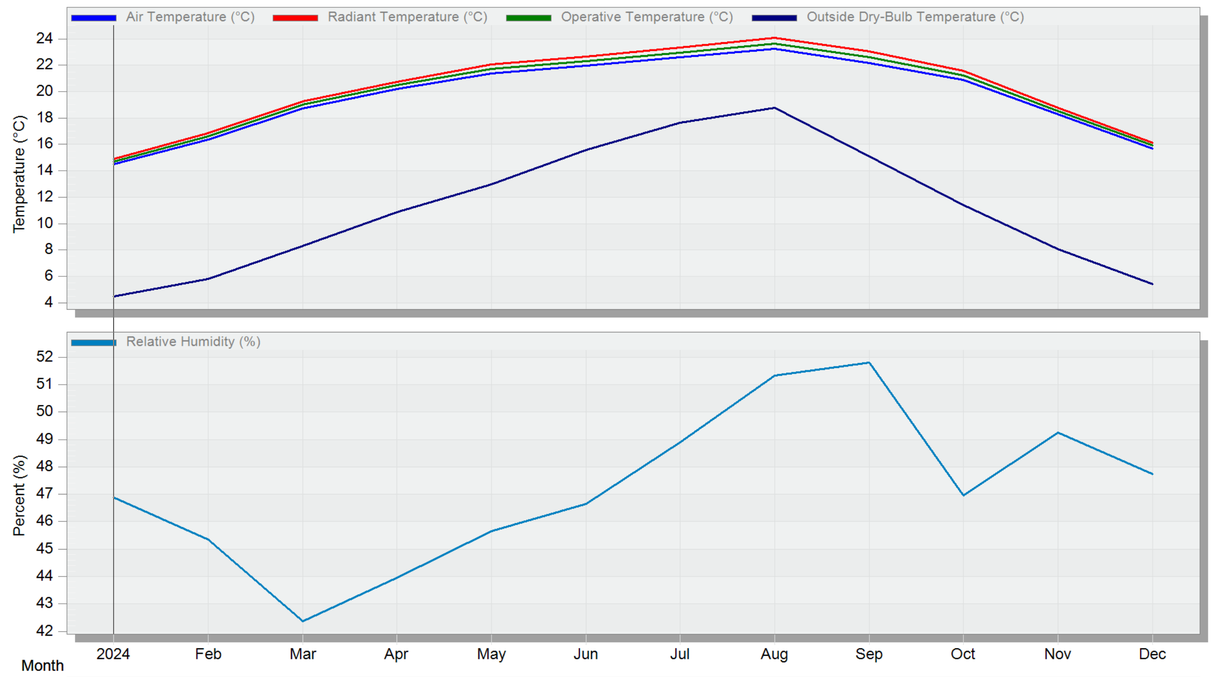
Air Temperature (°C)	19.68
Radiant Temperature (°C)	20.31
Operative Temperature (°C)	20.00
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	47.24

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output — 1 Jan - 31 Dec, Monthly — Student

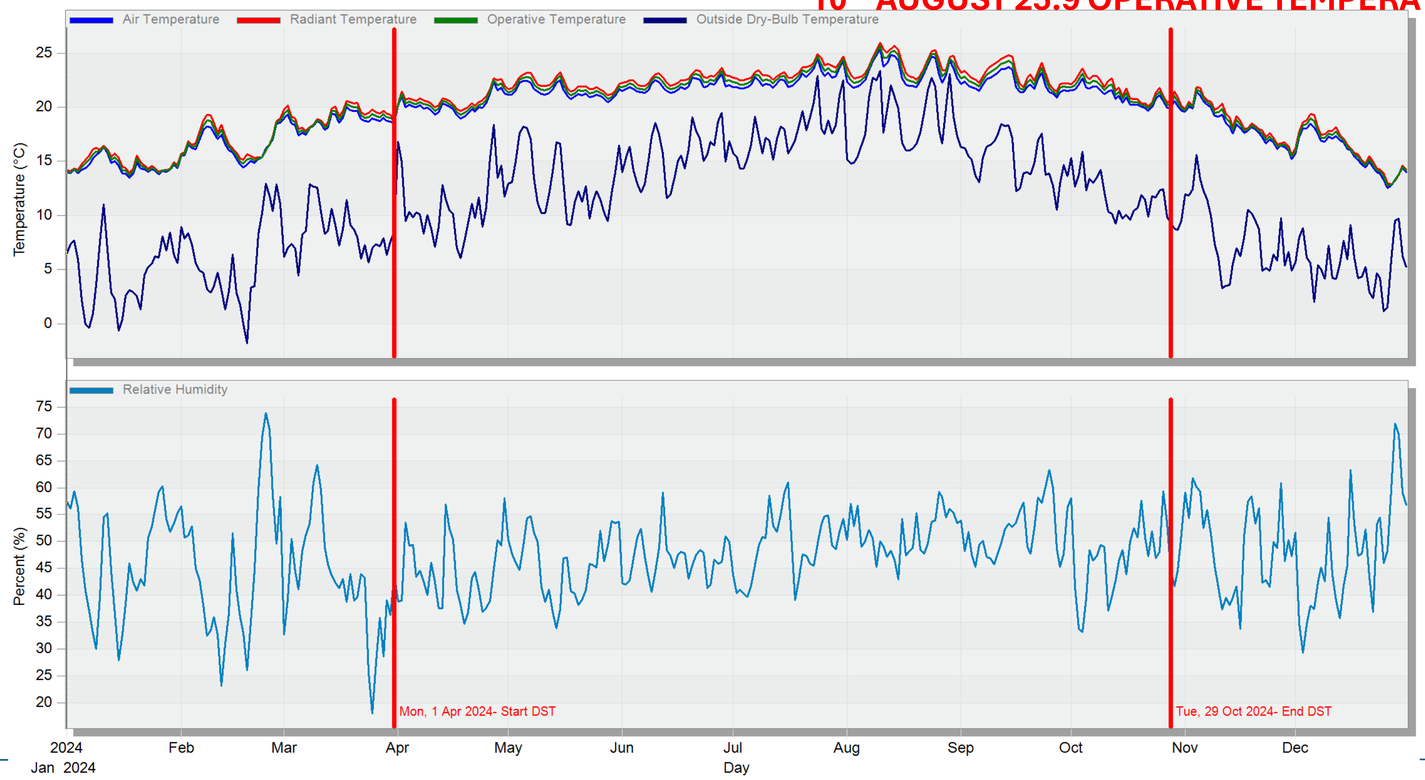


	2024	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Air Temperature (°C)	14.49	16.37	18.72	20.18	21.40	21.95	22.59	23.25	22.18	20.88	18.28	15.67
Radiant Temperature (°C)	14.89	16.87	19.29	20.76	22.07	22.65	23.34	24.06	23.06	21.60	18.80	16.13
Operative Temperature (°C)	14.69	16.62	19.01	20.47	21.74	22.30	22.97	23.65	22.62	21.24	18.54	15.90
Outside Dry-Bulb Temperature (°C)	4.50	5.82	8.33	10.85	12.97	15.59	17.66	18.80	15.09	11.43	8.07	5.42
Relative Humidity (%)	46.88	45.35	42.38	43.95	45.66	46.65	48.89	51.35	51.80	46.96	49.25	47.73

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output — 1 Jan - 31 Dec, Daily — Student

10th AUGUST 25.9 OPERATIVE TEMPERATURE



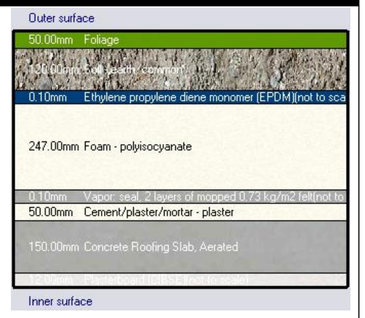
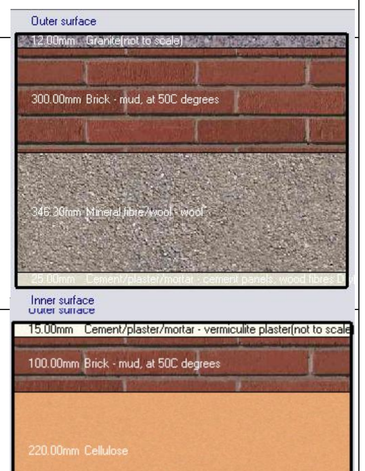
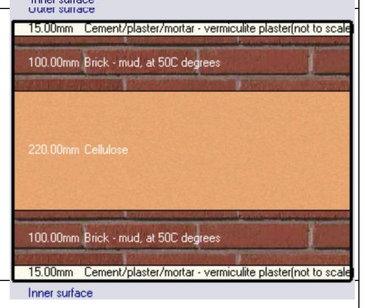
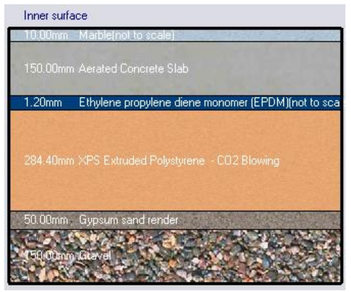
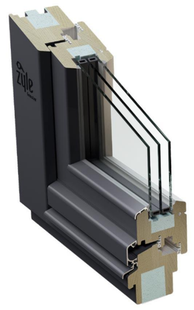
Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

ITERATION 2 – MATERIALS

ELEMENT(mm)	CROSS-SECTION
Roof (629mm) 50mm vegetation 120mm soil Waterproofing 247mm polysiocyanate insulation Vapour seal 50mm screed 150mm concrete 12mm plasterboard **U-value(0.1)**	
External wall (683mm) glass-fiber-reinforced concrete panels 300mm brick mud 346.3mm mineral fibre wool 12mm white granite **U-value(0.1)**	
Internal wall (450mm) 15mm plaster 100mm brick-mud 220mm cellulose 100mm brick-mud 15mmplaster **U-value(1.64)**	
Ground floor slab (540mm) 10mm marble 150mm aerated concrete 1.2mm damp proof 284.4mm extruded polystrene 50mm blinding 150mm hardcore **U-value(0.149)**	
Window: Triple lowE clear 3mm/13mm Argon **Frame: Wood;ext.aluminum** G-value: 0.47 VT: 0.66 **U-value(0.78)**	

ITERATION 2 - OVERHEATING CRITERIA – CIBSE TM 59

Zone	Criterion A (%)	Criterion B (hr)	Pass/Fail
MANAGERSOFFICE	0.00	N/A	Pass
RECEPTION	0.99	N/A	Pass
SOCIALISINGAREA	0.10	N/A	Pass
READINGROOM	0.00	0.00	Pass
STUDIO1	0.00	0.00	Pass
STUDIO14	0.00	0.00	Pass

ITERATION 2 - SPATIAL DAYLIGHT AUTONOMY AND ANNUAL SUNLIGHT EXPOSURE

Eligible zones for daylighting

Block	Zone	Floor area (m2)	sDA area in range (m2)	sDA % in Range%	ASE area in range (m2)	ASE % in range
Block 2	READING ROOM	14.0	7.36	52.52	14.02	100.0
Block 2	STUDIO 1	10.2	5.79	56.84	9.64	94.7
Block 2	STUDIO 14	8.7	4.40	50.62	8.49	97.5
Block 1	RECEPTION	103.2	83.46	80.88	101.19	98.1
Block 1	MANAGERS OFFICE	4.7	2.62	56.10	4.68	100.0
Block 1	SOCIALISING AREA	85.6	58.38	68.22	80.94	94.6
Total		**226.3**	**162.02**	**71.58**	**218.95**	**96.7**

ALL SOUTH WINDOWS SHADED – 1M OVERHANG
MANAGER'S OFFICE WINDOW LENGTH CHANGED FROM 1.7m to 2m
SUN SPACE GLASS TYPE WAS CHANGED TO DOUBLE
HEIGHT OF SOCIALISING AREA WINDOW CHANGED FROM 1.7m to 1.65m

ZONE	WINDOW SIZE	NUMBER	LOCATION
READING ROOM	1mx0.9	4	North
STUDIO 1	1.13mx3.1m	1	South
STUDIO 14	1.13mx3.1m	1	South
RECEPTION	(S)1.95mx1.65m ; 1.9mx1.65m(E)	(S)8 ; (E)2	South & East
MANAGER'S OFFICE	1.7m x 0.9m	1	North
SOCIALISING AREA	1.9mx1.65m	10	South

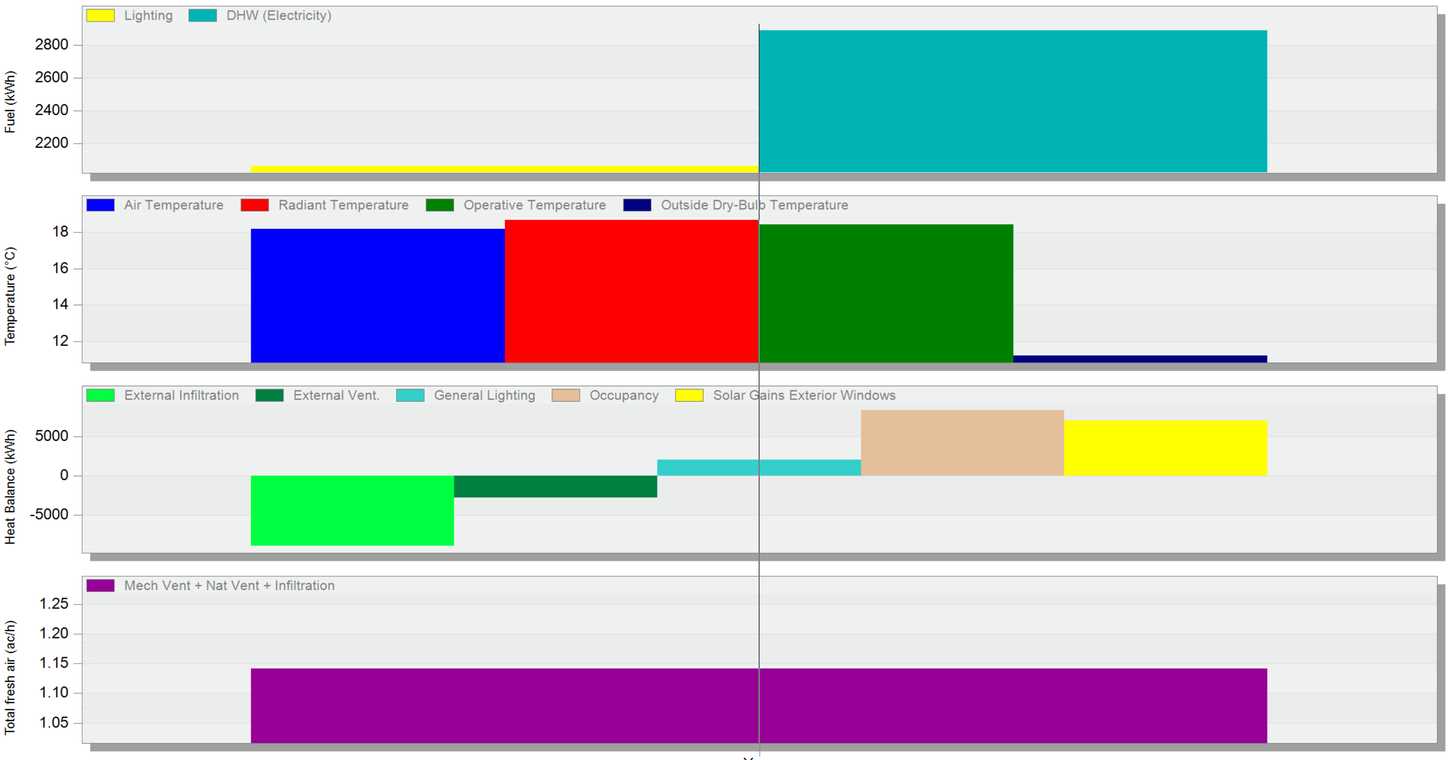
CALCULATIONS FOR ZONES ONLY

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Run period | Student



Temperatures, Heat Gains and Energy Consumption - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Run period | Student

Year

Lighting (kWh)	2065.32
DHW (Electricity) (kWh)	2889.01
Air Temperature (°C)	18.17
Radiant Temperature (°C)	18.65
Operative Temperature (°C)	18.41
Outside Dry-Bulb Temperature (°C)	11.24
External Infiltration (kWh)	-8926.67
External Vent. (kWh)	-2744.85
General Lighting (kWh)	2065.32
Occupancy (kWh)	8365.08
Solar Gains Exterior Windows (kWh)	7082.79
Mech Vent + Nat Vent + Infiltration (ac/h)	1.14

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Run period

Year

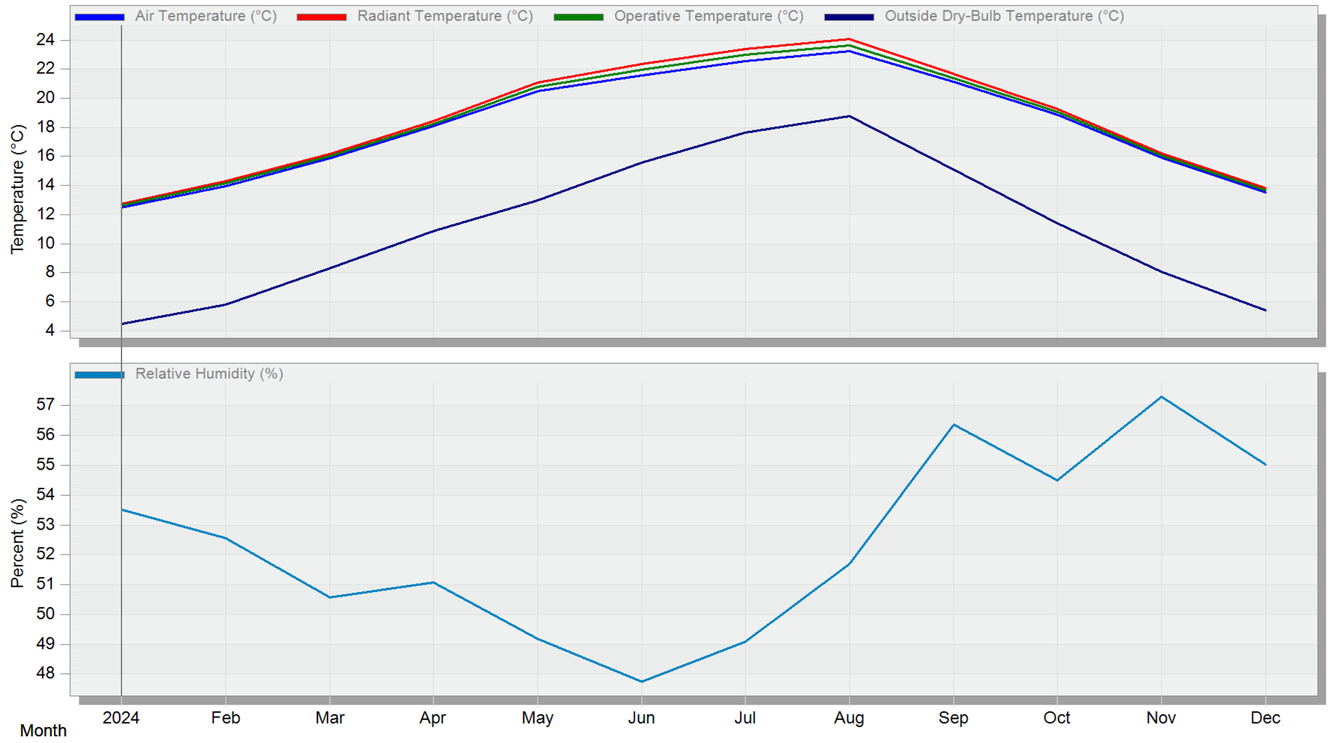
Air Temperature (°C)	18.17
Radiant Temperature (°C)	18.65
Operative Temperature (°C)	18.41
Outside Dry-Bulb Temperature (°C)	11.24
Relative Humidity (%)	52.37

Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Comfort - STUDENTS ACCM., Building 1

EnergyPlus Output | 1 Jan - 31 Dec, Monthly | Student

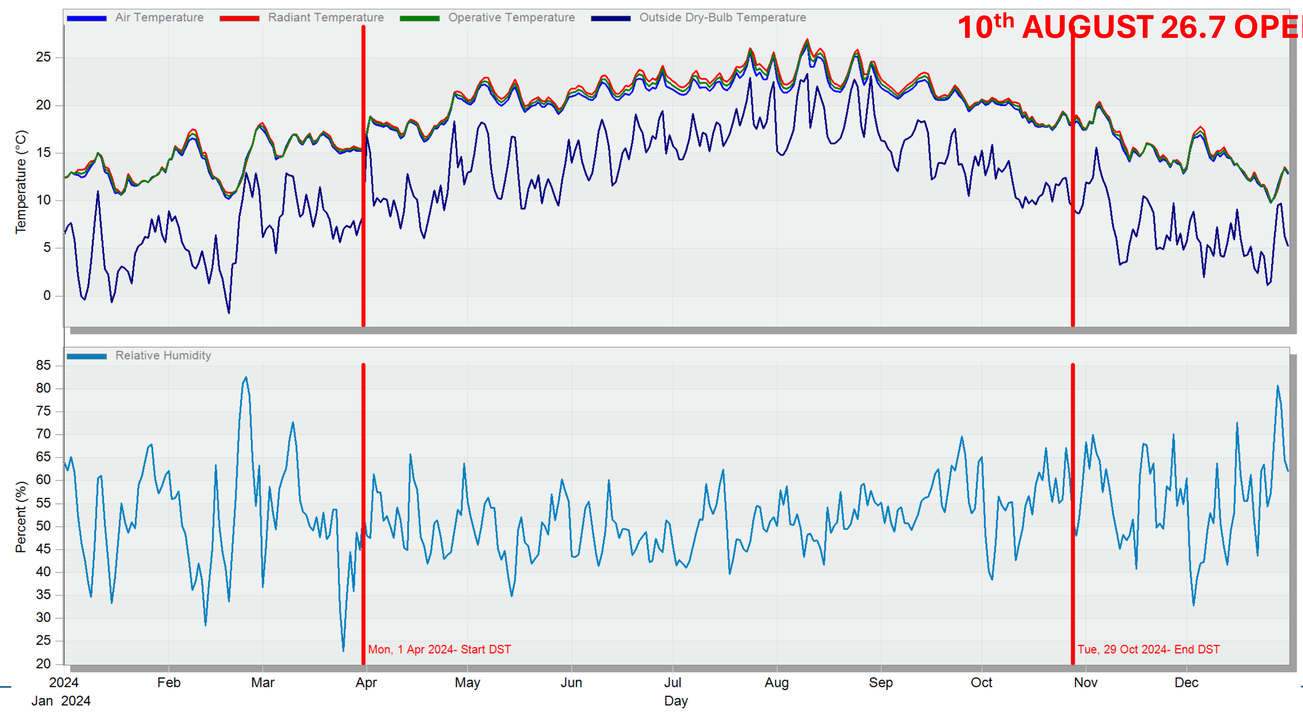


Month	2024	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Air Temperature (°C)	12.49	13.98	15.89	18.08	20.48	21.59	22.58	23.26	21.12	18.89	15.92	13.53
Radiant Temperature (°C)	12.76	14.30	16.15	18.41	21.08	22.39	23.42	24.06	21.68	19.25	16.23	13.83
Operative Temperature (°C)	12.62	14.14	16.02	18.25	20.78	21.99	23.00	23.66	21.40	19.07	16.07	13.68
Outside Dry-Bulb Temperature (°C)	4.50	5.82	8.33	10.85	12.97	15.59	17.66	18.80	15.09	11.43	8.07	5.42
Relative Humidity (%)	53.51	52.56	50.57	51.08	49.19	47.76	49.09	51.70	56.36	54.49	57.29	55.02

Comfort - STUDENTS ACCM., Building 1

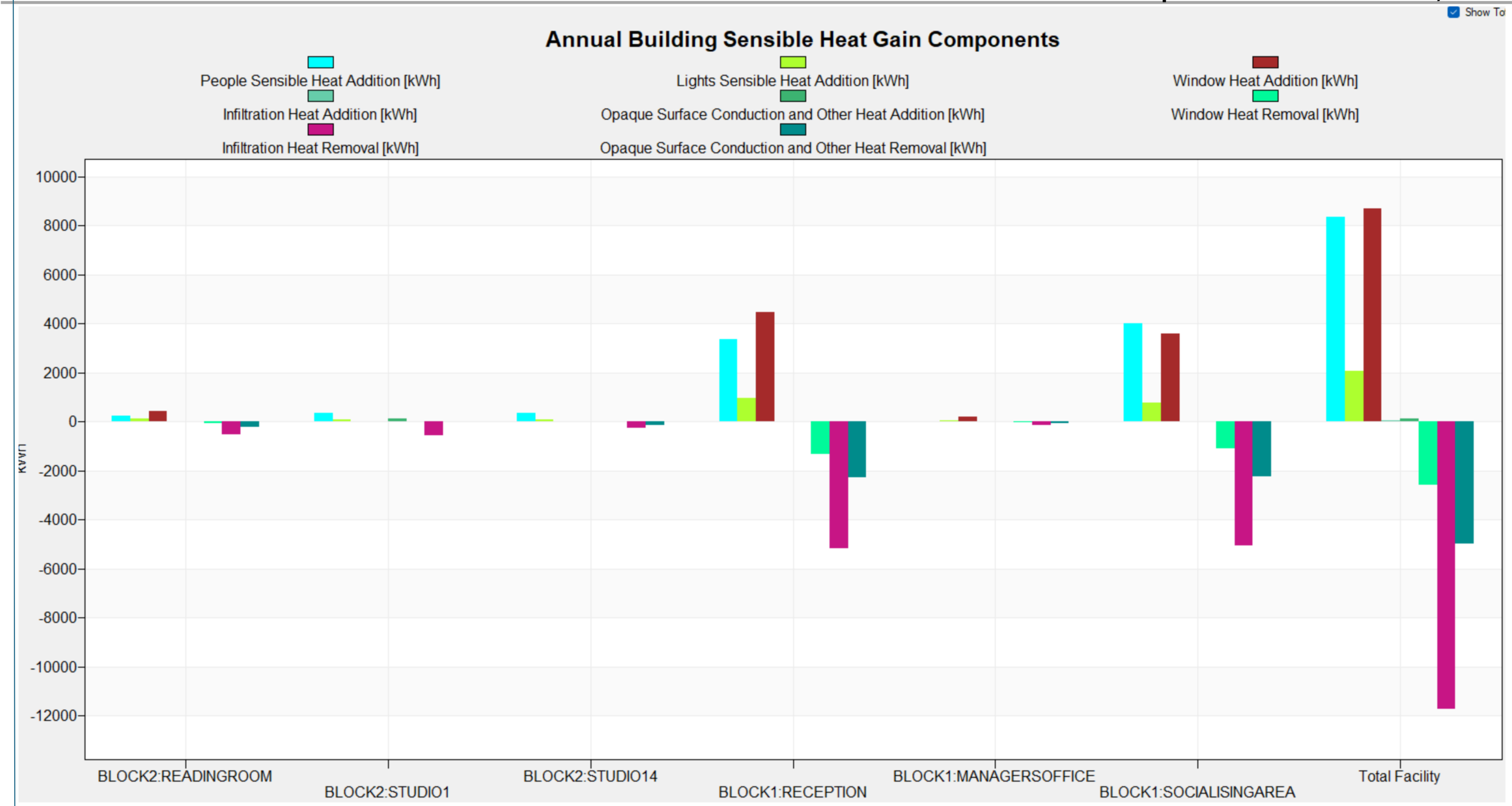
EnergyPlus Output | 1 Jan - 31 Dec, Daily | Student

10th AUGUST 26.7 OPERATIVE TEMPERATURE



Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)



Time Not Meeting the Adaptive Comfort Models during Occupied Hours

	ASHRAE55 90% Acceptability Limits [Hours]	ASHRAE55 80% Acceptability Limits [Hours]	CEN15251 Category I Acceptability Limits [Hours]	CEN15251 Category II Acceptability Limits [Hours]	CEN15251 Category III Acceptability Limits [Hours]
PEOPLE BLOCK2:READINGROOM	1285.50	976.00	4346.00	2485.50	1735.50
PEOPLE BLOCK2:STUDIO1	231.50	32.00	4338.50	1773.50	998.00
PEOPLE BLOCK2:STUDIO14	2239.50	1504.00	5054.00	3704.00	2711.50
PEOPLE BLOCK1:RECEPTION	608.50	293.50	1533.50	796.50	585.00
PEOPLE BLOCK1:MANAGERSOFFICE	925.50	660.50	2197.00	1468.50	1118.50
PEOPLE BLOCK1:SOCIALISINGAREA	247.50	67.00	1454.00	604.50	325.00

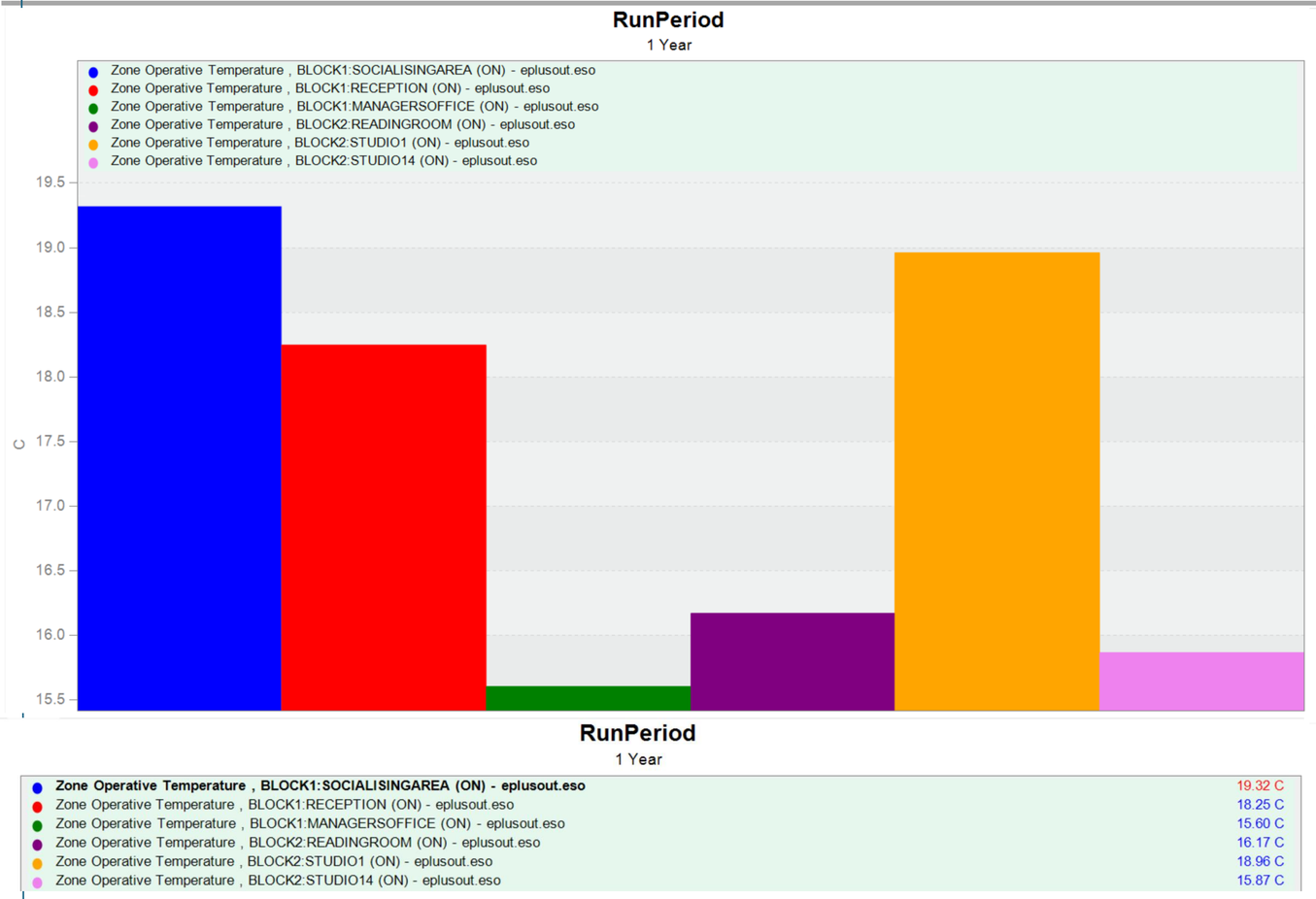
Time Not Comfortable Based on Simple ASHRAE 55-2004

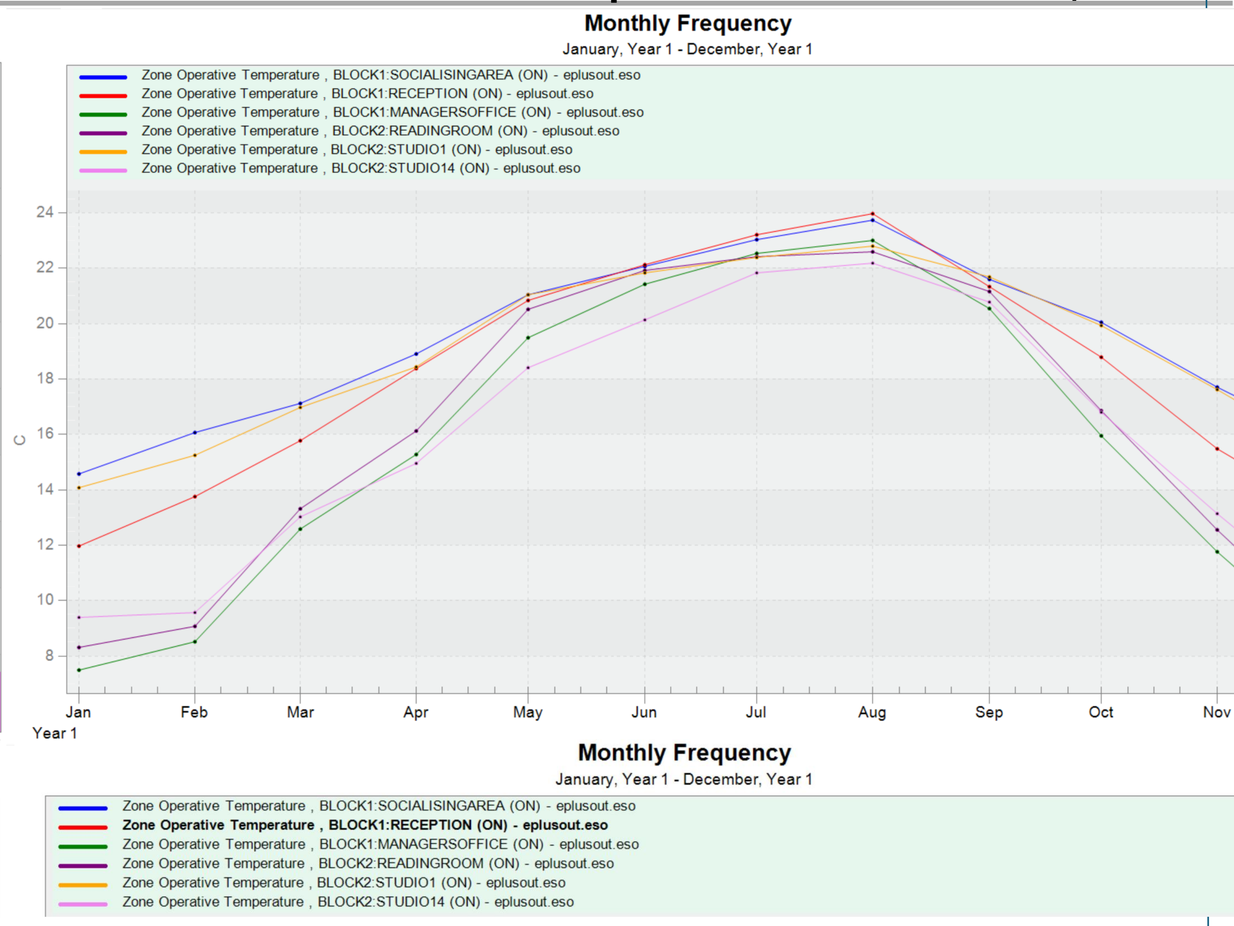
	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	5635.00	8674.00	5603.50
BLOCK2:STUDIO1	4906.00	8651.00	4854.50
BLOCK2:STUDIO14	6341.50	8746.50	6338.50
BLOCK1:RECEPTION	2966.00	4380.00	2702.00
BLOCK1:MANAGERSOFFICE	3289.00	4595.50	3208.00
BLOCK1:SOCIALISINGAREA	2544.00	4431.00	2331.50
Facility	6724.50	8759.00	6482.50

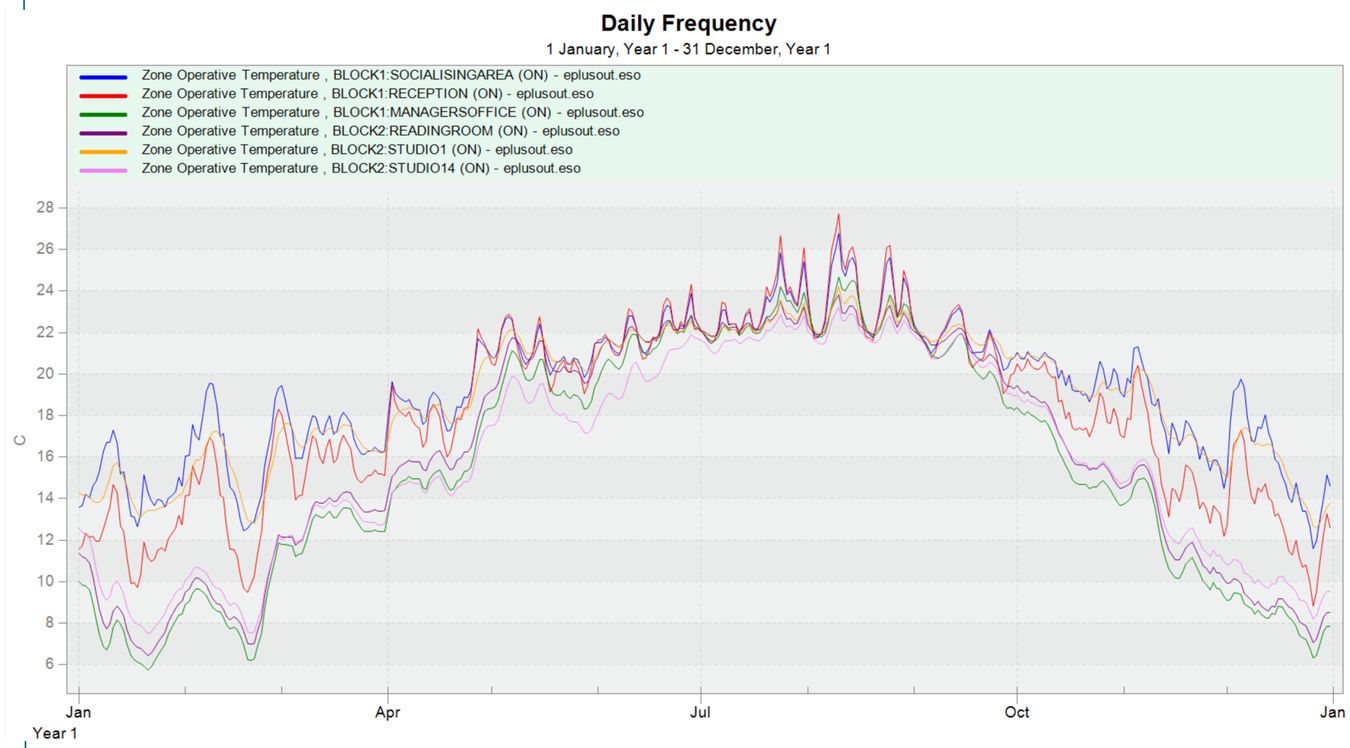
Aggregated over the RunPeriods for Weather

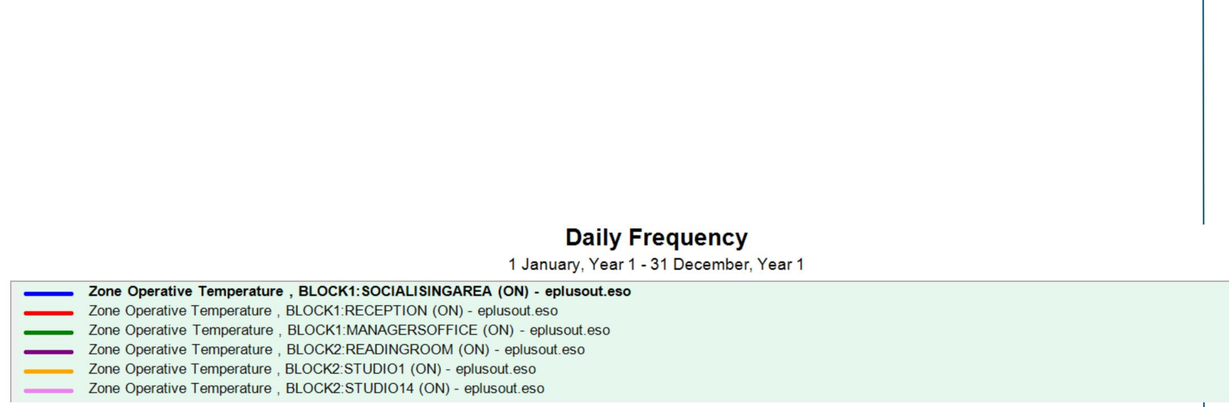
Sustainable Design of Students' Accommodation Block

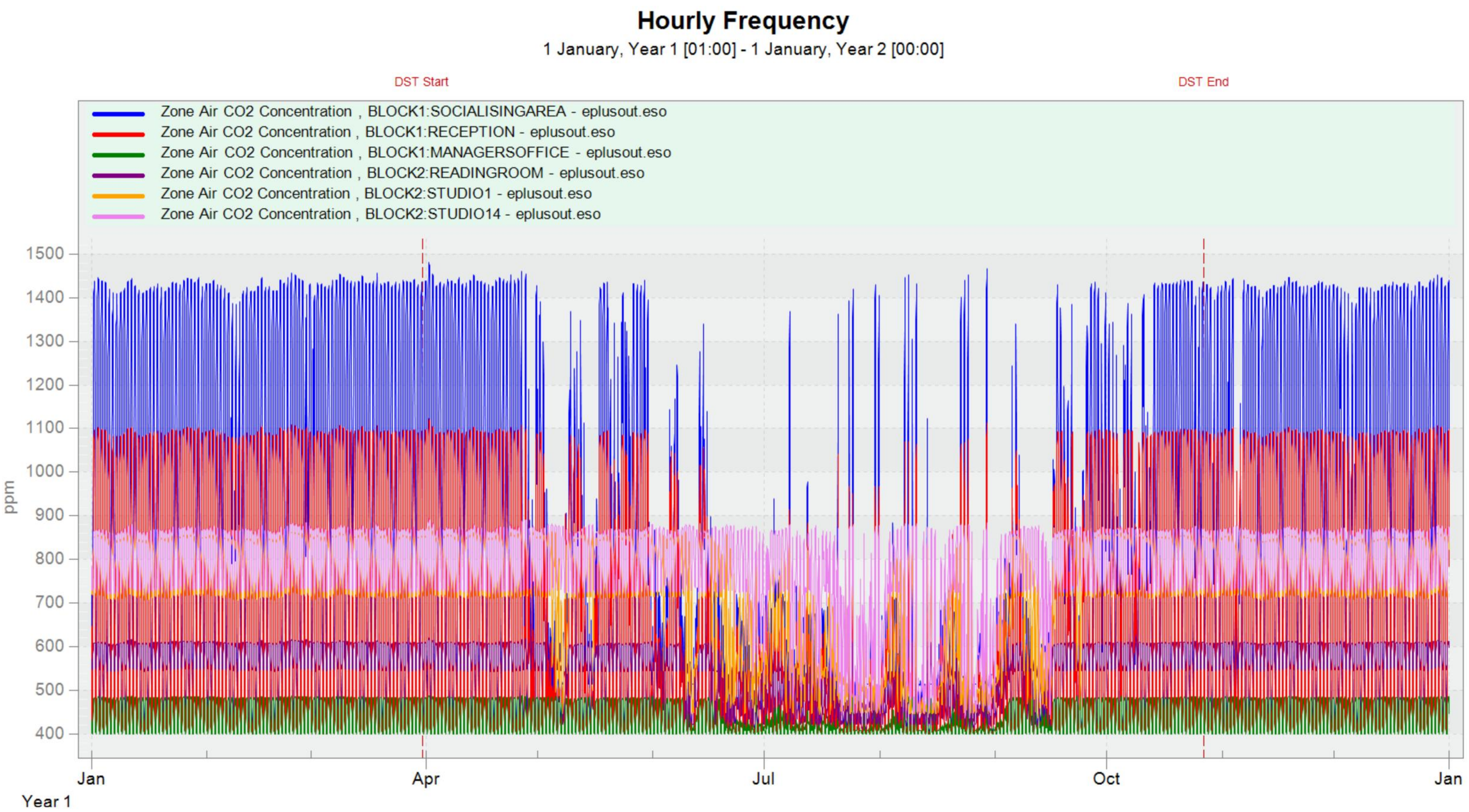
Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)










Hourly Frequency
1 January, Year 1 [01:00] - 1 January, Year 2 [00:00]
DST Start
DST End
Zone Air CO2 Concentration , BLOCK1:SOCIALISINGAREA - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:RECEPTION - eplusout.eso
Zone Air CO2 Concentration , BLOCK1:MANAGERSOFFICE - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:READINGROOM - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO1 - eplusout.eso
Zone Air CO2 Concentration , BLOCK2:STUDIO14 - eplusout.eso
ppm
1500
1400
1300
1200
1100
1000
900
800
700
600
500
400
Jan
Apr
Jul
Oct
Jan
Year 1

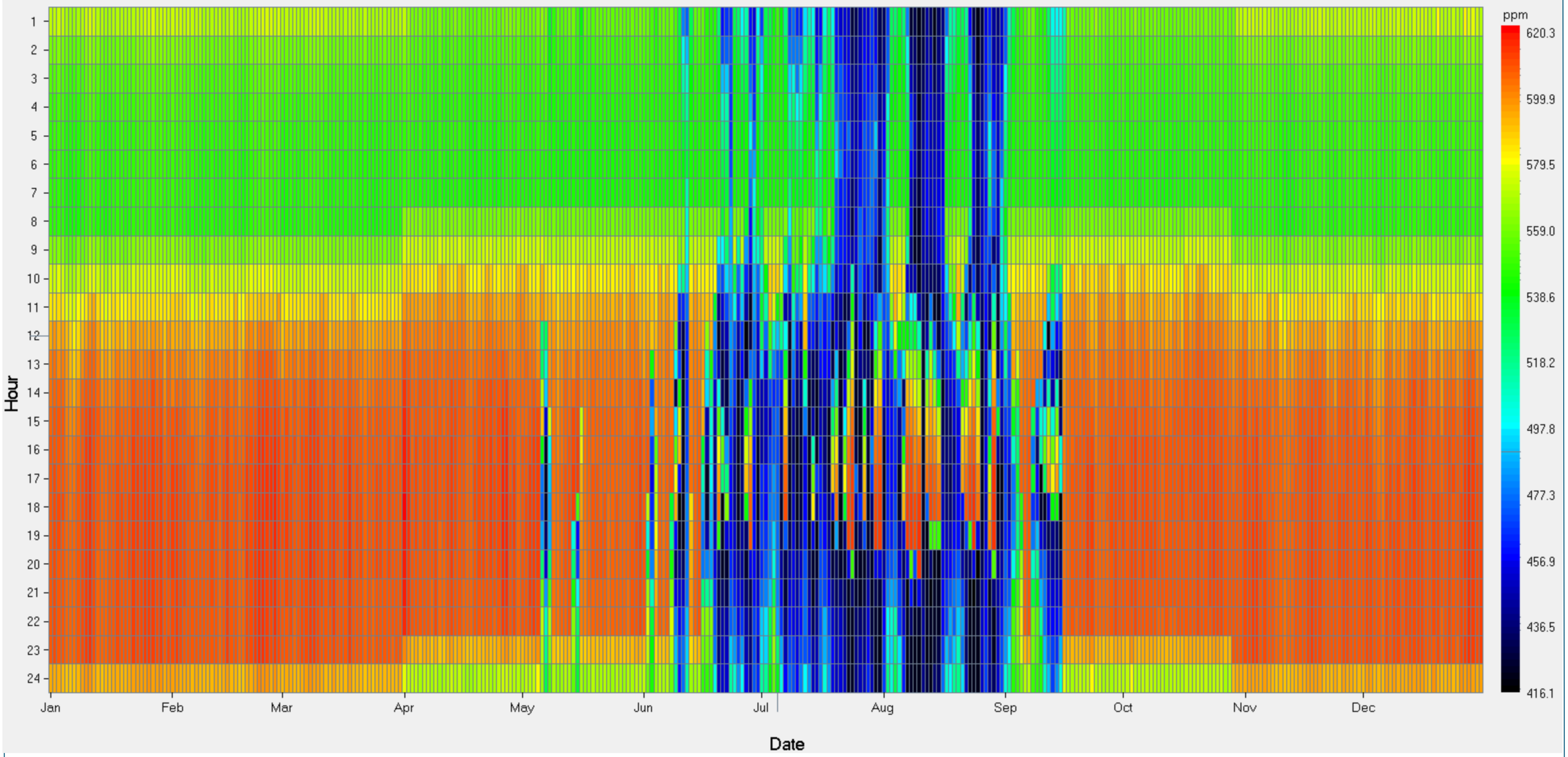
Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

Hourly | Daily | Monthly | RunPeriod | CIBSE TM59 | Hourly Heat Map | Summary | Summary Charts

Area: BLOCK2:READINGROOM Schedule:

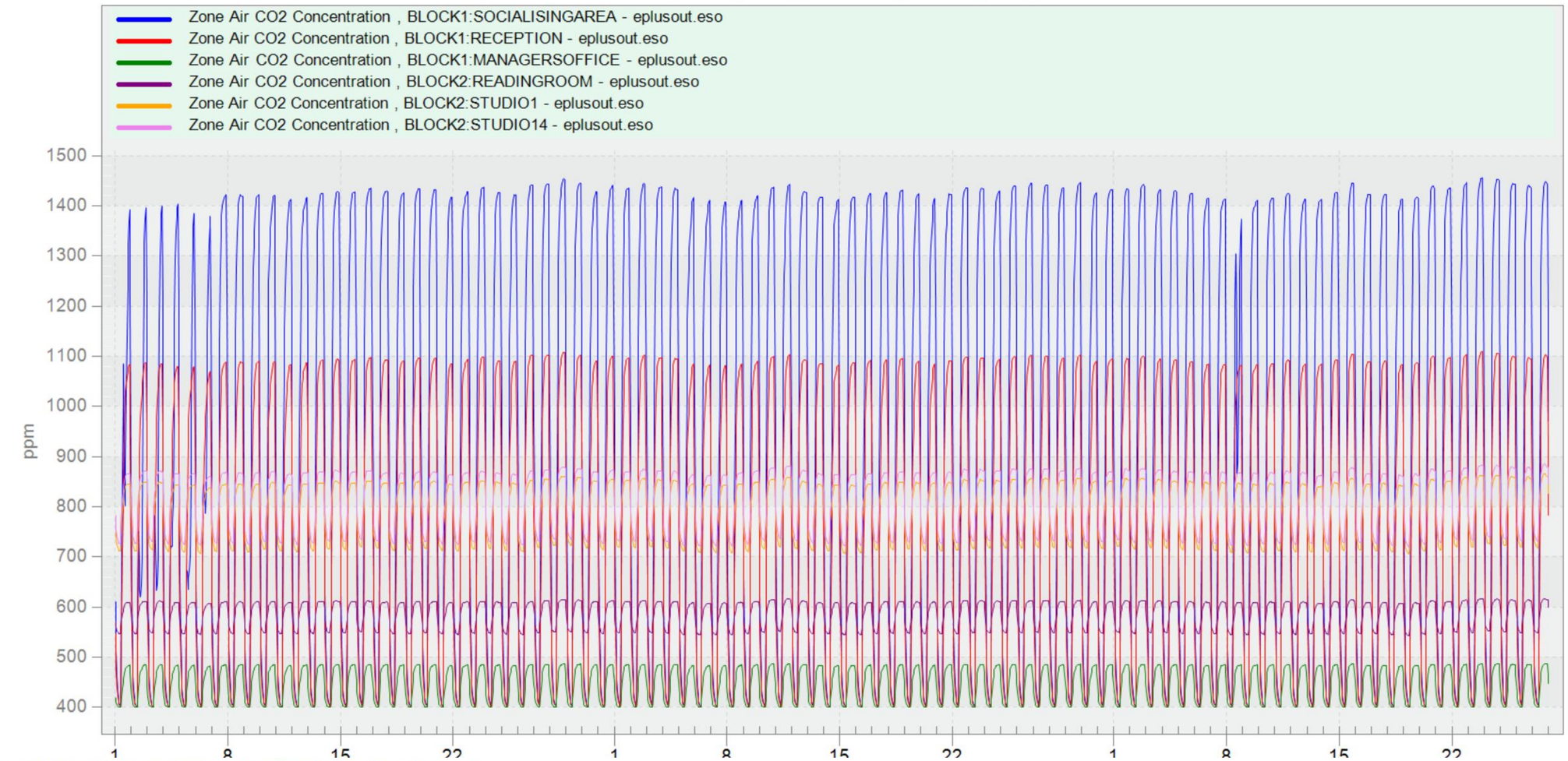
Report Type: Zone Air CO2 Concentration



Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design (ARCH 734)
Afua Ampomah-Amoako (201838915)

WINTER CO2 ONLY



Time Not Meeting the Adaptive Comfort Models during Occupied Hours

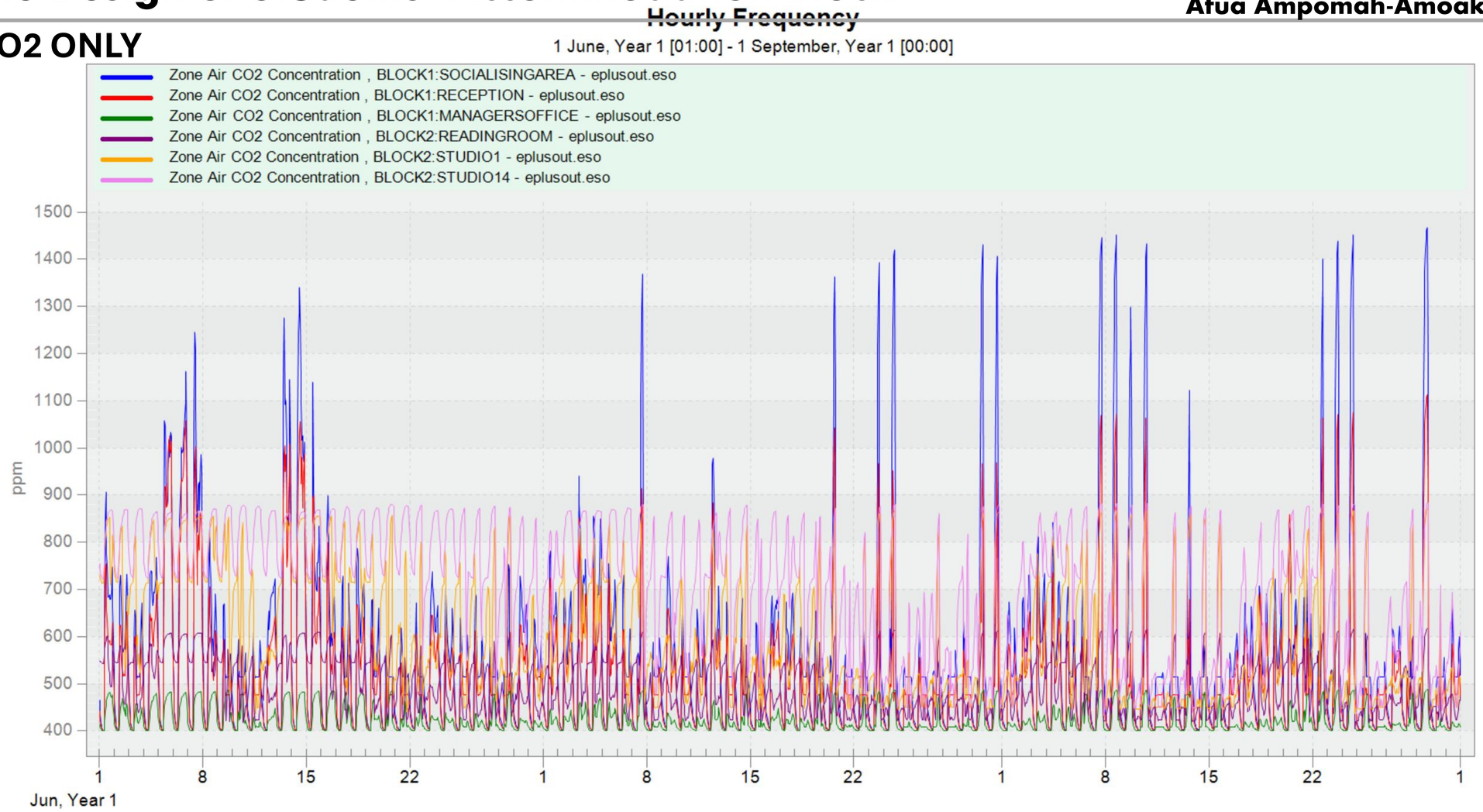
	ASHRAE55 90% Acceptability Limits [Hours]	ASHRAE55 80% Acceptability Limits [Hours]	CEN15251 Category I Acceptability Limits [Hours]	CEN15251 Category II Acceptability Limits [Hours]	CEN15251 Category III Acceptability Limits [Hours]
PEOPLE BLOCK2:READINGROOM	0.00	0.00	96.00	96.00	96.00
PEOPLE BLOCK2:STUDIO1	0.00	0.00	96.00	96.00	96.00
PEOPLE BLOCK2:STUDIO14	0.00	0.00	96.00	96.00	96.00
PEOPLE BLOCK1:RECEPTION	0.00	0.00	52.00	52.00	49.00
PEOPLE BLOCK1:MANAGERSOFFICE	0.00	0.00	52.00	52.00	52.00
PEOPLE BLOCK1:SOCIALISINGAREA	0.00	0.00	52.00	48.50	42.00

Time Not Comfortable Based on Simple ASHRAE 55-2004

	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	2136.00	2136.00	2136.00
BLOCK2:STUDIO1	2136.00	2136.00	2136.00
BLOCK2:STUDIO14	2136.00	2136.00	2136.00
BLOCK1:RECEPTION	1139.00	1157.00	1139.00
BLOCK1:MANAGERSOFFICE	1157.00	1157.00	1157.00
BLOCK1:SOCIALISINGAREA	1057.50	1157.00	1057.50
Facility	2136.00	2136.00	2136.00

Aggregated over the RunPeriods for Weather

SUMMER CO2 ONLY



Time Not Meeting the Adaptive Comfort Models during Occupied Hours

	ASHRAE55 90% Acceptability Limits [Hours]	ASHRAE55 80% Acceptability Limits [Hours]	CEN15251 Category I Acceptability Limits [Hours]	CEN15251 Category II Acceptability Limits [Hours]	CEN15251 Category III Acceptability Limits [Hours]
PEOPLE BLOCK2:READINGROOM	0.00	0.00	1427.50	240.00	1.50
PEOPLE BLOCK2:STUDIO1	0.00	0.00	1500.00	141.00	0.00
PEOPLE BLOCK2:STUDIO14	3.00	0.00	2106.50	1022.50	113.00
PEOPLE BLOCK1:RECEPTION	206.00	94.00	311.50	26.50	1.00
PEOPLE BLOCK1:MANAGERSOFFICE	0.00	0.00	574.50	31.50	0.00
PEOPLE BLOCK1:SOCIALISINGAREA	145.00	44.50	300.00	15.50	0.00

Time Not Comfortable Based on Simple ASHRAE 55-2004

	Winter Clothes [hr]	Summer Clothes [hr]	Summer or Winter Clothes [hr]
BLOCK2:READINGROOM	31.00	2122.50	0.00
BLOCK2:STUDIO1	63.00	2098.50	12.00
BLOCK2:STUDIO14	192.50	2197.00	189.50
BLOCK1:RECEPTION	328.00	882.50	87.50
BLOCK1:MANAGERSOFFICE	80.50	1048.50	0.50
BLOCK1:SOCIALISINGAREA	278.50	914.50	81.00
Facility	503.50	2208.00	277.50

END

Small scale housing

Operational energy

Implement the following indicative design measures:

Fabric U-values (W/m².K)

Walls	0.13 - 0.15
Floor	0.08 - 0.10
Roof	0.10 - 0.12
Exposed ceilings/floors	0.13 - 0.18
Windows	0.80 (triple glazing)
Doors	1.00

Efficiency measures

Air tightness	<1 (m³/h. m²@50Pa)
Thermal bridging	0.04 (y-value)
G-value of glass	0.6 - 0.5
MVHR	90% (efficiency) ≤2m (duct length from unit to external wall)

Window areas guide (% of wall area)

North	10-15%
East	10-15%
South	20-25%
West	10-15%

Balance daylight and overheating

Include external shading

Include openable windows and cross ventilation

Reduce energy consumption to:

35 kWh/m².yr

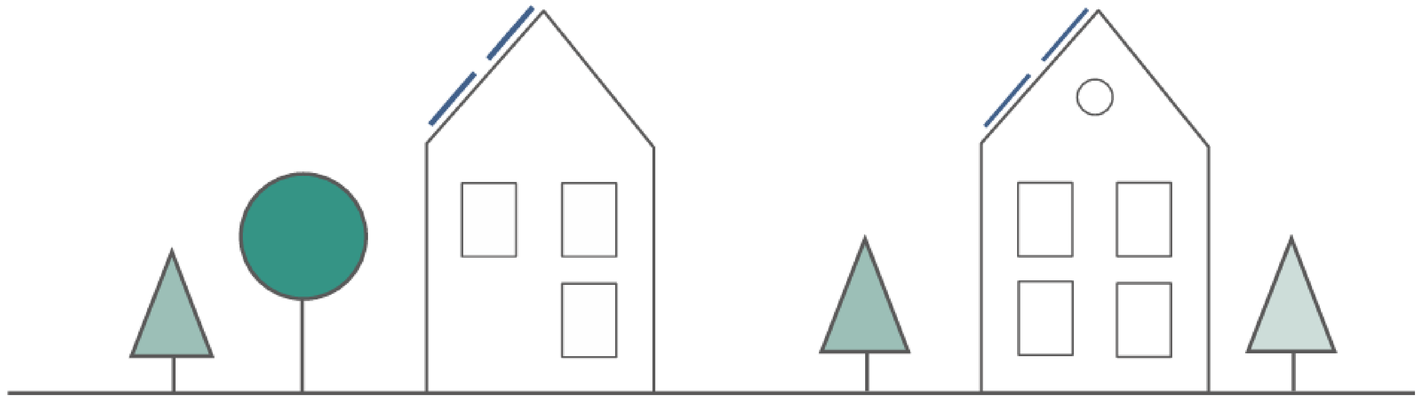
Energy Use Intensity (EUI) in GIA, excluding renewable energy contribution

Reduce space heating demand to:

15 kWh/m².yr

Maximise renewables so that 100% of annual energy requirement is generated on-site

Form factor of 1.7 - 2.5



END

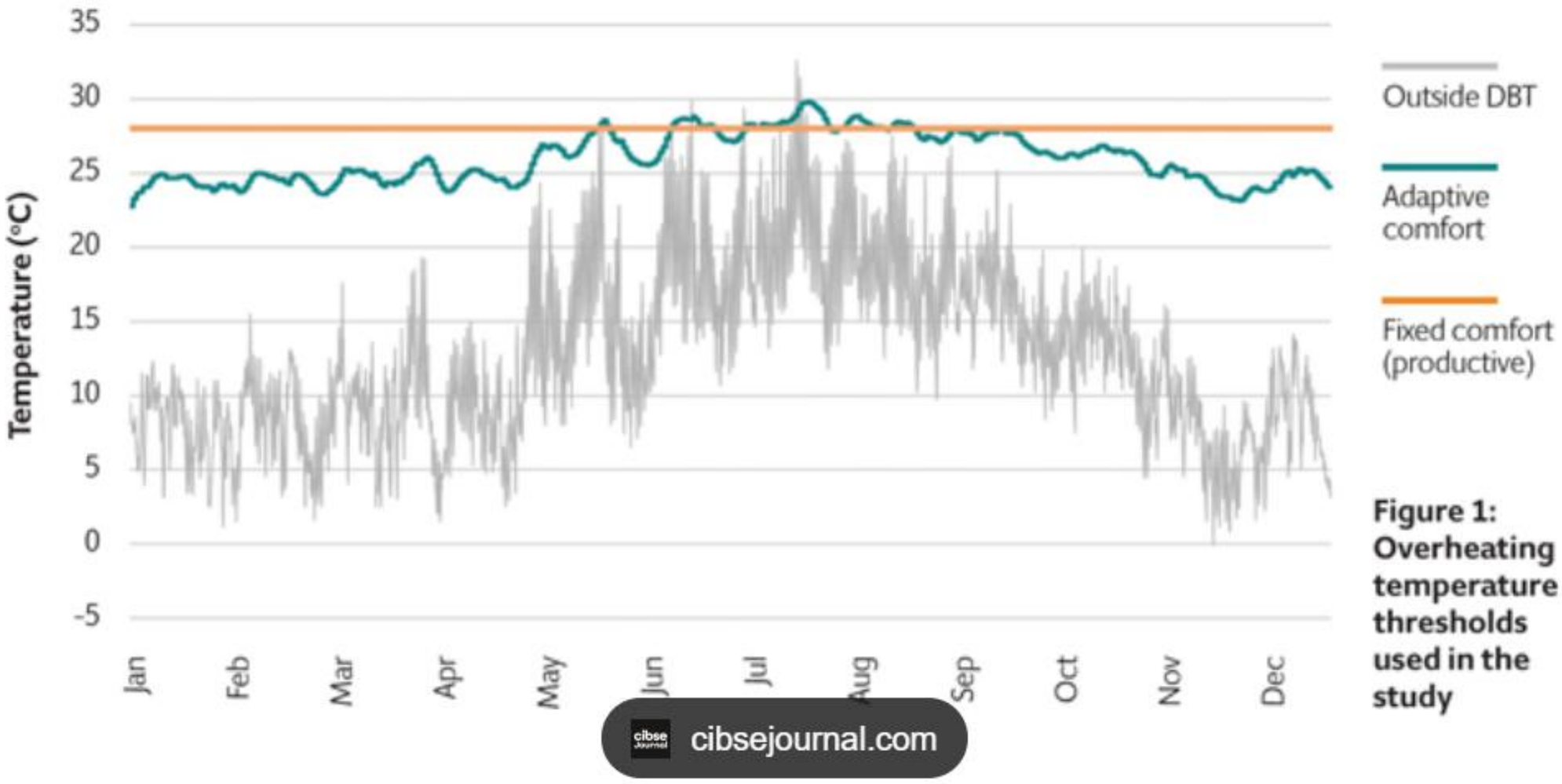


Figure 1: Overheating temperature thresholds used in the study

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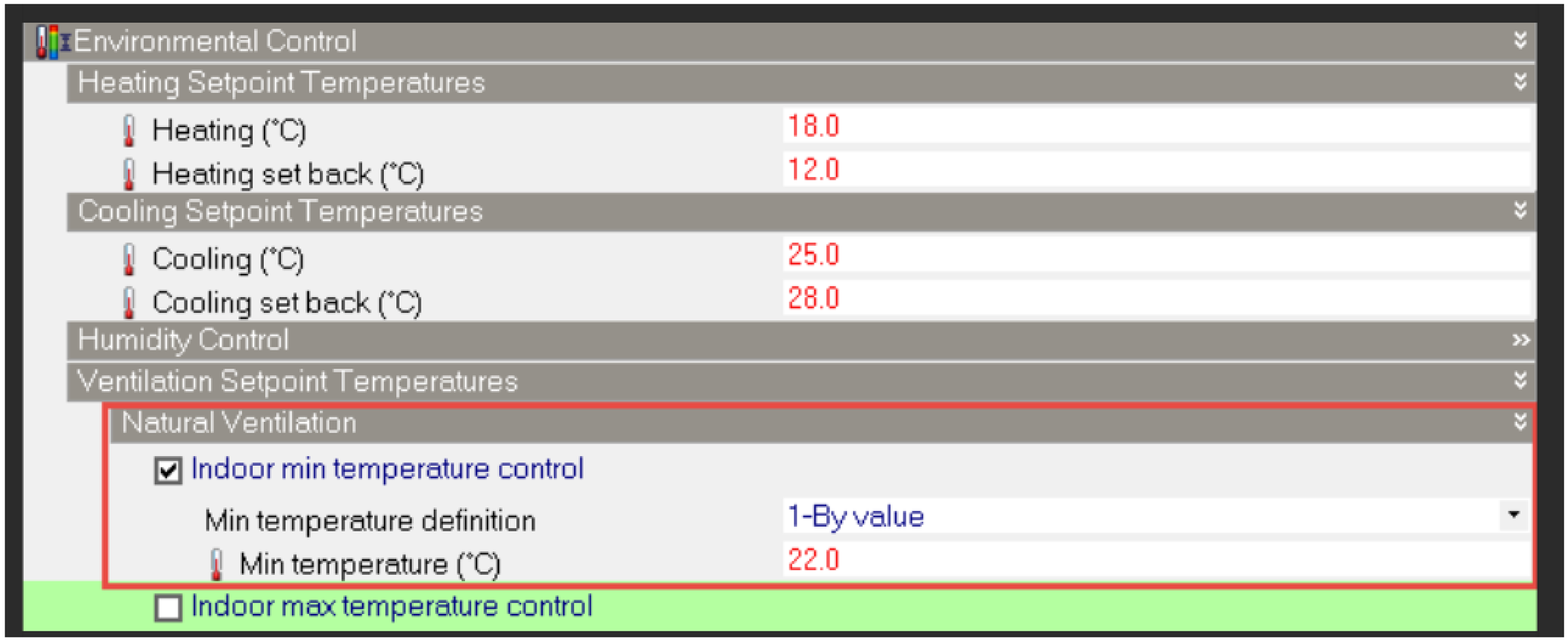
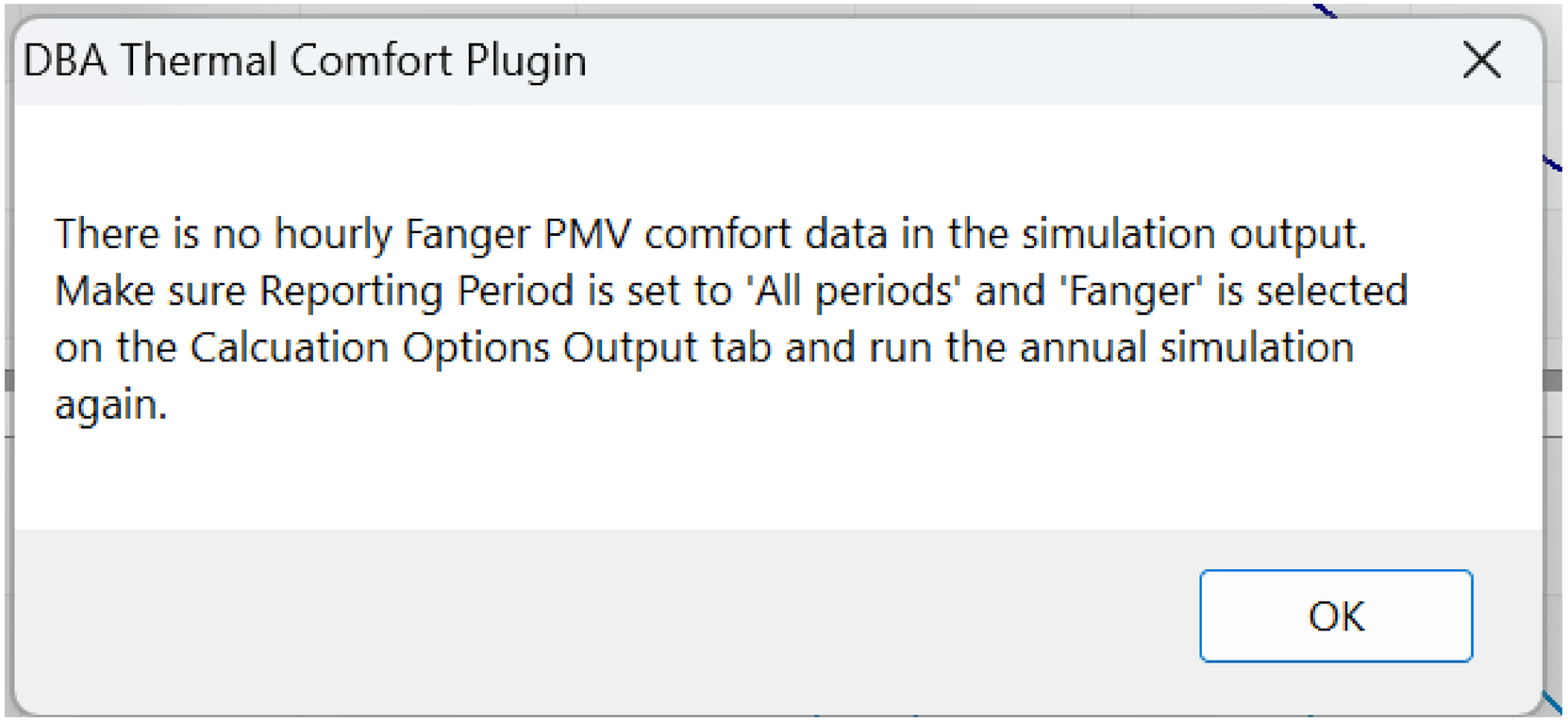
END


Environmental Control
Heating Setpoint Temperatures
Heating (°C)
18.0
Heating set back (°C)
12.0
Cooling Setpoint Temperatures
Cooling (°C)
25.0
Cooling set back (°C)
28.0
Humidity Control
Ventilation Setpoint Temperatures
Natural Ventilation
Indoor min temperature control
Min temperature definition
1-By value
Min temperature (°C)
22.0
Indoor max temperature control

END


DBA Thermal Comfort Plugin
There is no hourly Fanger PMV comfort data in the simulation output.
Make sure Reporting Period is set to 'All periods' and 'Fanger' is selected on the Calcuation Options Output tab and run the annual simulation again.
OK

END

I have comfort temperature range set between 18 degrees and 25 degrees for a naturally ventilated building. How can I know the operation hours below 18 degrees, the operation hours between 18 and 25 degrees and the hours above 25 degrees.

END